

TI Designs: TIDA-00204
EMI/EMC-Compliant Industrial Temp Dual-Port Gigabit Ethernet Reference Design



Description

TI Designs provide the foundation that you need including methodology, testing and design files to quickly evaluate and customize the system. TI Designs help you accelerate your time to market.

Resources

TIDA-00204	Design Folder
DP83867IR	Product Folder
LM46002	Product Folder
AM3359	Product Folder
TPS65910A3	Product Folder
TPS51200	Product Folder
LMZ10501	Product Folder
TPS720	Product Folder
TPS737	Product Folder
TPD4E05U06	Product Folder
TPS717	Product Folder
TPD4S012	Product Folder
CDCE913	Product Folder



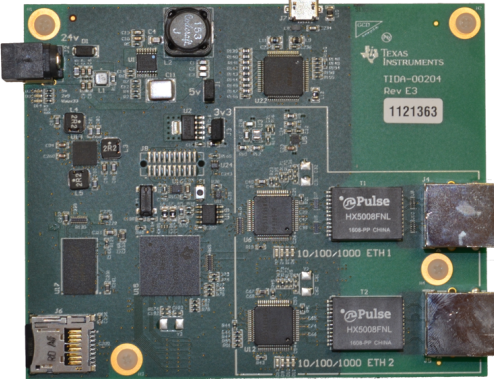
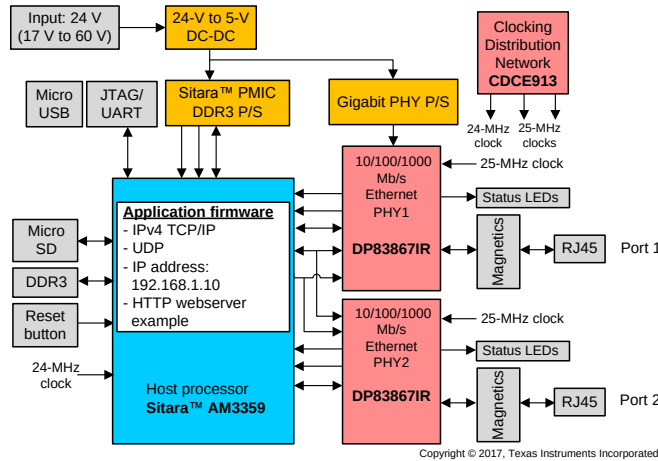
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Features

- EMI- and EMC-Compliant Design With Wide Input Voltage Range (17 to 60 V) Using Two DP83867IR Gigabit Ethernet PHYs and AM3359 Sitara™ Processor to Work in Harsh Industrial Environments
- Exceeds CISPR 11 / EN55011 Class A Radiated Emission Requirement by > 4.3 dB
- Exceeds IEC61800-3 EMC Immunity Requirements:
 - ±6-kV ESD CD per IEC 61000-4-2
 - ±4-kV EFT per IEC 61000-4-4
 - ±2-kV Surge per IEC 61000-4-5
- Sitara AM3359 Firmware, Including UDP and TCP/IP Stack and HTTP Web Server Examples, Boots From Onboard SD-Card Allowing Easy Standalone Operation
- Access to DP83867IR Registers Through USB Virtual COM Port Allows for Custom Specific PHY Configurations Such as RGMII Delay Mode
- Reduces Jitter and Phase Shift From Clocks by Using Clock Synthesizer to Generate System Clocks

Applications

- Industrial Drives
- Factory Automation and Control
- Industrial Networks
- Test and Measurement



TI 设计：TIDA-00204 EMI/EMC 合规工业温度双口千兆以太网参考设计



描述

TI 设计提供您所需的基础，包括方法、测试和设计文件，以便快速评估和定制系统。TI 设计可帮助您加速上市时间。

资源

TIDA-00204	设计文件夹
DP83867IR	产品文件夹
LM46002	产品文件夹
AM3359	产品文件夹
TPS65910A3	产品文件夹
TPS51200	产品文件夹
LMZ10501	产品文件夹
TPS720	产品文件夹
TPS737	产品文件夹
TPD4E05U06	产品文件夹
TPS717	产品文件夹
TPD4S012	产品文件夹
CDCE913	产品文件夹



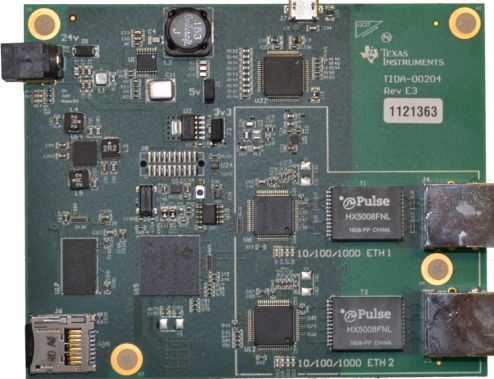
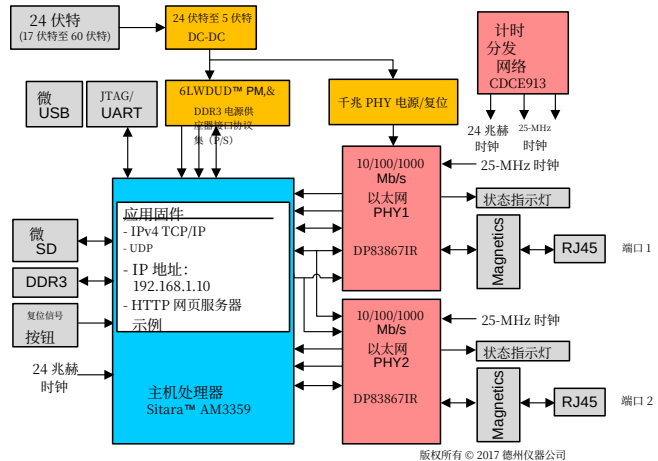
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功能

- 采用双 DP83867IR 千兆以太网 PHY 和 AM3359 Sitara™ 处理器，配合宽输入电压范围（17 至 60 V），实现符合 EMI 和 EMC 标准的设计，适用于恶劣工业环境
- 超过 CISPR 11 / EN55011 类 A 辐射发射要求 > 4.3 dB
- 超过 IEC61800-3 标准 EMC 抗扰度要求：
 - ±6-kV 静电放电 CD 符合 IEC 61000-4-2
 - ±4-kV EFT 根据 IEC 61000-4-4
 - ±2-kV 浪涌（符合 IEC 61000-4-5）
- Sitara AM3359 固件，包括 UDP 和 TCP/IP 堆栈以及 HTTP Web 服务器示例，可从板载 SD 卡启动，支持便捷的独立运行
- 通过 USB 虚拟 COM 端口访问 DP83867IR 寄存器，可实现自定义特定 PHY 配置，例如 RGMII 延迟模式
- 通过使用时钟合成器生成系统时钟，减少时钟的抖动和相位偏移

应用

- 工业驱动
- 工厂自动化与控制
- 工业网络
- 测试与测量





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1 System Description

This TI design supports dual-port Gigabit Ethernet communication through twisted pair copper cable as defined in IEEE 802.3ab. It is designed to be evaluated for harsh industrial environment with regards to standard compliance to CISPR 11 / EN55011 Class A radiated immunity requirements and EMC immunity requirements for ESD according to 61000-4-2, fast transient burst (EFT) according to IEC61000-4-4, and surge according to IEC61000-4-5.

The design implements dual-port Gigabit Ethernet using two DP83867IR Gigabit Ethernet PHYs, which are connected through the Reduced Gigabit Media Independent Interface (RGMII) to AM3359 Sitara processor with integrated Ethernet MAC and Switch. It offers a wide input voltage range from 17 to 60 V with nominal 24-V input voltage and meets industrial requirements for EMI and EMC.

For easy standalone operation, the host processor is configured to boot the pre-installed application firmware from an onboard SD-Card. The application firmware implements a driver for the DP83867IR, UDP and TCP/IP stack, and HTTP web server examples, based on TI's SYS/BIOS Industrial SDK and TI's Networking Development Kit NDK. A USB virtual COM port offers optional user access to read or write to DP83867IR registers for custom configurations like RGMII Delay Mode, if required. A JTAG interface on the AM3359 provides the option for custom software development, test, and debug.

Therefore, this design allows for performance evaluation of two DP83867IR Gigabit Ethernet PHYs and AM3359 Sitara processor with integrated Ethernet MAC and Switch.

1.1 Key System Specifications

This design allows for performance evaluation of two DP83867IR Gigabit Ethernet PHYs and AM3359 Sitara™ processors with an integrated Ethernet MAC and Switch. It meets industrial requirements for EMI and EMC, supports industrial temperature grade, and offers a wide input voltage range with a default of 24 V.

To allow for easy standalone operation, the AM3359 Sitara boots the application firmware from SD-Card. The application firmware implements the driver for the DP83867IR, UDP and TCP/IP stack, and HTTP web server examples and is based on TI's SYS/BIOS Industrial SDK. An additional option is provided to access the DP83867IR Gigabit Ethernet PHY registers through USB virtual COM port. A JTAG interface option is also provided to allow for custom software development with that board.

Table 1. TIDA-00204 Hardware Specification

FUNCTION	INFO	SPECIFICATION	COMMENT
Gigabit Ethernet IEEE 802.3ab	Number of ports	2	—
	MDI	1000BASE-T (copper)	—
	MAC interface	RGMII	—
	EMAC and switch	Y	Integrated with Sitara AM3359
	Status LED	Y	—
	IEEE 1588v2	Y	Hardware enabled but not tested
	Separate transformer and RJ45 jack	Y	Option to place protection diodes on either side of the magnetics, used for test purpose
	SMI	Y (2.5 MHz)	—
	Configurable PHY address (SMI)	Y	Through strap resistors
DP83867IR	Low power	565 mW	—
	Integrated termination resistors	Y	—
	RGMII Delay Mode on RX/TX	Programmable	—
	Clock	25 MHz (< 50 ppm)	—



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1 系统描述

该 TI 设计支持通过双端口千兆以太网通信，符合 IEEE 802.3ab 定义的双绞线铜缆标准。该设计旨在评估其在恶劣工业环境下的合规性，包括 CISPR 11 / EN55011 类 A 辐射抗扰度要求，以及根据 IEC 61000-4-2 的静电放电（ESD）、根据 IEC 61000-4-4 的瞬态脉冲群（EFT）和根据 IEC 61000-4-5 的浪涌的电磁兼容抗扰度要求。

该设计采用两个 DP83867IR 千兆以太网 PHY 实现双端口千兆以太网，通过简化千兆媒体独立接口（RGMII）连接到集成以太网 MAC 和交换机的 AM3359 Sitara 处理器。其输入电压范围宽，从 17 到 60 V，标称 24-V 输入电压，并满足工业环境下的电磁干扰（EMI）和电磁兼容（EMC）要求。

为便于独立运行，主机处理器配置为从板载 SD 卡启动预装的应用固件。该应用固件基于 TI 的 SYS/BIOS 工业 SDK 和 TI 的网络开发套件 NDK，实现了 DP83867IR、UDP 和 TCP/IP 栈以及 HTTP 网络服务器示例的驱动程序。USB 虚拟 COM 端口提供可选的用户访问权限，用于读取或写入 DP83867IR 寄存器，以实现如 RGMII 延迟模式等自定义配置（如需要）。AM3359 上的 JTAG 接口支持自定义软件开发、测试和调试。

因此，该设计允许对两个 DP83867IR 千兆以太网 PHY 以及集成以太网 MAC 和开关电容滤波器的 AM3359 Sitara 处理器进行性能评估。

1.1 关键系统规范

该设计支持对两个集成以太网 MAC 和交换机的 DP83867IR 千兆以太网 PHY 以及 AM3359 Sitara™ 处理器的性能进行评估。该设计符合工业级 EMI 和 EMC 要求，支持工业温度等级，并提供宽输入电压范围，默认值为 24 V。

为便于独立运行，AM3359 Sitara 从 SD 卡启动应用程序固件。该应用程序固件实现了 DP83867IR、UDP 和 TCP/IP 栈以及 HTTP Web 服务器示例的驱动程序，并基于 TI 的 SYS/BIOS 工业 SDK。还提供一种选项，可通过 USB 虚拟 COM 端口访问 DP83867IR 千兆以太网 PHY 寄存器。此外，还提供 JTAG 接口选项，以便在该板上进行自定义软件开发。

表 1. TIDA-00204 硬件规格

功能	INFO	规格	注释
千兆以太网 IEEE 802.3ab	端口数量	2	—
	MDI	1000BASE-T（铜缆）	—
	MAC 接口	RGMII	—
	EMAC 和开关电容滤波器	Y	集成 Sitara AM3359
	状态 LED	Y	—
	IEEE 1588v2	Y	硬件已启用但未测试
	分离变压器和 RJ45 连接器	Y	将保护二极管放置在磁性元件两侧以用于测试目的的选项
	SMI	Y (2.5 MHz)	—
	可配置的 PHY 地址（SMI）	Y	通过拉紧器 (GPIO) 电阻
DP83867IR	低功耗	565 毫瓦	—
	集成终止电阻器	Y	—
	RX/TX 的 RGMII 延迟模式	可编程	—
	时钟	25 MHz (< 50 ppm)	—

Table 1. TIDA-00204 Hardware Specification (continued)

FUNCTION	INFO	SPECIFICATION	COMMENT
Power	Input voltage	24 V (17 to 60 V)	—
	Output voltage	5 V	Intermediate voltage
	Output current	850 mA (nominal), 1.2 A (maximum)	—
	Indicator LED	Y	For 5 V, 3.3 V and 2.5 V
	Point of load for all ICs	Y	PMIC for AM3359, 3.3 V (I/O), 2.5 V and 1.1 V for PHYs
Standalone operation	Boot from SD-Card interface	Y	—
USB virtual COM port	Access to PHY registers	Y	Read and write operation to both registers of PHY supported
JTAG	JTAG header	Y	—
Temperature range	Industrial	–40°C to 85°C	Selected devices support industrial temperature range
EMI	CISPR 11 / EN55011	Class A radiated emissions	—
EMC	IEC61800-3	IEC61000-4-2 ±4-kV ESD CD, Criterion B	Shielded Ethernet cable
		IEC61000-4-4 ±2-kV EFT, Criterion B	Shielded Ethernet cable
		IEC61000-4-5 ±1-kV Surge, Criterion B	Shielded Ethernet cable, min 20 m

Table 2. TIDA-00204 Firmware Specification (AM3359)

FUNCTION	INFO	SPECIFICATION	COMMENT
Boot loader	Boot application firmware from SD-Card	Y	—
DDR3 memory	Driver for DDR3	Y	—
USB virtual COM port	Driver for UART	Y	—
PHY	Driver for DP83867IR	Y	—
RTOS	TI SYS/BIOS	Y	—
UDP and TCP/IP	IPv4 protocol support	Y	Based on TI's AM335x industrial SDK and NDK
	Fixed IP	192.168.1.10	—
HTTP	Web server example	Y	Based on TI's AM335x industrial SDK and NDK

表 1。TIDA-00204 硬件规格（续）

功能	INFO	规格	注释
电源	输入电压	24 V（17 至 60 V）	—
	输出电压	5 V	中间电压
	输出当前	850 毫安（标称值），1.2 安培（最大值）	—
	指示 LED	Y	对于 5 V、3.3 V 和 2.5 V
	所有 IC 的负载点	Y	AM3359 的 PMIC，3.3 V（I/O）、2.5 V 和 1.1 V 用于 PHY
独立运行	从 SD 卡接口启动	Y	—
USB 虚拟串口	访问 PHY 寄存器	Y	支持对 PHY 两个寄存器进行读写操作
JTAG	JTAG 接口	Y	—
温度范围	工业	–40°C 至 85°C	选定的设备支持工业温度范围
EMI	CISPR 11 / EN55011	A 类辐射发射	—
EMC	IEC61800-3 标准	IEC61000-4-2 ±4-kV 静电放电 CD，准则 B	屏蔽以太网电缆
		IEC61000-4-4 ±2-kV EFT，B 准则	屏蔽以太网电缆
		IEC61000-4-5 ±1-kV 浪涌，B 准则	屏蔽以太网电缆，最小 20 米

表 2. TIDA-00204 固件规范 (AM3359)

功能	INFO	规格	注释
引导加载程序	从 SD 卡启动应用程序固件	Y	—
DDR3 内存	DDR3 驱动程序	Y	—
USB 虚拟串口	UART 驱动程序	Y	—
PHY	DP83867IR 驱动程序	Y	—
RTOS	德州仪器 SYS/BIOS	Y	—
UDP 和 TCP/IP	IPv4 协议支持	Y	基于 TI 的 AM335x 工业 SDK 和 NDK
	固定 IP	192.168.1.10	—
HTTP	Web 服务器示例	Y	基于 TI 的 AM335x 工业 SDK 和 NDK

2 System Overview

2.1 Block Diagram

Figure 1 shows the system block diagram. The major building blocks are the DP83867IR Gigabit Ethernet PHY, the AM3359 Sitara Host Processor, and the power supplies.

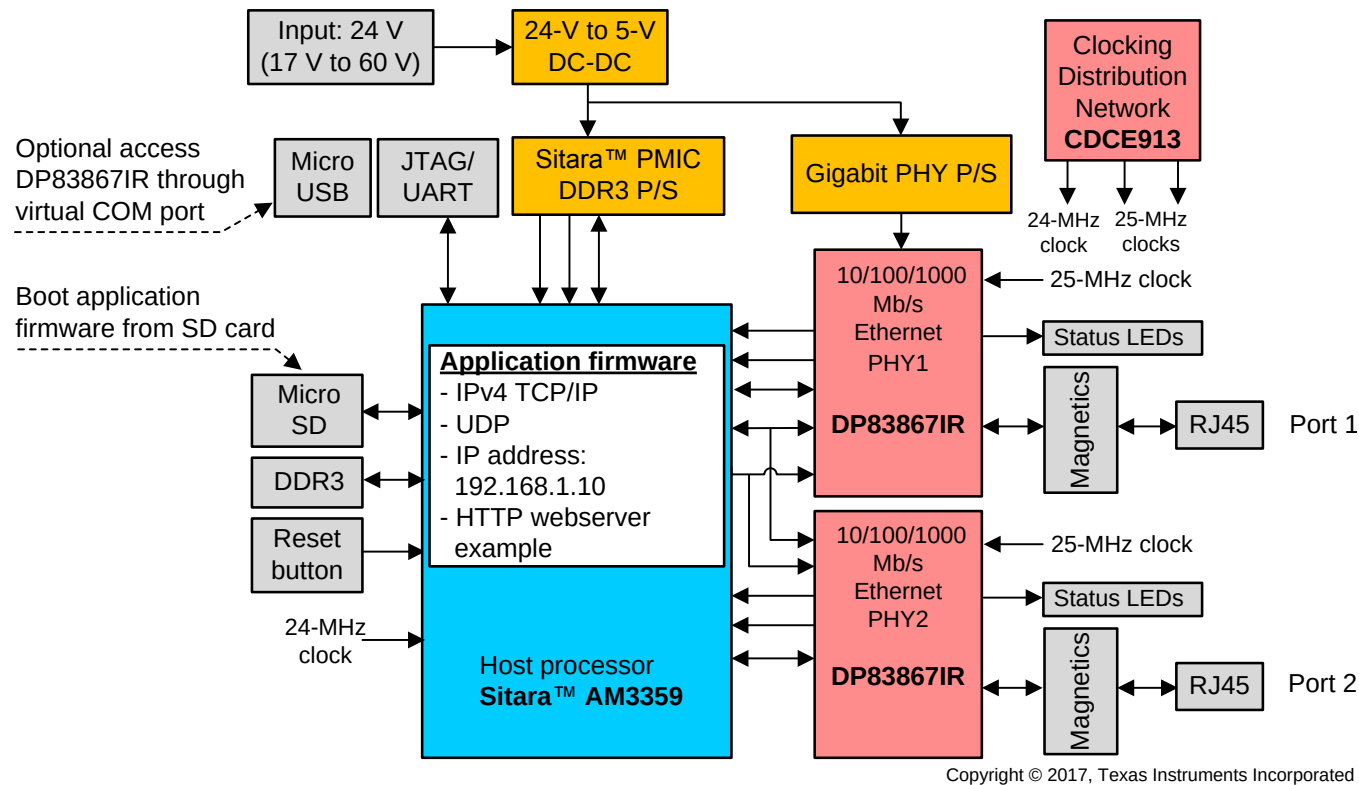


Figure 1. System Block Diagram of TIDA-00204

2.2 Design Considerations

2.2.1 Gigabit Ethernet Overview

Ethernet has heavily expanded usage over the years. Ethernet became an attractive option for industrial networking applications. The opportunity to use open protocols (such as TCP/IP over Ethernet networks) help replace proprietary communications in industrial control and factory automation applications. When using Ethernet, there are several speeds available today: 10 Mb/s, 100 Mb/s (Fast Ethernet), 1000 Mb/s (Gigabit Ethernet), and 10 Gb/s (10-Gigabit Ethernet).

Gigabit Ethernet uses the extended Ethernet MAC layer interface, connected through a Gigabit Media Independent Interface (GMII) layer to physical layer entities (PHY sublayers) such as 1000BASE-LX, 1000BASE-SX, 1000BASE-CX, and 1000BASE-T. The topology for a 1000-Mb/s full-duplex operation is comparable to the 100BASE-T full-duplex mode, and the minimum packet transmission time has been reduced by a factor of ten. The resulting achievable topologies for the half-duplex 1000-Mb/s CSMA/CD MAC are similar to those found in half duplex 100BASE-T.

2 系统概述

2.1 框图

图 1 显示了系统框图。主要组成部分包括 DP83867IR 千兆以太网 PHY、AM3359 Sitara 主机处理器以及电源。

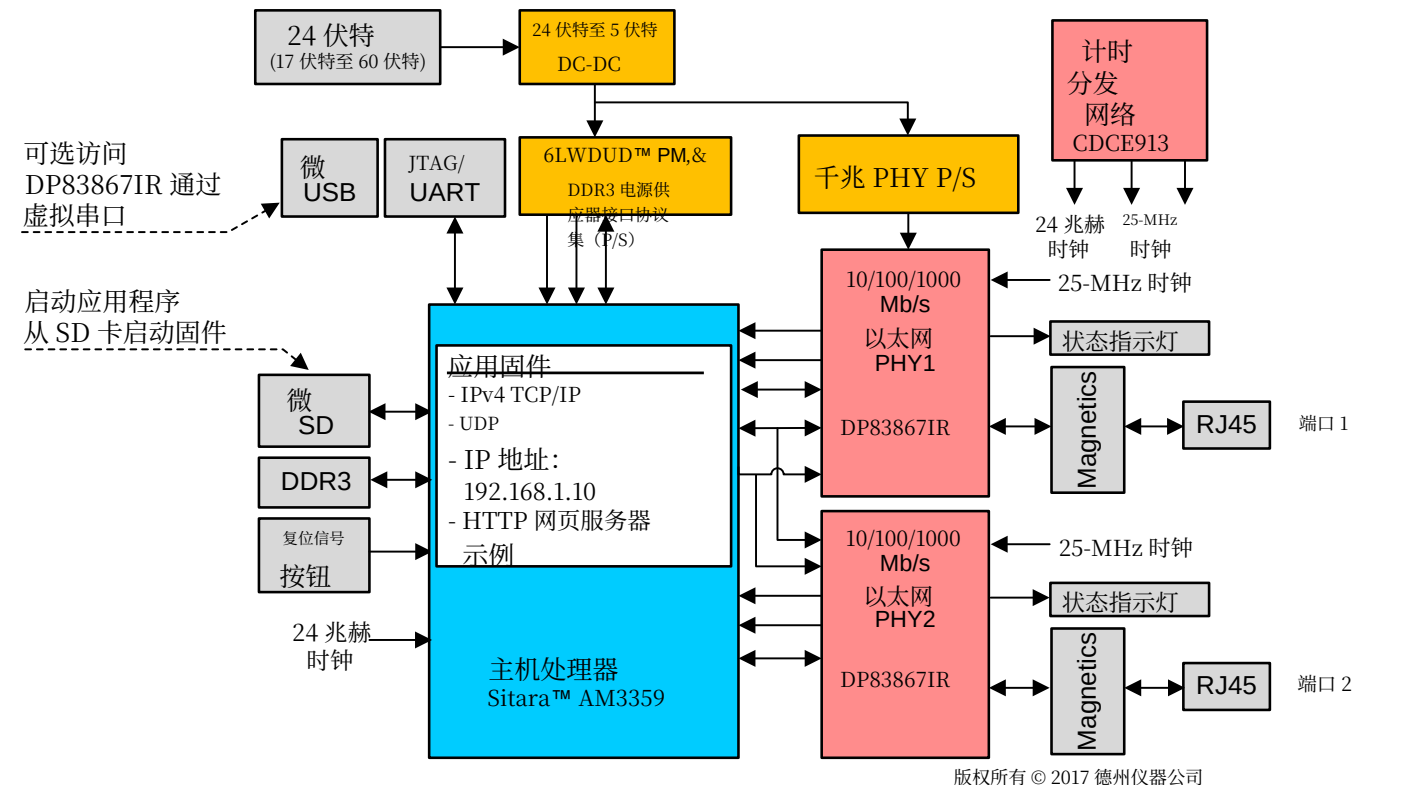


图 1。TIDA-00204 系统框图

2.2 设计考量

2.2.1 千兆以太网概述

以太网多年来得到了广泛应用。以太网已成为工业网络应用的理想选择。使用开放协议（如以太网网络上的 TCP/IP）的机会将有助于取代工业控制和工厂自动化应用中的专有通信。使用以太网时，目前提供多种速率：10 Mb/s、100 Mb/s（快速以太网）、1000 Mb/s（千兆以太网）和 10 Gb/s（10 千兆以太网）。

千兆以太网使用扩展的以太网 MAC 层接口，通过千兆媒体独立接口（GMII）层连接到物理层实体（PHY 子层），例如 1000BASE-LX、1000BASE-SX、1000BASE-CX 和 1000BASE-T。1000-Mb/s 全双工操作的拓扑结构与 100BASE-T 全双工模式相当，最小数据包传输时间减少了十倍。由此实现的半双工 1000-Mb/s CSMA/CD MAC 拓扑结构与半双工 100BASE-T 中的拓扑结构相似。

Table 3. Gigabit Ethernet Standards

IEEE STANDARD	NAME	MEDIUM
IEEE 802.3ab	1000BASE-T	Twisted-pair copper cable
IEEE 802.3z	1000BASE-SX	Fiber optic cable
	1000BASE-LX	Fiber optic cable
	1000BASE-CX	Twinax

As previously mentioned, this TI Design uses twisted-pair copper cables as defined IEEE 802.3ab.

2.2.2 Gigabit Ethernet PHY Interfaces

2.2.2.1 Medium Dependent Interface: PHY Layer 1000BASE-T

The 1000BASE-T Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA), and baseband medium specifications are intended for users who want a 1000-Mb/s performance over balanced twisted-pair cabling systems.

The 1000BASE-T PHY supports full-duplex baseband transmission through four pairs of minimum CAT5 balanced cables. The 1000 Mb/s is achieved by transmitting through four wires pairs, each at 250 Mb/s. Using hybrids and cancellers enable full duplex transmission by allowing symbols to be transmitted and received on the same wire pairs at the same time. Baseband signaling with a modulation rate of 125 MBd is used on each of the wire pairs. The transmitted symbols are selected from a four-dimensional five-level symbol constellation (4D-PAM5). In the absence of data, idle symbols are transmitted.

The IEEE 802.3 specifies specify that a PHY with a MDI that is not a power interface (PI) should provide electrical isolation between the port device circuits, including frame ground (if any) and all MDI leads. This electrical isolation shall withstand at least one of the following electrical strength tests:

- 1500 V_{RMS} at 50 to 60 Hz for 60 s
- 2250-V DC for 60 s
- A sequence of ten 2400-V impulses of alternating polarity, applied at intervals of not less than 1 s

To meet this requirement transformers are typically used for isolation. The typical transformer configuration for 1000BASE-T can be seen in Figure 2 for a one differential pair.

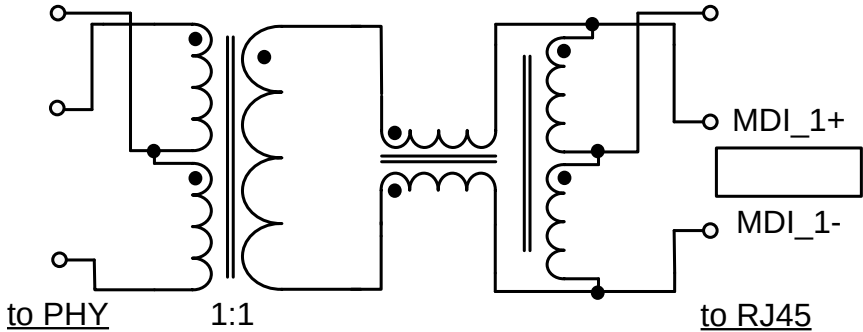


Figure 2. Gigabit Ethernet Transformer (One out of Four Pairs)

Depending on the PHY device, parallel termination might be needed for each of the MDI differential signal pair. The termination impedance is typically 100 Ω differentially. Newer devices like the DP83867IR include integrated termination so no external termination resistors are required.

表 3. 千兆以太网标准

IEEE 标准	NAME	中等
IEEE 802.3ab	1000BASE-T	双绞铜缆
IEEE 802.3z	1000BASE-SX	光纤电缆
	1000BASE-LX	光纤电缆
	1000BASE-CX	双轴电缆

如前所述，本 TI 设计采用符合 IEEE 802.3ab 标准的双绞铜缆。

2.2.2 千兆以太网物理层接口

2.2.2.1 中速依赖接口：物理层 1000BASE-T

1000BASE-T 物理编码子层（PCS）、物理介质附件（PMA）和基带介质规范旨在为希望在平衡双绞线系统中获得 1000-Mb/s 性能的用户提供标准。

1000BASE-T 物理层通过四对最小 CAT5 平衡电缆支持全双工基带传输。通过四对导线，每对以 250 Mb/s 的速率传输，实现 1000 Mb/s 的速率。使用混合器和消声器，允许在同一对导线上同时发送和接收符号，从而实现全双工传输。每对导线均使用 125 MBd 调制率的基带信号。传输的符号选自四维五电平符号星座（4D-PAM5）。在无数据时，传输空闲符号。

IEEE 802.3 规定，若 MDI 不是电源接口（PI），则物理层应提供端口设备电路之间的电气隔离，包括地线（如有）和所有 MDI 引脚。该电气隔离应至少能承受以下电气强度测试之一：

- 1500伏特_{RMS} 50 至 60 赫兹下持续 60 秒
- 2250-V DC 60 秒
- 一串极性交替的 2400-V 脉冲，间隔不小于 1 秒

为满足此要求，通常使用变压器进行隔离。典型的变压器 1000BASE-T 的配置可见于图 2 单差分对的。

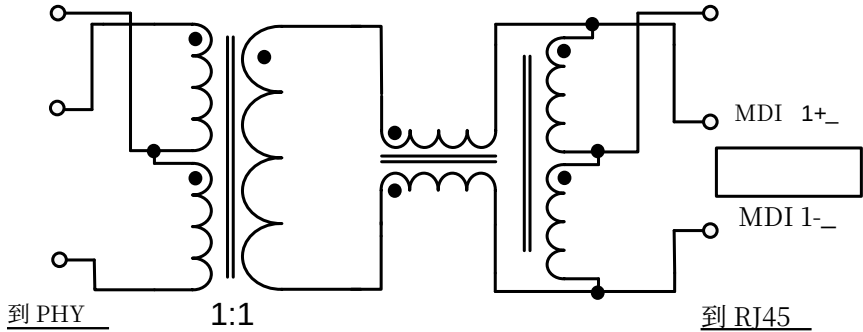


图 2. 千兆以太网变压器（四对之一）

根据物理层设备，MDI 差分信号对中的每一个可能需要并联终端。终端阻抗通常为 100 Ω 伏特差分。较新的设备（如 DP83867IR）包含集成终端，因此不需要外部终端电阻。

2.2.2.2 Medium Independent Interface: MAC Layer Interface

For the MAC layer interface to the Gigabit PHY, there are three different options defined in the IEEE 802.3ab standard: The standard Media Independent Interface (MII), the GMII, SGMII, or the RGMII.

2.2.2.2.1 GMII

The purpose of GMII is to make various physical media transparent to the MAC layer. The GMII accepts either GMII or MII data, control, and status signals and routes them either to the 1000BASE-T, 100BASE-TX, or 10BASE-T modules, respectively.

The GMII provides full-duplex operation and is an 8-bit wide transmit and receive data path interface clocked at 125 MHz defining speeds up to 1000 Mb/s. GMII is backwards compatible with the MII specification, thereby supporting 10 (2.5 MHz) and 100 (25 MHz) Mb/s speeds. Data and delimiters are synchronous to clock references. It also provides a simple management interface.

The transmit signals are GTX_CLK, TX_CLK, TX_D[7:0], TX_EN, and TX_ER. The GTX_CLK signal is supplied to the PHY when it is operating in Gigabit mode. When this is done, the TX_D, TX_EN, and TX_ER are synchronized to the GTX_CLK signal. For a 10/100-Mb operation, the TX_CLK is supplied to the MAC, and the TX_CLK signal is used to synchronize the signals (TX_D, TX_EN, and TX_ER). The receiver signals are RX_CLK, RX_D[7:0], RX_DV, RX_ER, COL, and CS. The GMII uses in total a maximum of 25 pins.

2.2.2.2.2 RGMII

The RGMII is designed to reduce the number of pins required to interconnect the MAC and PHY (12 pins for RGMII relative to 24 pins for GMII). With this optimization, the RGMII consists of 12 signals: 6 signals for receive, which are RX_CTL, RX_CLK, RX_D[3:0], and 6 for transmit, which are TX_CTL, TX_CLK, TX_D[3:0]. To accomplish this, the data paths and all associated control signals are reduced and are multiplexed. Both rising and trailing edges of the clock are used. The TX_CTL and RX_CTL signal carries data valid (DV) on the rising edge and DV XOR ERROR on the falling edge, for CTL signals there is no change between 10/100Mb/s or 1000Mb/s operation.

For a Gigabit operation, the GTX_CLK and RX_CLK clocks are 125 MHz, and for 10- and 100-Mb/s operation, the clock frequencies are 2.5 MHz and 25 MHz, respectively.

2.2.2.2.3 SGMII

The SGMII differs from the GMII and RGMII by having a higher clock frequency and the 8b/10b (SerDes) coded interface. It uses differential pairs at a 625-MHz clock frequency double data rate (DDR) for TX and RX data and for TX and RX clock. The transmit and receive path uses one differential pair for data and in some cases one for clock. TX and RX clocks must be generated on device output but are optional on device input. With revision 1.8 of the SGMII standard, clock recovery can be used, removing the need for the clock signal. This means that the SGMII can be a 4- to 8-pin solution.

2.2.2.3 Serial Management Interface

The serial management interface (SMI) consists of a Management Data Clock (MDC) and a Management Data Input/Output (MDIO) signal. It provides access to the PHY's internal register space for status information and configuration. The MDC and MDIO signals can be shared amongst several PHYs due to the serial communication protocol, where an address is used to identify the corresponding PHY slave. The MDIO has a standard set of registers from 0 to 31 each containing 16 bits. In IEEE 802ah clause 45, an extended register set was defined for extra functionality of the PHYs. The SMI is initially specified at a 2.5-MHz clock.

2.2.2.2 独立于媒介的接口：MAC 层接口

对于 MAC 层与千兆物理层之间的接口，IEEE 802.3ab 标准定义了三种不同的选项：标准媒体独立接口（MII）、GMII、SGMII 或 RGMII。

2.2.2.2.1 GMII

GMII 的目的是使各种物理介质对 MAC 层透明。GMII 接受 GMII 或 MII 的数据、控制和状态信号，并将其分别路由到 1000BASE-T、100BASE-TX 或 10BASE-T 模块。

GMII 提供全双工操作，是一个 8 位宽度的发送和接收数据路径接口，时钟频率为 125 MHz，定义最高可达 1000 Mb/s 的速率。GMII 与 MII 规范向后兼容，从而支持 10 (2.5 MHz) 和 100 (25 MHz) Mb/s 的速率。数据与分隔符与时钟参考同步。它还提供了一个简单的管理接口。

发送信号包括 GTX_CLK、TX_CLK、TX_D[7:0]、TX_EN 和 TX_ER。当 PHY 以千兆模式运行时，GTX_CLK 信号被提供给 PHY。在此模式下，TX_D、TX_EN 和 TX_ER 与 GTX_CLK 信号同步。对于 10/100-Mb 操作，TX_CLK 被提供给 MAC，并使用 TX_CLK 信号来同步信号（TX_D、TX_EN 和 TX_ER）。接收信号包括 RX_CLK、RX_D[7:0]、RX_DV、RX_ER、COL 和 CS。GMII 总共最多使用 25 个引脚。

2.2.2.2.2 RGMII

RGMII 旨在减少连接 MAC 和 PHY 所需的引脚数量（RGMII 需要 12 个引脚，而 GMII 需要 24 个引脚）。通过这一优化，RGMII 包含 12 个信号：6 个用于接收，即 RX_CTL、RX_CLK、RX_D[3:0]，以及 6 个用于发送，即 TX_CTL、TX_CLK、TX_D[3:0]。为实现此目标，数据路径及相关控制信号均被精简并复用。时钟信号的上升沿和下降沿均被使用。TX_CTL 和 RX_CTL 信号在上升沿携带数据有效（DV）信号，在下降沿携带 DV XOR ERROR 信号；对于 CTL 信号，在 10/100Mb/s 或 1000Mb/s 操作模式下无变化。

对于千兆位操作，GTX_CLK 和 RX_CLK 时钟为 125 MHz，对于 10 和 100-Mb/s 操作，时钟频率分别为 2.5 MHz 和 25 MHz。

2.2.2.2.3 SGMII

SGMII 与 GMII 和 RGMII 不同之处在于其具有更高的时钟频率以及 8b/10b（SerDes）编码接口。它使用差分对在 625-MHz 时钟频率下以双倍数据速率（DDR）传输 TX 和 RX 数据，以及 TX 和 RX 时钟。发送和接收路径各使用一对差分对用于数据，在某些情况下还使用一对用于时钟。TX 和 RX 时钟必须在设备输出端生成，但在设备输入端为可选。在 SGMII 标准 1.8 版本中，时钟恢复功能可用，从而无需时钟信号。这意味着 SGMII 可成为一种 4 至 8 引脚的解决方案。

2.2.2.3 串行管理接口

串行管理接口（SMI）由管理数据时钟（MDC）和管理数据输入/输出（MDIO）信号组成。它提供对 PHY 内部寄存器空间的访问，用于状态信息和配置。由于串行通信协议，MDC 和 MDIO 信号可在多个 PHY 之间共享，其中地址用于标识对应的 PHY 从机。MDIO 具有一组从 0 到 31 的标准寄存器，每个寄存器包含 16 位。在 IEEE 802ah 第 45 条中，为 PHY 的额外功能定义了一组扩展寄存器。SMI 最初指定为 2.5-MHz 时钟。

2.2.3 IEEE 1588v2

Precise time information is important, especially for distributed systems like in factory automation. The IEEE 1588v2 is an IEEE standard for precision clock synchronization protocol for networked measurement and control systems. The protocol is used to synchronize the time and clock frequency. It defines a way to provide sub-microsecond precision synchronization.

To define this synchronization, the start of package detection is needed. Depending on which application layer this detection happens, the timing error varies. The lower the layer, the smaller the error.

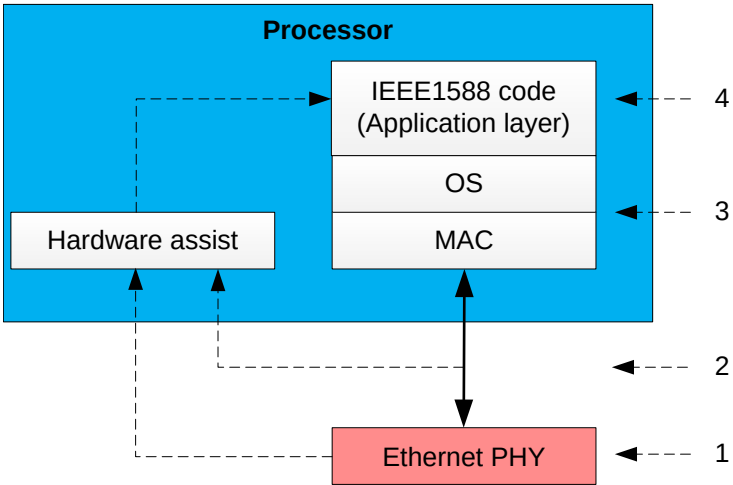


Figure 3. Options 1 to 4 for Time Stamp versus Layer

The closer to the PHY the time stamp is set the better the time reference. The time stamp consists of two signals the Ingress (RX) and Egress (TX) time stamp. Depending on when the time stamp is done, the delay can vary from nanoseconds to microseconds.

2.3 System Design Theory

2.3.1 Circuit Design and Component Selection

2.3.1.1 DP83867IR 10/100/1000-Mb/s Gigabit Ethernet PHY

The DP83867IR was selected due to following features:

- IEEE 802.3ab 1000BASE-T compliant
- Operating temperature range -40°C to 85°C
- 8-kV IEC 61000-4-2 ESD protection (direct contact)
- RGMII with software (register) programmable and hardware configurable (strap resistors) clock skew
- Integrated termination resistors
- Low power: 565 mW
- SOF detect for IEEE 1588 time stamp

In addition to the above features, the DP83867IR also offers the following features:

- Low deterministic TX and RX latency
- Wake on LAN (WoL)
- Synchronized clock output to synchronize multiple PHYs using one crystal (or clock)

2.2.3 IEEE 1588v2

精确的时间信息非常重要，尤其是在工厂自动化等分布式系统中。IEEE 1588v2 是 IEEE 为网络测量和控制系统的精密时钟同步协议制定的标准。该协议用于同步时间和时钟频率。它定义了一种提供亚微秒级精度同步的方法。

要定义此同步，需要包检测的起始点。根据检测发生的应用层不同，时序误差会发生变化。层级越低，误差越小。

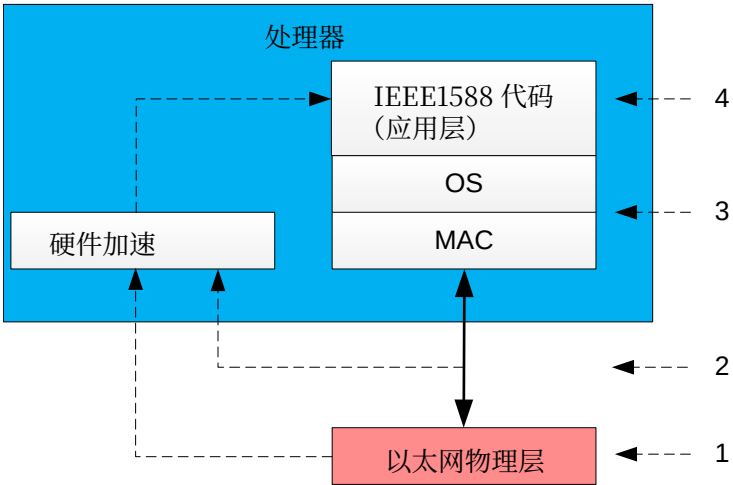


图 3. 时间戳与层之间的选项 1 至 4

越靠近 PHY 设置时间戳，时间参考就越准确。时间戳由两个信号组成：进站（RX）和出站（TX）时间戳。根据时间戳完成的时间不同，延迟可从纳秒级变化到微秒级。

2.3 系统设计理论

2.3.1 电路设计与组件选择

2.3.1.1 DP83867IR10/100/1000-Mb/s 千兆以太网 PHY

选择 DP83867IR 的原因如下：

- IEEE 802.3ab 1000BASE-T 兼容
- 工作温度范围 -40°C 至 85°C
- 8-kV IEC 61000-4-2 静电放电保护（直接接触）
- RGMII 软件（寄存器）可编程和硬件可配置（拉紧器电阻）时钟偏移
- 集成终止电阻器
- 低功耗：565 毫瓦
- IEEE 1588 时间戳 SOF 检测

除上述功能外，DP83867IR 还提供以下功能：

- 低确定性发送和接收延迟
- 唤醒局域网 (WoL)
- 同步时钟输出，使用单个晶振（或时钟）同步多个 PHY

2.3.1.1.1 DP83867IR Gigabit Ethernet PHY Configuration

When configuring the DP83867IR, this can be done using either a SMI or a strap configuration through the four-level strap pins. A pull-up resistor and a pull-down resistor of suggested values may be used to set the voltage ratio of the four-level strap pin input and the supply to select one of the possible selected modes. The device should feature four-level strap pins, each supporting at least four selectable options, as shown in Figure 4 and Table 4. Strap resistors with 1% tolerance are recommended.

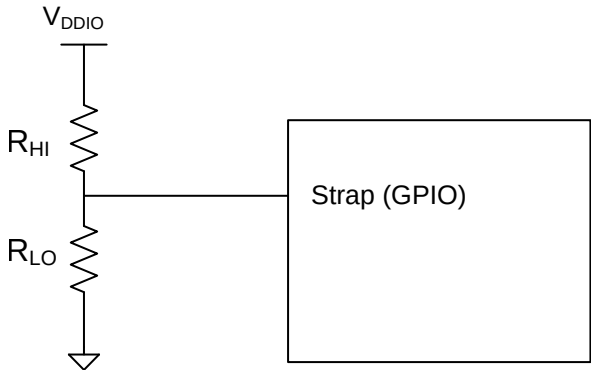


Figure 4. Strap Circuit

Table 4. Four-Level Strap Resistor Ratios

MODE	RESISTOR R _{HI} (kΩ)	RESISTOR R _{LO} (kΩ)
1	OPEN	OPEN
2	11	2.49
3	6.04	2.49
4	2.49	OPEN

Because this design employs two DP83867IR, the SMI will have two DP83867IR slaves addresses. This means that the SMI address of the two DP83867IR have to differ to ensure a valid communication. To set the DP83867IR SMI address a strap configuration option can be used on one DP83867IR to change the default address from 0x00. In this design, it was chosen to use the pin RX_D4 to set the SMI address, as this pin is not used for the RGMII.

A strap configuration option was chosen so the RGMII is always enabled to ensure the RGMII is enabled when as the RX_D6 pin is used as input pin on the AM3359 Sitara processor.

As the clock out option of the device is not used by default, RX_D7 strap configuration option was done to disable the clock out of the device. This feature is default on if this strap on is not done and if it needs to be disabled without strap configuration. It would have to be done using the SMI.

2.3.1.1.1 DP83867IR 千兆以太网 PHY 配置

配置 DP83867IR 时，可通过四个级联的拉紧器引脚使用 SMI 或拉紧器配置。可使用建议值的上拉电阻和下拉电阻来设置四个级联拉紧器引脚输入与电源之间的电压比，以选择可能的模式之一。该设备应配备四个级联拉紧器引脚，每个引脚至少支持四个可选选项，如图 4 所示。图 4 和表 4。建议使用公差为 1% 的拉紧器电阻。

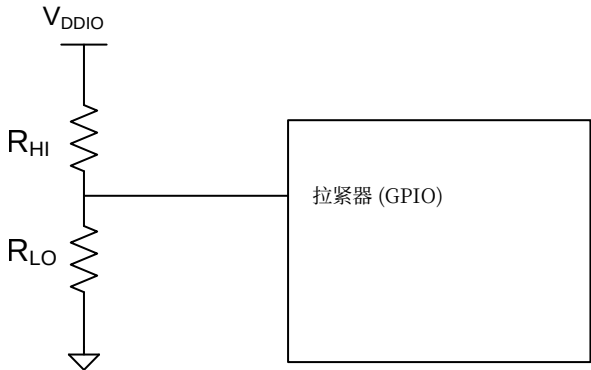


图 4. 拉紧器 (GPIO) 电路

表 4. 四级拉紧器电阻比率

MODE	电阻 R _{HI} (kΩ)	电阻 R _{LO} (kΩ)
1	OPEN	OPEN
2	11	2.49
3	6.04	2.49
4	2.49	OPEN

由于该设计采用两个 DP83867IR，SMI 将拥有两个 DP83867IR 从机地址。这意味着两个 DP83867IR 的 SMI 地址必须不同，以确保有效通信。要设置 DP83867IR 的 SMI 地址，可以在一个 DP83867IR 上使用拉紧器 (GPIO) 配置选项，将默认地址从 0x00 更改。在此设计中，选择使用引脚 RX_D4 来设置 SMI 地址，因为该引脚未用于 RGMII。

选择了拉紧器配置选项，以确保 RGMII 始终启用，从而保证在 AM3359 Sitara 处理器上，当 RX_D6 引脚用作输入引脚时，RGMII 处于启用状态。

由于设备的时钟输出选项默认未使用，因此通过配置 RX_D7 拉紧器 (GPIO) 来禁用设备的时钟输出。若未进行此拉紧器配置，该功能将默认启用；若需在不进行拉紧器配置的情况下禁用该功能，则必须通过 SMI 完成。

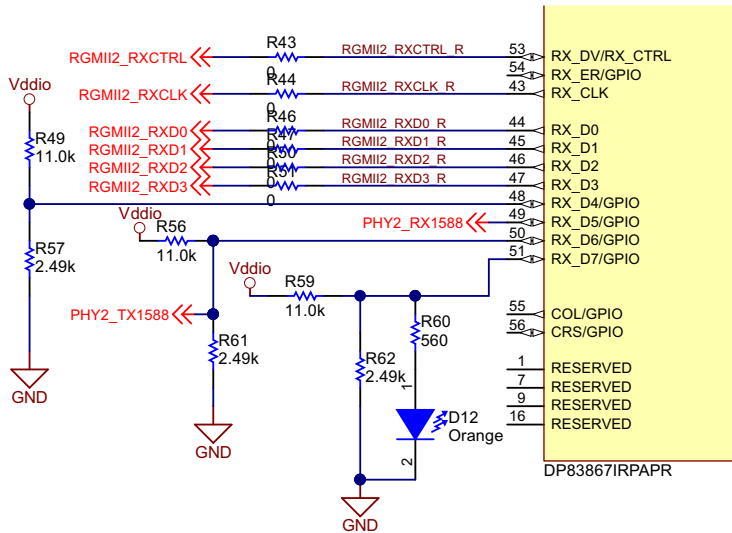


Figure 5. Schematic for DP83867IR Strap Configuration on ETH2

Table 5. DP83867IR Strap on Resistor Chosen

DP83687IR PIN NAME	CONFIGURATION	MODE
RX_D4	Strap resistors	2 (PHY_ADD4 = 1, only ETH2)
RX_D6	Strap resistors	2 (RGMII enable)
RX_D7	Strap resistors	2 (Clock out disable)

When using the strap configuration on a specific pin, ensure that the additional function mapped to this pin is applicable. For example, due to this the TX_D0 to TX_D3 and RX_D0 to RX_D3 pins, which are used by the RGMII, have not been used for strap configuration. This is because this strap resistor configuration would have needed to be compensated with trace matching from the other RGMII pins used. Another example is the LED indicator pins: When used with both LED functionality and strap configuration, take caution on how to connect the LED with regards to the strap resistors. An example for strap mode 0 and strap mode 4 with indicator LED is shown in Figure 6.

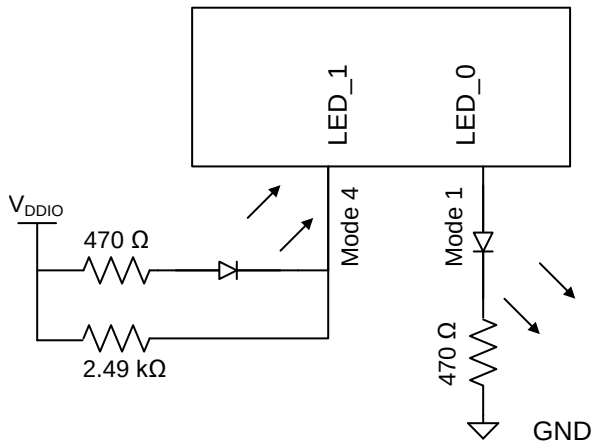


Figure 6. Example Strap Connections With Indicator LEDs for Mode 1 and Mode 4

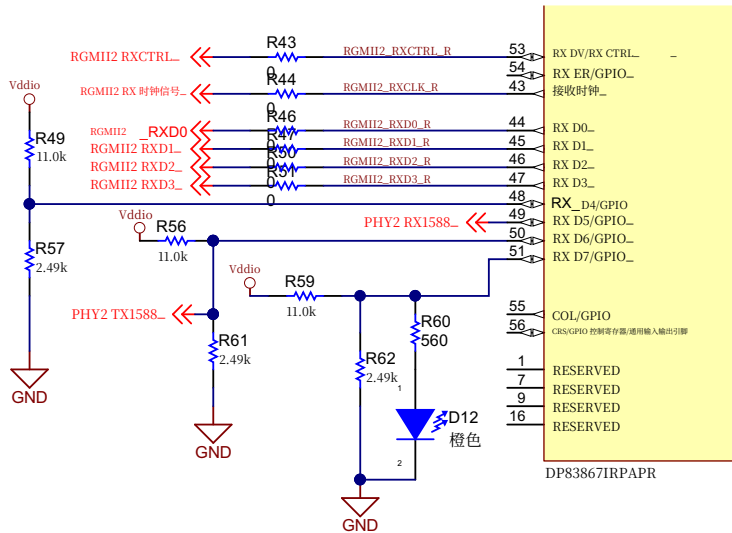


图 5。DP83867IR 拉紧器 (GPIO) 配置在 ETH2 上的原理图

表 5. DP83867IR 拉紧器 (GPIO) 电阻选择

DP83687IR 引脚名称	配置	MODE
RX_D4	拉紧器 (GPIO) 电阻器	2 (PHY_ADD4 = 1, 仅 ETH2)
RX_D6	拉紧器 (GPIO) 电阻器	2 (启用 RGMII)
RX_D7	拉紧器 (GPIO) 电阻器	2 (时钟输出禁用)

当在特定引脚上使用拉紧器配置时，请确保映射到此引脚的附加功能适用。例如，由于此原因，TX_D0 到 TX_D3 和 RX_D0 到 RX_D3 引脚（由 RGMII 使用）未用于拉紧器配置。这是因为该拉紧器电阻配置需要从其他 RGMII 引脚进行迹线匹配补偿。另一个例子是 LED 指示引脚：当同时使用 LED 功能和拉紧器配置时，请小心连接 LED 与拉紧器电阻的关系。拉紧器模式 0 和拉紧器模式 4 带指示 LED 的示例显示在图 6。

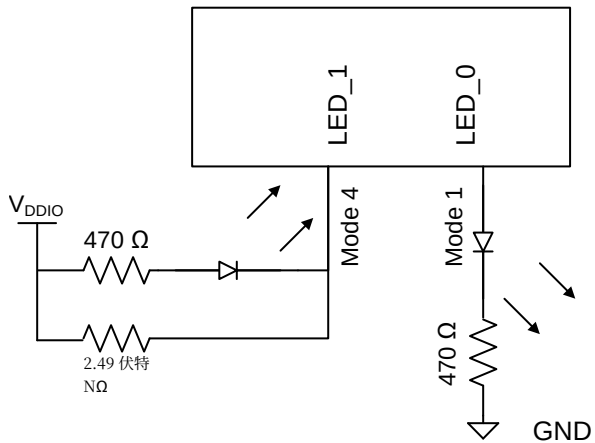


图 6. 模式 1 和模式 4 的示例拉紧器 (GPIO) 连接及指示 LED

2.3.1.1.2 MAC Layer Interface

The MAC interface of this design was chosen according to minimum number of pins at lowest possible clock frequency. See [Table 6](#) of the pin definitions.

Table 6. MAC Interface Option Pinout for the Different Standards

DECISION CRITERION	GMII	RGMII	SGMII
Number of MAC interface signals	25	12	4 or 6
Clock frequency	125 MHz	125 MHz	625 MHz

With those considerations, the choice was to use the RGMII as it is running at a lower frequency than the SGMII. Therefore, the RGMII allows for easier PCB routing of the signals, and the number of signals is acceptable.

The RGMII signals are not differential but single-ended, and they are referenced to a clock signal of 125 MHz. It is crucial that length matching is done properly for all signals to avoid skew due to different propagation delay between the clock and the data signals.

For this design, the following rules were followed on the RGMII signals.

Table 7. DP83867 RGMII Design Rules on TIDA-00204 PCB

RULES	DISTANCE
RGMII TX length matching	0.254 mm
RGMII RX length matching	0.254 mm
RGMII data-to-data distance separation	0.762 mm
RGMII clock-to-data distance separation	1.27 mm
RGMII clock-to-clock distance separation	1.27 mm
RGMII TX length matching	0.254 mm
RGMII RX length matching	0.254 mm
Max total trace length	63.5 mm

These rules are referenced from [High-Speed Interface Layout Guidelines](#) (SPRAAR7).

Additional considerations are to place each PHYs RGMII TX and RX on the same layer and add series termination resistors close to the corresponding output pins. Because the DP83867IR has an integrated 50-Ω series impedance, a 0-Ω series termination has been placed at the DP83867 RGMII RX output pins. This was a test and debug option and the termination resistor (array) can be removed in a production design.

A 22-Ω series termination resistor was placed close to the RGMII1 and RGMII2 TX pins of the AM3359 Sitara.

2.3.1.1.2 MAC 层接口

本设计所选用的 MAC 接口是在最低可能时钟频率下引脚数量最少的方案。参见[表 6](#)引脚定义。

表 6. 不同标准的 MAC 接口选项引脚定义

决策标准	GMII	RGMII	SGMII
MAC 接口信号数量	25	12	4 或 6
时钟频率	125 MHz	125 MHz	625 MHz

基于上述考量，选择采用 RGMII，因为其运行频率低于 SGMII。因此，RGMII 允许信号布线更易于在 PCB 板上进行，且信号数量可接受。

RGMII 信号为非差分单端信号，并参考 125 MHz 的时钟信号。必须确保所有信号的等长匹配正确，以避免因时钟与数据信号之间的传播延迟不同而产生的时序偏差。

本设计中，对 RGMII 信号遵循以下规则。

表 7. TIDA-00204 PCB 上 DP83867 RGMII 设计规则

规则	距离
RGMII TX 长度匹配	0.254 毫米
RGMII RX 长度匹配	0.254 毫米
RGMII 数据到数据距离分离	0.762 毫米
RGMII 时钟到数据距离分离	1.27 毫米
RGMII 时钟到时钟距离分离	1.27 毫米
RGMII TX 长度匹配	0.254 毫米
RGMII RX 长度匹配	0.254 毫米
最大总走线长度	63.5 毫米

这些规则参考自[高速接口布局指南](#)(SPRAAR7)。

额外的考虑是将每个 PHY 的 RGMII TX 和 RX 放置在同一层，并在靠近相应输出引脚处添加串联终止电阻。由于 DP83867IR 具有集成的 50-Ω串联阻抗，因此在 DP83867 RGMII RX 输出引脚处放置了 0-Ω 串联终止电阻。这是一个测试和调试选项，在量产设计中可以移除该终止电阻（阵列）。

在 AM3359 Sitara 的 RGMII1 和 RGMII2 TX 引脚附近放置了一个 22-ΩΩ 的终端电阻。

2.3.1.1.3 Interface From PHY to RJ-45

The transformer used in the MDI connection provides DC isolation between local circuitry and the network cable. The center tap of the isolated winding has a "Bob Smith" termination through a 75-Ω and a 1000-pF capacitor-to-chassis ground. The termination capacitor should be voltage rated to at least 2 kV. The Bob Smith termination reduces noise resulting from common mode current flows.

NOTE: A TVS diode has been placed each between the DP83867IR and the transformer and the transformer and the RJ45 jack. This was a test and debug option only and the position between the DP83867IR and the magnetics showed slightly better results in EMC and EMI, as shown in [Figure 7](#).

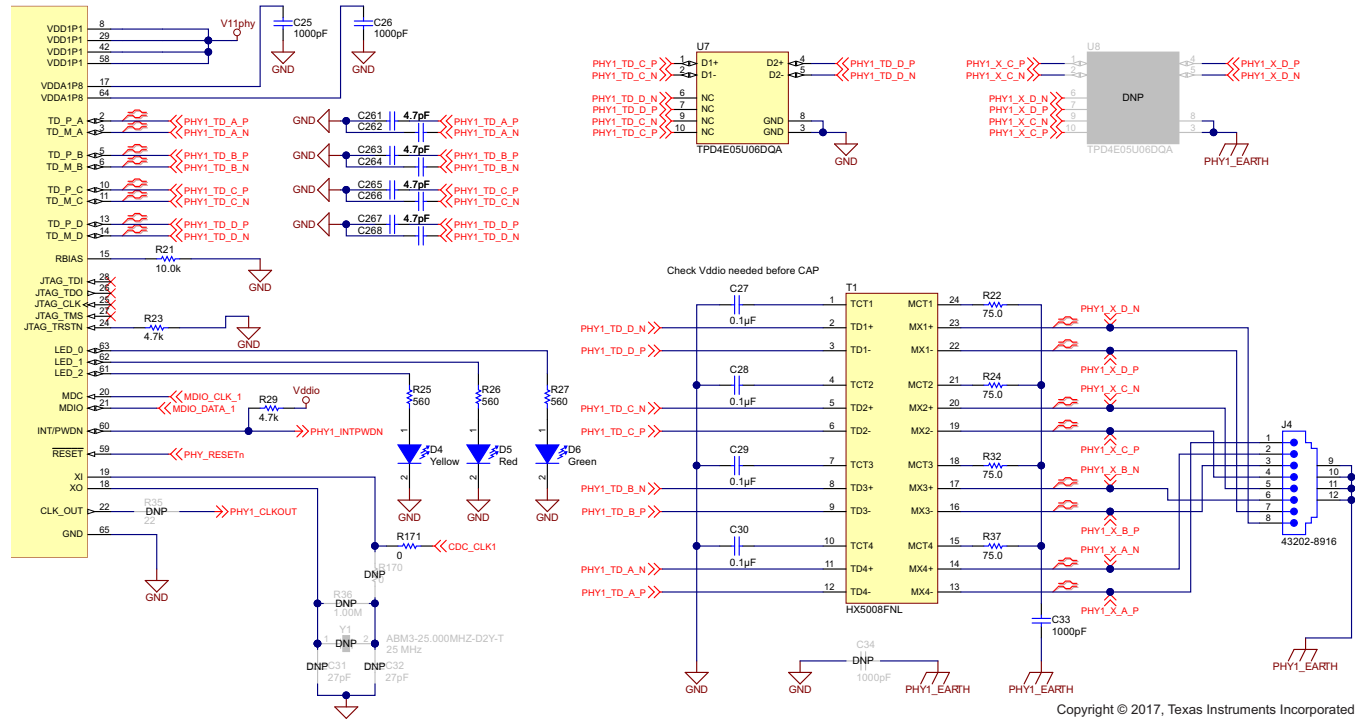


Figure 7. Schematic of Media Dependent Interface of ETH1

The trace length that needs to be considered is from RJ45 to magnetic and from magnetic to DP8867IR, for these signals they are differential pairs. This means that the signals needs to be routed differentially as long as possible without making the traces longer than necessary. To ensure data integrity, the trace's difference should be below 10 mil (0.254 mm).

Table 8. Differential Signal Trace Length From PHY to Magnetic and Magnetic to RJ45 Jack

MEDIA DEPENDENT INTERFACE	RJ45 TO MAGNETIC (mm)	MAGNETIC TO DP8867IR (mm)
PHY1 Differential pair A (N,P)	13.0123 (N), 13.0708 (P)	10.0302 (N), 10.0302 (P)
PHY1 Differential pair B (N,P)	13.1709 (N), 13.3115 (P)	10.0223 (N), 10.0223 (P)
PHY1 Differential pair C (N,P)	13.1321 (N), 13.0146 (P)	10.0175 (N), 10.0082 (P)
PHY1 Differential pair D (N,P)	13.0804 (N), 13.0003 (P)	10.0163 (N), 10.0163 (P)
PHY2 Differential pair A (N,P)	12.9469 (N), 13.0056 (P)	13.3910 (N), 13.4938 (P)
PHY2 Differential pair B (N,P)	12.9398 (N), 13.0815 (P)	13.1988 (N), 13.1047 (P)
PHY2 Differential pair C (N,P)	13.0899 (N), 13.0409 (P)	13.0679 (N), 13.1622 (P)
PHY2 Differential pair D (N,P)	13.1864 (N), 13.1063 (P)	13.3252 (N), 13.2272 (P)

2.3.1.1.3 从 PHY 到 RJ-45 的接口

MDI 连接中使用的变压器在本地电路和网线之间提供直流隔离。隔离绕组的中心抽头通过 75- Ω 和 1000-pF 电容至机箱地实现"Bob Smith"终端。终端电容的电压额定值应至少为 2 kV。Bob Smith 终端可减少由共模电流流动引起的噪声。

注意：在 DP83867IR 与变压器之间、变压器与 RJ45 连接器之间各放置了一个 TVS 二极管。这是一个测试和调试选项，DP83867IR 与磁性元件之间的位置在 EMC 和 EMI 方面显示出略好的结果，如图 7 所示。[图 7](#)。

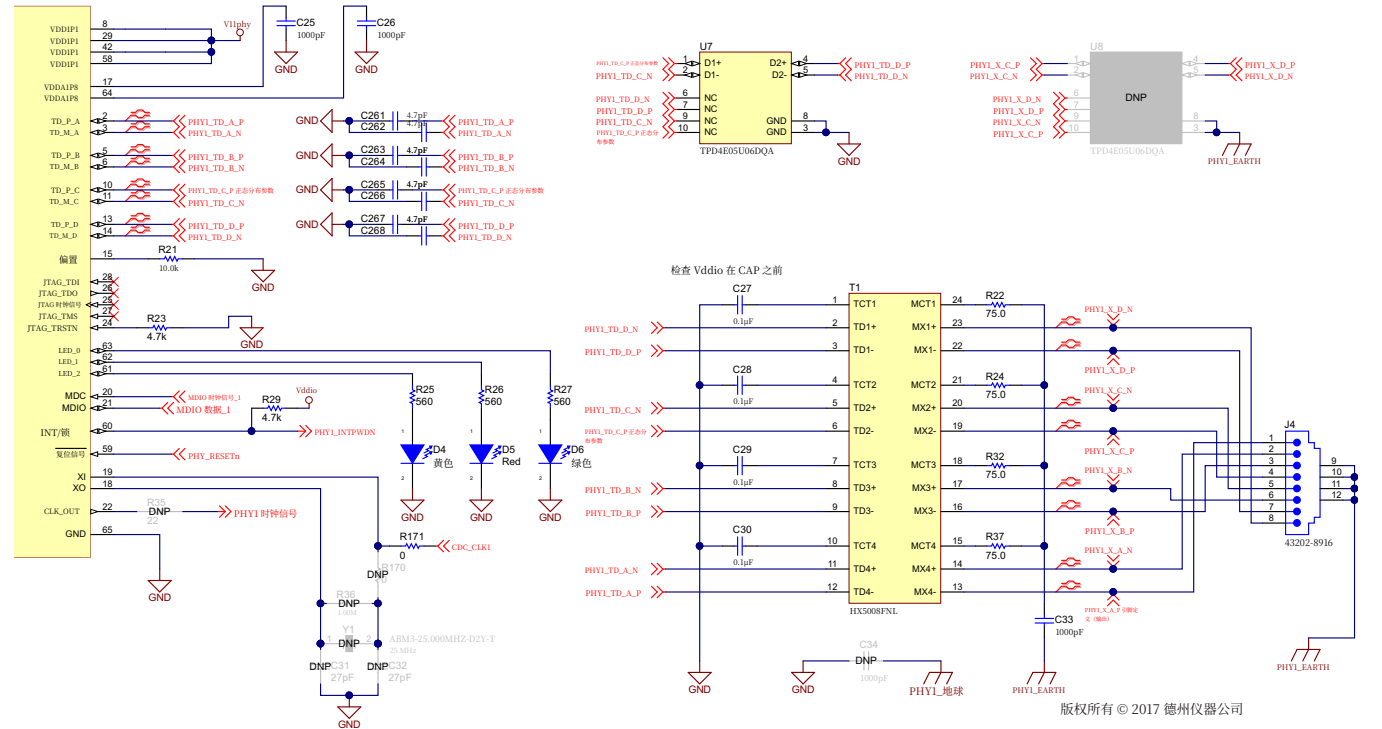


图 7。ETH1 的媒体相关接口示意图

需要考虑的走线长度是从 RJ45 到磁环, 以及从磁环到 DP8867IR。对于这些信号, 它们是差分对。这意味着信号应尽可能以差分方式布线, 同时避免走线过长。为确保数据完整性, 走线间的差分应小于 10 mil (0.254 mm)。

表 8. 从 PHY 到磁感应线圈及从磁感应线圈到 RJ45 连接器的差分信号走线长度

依赖媒体接口	RJ45 转磁头 (mm)	MAGNETIC TO DP88671R (mm)
PHY1 差分对 A (N,P)	13.0123 (N), 13.0708 (P)	10.0302 (N), 10.0302 (P)
PHY1 差分对 B (N,P)	13.1709 (N), 13.3115 (P)	10.0223 (N), 10.0223 (P)
PHY1 差分对 C (N,P)	13.1321 (N), 13.0146 (P)	10.0175 (N), 10.0082 (P)
PHY1 差分对 D (N,P)	13.0804 (N), 13.0003 (P)	10.0163 (N), 10.0163 (P)
PHY2 差分对 A (N,P)	12.9469 (N), 13.0056 (P)	13.3910 (N), 13.4938 (P)
PHY2 差分对 B (N,P)	12.9398 (N), 13.0815 (P)	13.1988 (N), 13.1047 (P)
PHY2 差分对 C (N,P)	13.0899 (N), 13.0409 (P)	13.0679 (N), 13.1622 (P)
PHY2 差分对 D (N,P)	13.1864 (N), 13.1063 (P)	13.3252 (N), 13.2272 (P)

To ensure no unnecessary stubs, all traces are routed on the top layer. The only exception is the pins D2_P and D2_N. Due to the RJ45 jack pin assignment without integrated transformer, this differential pair changed the layer to minimize the overall differential trace length of the four pairs.

Table 9. RJ45 Connector Pinout

PIN	SIGNAL	PIN	SIGNAL
1	D1_P	5	D3_N
2	D1_N	6	D2_N
3	D2_P	7	D4_P
4	D3_P	8	D4_N

2.3.1.1.4 DP83867IR Input Clock Selection

For the input clock, either a crystal or an external clock source can be used. For both cases, the clock needs to be 25 MHz with a tolerance of less than ± 50 ppm. For more details, refer to the DP83867IR data sheet, Sections 8.2.1.2 and 8.2.1.3[1]. For this design, a 25-MHz crystal was chosen (see Section 2.3.1.4 for more details).

The DP83867IR has a clock out pin CLK_OUT too. This allows to route the 25-MHz clock from one DP83867IR PHY to the second DP83867IR PHY and eliminates the need for a crystal at the second PHY, reducing costs. The TIDA-00204 design has been prepared for this configuration by adding series 0- Ω resistors.

The preferred option, as realized with revision E3 of this design, is to have all system clocks in the design synced to a single reference clock. This consolidation helps to reduce jitter between the individual clocks on the board. This option is a key feature for system performance between the PHY and the MAC layer communication, especially in real-time Ethernet systems. See Section 2.3.1.4.2 for more details.

The CDCE913 programmable 1-PLL VCXO clock synthesizer is used to generate the two 25-MHz clocks for the DP83867IR.

2.3.1.1.5 Power and Ground Pins

For each of the three power rails on the DP83867IR, decoupling capacitors are recommended as follows. A 1-nF capacitor is recommended be placed close to each supply pin of the DP83867IR. A 10-nF and a 10- μ F capacitor are recommended per supply rail.

2.3.1.1.6 PHY RESET

The DP83867IR PHYs are automatically reset after power-up, through signal SYS_RESETh. Additionally, the DP83867IR hardware reset signal PHY_RESETh can be issued by the Sitara AM3359 GPIO pin with the signal GPIO_PHY_RESETh, which offers a software option to reset the PHYs if needed.

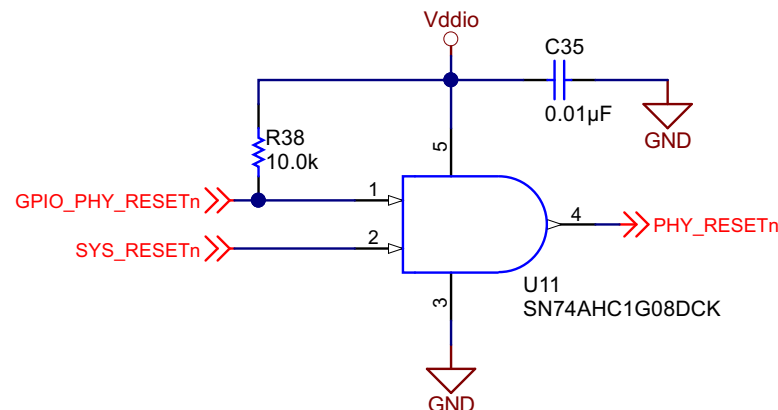


Figure 8. PHY Reset Option During Power Up (SYS_RESETh) and Software Reset (GPIO_PHY_RESETh)

为确保无不必要的过孔，所有走线均布于顶层。唯一例外是引脚 D2_P 和 D2_N。由于 RJ45 连接器引脚分配未集成变压器，该差分对将层切换，以最小化四对差分走线的总长度。

表 9. RJ45 连接器引脚定义

PIN	信号	PIN	信号
1	D1_P	5	D3_N
2	D1_N	6	D2_N
3	D2_P	7	D4_P
4	D3_P	8	D4_N

2.3.1.1.4 DP83867IR 输入时钟选择

对于输入时钟，可以使用晶振或外部时钟源。对于这两种情况，时钟频率需要为 25 MHz，容差小于 ± 50 ppm。有关更多详细信息，请参阅 DP83867IR 数据手册的第 8.2.1.2 节和第 8.2.1 节。3[1]。本设计中选择了 25-MHz 晶振（有关更多详细信息，请参阅第 2.3.1.4 节）。

DP83867IR 还有一个 CLK_OUT 时钟输出引脚。这允许将 25-MHz 时钟信号从一个 DP83867IR PHY 路由到第二个 DP83867IR PHY，从而消除了第二个 PHY 处对晶振的需求，降低了成本。TIDA-00204 设计已通过添加串联 0- Ω 电阻为这种配置做好了准备。

本设计首选方案（在 E3 版本中实现）是将设计中的所有系统时钟与单一参考时钟同步。这种整合有助于减少板载各个时钟之间的抖动。该选项是 PHY 与 MAC 层通信之间系统性能的关键特性，特别是在实时以太网系统中。见第 2.3.1.4.2 节了解更多详情。

CDCE913 可编程 1-PLL VCXO 时钟合成器用于为 DP83867IR 生成两个 25-MHz 时钟。

2.3.1.1.5 电源和地引脚

对于 DP83867IR 的三个电源轨，推荐的去耦电容配置如下。建议在 DP83867IR 的每个电源引脚附近放置一个 1-nF 电容。每个电源轨推荐配置一个 10-nF 电容和一个 10- μ F 电容。

2.3.1.1.6 PHY 复位信号

DP83867IR PHY 上电后会自动通过信号 SYS_RESETh 进行复位。此外，Sitara AM3359 的 GPIO 引脚可通过信号 GPIO_PHY_RESETh 发出 DP83867IR 硬件复位信号 PHY_RESETh，这为必要时复位 PHY 提供了软件选项。

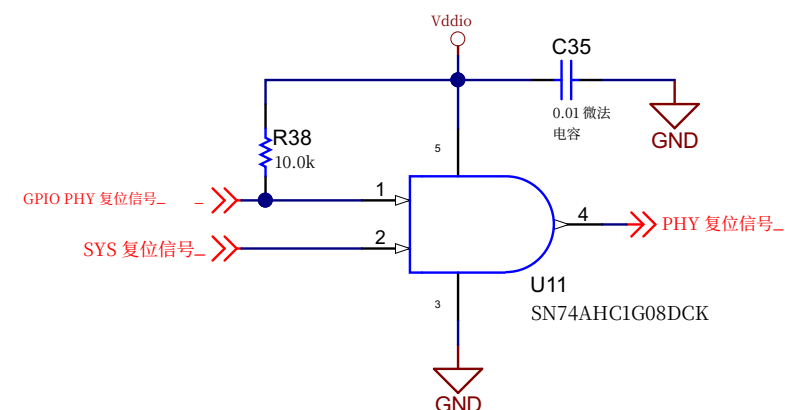


图 8. 上电期间 (SYS_RESETh) 和软件复位 (GPIO_PHY_RESETh) 的 PHY 复位选项

2.3.1.1.7 Not Connected Pins on DP83867

Table 10. DP83867IR Not Connected Pins

PINS	REASON
TXD4:7	Internal pull-down
TX_ER	Internal pull-down
TX_CLK	Output
COL	Internal pull-down
CRS	Internal pull-down
Reserved	Reserved
JTAG (TDI, TMS, TCLK)	Internal pull-up
JTAG (TDO)	Output

2.3.1.2 Host Processor

The Sitara AM3359 was chosen as it has a 2-Gb Ethernet MAC with switch layer supporting RGMII to both DP83867IRs. The MAC switch layer inside the AM3359 is supporting several features of passing messages from one PHY to the other without the use of the core. To achieve deterministic and very low transmit and receive latency, the ICSS/PRU subsystem can be used to capture the start of frame for IEEE 1588 time stamp. The TIDA-00204 hardware is provisioned for this feature but not tested.

2.3.1.2.1 AM3359 Boot Mode Configuration

The AM3359 internal ROM code selects the corresponding peripheral based on the level of the SYSBOOT configuration at power on reset of the device. For the full details of this boot procedure, read Sections 26.1.5 to 26.1.8 of the Technical Reference Manual of the AM335x Sitara processors.

For easy standalone operation, the AM3359 has been configured to boot from MMC1/SD-Card. This configuration requires the SYSBOOT[15:0] pins to be "0100 0000 0001 1100". This is achieved by setting the corresponding pull-up and pull-down resistors as per the schematics in Figure 9.

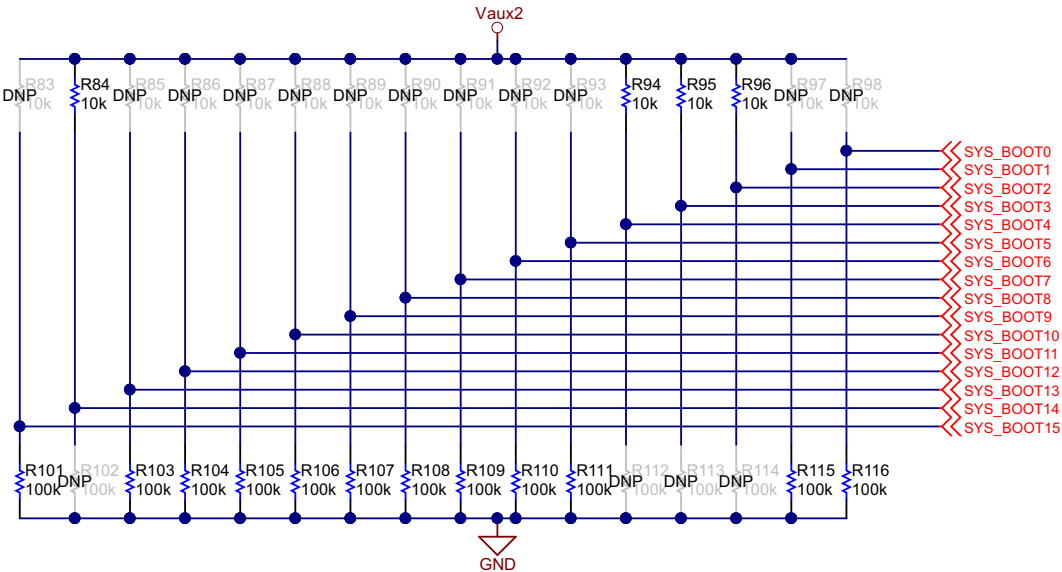


Figure 9. AM3359 SYSBOOT Pin Configuration for MMC1/SD-Card Boot Mode

2.3.1.1.7 未连接 PinsonDP83867

表 10. DP83867IR 未连接引脚

PINS	原因
TXD4:7	内部下拉菜单
TX_ER	内部下拉菜单
TX 时钟信号	输出
COL	内部下拉菜单
CRS	内部下拉菜单
保留	保留
JTAG (TDI, TMS, 时钟信号)	内部上拉
JTAG (TDO)	输出

2.3.1.2 主机处理器

选择 Sitara AM3359，因为它具有 2-Gb 以太网 MAC，其交换层支持向两个 DP83867IR 发送 RGMII。AM3359 内部的 MAC 交换层支持在不使用核心（core）的情况下，在两个 PHY 之间传递消息的若干功能。为实现确定性的极低发送和接收延迟，可使用 ICSS/PRU 子系统捕获帧起始以进行 IEEE 1588 时间戳。TIDA-00204 硬件已为此功能预留，但尚未测试。

2.3.1.2.1 AM3359 启动模式配置

AM3359 内部 ROM 代码根据设备上电复位时 SYSBOOT 配置电平选择相应的外设。有关此启动过程的完整详情，请参阅 AM335x Sitara 处理器技术参考手册的第 26.1.5 节至第 26.1.8 节。

为便于独立运行，AM3359 已配置为从 MMC1/SD 卡启动。此配置需要 SYSBOOT[15:0] 引脚设置为"01000000 0001 1100"。这是通过将相应的上拉和下拉电阻设置为与图 9 中所示原理图一致来实现的。

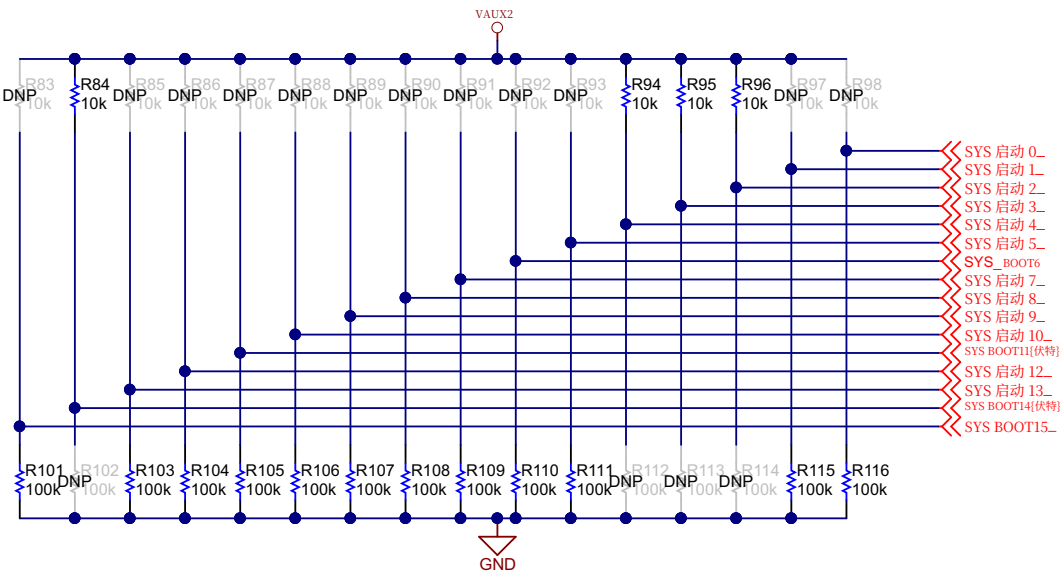


图 9. AM3359 用于 MMC1/SD 卡启动模式的 SYSBOOT 引脚配置

2.3.1.2.2 AM3359 GPIO Pin Assignment

When using the Sitara before starting the design, ensure that the peripherals functions required for the design can be assigned to the specific GPIO pins without a mux conflict. For this purpose, there is a pin mux tool utility for TI processors called PINMUXTOOL-V3. The tool can be downloaded from the TI website (www.ti.com). There are specific versions pending the device, part, and package. This tool is a huge help for AM3359 pin assignment.

This design uses the following peripherals: EMAC with RGMII1 and RGMII2, UART, MMC0 and MMC1, MDIO, SDRAM/DDR3, I²C, and ECAP.

The pin mux tool was used to find the optimum pin assignment for the different peripherals used in the TIDA-00204 reference design.

Table 11. AM3359 Pin Assignment

SIGNAL	PINS (GPIO MODE)	AMOUNT OF PINS
RGMII1	J16-J18, K15-K18, L15-L18, M16 (Mode 2)	12
RGMII2	R13-14, V14-V17, U14-U16, T14-T16 (Mode 2)	12
UART	G15-G16(Mode 3)	2
MMC	U9, V9 (Mode 2), U7, V7, R8, T8 (Mode 1)	6
MDIO	M17-M18 (Mode 0)	2
JTAG	Fixed pins not available with mux	8 to 11
SDRAM (DDR3)	Fixed pins not available with mux	52
I ² C	C16-C17 (Mode 0)	2
ECAP	U13 (Mode 5), C15 (Mode 2), C18 (Mode 0), E15 (Mode 4)	4

The remaining pins could be used in GPIO mode or assigned to unused peripherals on the AM3359 device.

2.3.1.2.2 AM3359GPIO 引脚分配

在设计开始前使用 Sitara 时，请确保设计所需的外设功能可以分配给特定的 GPIO 引脚，而不会发生复用冲突。为此，德州仪器（TI）处理器提供了一款名为 PINMUXTOOL-V3 的引脚复用工具。该工具可从 TI 网站下载（www.ti.com）。该工具有针对特定设备、型号和封装的版本。该工具对 AM3359 引脚分配提供了巨大帮助。

本设计使用以下外设：带有 RGMII1 和 RGMII2 的 EMAC、UART、MMC0 和 MMC1、MDIO、SDRAM/DDR3、I²C 以及 ECAP。

使用引脚复用工具为 TIDA-00204 参考设计中使用的不同外设找到最佳引脚分配。

表 11. AM3359 引脚定义

信号	PINS (GPIO 模式)	引脚数量
RGMII1	J16-J18, K15-K18, L15-L18, M16（模式 2）	12
RGMII2	R13-14, V14-V17, U14-U16, T14-T16（模式 2）	12
UART	G15-G16(模式 3)	2
MMC	U9, V9（模式 2），U7, V7, R8, T8（模式 1）	6
MDIO	M17-M18（模式 0）	2
JTAG	固定引脚与多路复用器不兼容	8 到 11
SDRAM (DDR3)	固定引脚与多路复用器不兼容	52
I ² C	C16-C17（模式 0）	2
ECAP	U13（模式 5），C15（模式 2），C18（模式 0），E15（模式 4）	4

剩余的引脚可用于 GPIO 模式，或分配给 AM3359 设备上未使用的外设。

2.3.1.2.3 MMC1 Micro SD-Card

Figure 10 shows the micro SD-Card interface including ESD protection device on the TIDA-00204. The TPD6E001 is a low-capacitance ± 15 -kV ESD-protection diode array designed to protect sensitive electronics attached to communication lines. Each channel consists of a pair of diodes that steer ESD current pulses to VCC or GND. The TPD6E001 protects against ESD pulses up to ± 15 -kV human-body model (HBM), ± 8 -kV contact discharge (CD), and ± 15 -kV air-gap discharge, as specified in IEC 61000-4-2.

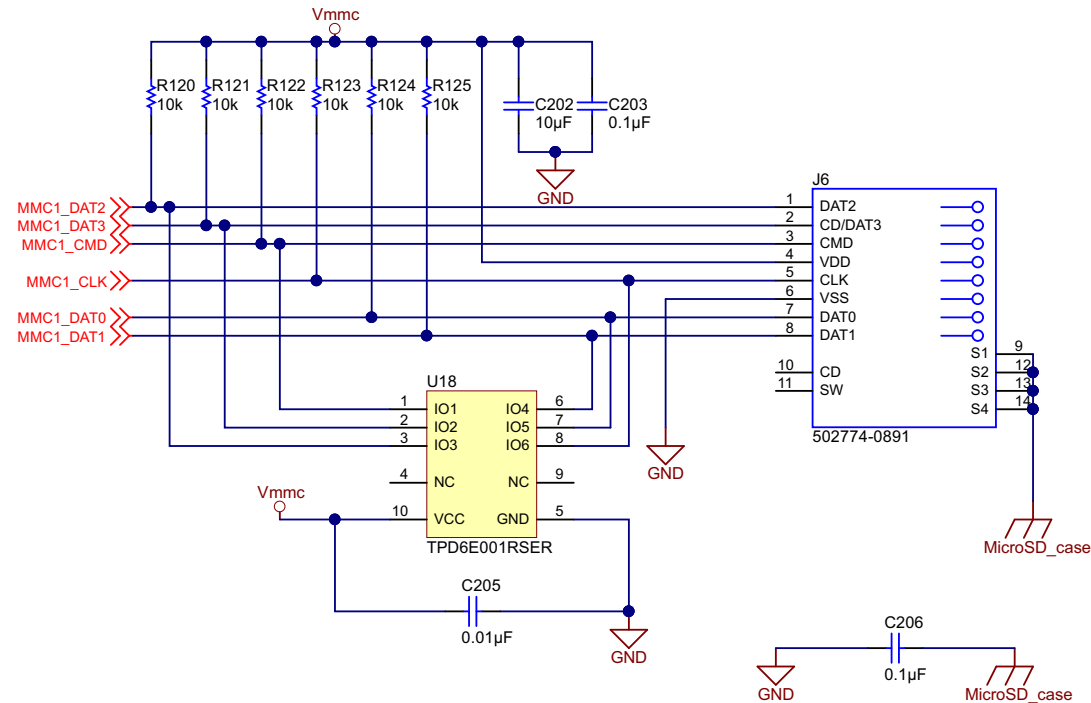


Figure 10. MMC1 Micro SD-Card Interface With ESD Protection

2.3.1.2.4 I²C Communication to PMIC and EEPROM

The I²C module 0 is used for communicating with both, the EEPROM, and the Power Management IC (PMIC). The PMIC I²C is hardcoded to the I²C address equal to 0101101 binary or 0x2D in hex.

The EEPROM was given the binary address 1010000 equal to 0x50 in hex. This is done by setting the A₂, A₁, and A₀ pin. This then forms the 7-bit address as "1010A₂A₁A₀".

2.3.1.2.3 MMC1 Micro SD 卡

图 10 展示了 TIDA-00204 上的 Micro SD 卡接口，包括静电放电（ESD）保护器件。TPD6E001 是一款低电容的 ± 15 kV 静电放电保护二极管阵列，旨在保护连接到通信线路的敏感电子设备。每个通道由一对二极管组成，用于将静电放电电流脉冲引导至 VCC 或 GND。TPD6E001 可防护高达 ± 15 kV 人体模型（HBM） ± 8 kV 接触放电（CD），以及 ± 15 kV 空气间隙放电，如 IEC 61000-4-2 标准所述。

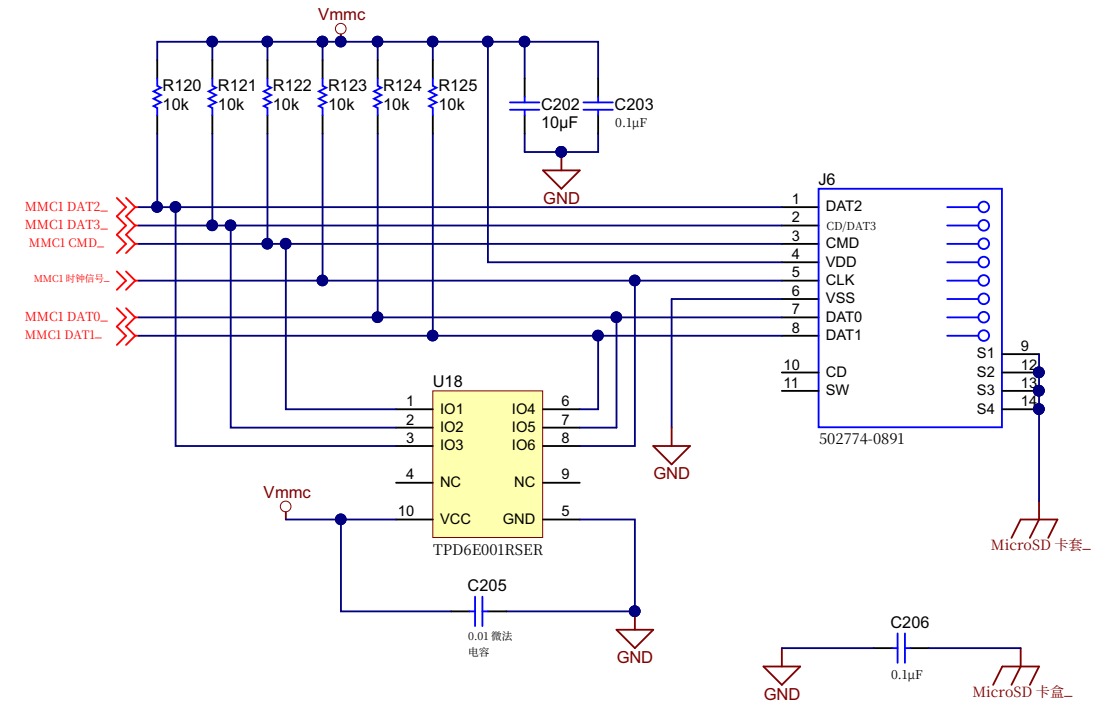


图 10. MMC1 Micro SD 卡接口与静电放电保护

2.3.1.2.4 I²C 与 PMIC 和 EEPROM 的通信

I²C 模块 0 用于与 EEPROM 和电源管理集成电路（PMIC）通信。PMIC I²C 已硬编码为 I²C 地址等于二进制 0101101 或十六进制 0x2D。

EEPROM 被分配了二进制地址 1010000，即十六进制的 0x50。这是通过将 A₂, A₁, 和 A₀ 引脚设置为 1010A₂A₁A₀ 来实现的。

2.3.1.2.5 DDR3 RAM

For this design, the MT41J128M16JT-125K TR DDR3 RAM was chosen to leverage experience from the TI AM3359 Industrial Communications Engine development platform with part number TMD5ICE3359. This memory requires a 1.5-V supply with a rise time better than 200 ms. During this time a V_{TT} supply needs to be provided as this rail is a system supply for signal termination resistors. Once powered, the device needs to be reset before going into the initialization state. Then the memory is ready for operation.

Due to the speed of the signals to the DDR3 RAM, length matching and termination are important to ensure performance. For the data lines, length matching was done by defining different net classes with specific considerations. The concept of defining different net classes is important when routing high speed signals, as they can insure timing constraints or timing relationships. When defining these groups, it is important to understand the basics of the DDR3 memory signals. For DDR3, the signals can be divided into four different groups that have similar requirements.

Table 12. DDR3 Net Class Definitions

NET CLASS	SIGNALS
Data	DDR_D[15:0],DDR_DQM[1:0], DDR_DQS0[_P and _N], DDR_DQS1[_P and _N]
Address/Command	DDR_A[13:0],DDR_BA[2:0], DDR_CASN, DDR_RASN, DDR_WEN
Control	DDR_CKE, DDR_CSN, DDR_ODT, DDR_RESETN
Clock	DDR_CLK_N,DDR_CLK_P

For the different groups, it is possible with either software or hardware to shift the sampling point. Even with this feature of DDR3 RAM, it is still needed to do a timing budget calculation. With this calculation, it is possible to have an estimation of how the length matching needs to be. Length matching is needed due to the fact that the typical propagation delay for the FR4 PCB is approximately 6.5 ps/mm and so a length mismatch would lead to a delay of the signal compared to the other signal lines. As an example the net class data has been matched as seen in [Table 13](#).

Table 13. DDR3 Trace Length on PCB for Net Class Data

SIGNAL	TRACE LENGTH (mm)	SIGNAL	TRACE LENGTH (mm)
DDR_D0	27.4776	DDR_D8	25.3669
DDR_D1	27.6777	DDR_D9	25.2680
DDR_D2	27.4567	DDR_D10	25.4147
DDR_D3	27.2947	DDR_D11	25.6113
DDR_D4	27.4070	DDR_D12	25.6584
DDR_D5	27.5413	DDR_D13	25.3574
DDR_D6	27.4398	DDR_D14	25.5648
DDR_D7	27.3044	DDR_D15	25.4486
DDR_DQM0	27.3479	DDR_DQM1	25.4907
DDR_DQS0_N	27.4755	DDR_DQS1_N	25.4209
DDR_DQS0_P	27.4726	DDR_DQS1_P	25.4285

2.3.1.2.5 DDR3 内存

本设计中，选用 MT41J128M16JT-125KTRDDR3 RAM，以利用 TI AM3359 工业通信引擎开发平台（型号 TMD5ICE3359）的经验。该存储器需要 1.5 V 供电，上升时间优于 200 ms。在此期间，需提供 V_{TT} 供电，因为该轨是信号终止电阻的系统供电。上电后，设备需在进入初始化状态前进行复位。随后，存储器即可投入使用。

由于 DDR3RAM 信号的传输速度，长度匹配和端接对于确保性能至关重要。对于数据线，通过定义具有特定考量的不同网类来实现长度匹配。在高速信号布线时，定义不同网类的概念至关重要，因为它们可以确保时序约束或时序关系。在定义这些组时，必须理解 DDR3 内存信号的基础知识。对于 DDR3，信号可分为四个具有相似要求的不同组。

表 12. DDR3 Net 类定义

NET 类	信号
Data	DDR_D[15:0],DDR_DQM[1:0], DDR_DQS0[_P 和 _N], DDR_DQS1[_P 和 _N]
地址/命令	DDR_A[13:0],DDR_BA[2:0],DDR_CASN, DDR_RASN, DDR_WEN
控制	DDR_CKE, DDR_CSN, DDR_ODT, DDR_RESETN
时钟	DDR 时钟信号 N, DDR 时钟信号 P

对于不同的组，可以通过软件或硬件来调整采样点。即使具备 DDR3 RAM 的这一特性，仍然需要进行时序预算计算。通过该计算，可以估算出长度匹配的需求。由于 FR4 PCB 的典型传播延迟约为 6.5 ps/mm，因此长度不匹配会导致信号相对于其他信号线产生延迟。例如，网类数据已进行匹配，如[表 13](#)。

表 13. PCB 板上 Net 类数据的 DDR3 走线长度

信号	轨迹长度 (mm)	信号	轨迹长度 (mm)
DDR_D0	27.4776	DDR_D8	25.3669
DDR_D1	27.6777	DDR_D9	25.2680
DDR_D2	27.4567	DDR_D10	25.4147
DDR_D3	27.2947	DDR_D11	25.6113
DDR_D4	27.4070	DDR_D12	25.6584
DDR_D5	27.5413	DDR_D13	25.3574
DDR_D6	27.4398	DDR_D14	25.5648
DDR_D7	27.3044	DDR_D15	25.4486
DDR_DQM0	27.3479	DDR_DQM1	25.4907
DDR_DQS0_N	27.4755	DDR_DQS1_N	25.4209
DDR_DQS0_P	27.4726	DDR_DQS1_P	25.4285

Termination was done with parallel termination of 47 Ω to improve performance. Arrays were chosen here to minimize size.

Further general considerations are:

- All nets in the address and command fly-by groups shall have the same number of vias in each length-matched segment. Ground vias are placed to ensure a proper current return path. Minimize use of vias on signal traces as they negatively impact signal integrity.
- The single-ended Address/Command net class and the control net class needs external termination to V_{TT} .
- For the data net class within a byte lane, the data bits can be swapped to simplify routing.
- Organize the power, ground, and signal planes to eliminate or significantly reduce the number of split or cut planes present in the design (no splits are allowed under any DDR3 routes).
- Maintain an acceptable level of skew across the entire DDR3 interface (by net class).
- It is strongly recommended that all nets be simulated to assure proper design, performance, and signal integrity.
- Take into account the differences in propagation delays between microstrip and stripline nets when evaluating timing constraints. All long routes should be stripline to reduce EMI and timing skew, and any microstrip routed for BGA breakouts should be as short as possible.
- Routes along the same path and routing segment must have the same number of vias. Vias can be blind, buried, or HDI microvia for improved signal integrity, but are not required for standard data rates. Similarly, back drilling vias is not required for standard data rates but can be used to eliminate via stubs.
- For this design, the DDR3 layout was copied from the TI AM3359 Industrial Communications Engine development platform with part number TMDSICE3359 to leverage a working and fully tested design with the peripheral settings for the DDR3 interface.

2.3.1.2.6 AM3359 Clocking Options

For the clock, either a crystal or a digital clock source can be used. For both cases, the clock needs to be 19.2, 24, 25, or 26 MHz with a tolerance of ± 50 ppm. For more details, see Section 6.2.2 of the AM3359 data sheet[5]. For this design, a 24-MHz crystal was chosen as used with the TMDSICE3359 AM3359 Industrial Communications Engine development platform.

The preferred option, as realized with revision E3 of this design, is to have all system clocks in the design synced to a single reference clock. This consolidation helps to reduce jitter between the individual clocks on the board. This option is a key feature for system performance between the PHY and the MAC layer communication, especially in real-time Ethernet systems. See Section 2.3.1.4.2 for more details.

The CDCE913 programmable 1-PLL VCXO clock synthesizer is used to generate the 24-MHz clock for the AM3359.

2.3.1.2.7 Power Supply Pins

For the decoupling capacitors on the different power rails of the AM3359, see Section 5.9 of the AM3359 data sheet[5].

使用 47 Ω 的并行终止完成终止，以提高性能。此处选择数组以最小化大小。

其他一般性考虑如下：

- 地址和控制飞越组中的所有网络在每个等长段中必须具有相同数量的过孔。放置地过孔以确保正确的电流返回路径。尽量减少信号走线上使用过孔，因为它们会对信号完整性产生负面影响。

- 单端地址/命令网类和控网类需要外部终端到 V_{TT} 。

- 对于字节通道内的数据网类，数据位可以交换以简化布线。
- 组织电源、地和信号层，以消除或显著减少设计中存在的分割或切割层（在 DDR3 走线中不允许任何分割）。

- 在整个 DDR3 接口（按网络类）上保持可接受的 skew 水平。
- 强烈建议对所有网络（电路）进行仿真，以确保设计正确、性能良好以及信号完整性。

- 在评估时序约束时，请考虑微带线和带状线网络之间的传播延迟差异。所有长路径应使用带状线以减少电磁干扰和时序偏差，任何用于 BGA breakout 的微带线路由应尽可能短。

- 同一路径和路由段上的走线必须具有相同数量的过孔。过孔可以是盲孔、埋孔或 HDI 微孔，以改善信号完整性，但标准数据速率下并非必需。
同样，标准数据速率下不需要背钻过孔，但可以使用背钻过孔来消除过孔盲端。

- 本设计中的 DDR3 布局复制自德州仪器（TI）AM3359 工业通信引擎开发平台（型号 T MDSICE3359），以利用经过验证且完全测试的设计，并配置 DDR3 接口的周边设置。

2.3.1.2.6 AM3359 时钟选项

对于时钟，可以使用晶振或数字时钟源。对于这两种情况，时钟频率需为 19.2、24、25 或 26 MHz，容差为 ± 50 ppm。有关详细信息，请参阅 AM3359 数据手册第 6.2.2 节[5]。本设计选用 24-MHz 晶振，与 TMDSICE3359 AM3359 工业通信引擎开发平台所用相同。

本设计首选方案（在 E3 版本中实现）是将设计中的所有系统时钟与单个参考时钟同步。这种整合有助于减少板载各个时钟之间的抖动。该选项是物理层与 MAC 层通信之间系统性能的关键特性，特别是在实时以太网系统中。请参阅第 2.3.1.4.2 节以获取更多信息。

CDCE913 可编程 1-PLL VCXO 时钟合成器用于为 AM3359 生成 24 兆赫时钟。

2.3.1.2.7 电源供应器引脚

对于 AM3359 不同电源轨上的去耦电容，请参阅 AM3359 数据手册[5]的第 5.9 节。

2.3.1.2.8 USB Virtual COM Port and JTAG Interface

An FTDI chip connects to a micro USB port. This FTDI chip can be used for both JTAG (XDS100) and Virtual COM port functionality. A separate JTAG header is available too.

To use the FTDI chip, the onboard EEPROM, which is connected to the FTDI chip, needs to be programmed as follows:

- To program the EEPROM connected to the FTDI chip, get the tool "Mprog". To find this tool, go to the following webpage for the download link.
http://processors.wiki.ti.com/index.php/XDS100#How_to_make_an_XDS100
- The EEPROM needs to be programmed in a specific way. To find an example image also called a *.ept file to write in the EEPROM, go to the following webpage:
<http://processors.wiki.ti.com/index.php/XDS100>

A .ept file for the Mprog tool is available for the specific XDS100 version needed. Download a .ept file, which includes both JTAG and Virtual COM port functionality.

To verify the image is programmed correctly, power the board and connect the micro USB to the PC. Now go into the start menu and open the menu *Devices and Printers* for Windows® 7. The following connection should now be found (see [Figure 11](#)).

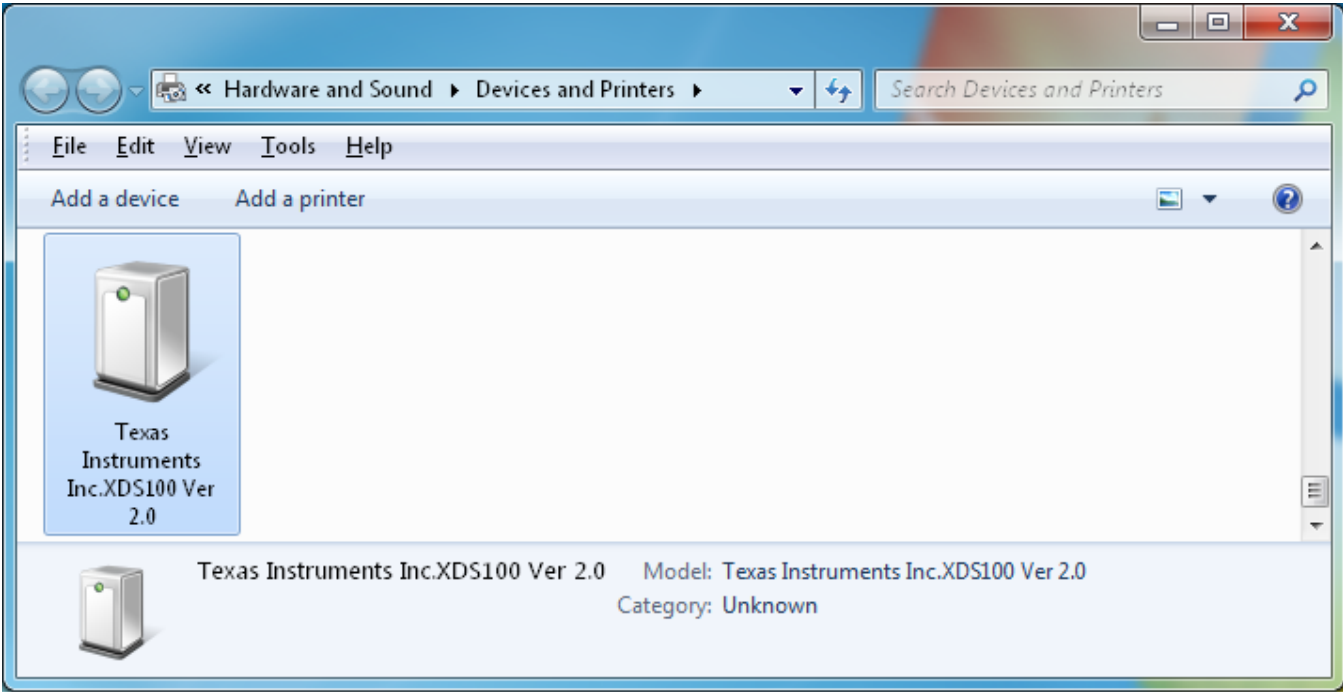


Figure 11. XDS100v2 Driver in Windows 7 Devices and Printers Menu

If [Figure 11](#) does not appear, retry the previous steps.

When this is done properly, Code Composer Studio™ v6 can be used to connect to the AM3359 core through JTAG using the XDS100v2 driver.

2.3.1.2.8 USB 虚拟 COM 端口和 JTAG 接口

一个 FTDI 芯片连接到微 USB 端口。该 FTDI 芯片可用于 JTAG (XDS100) 和虚拟 COM 端口功能。此外，还有一个独立的 JTAG 连接器。

要使用 FTDI 芯片，需要对其连接的板载 EEPROM 进行如下编程：

- 要编程连接到 FTDI 芯片的 EEPROM，请获取工具"Mprog"。要找到该工具，请访问以下网页以获取下载链接。
http://processors.wiki.ti.com/index.php/XDS100#如何制作_XDS100
- EEPROM 需要以特定方式进行编程。要查找示例图像（也称为 *.ept 文件写入 EEPROM，请访问以下网页：
<http://processors.wiki.ti.com/index.php/XDS100>

针对特定 XDS100 版本，Mprog 工具提供 .ept 文件。下载 .ept 文件，该文件包含 JTAG 和虚拟 COM 端口功能。

为验证图像是否正确编程，请上电并将微 USB 连接到 PC。现在进入开始菜单并打开菜单 *设备和打印机*（适用于 Windows® 7）。现在应该能找到以下连接（参见 [图 11](#)）。

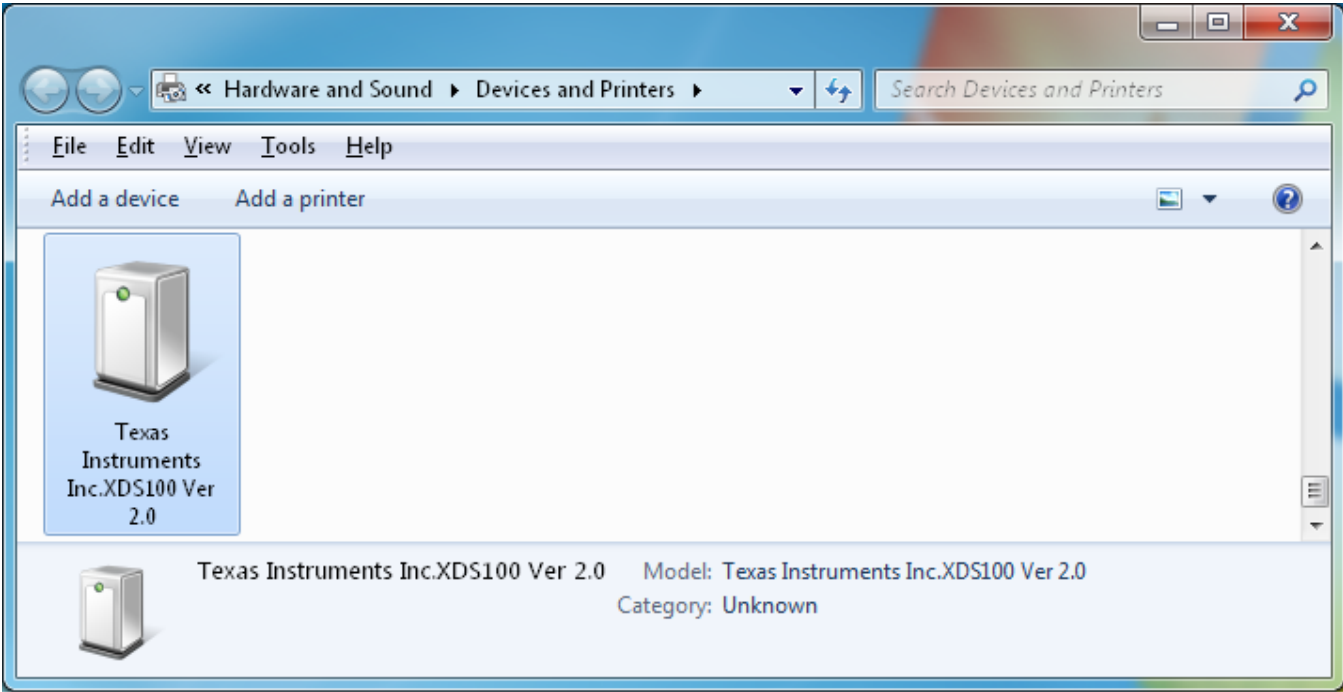


图 11. Windows 7 设备和打印机菜单中的 XDS100v2 驱动程序

如果 [图 11](#)未出现，请重试前面的步骤。

完成此步骤后，可使用 Code Composer Studio™ v6 通过 JTAG 连接至 AM3359 核心，并使用 XDS100v2 驱动程序。

2.3.1.3 Power Supplies

The power supplies are realized in three stages. The first stage is a 24-V to 5-V DC/DC to provide an intermediate 5-V rail to the other two stages. The second stage is a PMIC supplying the host processor, and the last stage is supplying the two Gigabit PHYs.

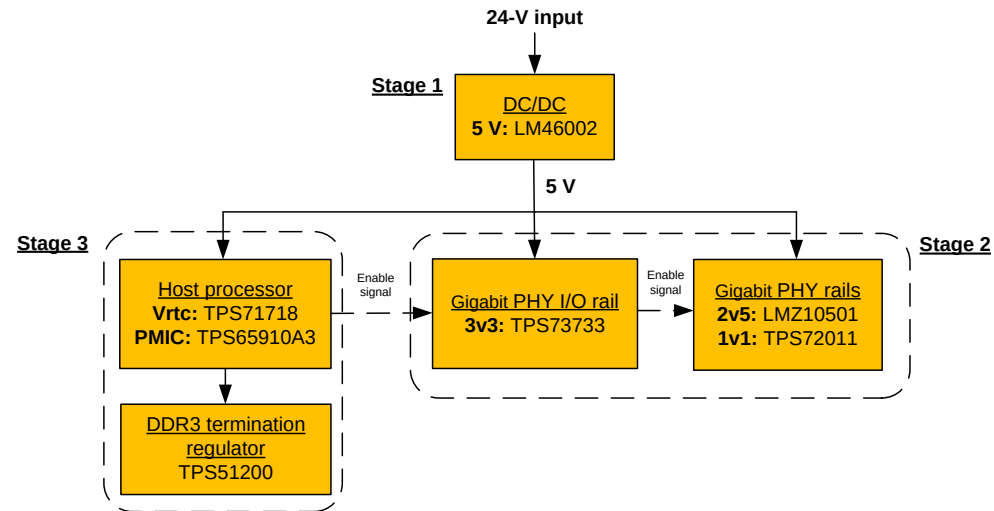


Figure 12. TIDA-00204 Power Supply Stages

2.3.1.3.1 24-V to 5-V DC/DC Buck Converter

For this stage, the LM46002 SIMPLE SWITCHER® regulator was chosen. It is an easy-to-use synchronous step-down DC-DC converter capable of driving up to 2 A of load current from an input voltage ranging from 3.5 to 60 V. The LM46002 provides exceptional efficiency, output accuracy, and drop-out voltage in a very small solution size. The family requires few external components. Pin arrangement allows for a simple, optimum PCB layout.

An extended family is available in various load current options with pin-to-pin compatible packages, including LM46001.

The schematic of the LM46002 is shown in Figure 13.

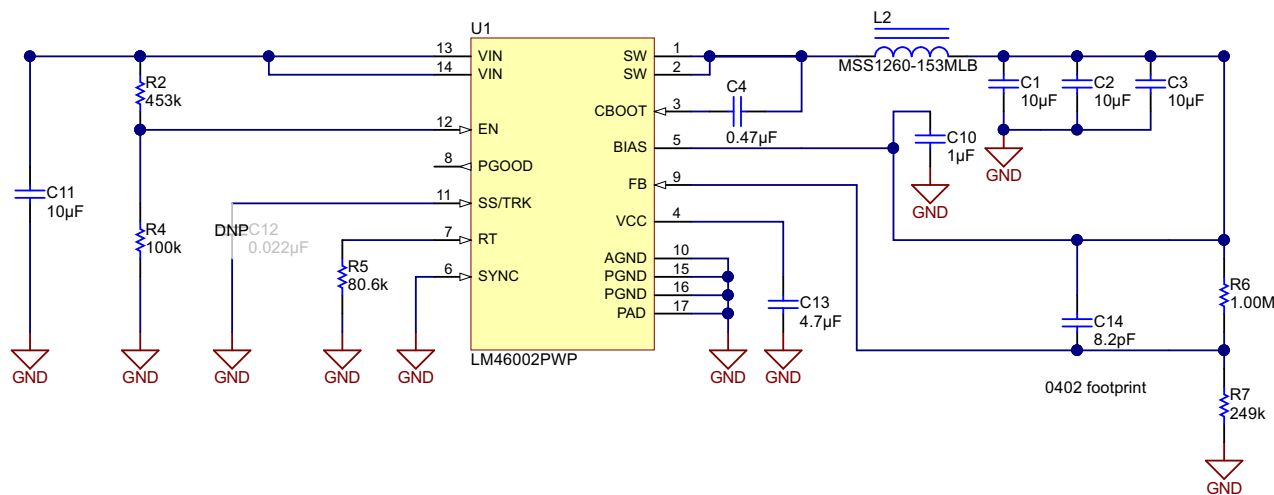


Figure 13. Schematic of 24-V to 5-V DC/DC Buck Converter With LM46002

2.3.1.3 电源

电源供应分为三个阶段。第一阶段是 24 伏至 5 伏的 DC/DC 转换器，为另外两个阶段提供中间 5 伏轨。第二阶段是向主机处理器供电的电源管理集成电路，最后阶段是为两个千兆位物理层芯片供电。

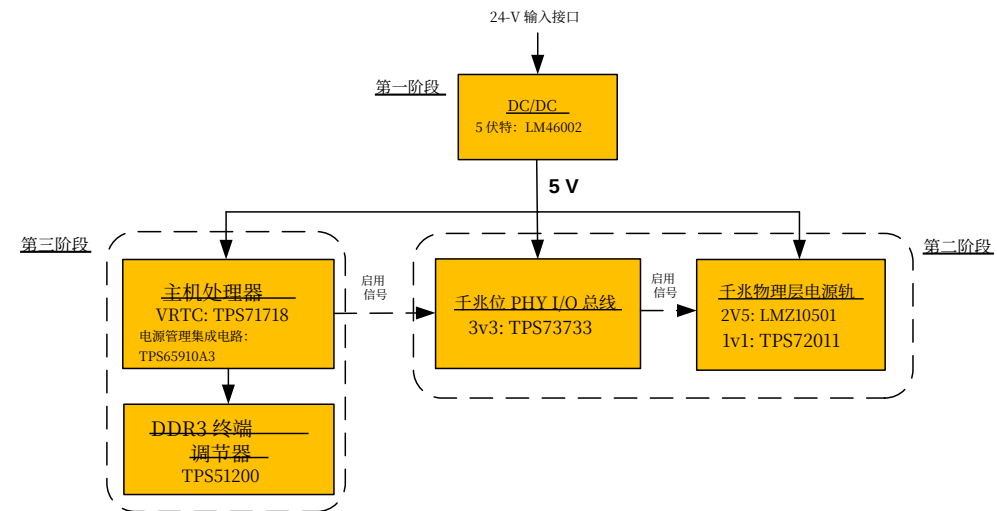


图 12. TIDA-00204 电源供应器级

2.3.1.3.1 24 伏转 5 伏 DC/DC 降压转换器

本阶段选用 LM46002 SIMPLE SWITCHER® 调节器。它是一款易于使用的同步降压 DC-DC 转换器，可从 3.5 至 60 V 的输入电压驱动高达 2 A 的负载电流。LM46002 在非常小的封装尺寸下提供卓越的效率、输出精度和关断电压。该系列仅需少量外部组件。引脚排列支持简单且优化的 PCB 板布局。

提供具有引脚对引脚兼容封装的多种负载电流选项的扩展型器件，包括 LM46001。

LM46002 的电路原理图如图 13 所示。

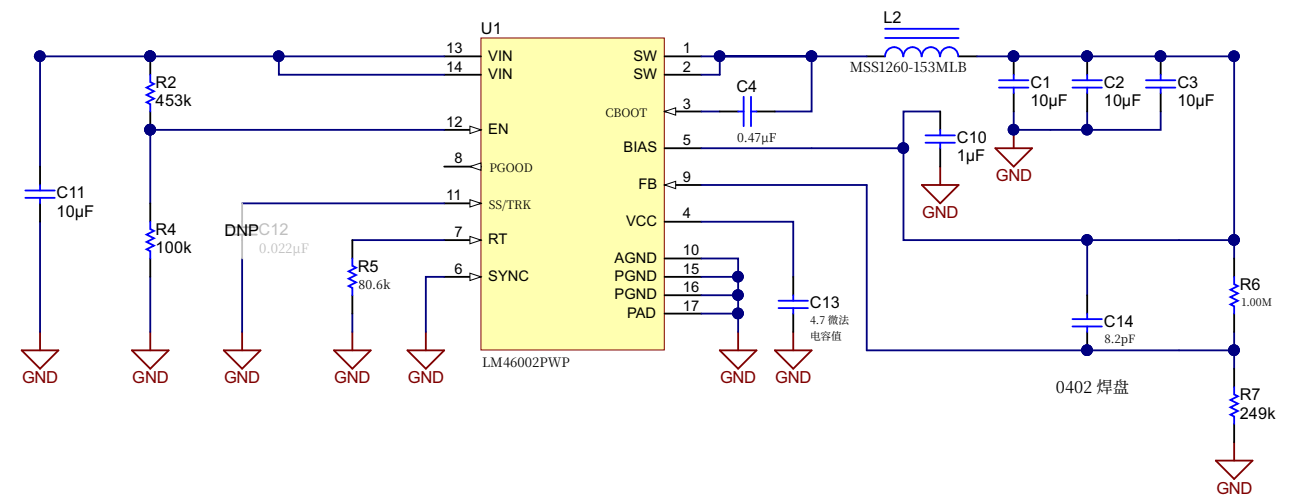


图 13. 带有 LM46002 的 24 V 至 5 V DC/DC 降压转换器原理图

2.3.1.3.1.1 Passive Components Calculations

The first step is to decide and set the switching frequency of the regulator. In the TIDA-00204, the switching frequency (F_{sw}) is set to 500 kHz thanks to R_5 . [Equation 1](#) indicates that R_5 should be 79.8 k Ω . 80.6 k Ω is then used as value for R_5 .

$$R_5 = \frac{40200}{F_{sw}} - 0.6 \quad (1)$$

with

- F_{sw} in kHz and R_5 in k Ω

The default frequency is also 500 kHz, due to the internal oscillator precision of 20% it was chosen to use a resistor for the 500-kHz frequency. This gives a 10% precision. Secondly, during EMI test it can be useful for fine tuning to move the frequency up or down to change the harmonic spectrum of the converter.

Once the switching frequency is set, the size of inductance needed it chosen. For most buck converters, this value is chosen to achieve the wanted peak-to-peak ripple current that flows in the inductor along with the DC load current. A higher inductance gives lower ripple current, which then gives a lower output voltage ripple with the same output capacitors. An inductance that gives a ripple current of 20% to 40% at the maximum current is normally a good starting point. The minimum inductor value is calculated based on input voltage range (17 to 60 V), output voltage (5 V), maximum output current (2 A) and the switching frequency (500 kHz) as set in the previous step. The duty cycle (D) can be approximated as $D = V_{OUT}/V_{IN}$, assuming no loss power conversion.

$$\frac{(V_{IN} - V_{OUT}) \times D}{0.4 \times F_{sw} \times I_{OUT}} \leq L_{min} \leq \frac{(V_{IN} - V_{OUT}) \times D}{0.2 \times F_{sw} \times I_{OUT}} \quad (2)$$

The calculation gives that the inductor needs to be between 9 and 23 μ H; for this design, 15 μ H was chosen.

Besides the inductor size, the rated saturation current needs to be specified as per [Equation 4](#).

$$I_{RIPPLE} = \frac{(V_{IN} - V_{OUT}) \times D}{L_2 \times F_{sw}} \quad (3)$$

$$I_{L-peak} = I_{OUT} + \frac{I_{RIPPLE}}{2} \quad (4)$$

The inductor peak current is 2.305 A according to [Equation 3](#) and [Equation 4](#). A good practice is to use an inductor with the saturation current around 1.5 to 2 times higher than calculated.

The LM46002 has both valley current limit and peak current limit. During an instantaneous short, the peak inductor current can be high due to a momentary increase in duty cycle. The inductor current rating should be higher than the peak current limit. Select an inductor with a larger core saturation margin and preferably a softer roll off of the inductance value over load current.

In general, it is preferable to choose lower inductance in switching power supplies, because it usually corresponds to faster transient response, smaller DCR, and reduced size for more compact designs. But a too low inductance can generate too large inductor current ripple such that over current protection at the full load could be falsely triggered. It also generates more conduction loss because the RMS current is slightly higher relative that with lower current ripple at the same DC current. Larger inductor current ripple also implies larger output voltage ripple with the same output capacitors. With peak current mode control, it is not recommended to have too small of an inductor current ripple. Enough of an inductor current ripple improves the signal-to-noise ratio on the current comparator and makes the control loop more immune to noise.

2.3.1.3.1.1 无源组件计算

第一步是确定并设置调节器的开关频率。在 TIDA-00204 中，开关频率 (F_{sw}) 设定为 500 kHz，这得益于 R_5 。 [公式 1](#)表明 R_5 应为 79.8 k Ω . 80.6 k Ω ， 然后将其用作 R_5 的值。

$$R_5 = \frac{40200}{F_{sw}} - 0.6 \quad (1)$$

with

- F_{sw} 开关电容滤波器 in kHz 和 R_5 in k Ω

默认频率也为 500 kHz， 由于内部振荡器的精度为 20%， 因此选择使用电阻来实现 500 kHz 的频率。这提供了 10% 的精度。其次， 在 EMI 测试期间， 它可以用于微调， 将频率向上或向下移动， 以改变转换器的谐波谱。

一旦设定了开关频率， 所需电感值即可确定。对于大多数降压转换器， 该值通常被选择为在电感中实现所需的峰峰值纹波电流， 该纹波电流与直流负载电流一同流过电感。较高的电感值会产生较低的纹波电流， 从而在相同的输出电容下获得更低的输出电压纹波。在最大电流下产生 20% 至 40% 纹波电流的电感值通常是一个良好的起点。最小电感值是根据输入电压范围（17 至 60 V）、输出电压（5 V）、最大输出电流（2 A）以及在上一步中设定的开关频率（500 kHz）计算得出的。占空比 (D) 可近似为 $D = V_{OUT}/V_{IN}$ ， 假设无损耗功率转换。

$$\frac{(V_{IN} - V_{OUT}) \times D}{0.4 \times F_{sw} \times I_{OUT}} \leq L_{min} \leq \frac{(V_{IN} - V_{OUT}) \times D}{0.2 \times F_{sw} \times I_{OUT}} \quad (2)$$

计算表明， 电感线圈需要在 9 到 23 μ H 之间； 为此设计， 选择了 15 μ H。

除了电感线圈尺寸外， 额定饱和电流也需根据 [公式 4](#)。

$$I_{RIPPLE} = \frac{(V_{IN} - V_{OUT}) \times D}{L_2 \times F_{sw}} \quad (3)$$

$$I_{L-peak} = I_{OUT} + \frac{I_{RIPPLE}}{2} \quad (4)$$

电感线圈的峰值电流为 2.305 A， 根据 [公式 3](#) 和 [公式 4](#)。 良好的做法是使用饱和电流约为计算值 1.5 到 2 倍的电感线圈。

LM46002 具有谷值电流限制和峰值电流限制。在瞬时短路期间， 由于占空比暂时增加， 电感线圈的峰值电流可能很高。电感线圈的额定电流应高于峰值电流限制。请选择具有较大磁芯饱和余量且电感值随负载电流变化时滚降更平缓的电感线圈。

一般来说， 在开关电容电源中， 选择较小的电感通常更优， 因为它通常对应更快的瞬态响应、 更小的 DCR 以及更小的尺寸， 从而适用于更紧凑的设计。但电感过小会产生过大的电感电流纹波， 导致满载时过流保护被错误触发。此外， 由于在相同的直流电流下， 较小的电流纹波会导致更大的 RMS 电流， 因此会产生更多的导通损耗。较大的电感电流纹波在相同的输出电容下也会意味着更大的输出电压纹波。在峰值电流模式控制中， 不建议电感电流纹波过小。足够的电感电流纹波可以提高电流比较器的信噪比， 并使控制环路对噪声具有更强的免疫力。

The maximum desired output voltage ripple and the transient response to load changes are the parameters to take into account for selecting the value of the output capacitors.

$$C_{OUT} > \frac{1}{F_{sw} \times r \times \Delta V_{OUT} / \Delta I_{OUT}} \times \left(\left(\frac{r}{12} \times (1 + (1 - D)) \right) + ((1 - D) \times (1 + r)) \right) \quad (5)$$

with

- ΔI_{OUT} equal load current (2 A)
- ΔV_{OUT} equal target output voltage undershoot (0.5 V)
- r the inductor ripple ratio ($\Delta I_L / I_{OUT}$)

Equation 5 indicates that C_{OUT} should be higher than 10.0 μF . A general guideline would be to pick C_{OUT} between the minimum required output capacitance, calculated in Equation 6, and multiply this minimum required capacitance by 10. Two 10.0 μF in parallel were chosen to fit the C_{OUT} requirements.

The second requirement is the maximum ESR of the capacitor, which can be found using Equation 6.

$$ESR < \frac{(1 - D)}{F_{sw} \times C_{OUT}} \times \left(\frac{1}{r} + 0.5 \right) \quad (6)$$

Equation 6 calculates the maximum ESR of the output capacitor to meet the maximum output ripple required. The equivalent ESR of the output capacitors C_3 and C_4 should be lower than 0.288 Ω .

The LM46002 requires high quality ceramic (X5R or X7R) input decoupling capacitors. The recommended value is between 4.7 to 10 μF . A good practice for ceramic capacitors is to choose a voltage rating of twice the maximum input voltage. If the placement of the LM46002 additional bulk capacitance is needed, this additional capacitor dampens voltage spikes caused by the lead inductance of the trace. The value is not critical but it must be rated to fit the voltage requirements.

The output voltage is externally adjusted with a resistor divider network. The divider network is comprised of top feedback resistor R_6 and bottom feedback resistor R_7 . Equation 7 is used to determine the output voltage of the converter.

$$R_7 = \frac{V_{FB}}{V_{OUT} - V_{FB}} \times R_6 \quad (7)$$

R_6 is chosen to be 1 M Ω to minimize quiescent current, which improves light load efficiency in this application. With the desired output voltage set to be 5 V and the V_{FB} is equal to 1.01 V, the R_7 value can then be calculated using Equation 7. The equation gives the value 253.4 Ω . The chosen value for this design is a 249-k Ω resistor. For more details on the adjustable output voltage, see the LM46002 data sheet[11].

The LM46002 is internally compensated and the internal R-C values are 400 k Ω and 50 pF, respectively. Depending on the V_{OUT} and the frequency F_S , if the output capacitor C_{OUT} is dominated by low ESR (ceramic types) capacitors, it could result in low-phase margin. To improve the phase boost, an external feed forward capacitor C_{14} can be added in parallel with R_6 . C_{14} is chosen such that the phase margin is boosted at the crossover frequency without C_{14} . A simple estimation for the crossover frequency without C_{14} (F_X) is shown in Equation 8.

$$f_X = \frac{4.35}{V_{OUT} \times C_{OUT}} \quad (8)$$

The following equation for C_{14} was tested:

$$C_{14} = \frac{1}{2 \times \pi \times F_X} \times \frac{1}{\sqrt{R_6 \times (R_6 \parallel R_7)}} \quad (9)$$

Equation 9 indicates that the crossover frequency is geometrically centered on the zero and pole frequencies caused by the C_{14} capacitor.

For designs with higher ESR, C_{14} is not needed. When C_{OUT} has medium ESR, C_{14} calculated from Equation 9 should be reduced. With Low ESR used the calculated value. Table 2 of the LM46002 data sheet can be used as a starting point. The calculated value is 8.2 pF, and an 8.2-pF COG capacitor was selected for C_{14} .

最大期望输出纹波电压和负载变化时的瞬态响应是选择输出电容值时需要考虑的参数。

$$C_{OUT} > \frac{1}{F_{sw} \times r \times \Delta V_{OUT} / \Delta I_{OUT}} \times \left(\left(\frac{r}{12} \times (1 + (1 - D)) \right) + ((1 - D) \times (1 + r)) \right) \quad (5)$$

with

- ΔI_{OUT} 等负载电流 (2 A)
- ΔV_{OUT} 等目标输出电压下陷 (0.5 伏特)
- 电感线圈纹波比 ($\Delta I_L / I_{OUT}$)

公式 5表明 C_{OUT} 应高于 10 伏特。0 μF 。一般指导原则是选择 C_{OUT} 在公式 6中计算的最小所需输出电容, 并将其乘以 10。两个 100 μF 并联以符合 C_{OUT} 要求。

电容的最大等效串联电阻是第二个要求, 可通过公式 6。

$$ESR < \frac{(1 - D)}{F_{sw} \times C_{OUT}} \times \left(\frac{1}{r} + 0.5 \right) \quad (6)$$

公式 6用于计算输出电容的最大等效串联电阻, 以满足所需的最大输出纹波。输出电容的等效 ESR C_3 和 C_4 应低于 0.288 Ω 。

LM46002 需要高质量的陶瓷 (X5R 或 X7R) 输入去耦电容。推荐值在 4.7 到 10 μF 之间。对于陶瓷电容的良好实践是选择额定电压为最大输入电压的两倍。如果 LM46002 需要额外的bulk 电容, 该额外电容可阻尼由走线电感引起的电压尖峰。该值并不关键, 但必须满足电压要求。

输出电压通过外部电阻分压网络进行调整。该分压网络由上反馈电阻 R_6 和下反馈电阻 R_7 组成。公式 7用于确定转换器的输出电压。

$$R_7 = \frac{V_{FB}}{V_{OUT} - V_{FB}} \times R_6 \quad (7)$$

R_6 被选为 1 M Ω 以最小化静态电流, 从而提高本应用中的轻载效率。将期望输出电压设置为 5 V, 且 V_{FB} 等于 1.01 V, 则 R_7 的值可使用 公式 7计算。该公式得出的值为 253.4 Ω 。本设计所选值为 249-k Ω 电阻。有关可调输出电压的更多详情, 请参阅 LM46002 数据手册[11]。

LM46002 内部已进行补偿, 内部 R-C 值分别为 400 k Ω 和 50 pF。根据 V_{OUT} 和频率 F_S , 如果输出电容 C_{OUT} 由低等效串联电阻 (ESR) (陶瓷类型) 电容主导, 可能会导致低相位余量。为了提高相位裕度, 可以在与 R_6 并联的外部前馈电容 C_{14} 。 C_{14} 的选择应使得在穿越频率处提升相位裕度, 而 C_{14} 。一个简单的穿越频率估算公式 (无 C_{14} (F_X) 在 公式 8中给出。

$$f_X = \frac{4.35}{V_{OUT} \times C_{OUT}} \quad (8)$$

对 C_{14} 进行了以下方程的测试:

$$C_{14} = \frac{1}{2 \times \pi \times F_X} \times \frac{1}{\sqrt{R_6 \times (R_6 \parallel R_7)}} \quad (9)$$

公式 9表明, 交叉频率在由 C_{14} 电容引起的零点和极点频率之间几何居中。

对于高 ESR 的设计, C_{14} 无需此元件。当 C_{OUT} 具有中等 ESR 时, C_{14} 应根据 公式 9 进行降额。使用低 ESR 时, 则采用计算值。LM46002 数据表第 2 表可作为起点。计算值为 8.2 pF, 并选用 8.2-pF COG 电容用于 C_{14} 。

The LM46002 needs a bootstrap capacitor C_4 . The recommended bootstrap capacitor is 0.47 μF and rated at 6.3 V or higher. The bootstrap capacitor must be a high quality ceramic type with X7R or X5R grade dielectric for temperature stability.

The V_{CC} pin is the output of an internal LDO for the LM46002. The input for this LDO comes from either V_{IN} or bias. To ensure stability of the part, place a minimum of a 2.2- μF , 10-V capacitor from this pin to ground.

For an output voltage of 3.3 V or higher, the BIAS pin can be connected to the output in order to increase efficiency. This pin is an input for the V_{CC} LDO. When BIAS is not connected, the input for the V_{CC} LDO will be internally connected into V_{IN} . Because this is an LDO, the voltage differences between the input and output will affect the efficiency of the LDO. If necessary, a capacitor with the value of 1 μF can be added close to the BIAS pin as an input capacitor for the LDO.

The soft start capacitor (C_{12}) determines the minimum amount of time it will take the output voltage to reach its programmed value during start up. This also allows limiting the inrush current in the LM46002 (current require to charge the output capacitors to the programmed value). If this pin is left unconnected, the soft start time is 4.1 ms typically. Longer soft start times can be set by an external soft start capacitor per [Equation 10](#).

$$C_{12} = I_{\text{SSC}} \times t_{\text{ss}} \quad (10)$$

with

- C_{12} equal the soft start capacitor value (μF)
- I_{SSC} the soft start charging current (μA)
- t_{ss} equals the desired soft start times

The I_{SSC} current is 2.2 μA , defining a soft start time of 10 ms would equal a soft start capacitor of 0.022 μF .

The undervoltage lockout (UVLO) is adjusted using the external voltage divider network of R_2 and R_4 . R_2 is connected between the VIN pin and the EN pin of the device. R_4 is connected between the EN pin and the GND pin. The UVLO has two thresholds: one for power up when the input voltage is rising, and one for power down or brown outs when the input voltage V_{IN} is falling. Use [Equation 11](#) to determine the $V_{\text{UVLO-Falling}}$ level.

$$R_2 = \left(\frac{V_{\text{UVLO-Falling}}}{V_{\text{ENL}}} - 1 \right) \times R_4 \quad (11)$$

The enable falling edge threshold (V_{ENL}) for the LM46002 is 1.8 V. R_4 was chosen to 100 k to minimize input current from the supply. If the desired VIN UVLO falling level is 10 V, the value of R_2 can be calculated using [Equation 12](#). That is equal to 455.6 k. R_2 was chosen to be 453 k.

$$V_{\text{UVLO-Rising}} = V_{\text{ENH}} \times \frac{(R_4 + R_2)}{R_4} \quad (12)$$

The enable rising edge threshold (V_{ENH}) for the LM46002 is 2.1 V. [Equation 12](#) can be used to calculate the required input voltage to meet the rising threshold to enable the converter. With the above chosen resistors, the LM46002 will enable at 11.6 V.

LM46002 需要一个自举电容 C_4 。推荐的自举电容为 0.47 μF ，额定电压为 6.3 V 或更高。自举电容必须是高质量陶瓷类型，具有 X7R 或 X5R 等级电介质，以确保温度稳定性。

V_{CC} 引脚是 LM46002 内部低压差线性稳压器的输出。该 LDO 的输入来自 V_{IN} 或偏置电压。为确保器件稳定，需在此引脚与地之间放置一个最小值为 2.2- μF 、额定电压为 10-V 的电容。

对于 3.3 V 或更高的输出电压，BIAS 引脚可连接到输出端以提高效率。该引脚是 V_{CC} LDO 的输入端。当 BIAS 未连接时， V_{CC} LDO 的输入将内部连接到 V_{IN} 。由于这是 LDO，输入与输出之间的电压差会影响 LDO 的效率。如有必要，可在 BIAS 引脚附近添加一个容量为 1 μF 的电容，作为 LDO 的输入电容。

软启动电容 (C_{12}) 决定了输出电压在启动期间达到其设定值所需的最小时间。这同时也限制了 LM46002 中的浪涌电流（即给输出电容充电至设定值所需的电流）。如果该引脚保持悬空，软启动时间通常为 4.1 ms。较长的软启动时间可通过外部软启动电容按 [公式 10](#)。设置。

$$C_{12} = I_{\text{SSC}} \times t_{\text{ss}} \quad (10)$$

with

- C_{12} 等于软启动电容值 (μF)
- I_{SSC} 软启动充电电流 (μA)
- t_{ss} 等于所需的软启动时间

The 电流为 2.2 μA ，设定软启动时间为 10 ms 将等于软启动电容为 0.022 μF 。欠压锁定 (UVLO) 使用外部电阻分压网络 R_2 和 R_4 。 R_2 连接在 VIN 引脚和设备的 EN 引脚之间。 R_4 连接在 EN 引脚和 GND 引脚之间。UVLO 有两个阈值：一个用于输入电压上升时的上电，另一个用于输入电压 V_{IN} 下降时的掉电或欠压。使用 [公式 11](#) 来确定 $V_{\text{UVLO-Falling}}$ 电平。

$$R_2 = \left(\frac{V_{\text{UVLO-Falling}}}{V_{\text{ENL}}} - 1 \right) \times R_4 \quad (11)$$

使能下降沿阈值 (V_{ENL}) 对于 LM46002 为 1.8 V。 R_4 被选为 100 k 以最小化来自电源的输入电流。如果所需的 VIN UVLO 下降电平为 10 V，则 R_2 的值可使用 [公式 12](#) 计算。即等于 455.6 k。 R_2 被选为 453 k。

$$V_{\text{UVLO-Rising}} = V_{\text{ENH}} \times \frac{(R_4 + R_2)}{R_4} \quad (12)$$

使能上升沿阈值 (V_{ENH}) 对于 LM46002 为 2.1 V。 [公式 12](#) 可用于计算满足使能转换器的上升阈值所需的输入电压。使用上述选定的电阻，LM46002 将在 11.6 V 时使能。

2.3.1.3.2 Power Supply for Two Gigabit Ethernet PHYs

The PHY power supplies need to provide three power rails. The 3.3-V V_{DDIO} power rail is sequenced from the 3.3-V PMIC rail (see [Section 2.3.1.3.2.1](#)). When the V_{DDIO} power rail is enabled, it will enable the remaining two rails 2.5 V and 1.1 V as seen in [Figure 14](#). The information on the power consumption of the Gigabit Ethernet PHY can be seen in Section 6.5 of the DP83867IR data sheet[\[1\]](#).

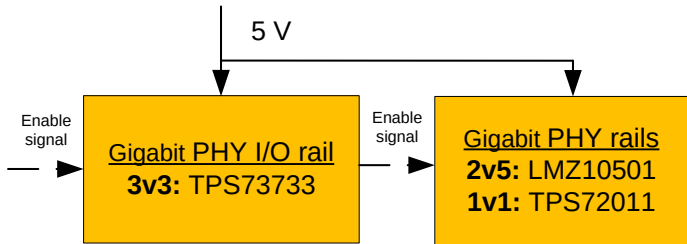


Figure 14. Power Stage for Two Gigabit Ethernet PHYs

2.3.1.3.2.1 DP83867IR Current Consumption

The DP83867IR has two power options. One option requires two rails for the core and one rail for the V_{DDIO} . In this setup, the PHY draws 565 mW. For the two-rail option, no sequencing is required. If power dissipation on the DP83867IR is critical, the second option uses three rails for core and one rail for V_{DDIO} . In this setup, the PHY draws 545 mW. V_{DDIO} can be 3.3, 2.5, or 1.8 V. See its data sheet for more details on the current consumption per rail and if a voltage other than 3.3 V is used in the V_{DDIO} rail[\[1\]](#).

Table 14. Power Consumption per Voltage Rail

RAIL	STANDARD POWER MODE (565 mW)	LOW POWER MODE (545 mW)
1.1 V	106 mA	106 mA
1.8 V	N/A	64 mA
2.5 V	157 mA	103 mA
V_{DDIO} (3.3, 2.5, or 1.8 V)	31 mA (1.8 V)	31 mA (1.8 V)

For this design, the power supply was designed for the two-rail standard power mode option to fit two DP83867IR PHYs.

The TPS72011 and LMZ105001 were chosen for the 1.1-V rail and the 2.5-V rail, respectively.

For the V_{DDIO} 3.3 V, the TPS73733 was chosen. This rail is also used as supply rail for other interfaces on board (for example, for the Gigabit Ethernet PHY LEDs and V_{DDIO} voltage rail, the EEPROM, the debug or JTAG circuitry, reset circuitry, and additional pull up resistors for data communication lines).

These parts were chosen because of their feature of industrial temperature range -40°C to 85°C and that they could dissipate the needed power at 85°C . Here it is important to choose the correct package of the device; several other part numbers could fit the electrical specification. However, their package would not be able to dissipate the needed power at the full current and temperature range. Knowing the voltage drop across the LDO and the power dissipation at 85°C , the maximum current that can be dissipated for the decided voltage rail can be calculated.

Table 15. 3.3-V and 1.1-V LDO Power Dissipation per Package

DEVICE	INPUT VOLTAGE	OUTPUT VOLTAGE	VOLTAGE DROP	MAX POWER DISSIPATION AT 85°C AMBIENT TEMPERATURE	ReJA	MAX CURRENT AT 85°C
TPS72011DRV	2.5 V	1.1 V	1.4 V	615 mW	65 K/W	439.29 mA
TPS73733DCQ	5 V	3.3 V	1.7 V	790 mW ⁽¹⁾	53.1 K/W	464.71 mA ⁽¹⁾

⁽¹⁾ Estimated power dissipation and max current at 85°C using junction-to-ambient thermal resistance.

2.3.1.3.2 两个千兆以太网 PHY 的电源

PHY 电源需要为三个电源轨供电。3.3 伏特 V_{DDIO} 电源轨由 3.3 伏特 PMIC 电源轨启动（参见[第 2.3.1.3.2.1 节](#)）。当 V_{DDIO} 电源轨启用时，它将启用剩余的 2.5 伏特和 1.1 伏特两个电源轨，如图 14 所示。DP83867IR 数据手册第 6.5 节包含千兆以太网 PHY 的功耗信息[\[1\]](#)。

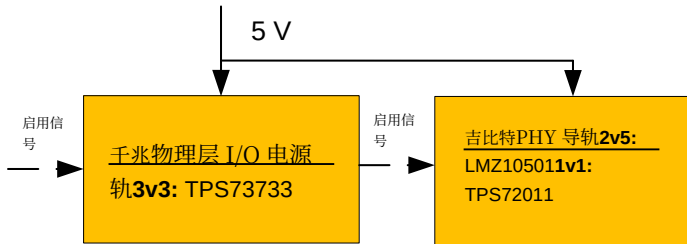


图 14. 两个千兆以太网 PHY 的电源级

2.3.1.3.2.1 DP83867IR 电流消耗

DP83867IR 提供两种电源选项。其中一种选项需要为核心提供两条电源轨，为 V_{DDIO} 提供一条电源轨。在此配置下，PHY 功耗为 565 mW。对于双电源轨选项，无需进行电源排序。如果 DP83867IR 的功耗是关键考虑因素，第二种选项为核心提供三条电源轨，为 V_{DDIO} 提供一条电源轨。在此配置下，PHY 功耗为 545 mW。 V_{DDIO} 可以是 3.3、2.5 或 1.8 V。有关每根电源轨的电流消耗以及当 V_{DDIO} rail[\[1\]](#) 使用非 3.3 V 电压时的详细信息，请参阅其数据手册。

表 14. 每电压轨的功耗

RAIL	标准功率模式（565 mW）	低功耗模式（545 mW）
1.1 伏特	106 毫安	106 毫安
1.8 伏特	N/A	64 毫安
2.5 伏特	157 毫安	103 毫安
V_{DDIO} (3.3, 2.5, 或 1.8 V)	31 mA (1.8 V)	31 mA (1.8 V)

本设计中，电源供应器采用双轨标准电源模式配置，以适配两个 DP83867IR PHY。

TPS72011 和 LMZ105001 分别被选为 1.1-V 轨和 2.5-V 轨的电源芯片。

对于 V_{DDIO} 3.3 V，选择了 TPS73733。该轨也用作板载其他接口的供电轨（例如，千兆以太网 PHY 的 LED 和 V_{DDIO} 电压轨、EEPROM、调试或 JTAG 电路、复位电路以及数据通信线的额外上拉电阻）。

选择这些器件是因为它们具备工业级温度范围 -40°C 至 85°C 的特性，并且能够在 85°C 下耗散所需的功率。在此，选择器件的正确封装至关重要；其他一些型号也能符合电气规格，但其封装无法在全电流和全温度范围内耗散所需的功率。已知 LDO 上的压降以及 85°C 时的功耗，即可计算出在选定电压轨下可耗散的最大电流。

表 15. 3.3 伏特和 1.1 伏特 LDO 每封装的功耗

设备	输入电压	输出电压	电压降	85°C 环境温度下的最大功耗	ReJA	85°C 时最大电流
TPS72011DRV	2.5 伏特	1.1 伏特	1.4 伏特	615 毫瓦	65 千瓦/瓦特	439.29 毫安
TPS73733DCQ	5 V	3.3 伏特	1.7 伏特	790 毫瓦 ⁽¹⁾	53.1 千瓦/瓦	464.71 毫安 ⁽¹⁾

⁽¹⁾ 使用结到环境的热阻估算 85°C 时的功耗和最大电流。

The 2.5-V rail at LMZ10501 was chosen due to the fact that an LDO with the current needed would be a very expensive device and would dissipate a significant amount of power. When two PHYs are powered approximately 526 mA is required, with margin the 1-A LMZ10501 was chosen.

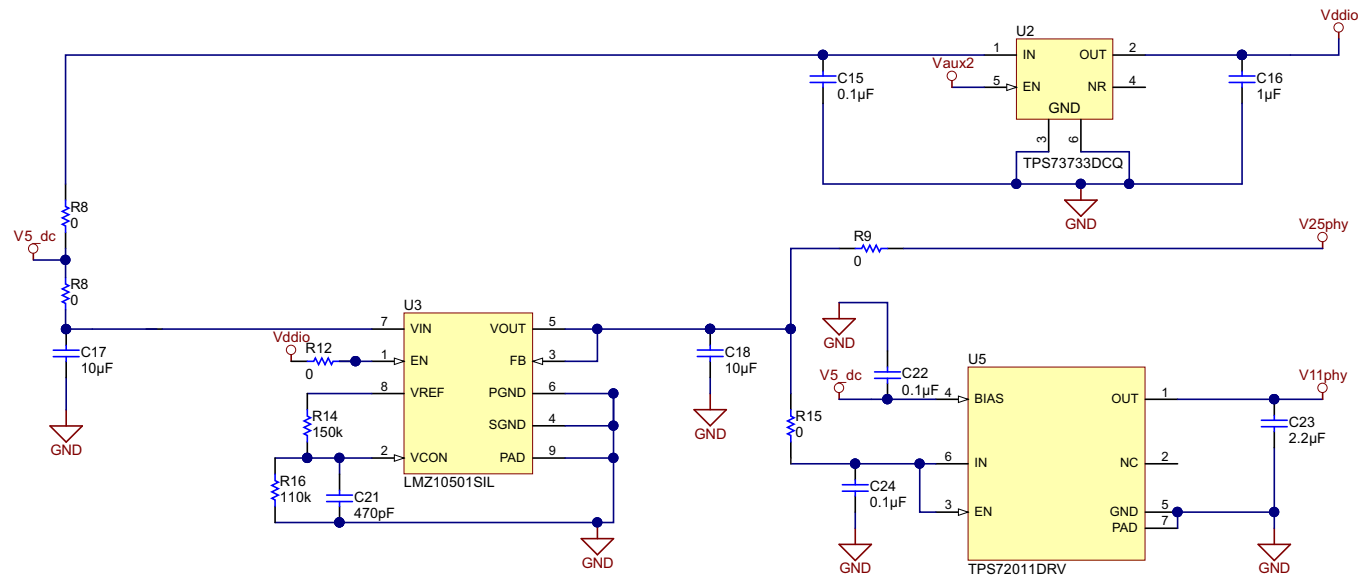


Figure 15. Schematic of Power Supply for Two DP83867IR Gigabit Ethernet PHYs

Use Equation 13 to calculate the output voltage for the LMZ10501.

$$R_{16} = \left(\frac{V_{OUT}}{5.875 \text{ V} - V_{OUT}} \right) \times R_{14} \quad (13)$$

R_{14} needs to be between 80 and 300 kΩ and R_{16} can be calculated. R_{14} has to be minimum 80 kΩ to ensure that the V_{REF} output current loading is not greater than 30 µA and the reference voltage is maintained. This design needed a 2.5-V output voltage and the R_{14} was decided to be 150 kΩ and the R_{16} is now calculated to 111.1 kΩ. A 110-kΩ resistor was chosen. A small filter capacitor of 470 pF with a voltage rating of 6.3 V or 10 V is connected in parallel to R_{16} . The input and output capacitors are recommended to be X5R or X7R.

2.5-V 轨在 LMZ10501 中被选中，因为具有所需电流的 LDO 将是非常昂贵的设备，并且会消耗大量功率。当两个 PHY 供电时，大约需要 526 mA，考虑到 MARGIN，选择了 1-A LMZ10501。

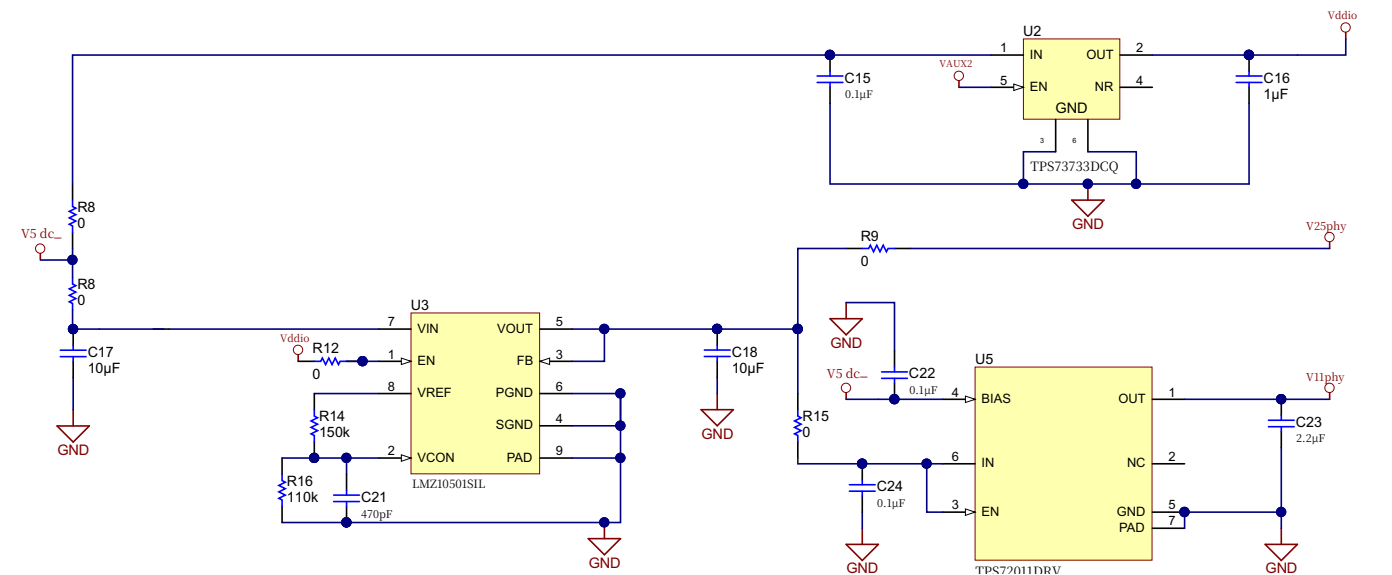


图 15. 两个 DP83867IR 千兆以太网 PHY 的电源供应器原理图

使用 公式 13 来计算 LMZ10501 的输出电压。

$$R_{16} = \left(\frac{V_{OUT}}{5.875 \text{ V} - V_{OUT}} \right) \times R_{14} \quad (13)$$

R_{14} 需要在 80 到 300 kΩ 之间， R_{16} 可以计算得出。 R_{14} 必须至少为 80 kΩ，以确保 V_{REF} 输出电流负载不超过 30 µA，并保持参考电压。该设计需要 2.5-V 输出电压， R_{14} 被确定为 150 kΩ， R_{16} 现在计算为 111.1 kΩ。选择了 110-kΩ 电阻。一个 470 pF 的小滤波电容，额定电压为 6.3 V 或 10 V，并联连接到 R_{16} 。输入和输出电容推荐采用 X5R 或 X7R 材质。

2.3.1.3.3 Power Stage for AM3359 Host Processor

The power stage for the AM3359 Sitara is realized using the PMIC TPS65910A3 and a DDR3 terminator regulator TPS51200. An additional TPS71718 LDO is used for the RTC circuit of the Sitara, as the TPS65910A3 RTC circuit has been turned off.

The TPS65910A3 can be configured through I²C, this interface can be used to set internal registers of the PMIC. For more details on these registers, see the TPS65910A3 user's guide. The TPS65910A3 have the following power sources that can be used, each of the power supplies can be programmed to work at several voltages. For the TPS65910A3, there are several boot modes depending on the device that needs to be powered. Because the AM3359 is used, the EEPROM sequence has been chosen boot[1:0] = "10". For more information on this topic, see [Section 8](#).

Table 16. TPS65910A3 Default Voltage Rails per "10" EEPROM Sequence

RAIL NAME	TYPE	VOLTAGE OPTIONS	DEFAULT VOLTAGE	POWER
VIO	SMPS	1.5, 1.8, 2.5, or 3.3 V	1.5 V	1000 mA
VDD1	SMPS	0.6 to 1.5 V in 12.5-mV steps Programmable multiplication factor: ×2, ×3 1.1 V	1.1 V	1500 mA
VDD2	SMPS	0.6 to 1.5 V in 12.5-mV steps Programmable multiplication factor: ×2, ×3 1.1 V	1.1 V	1500 mA
VDD3	LDO	5 V or OFF	N/A	100 mA
VDIG1	LDO	1.2, 1.5, 1.8, or 2.7 V	1.8 V	300 mA
VDIG2	LDO	1, 1.1, 1.2, or 1.8 V	1.8 V	300 mA
VPLL	LDO	1.0, 1.1, 1.8, or 2.5 V	1.8 V	50 mA
VDAC	LDO	1.8, 2.6, 2.8, or 2.85 V	1.8 V	150 mA
VAUX1	LDO	1.8, 2.5, 2.8, or 2.85 V	1.8 V	300 mA
VAUX2	LDO	1.8, 2.8, 2.9, or 3.3 V	3.3 V	150 mA
VAUX33	LDO	1.8, 2.0, 2.8, or 3.3 V	3.3 V	150 mA
VMMC	LDO	1.8, 2.8, 3.0, or 3.3 V	3.3 V	300 mA
VRTC	LDO	1.8 V or OFF	OFF	OFF

The TPS51200 is used for the DDR3 ram as the sink/source termination regulator. The DDR memory termination structure determines the main characteristics of the V_{TT} rail, which is to be able to sink and source current while maintaining acceptable V_{TT} tolerances. The V_{TT} accuracy has a direct impact on the memory signal integrity, it is imperative to understand the tolerance requirements on V_{TT}. The termination current demand of the DDR3 is less than 1 A of burst current. The TPS51200 ensures the regulator output voltage to be between V_{TTR}REF – 25 mV < V_{TT} < V_{TTR}REF + 25 mV.

2.3.1.3.3 AM3359 主机处理器电源级

AM3359 Sitara 的电源级采用 PMIC TPS65910A3 和 DDR3 终端稳压器 TPS51200 实现。Sitara 的 RTC 电路还使用额外的 TPS71718 LDO，因为 TPS65910A3 的 RTC 电路已关闭。

The TPS65910A3 可通过 I²C 配置，此接口可用于设置 PMIC 的内部寄存器。有关这些寄存器的详细信息，请参阅 TPS65910A3 用户指南。TPS65910A3 具有多种电源，每种电源均可编程以在多个电压下工作。对于 TPS65910A3，根据需要供电的设备，有多种启动模式。由于使用了 AM3359，因此选择了 EEPROM 序列启动[1:0] = "10"。有关此主题的更多信息，请参阅[第 8 节](#)。

表 16. TPS65910A3 的"10"EEPROM 序列对应的默认电压轨

RAIL 名称	TYPE	电压选项	默认电压值	电源
VIO	SMPS	1.5、1.8、2.5 或 3.3 伏特	1.5 伏特	1000 毫安
VDD1	SMPS	0.6 伏特至 1.5 伏特，步长为 12.5 毫伏 可编程乘法因子：×2，×3 1.1 伏特	1.1 伏特	1500 毫安
VDD2	SMPS	0.6 伏特至 1.5 伏特，步长为 12.5 毫伏 可编程乘法因子：×2，×31.1 V	1.1 伏特	1500 毫安
VDD3	LDO	5 伏特或关闭	N/A	100 毫安
VDIG1 引脚	LDO	1.2、1.5、1.8 或 2.7 伏特	1.8 伏特	300 毫安
VDIG2	LDO	1, 1.1, 1.2 或 1.8 伏特	1.8 伏特	300 毫安
VPLL	LDO	1.0、1.1、1.8 或 2.5 V	1.8 伏特	50 毫安
VDAC	LDO	1.8、2.6、2.8 或 2.85 V	1.8 伏特	150 毫安
VAUX1	LDO	1.8、2.5、2.8 或 2.85 伏特	1.8 伏特	300 毫安
VAUX2	LDO	1.8、2.8、2.9 或 3.3 伏特	3.3 伏特	150 毫安
VAUX33	LDO	1.8、2.0、2.8 或 3.3 伏特	3.3 伏特	150 毫安
VMMC	LDO	1.8、2.8、3.0 或 3.3 伏特	3.3 伏特	300 毫安
VRTC	LDO	1.8 伏特或关闭	OFF	OFF

TPS51200 用作 DDR3 内存的源/端匹配稳压器。DDR 内存的端接结构决定了 V_{TT} 轨的主要特性，使其能够在保持可接受的 V_{TT} 容差的同时进行电流吸收和输出。V_{TT} 精度直接影响内存信号完整性，因此必须理解 V_{TT} 的容差要求。DDR3 的端接电流需求小于 1 A 的突发电流。TPS51200 确保稳压器输出电压在 V_{TTR}REF – 25 mV < V_{TT} < V_{TTR}REF + 25 mV 之间。

2.3.1.4 System Clocks

There are several options to apply the clocks. For systems with distributed clocks, a clock distribution circuit can be used too. The initial design leverages a crystal on each DP83867IR as well as on AM3359 Sitara. This was due to initial test and debug to allow individual component testing.

2.3.1.4.1 Individual Crystals

The crystal manufacturers define the load capacitance of their crystal in their data sheet. This load capacitance C_L is the sum of the PCB parasitic stray capacitance C_S and the external load capacitors C_X and C_Y , respectively, as per Equation 14. Assuming that the external load capacitors are the same value, they can be calculated with Equation 14.

$$C_X = C_Y = 2 \times (C_L - C_S) \quad (14)$$

C_S is the stray capacitance (device and PCB) this value can be assumed to a few pF as a rule of thumb. The load capacitors need to be NPO/COG type. Table 17 shows the selected load capacitors per crystal and per device on the TIDA-00204.

Table 17. Parallel Capacitors Needed for Oscillation Circuit

CRYSTAL	DEVICE	C_L	C_S	$C_X = C_Y$
ABM3-25.000MHZ-D2W-T	DP83867IR	18 pF	4 pF	27 pF
ABM3-24.000MHZ-D2W-T	AM3359	18 pF	4 pF	27 pF
ABM3-12.000MHZ-D2Y-T	FT2232HL	18 pF	4 pF	27 pF
MC-306 32.7680K-A0:ROHS	AM3359 (RTC)	12.5 pF	3 pF	19 pF

In this design, the default configuration is one crystal at each DP83867IR PHY. However, the hardware is prepared thanks to 0-Ω resistors to clock the second DP83867IR PHY from the PHY 1, as shown in Figure 16. If desired, this configuration can be tested.

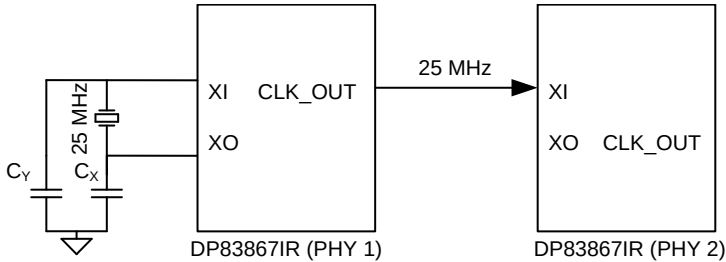


Figure 16. Crystal Configuration for DP83867IR and Option to Clock the Second DP83867IR PHY From the First DP83867IR

2.3.1.4 系统时钟

有多种方式应用时钟。对于具有分布式时钟的系统，也可以使用时钟分配电路。初始设计在每个 DP83867IR 以及 AM3359 Sitara 上均使用晶振。这是出于初始测试和调试的需要，以便对各个组件进行单独测试。

2.3.1.4.1 单晶

晶振制造商在其数据表中定义其晶振的负载电容。该负载电容 C_L 是 PCB 寄生杂散电容 C_S 和外部负载电容 C_X 和 C_Y 的总和，分别按照公式 14。假设外部负载电容的值为相同，它们可以通过公式 14 计算。

$$C_X = C_Y = 2 \times (C_L - C_S) \quad (14)$$

C_S 是寄生电容（器件和 PCB 板），该值可按经验法则假设为几皮法。负载电容需采用 NPO/COG 类型。表 17 列出了 TIDA-00204 上每个晶振和每个器件所选用的负载电容。

表 17. 振荡电路所需的并联电容

CRYSTAL	设备	C_L	C_S	$C_X = C_Y$
ABM3-25.000MHz-D2W-T 滤波器	DP83867IR	18 皮法	4 pF	27 皮法
ABM3-24.000MHZ-D2W-T	AM3359	18 皮法	4 pF	27 皮法
ABM3-12.000MHZ-D2Y-T	FT2232HL	18 皮法	4 pF	27 皮法
MC-306 32.7680K-A0:ROHS	AM3359（实时时钟）	12.5 皮法	3 pF	19 皮法

本设计中，默认配置是每个 DP83867IR PHY 一个晶振。但硬件已准备好，可通过 0-Ω电阻将第二个 DP83867IR PHY 从 PHY 1 进行时钟驱动，如图 16所示。如需，可测试此配置。

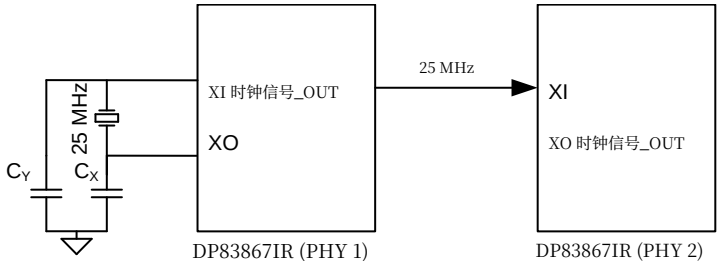


图 16. DP83876IR 的晶体配置以及从第一个 DP83867IR 为第二个 DP83867IR PHY 提供时钟的选项

2.3.1.4.2 Clock Distribution Circuits

Alternatively to the XTAL clocking as implemented on the TIDA-00204 reference design, a more advanced clocking solution is possible. The following section describes how this could be implemented.

One reason for implementing a clock distribution circuit is to achieve high timing accuracy, which is required for the real-time Ethernet protocols. A crystal can typically have around ± 50 ppm in combined tolerance and a stability specification at 25°C, which means that a 25-MHz crystal with 50 ppm can potentially run at 25 MHz ± 1250 Hz.

When running several crystals, the phase shift between the two crystals can potentially be maximum negative and maximum positive, which Figure 17 shows.

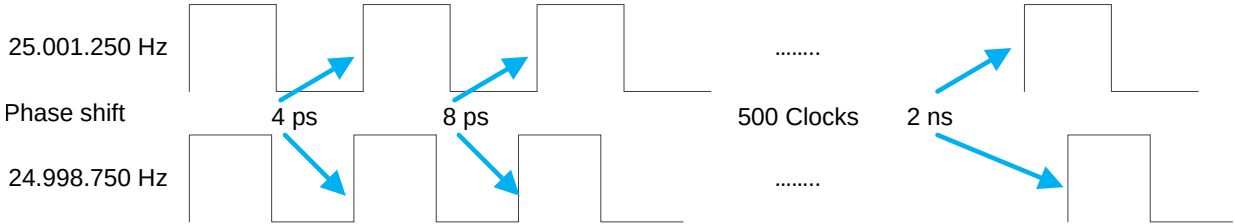


Figure 17. Crystal Phase Shift Diagram

This distribution means that, depending on how many clocks are seen on a signal package, the phase shift of the two clocks accumulates. Avoid this accumulation by adding a clocking distribution network. After this action, only the one clock cycle jitter requires consideration and the phase shift is no longer a concern.

The solution to this problem is to replace XTALs of the PHY and Sitara chips with a CDCE913 programmable 1-PLL VCXO clock synthesizer, with three outputs. If more than three clocks are required, the CDCE9xx device family is available with various output and maximum output clock frequency.

This solution yields several advantages in general:

- High accurate clocking with minimum frequency and phase shift
- Smaller component area used for clocking
- Potentially lower cost
- Possibility to extend easily

Figure 18 shows how a CDCE913 could be connected and used to provide three reference LVCMOS clocks.

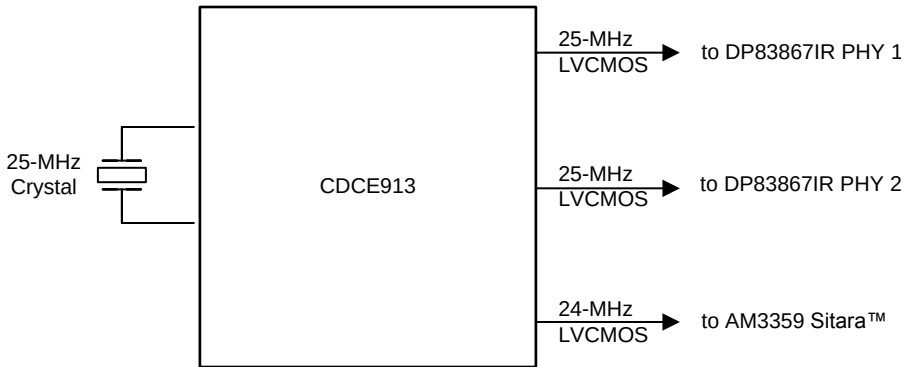


Figure 18. CDCE913 Clock Distribution Circuit Example

2.3.1.4.2 时钟分配电路

与 TIDA-00204 参考设计中实现的 XTAL 时钟驱动方式不同，可以实现更高级的时钟驱动方案。以下章节将介绍如何实现该方案。

实现时钟分配电路的一个原因是为了实现高精度定时，这对于实时以太网协议是必需的。晶体通常具有约 ± 50 ppm 的综合容差和 25°C 下的稳定性规格，这意味着一个具有 50 ppm 容差的 25-MHz 晶体可能以 25 MHz ± 1250 Hz 的频率运行。

当运行多个晶体时，两个晶体之间的相位差可能达到最大负值和最大值，图 17 所示。

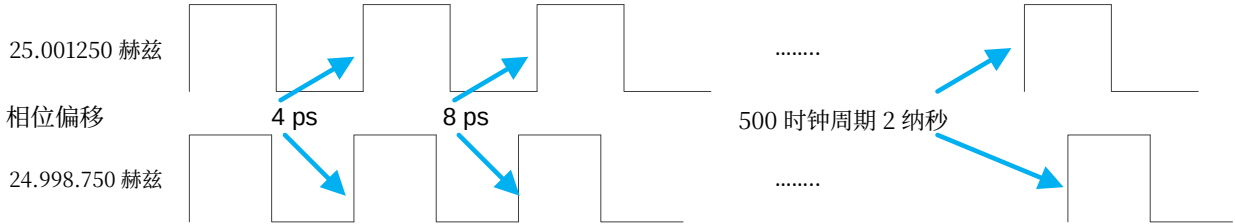


图 17. 晶体相移图

该分布意味着，根据信号包上可见时钟的数量，两个时钟的相位差会累积。通过添加时钟分配网络可避免这种累积。在此操作之后，仅需考虑单个时钟周期的抖动，相位差已不再重要。

该问题的解决方案是用 CDCE913 可编程 1-PLL VCXO 时钟合成器替换 PHY 和 Sitara 芯片中的 XTAL，该合成器具有三个输出。如果需要超过三个时钟，CDCE9xx 设备系列可提供不同的输出和最大输出时钟频率。

该方案在总体上具有多方面的优势：

- 高精度时钟，最小频率和相位偏移
- 用于时钟的更小组件面积
- 潜在降低成本
- 易于扩展

图 18 展示了 CDCE913 如何连接并使用以提供三个参考 LVCMOS 时钟。

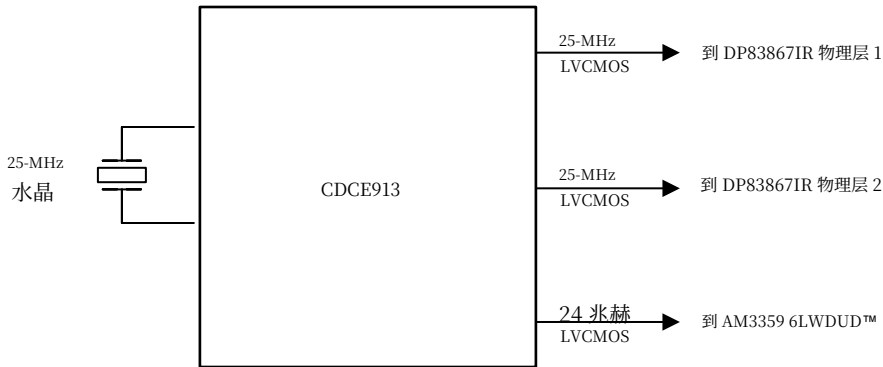
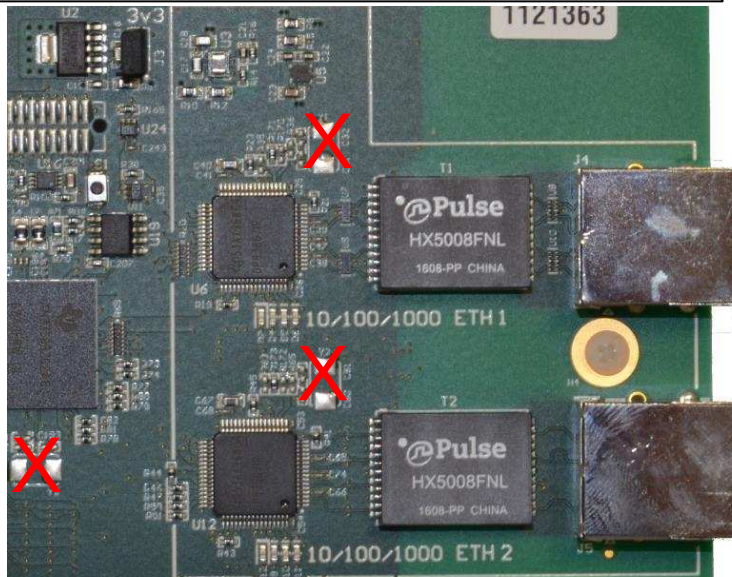


图 18. CDCE913 时钟分配电路示例

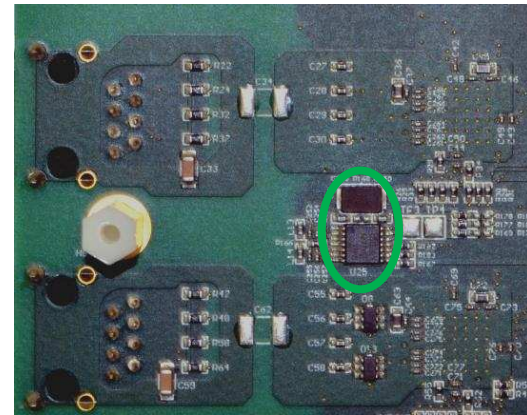
This new clocking source could be used to change the component floor plan as shown in Figure 19 with an Altium 3-D PCB simulation.

Step 1: Remove the oscillator circuitry used for the two DP83867 PHYs and the AM3359 Sitara as shown in the picture.

Step 2: Place CDCE913 and its supporting components on bottom side, between the PHYs and close to AM3359. This allows short routing distances.



PCB top view



PCB bottom view (flipped)

Figure 19. PCB Component Area Saving With CDCE913 Clock Generator

When using the CDCE913 part, programming the EEPROM of the part before soldering it may be an easier solution. TI provides the CDCEL9XXPROGEVM tool for this reason. This tool can be used to program the part defining its input clock and the output clocks desired as shown in Figure 20.

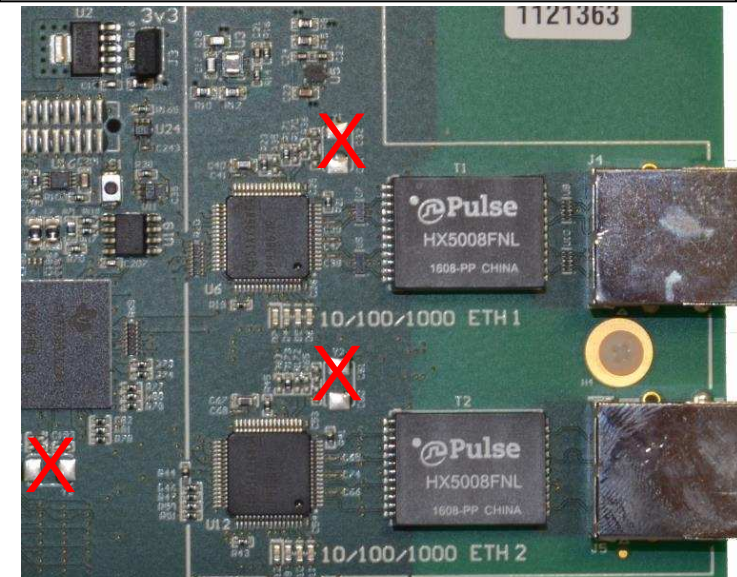
Use the following software to program using the EVM: <http://www.ti.com/tool/clockpro>.

Using this tool, the following setting was chosen for the CDCE913 for this design (see Figure 20).

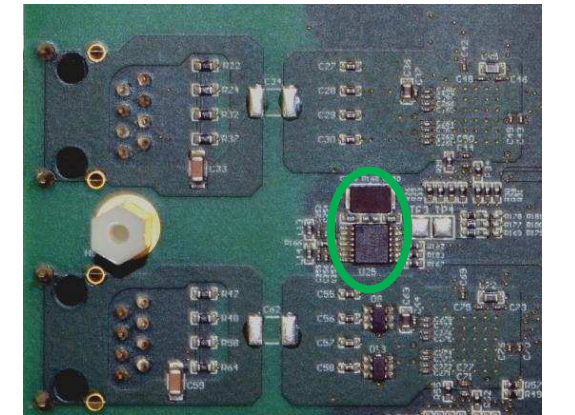
此新的时钟源可用于更改组件布局，如图 19 所示，并配合 Altium 3-D PCB 仿真。

步骤 1：如图所示，移除用于两个 DP83867 PHY 和 AM3359 Sitara 的振荡电路。

步骤 2：将 CDCE913 及其支持组件放置在 PHY 与 AM3359 之间且靠近 AM3359 的底部。这有助于缩短布线距离。



PCB 板俯视图



PCB 板底视图 (翻转)

图 19. 使用 CDCE913 时钟发生器节省 PCB 组件面积

在使用 CDCE913 器件时，在焊接前对器件的 EEPROM 进行编程可能是一种更简便的解决方案。TI 为此提供了 CDCEL9XXPROGEVM 工具。该工具可用于对器件进行编程，以定义其输入时钟和所需的输出时钟，如图 20 所示。

请使用以下软件通过 EVM 进行编程：<http://www.ti.com/tool/clockpro>。

使用此工具，为 CDCE913 选择了以下设置（参见 图 20）。

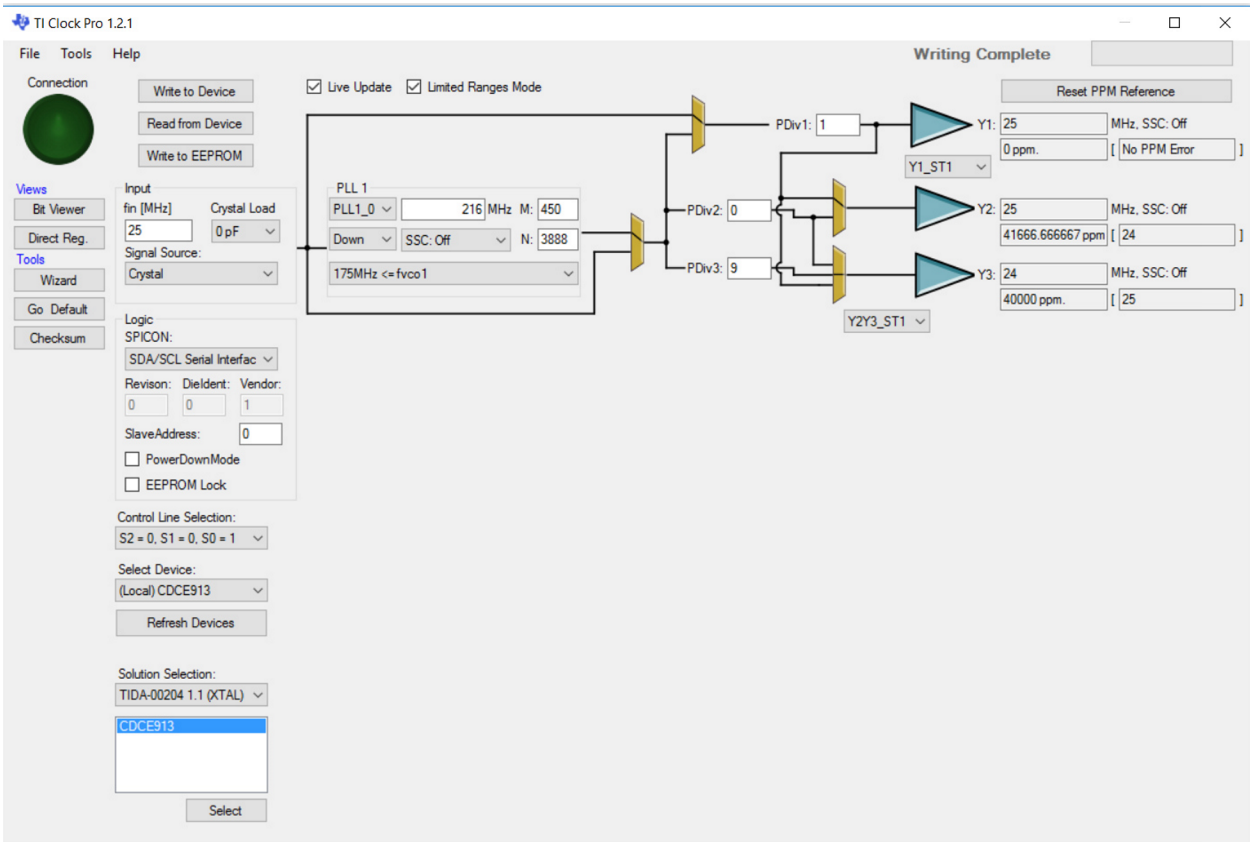


Figure 20. Settings Used to Program CDCE913 Device on TIDA-00204 Reference Design

The diagrams also shows that the load crystals can be removed and replaced by an internal crystal.

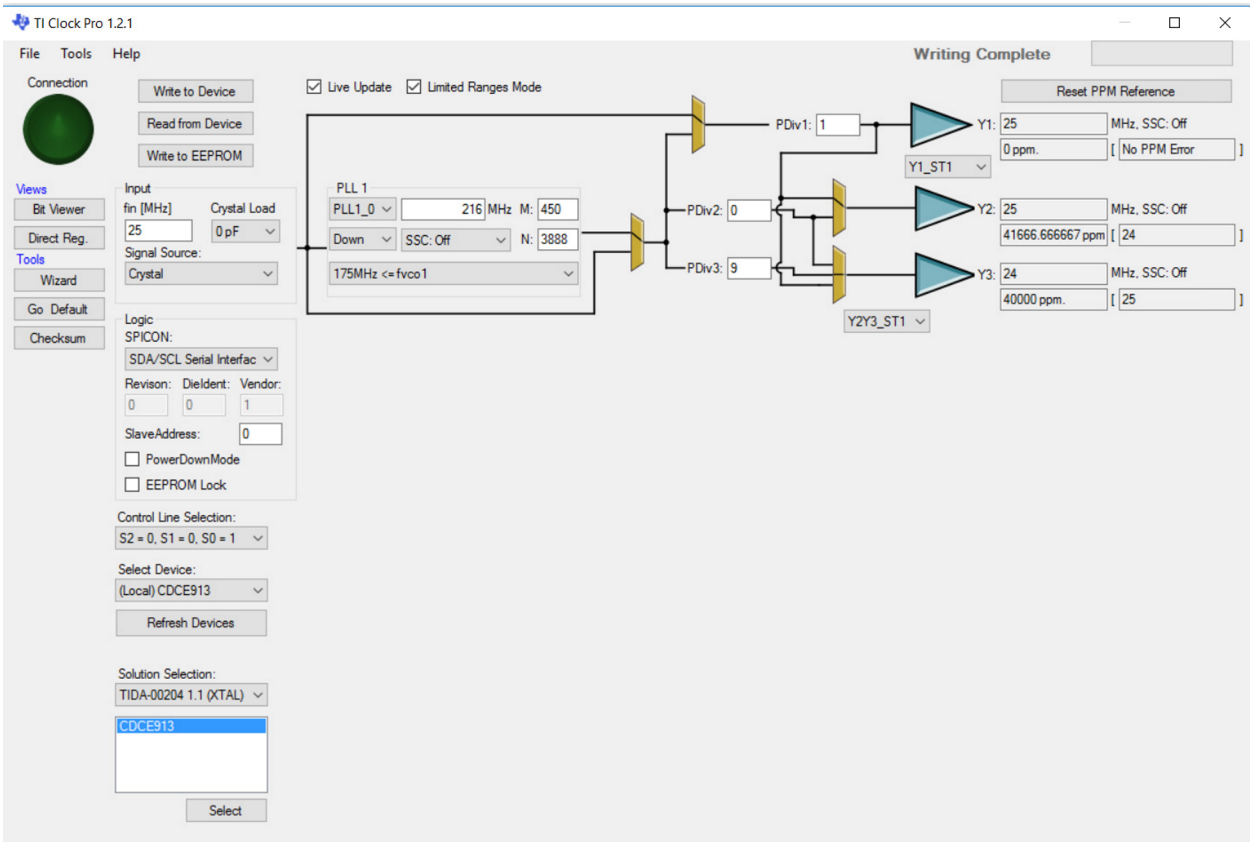


图 20. 在 TIDA-00204 参考设计中对 CDCE913 器件进行编程时使用的设置

图表还显示，负载晶体可以被移除并替换为内部晶体。

3 Software Design

3.1 Sitara AM3359 Firmware

The goal of the software design was to provide an example application firmware for evaluating Gigabit Ethernet, which boots from micro SD-Card automatically after the TIDA-00204 is powered. The application firmware implements a driver for the DP83867IR, UDP and TCP/IP stack, and HTTP web server examples. A USB virtual COM port offers optional user access to read or write to DP83867IR registers for custom configurations like RGMII Delay Mode, if required.

3.1.1 Functional Overview

The host processor Sitara AM3359 is configured to boot from MMC1/SD-Card. The AM3359 copies the TIDA-00204 example application binary file app from the SD-Card to DDR3 RAM and launches the example application. The flowchart of the SD-Card boot load process is shown in [Figure 21](#).

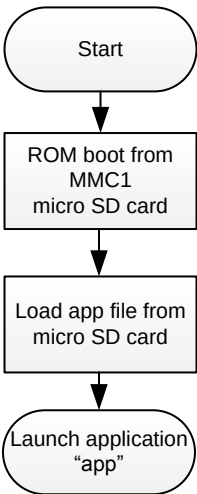


Figure 21. AM3359 Boot Load Process From MMC1 Micro SD-Card

3 软件设计

3.1 Sitara AM3359 固件

软件设计的目标是为评估千兆以太网提供一个示例应用程序固件，该固件在 TIDA-00204 上电后会自动从 Micro SD 卡启动。应用程序固件实现了 DP83867IR、UDP 和 TCP/IP 栈以及 HTTP Web 服务器的驱动程序。如果需要，USB 虚拟 COM 端口提供可选的用户访问权限，用于读取或写入 DP83867IR 寄存器，以实现如 RGMII 延迟模式等自定义配置。

3.1.1 功能概述

主机处理器 Sitara AM3359 配置为从 MMC1/SD 卡启动。AM3359 将 SD 卡中的 TIDA-00204 示例应用程序二进制文件 app 复制到 DDR3 RAM 并启动该示例应用程序。SD 卡启动加载过程的流程图如 [图 21](#)。

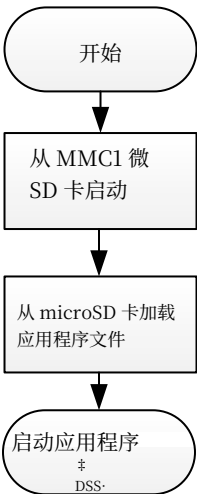


图 21. 从 MMC1 Micro SD 卡启动 AM3359 的加载过程

The TIDA-00204 example application app consists of three major functional blocks.

The first part initializes the AM3359 peripherals like the Ethernet MAC, Ethernet Switch (CPWS), and UART. Then, the app initializes SYS/BIOS and sets up a UART task thread as well as an Ethernet task thread. After that, the app starts SYS/BIOS. Then, both the UART thread and the Ethernet thread are scheduled and run accordingly.

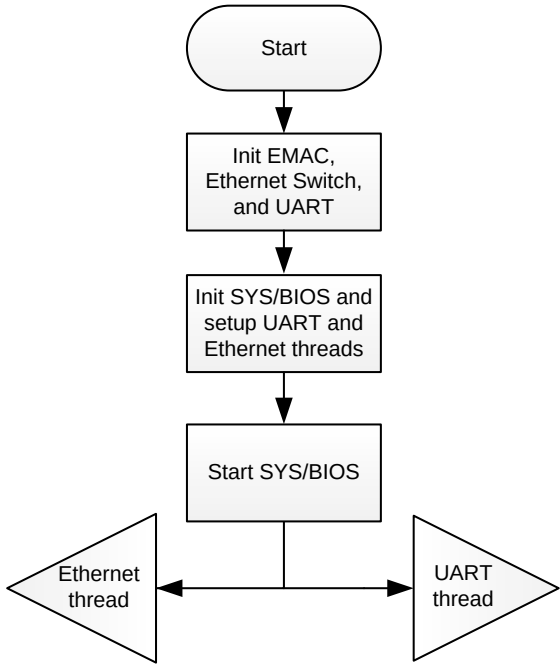


Figure 22. AM3359 Initialization

The Ethernet thread initializes the configuration registers of both DP83867IR Gigabit PHYs through SMI. For example, set RGMII mode, autonegotiation, indicator LEDs, and more. Then the thread enables the network layer with fixed IPv4 address 192.168.1.10 and TCP and UDP transport layer. Additionally, a HTTP web server example application is included.

The UART thread enables communication through virtual COM port and offers an additional option to read and write to the individual registers of the DP83867 Gigabit PHYs through SMI. For example, the user can adjust the RGMII Delay Mode according to the specific hardware, if desired.

TIDA-00204 示例应用程序包含三个主要功能模块。

第一部分初始化 AM3359 外设，如以太网 MAC、以太网交换机 (CPWS) 和 UART。随后，应用程序初始化 SYS/BIOS，并设置 UARTtask 线程以及以太网任务线程。之后，应用程序启动 SYS/BIOS。然后，UARTthread 和 Ethernet thread 被调度并相应地运行。

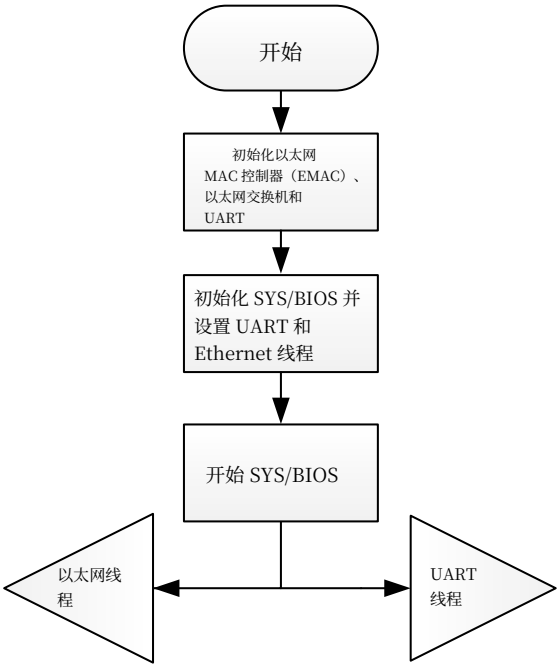


图 22。AM3359 初始化

以太网线程通过 SMI 初始化两个 DP83867IR 千兆 PHY 的配置寄存器。例如，设置 RGMII 模式、自动协商、指示灯 LED 等。然后，线程启用网络层，固定 IPv4 地址为 192.168.1.10，并启用 TCP 和 UDP 传输层。此外，还包含一个 HTTP 网络服务器示例应用程序。

UART 线程支持通过虚拟 COM 端口进行通信，并提供通过 SMI 读写 DP83867 千兆 PHY 各个寄存器的附加选项。例如，用户可根据具体硬件需求，选择性地调整 RGMII 延迟模式。

The flowcharts of both threads are shown in [Figure 23](#) and [Figure 24](#).

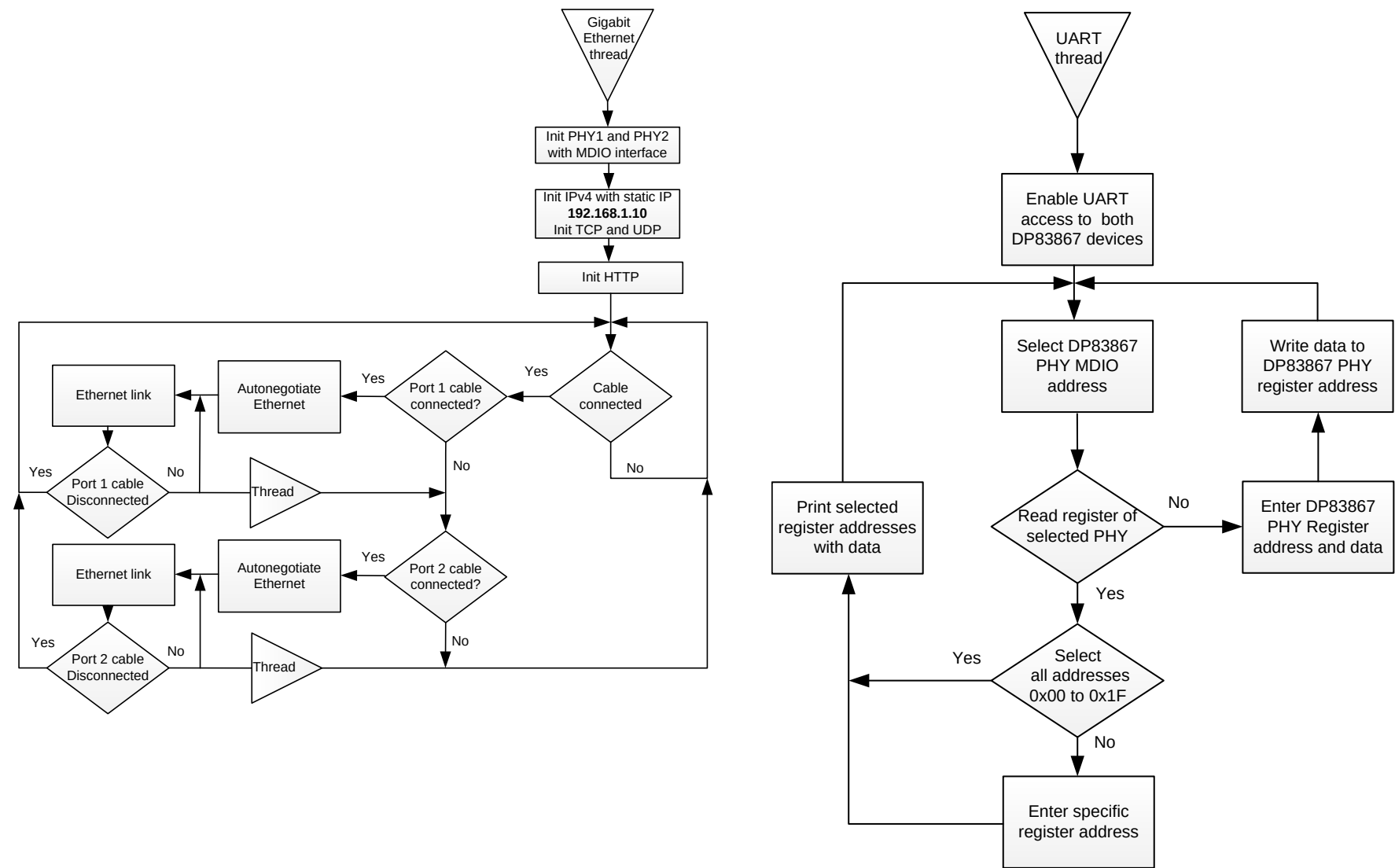


Figure 23. AM3359 Gigabit Ethernet Thread Flowchart

Figure 24. AM3359 UART Thread Flowchart

两个线程的流程图如图 23 所示和图 24。

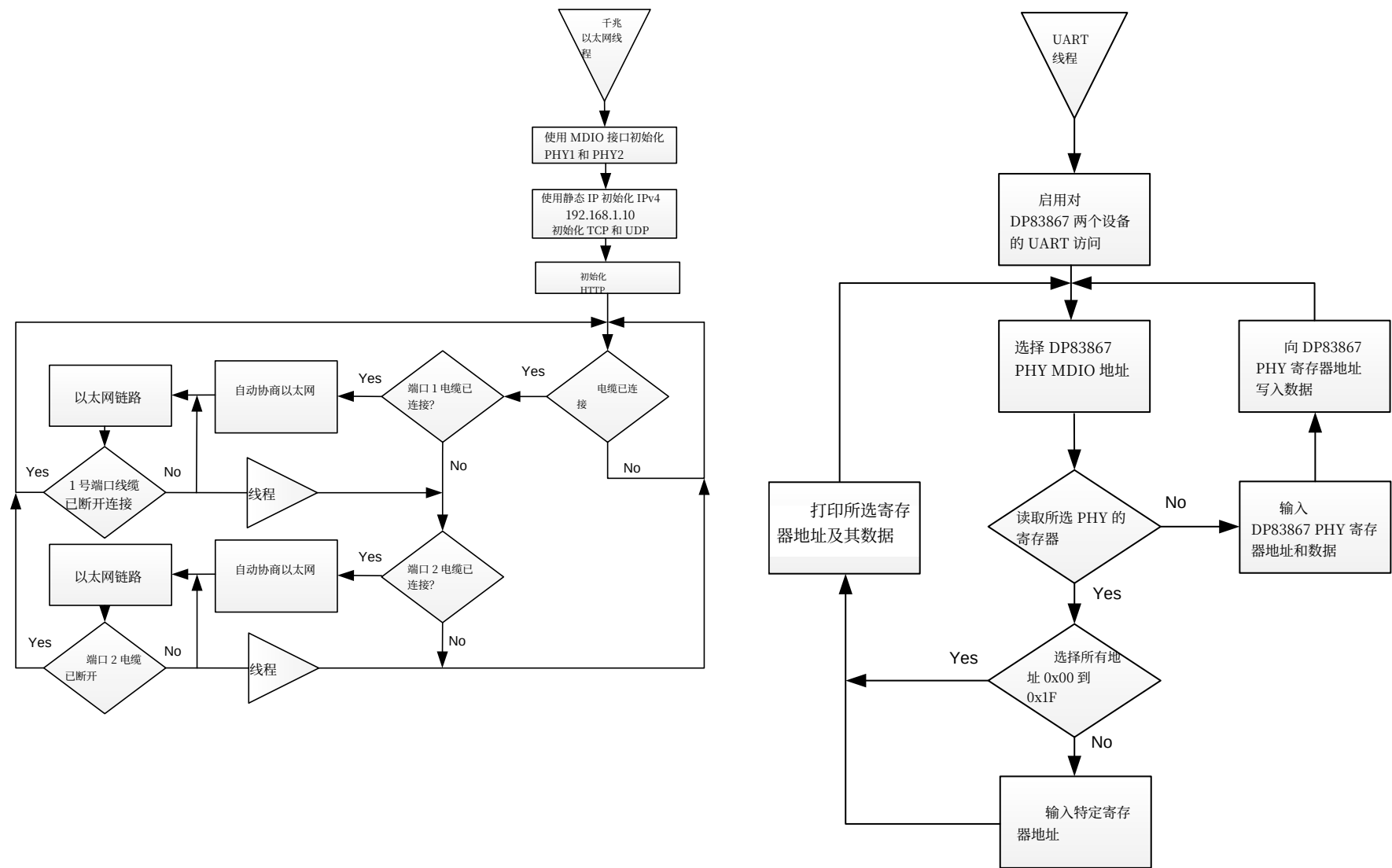


图 23. AM3359 千兆以太网线程流程图

图 24. AM3359 UART 线程流程图

3.1.2 Software Components

The "TIDA-00204 SD-Card Image" consists of two binary files. The AM3359 boot loader binary file called "MLO" and the example application binary file called "app", as described in [Section 3.1.1](#). Both binaries were compiled in separate CCS 6 projects.

3.1.2.1 "MLO" Binary File

The TIDA-00204 boot loader binary file "MLO" was developed with the Texas Instruments AM335x_SYSBIOS_Industrial_SDK_01_01_00_06 software package. This example project was modified to the TIDA-00204 hardware with boot from MMC1/SD-Card. The binary file was renamed after compilation to MLO to fit the AM3359 ROM boot loader and is part of the TIDA-00204_Firmware_AM3359_SD-Card_Image.

3.1.2.2 "app" Binary File

The TIDA-00204 example application binary file app was developed with the following Texas Instruments software packages:

- AM335x_SYSBIOS_Industrial_SDK 01_01_00_06: http://downloads.ti.com/sitara_indus/esd/AM335x_SYSBIOS_Industrial_SDK/latest/index_FDS.html
- SYS/BIOS 6_40_03_39: http://software-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/bios/sysbios/6_40_03_39/exports/bios_setupwin32_6_40_03_39.exe
- XDCtools 3_30_05_60: http://software-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/rtsc/3_30_05_60/index_FDS.html
- NDK 2_24_00_11: http://software-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/ndk/2_24_00_11/index_FDS.html
- NSP 1_10_02_09 Product Download Page: http://software-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/ndk/nsp_1_10_02_09/index_FDS.html

The example application uses only a few example projects including the source and header files.

3.1.2 软件组件

“TIDA-00204 SD-Card Image"由两个二进制文件组成。AM3359引导程序二进制文件名为"MLO", 示例应用程序二进制文件名为"app", 如[第 3.1.1 节](#)所述。两个二进制文件均在单独的 CCS 6 项目中编译。

3.1.2.1"MLO" 二进制文件

TIDA-00204 引导加载程序二进制文件"MLO"是使用德州仪器 AM335x_SYSBIOS_Industrial_SDK_01_01_00_06 软件包开发的。此示例项目已修改为适配 TIDA-00204 硬件, 并支持从 MMC1/SD 卡引导。编译后, 该二进制文件被重命名为 MLO, 以适配 AM3359 ROM 引导加载程序, 并包含在 TIDA-00204_Firmware_AM3359_SD-Card_Image 中。

3.1.2.2"app"二进制文件

TIDA-00204 示例应用程序二进制文件 app 是使用以下德州仪器软件包开发的:

- AM335x_SYSBIOS_Industrial_SDK 01_01_00_06: http://downloads.ti.com/sitara_indus/静电放电/AM335x_SYSBIOS_Industrial_SDK/latest/index_FDS.html
- SYS/BIOS 6_40_03_39:http://软件-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/bios/sysbios/6_40_03_39/exports/bios_setupwin32_6_40_03_39.exe
- XDCtools 3_30_05_60:http://软件-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/rtsc/3_30_05_60/index_FDS.html
- NDK 2_24_00_11:http://软件-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/ndk/2_24_00_11/index_FDS.html
- NSP 1_10_02_09 产品下载页面: http://软件-dl.ti.com/dsps/dsps_public_sw/sdo_sb/targetcontent/ndk/nsp_1_10_02_09/index_FDS.html

示例应用程序仅包含少量示例项目, 包括源文件和头文件。

3.1.3 Creating the Micro SD-Card Image

To create the TIDA-00204_Firmware_AM3359_SD-Card_Image to boot the demo application firmware from the micro SD-Card, the following equipment is required:

- A pre-formatted micro SD-Card with FAT file system
- A PC with a micro SD-Card reader
- Win32 Disk Imager installed on PC

Copy the MLO and app binary file to the micro SD-Card. After that, invoke the Win32 Disk Imager. See an example of the program shown in Figure 25.

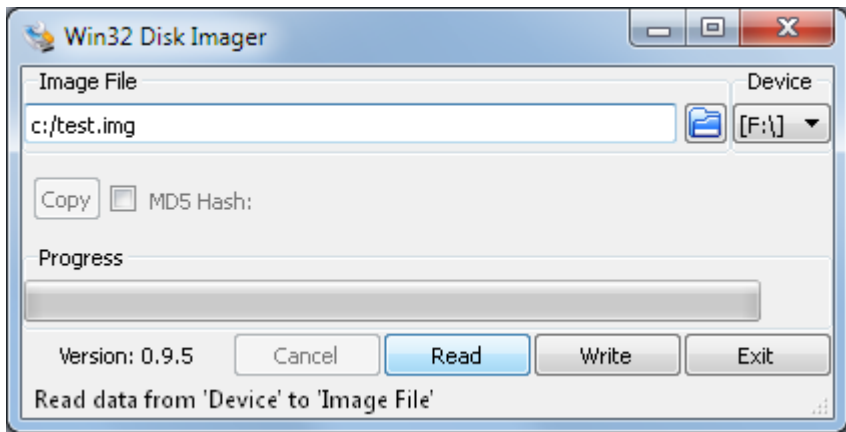


Figure 25. Procedure to Create SD-Card Image With Win32 Disk Imager

Select *Read* to write the current inserted micro SD-Card into the name image in this example called "test.img". This image was renamed to TIDA-00204_SD-Card_Image_rev1_0 and then included into the TIDA-00204 software package.

3.2 PC Test Software for UDP Packet Transfer Based on TI's NDK

The tests for the Gigabit Ethernet communication path developed with the Texas Instruments NDK 2_24_00_11 software package. From this software package, the following source file was used:

- c:\ti\ndk_2_24_00_11\packages\ti\ndk\winapps\testudp.c

For more information on the testudp files, see Section 2.2.3.3 of the NDK document[8].

With original executable testudp.exe, the utilization was around 1%, which was below the needs for this design; therefore, modification was done. The original source was modified adding hyper threading to enable higher utilization. To compile the source file, MinGW was used.

NOTE: To use the hyper threading on a Windows PC, add libwinpthread-1.dll in the same directory as the executable. The library file libwinpthread-1.dll can be downloaded from the internet.

With hyper threading, 25% utilization was achieved on a single Gigabit Ethernet port and 20% each in case of on dual Gigabit Ethernet port 1 and port 2.

Two variants were generated as explained in the following two sections.

3.1.3 创建 Micro SD 卡镜像

要启动演示应用程序固件，需使用 Micro SD 卡从 Micro SD 卡中引导 TIDA-00204_Firmware_AM3359_SD-Card_Image，所需设备如下：

- 预格式化 Micro SD 卡，采用 FAT 文件系统
- 带有 Micro SD 卡读取器的 PC
- Win32 磁盘映像器已安装在 PC 上

将 MLO 和应用程序二进制文件复制到 Micro SD 卡。之后，调用 Win32 Disk Imager。参见程序示例，如图 25 所示。图 25。

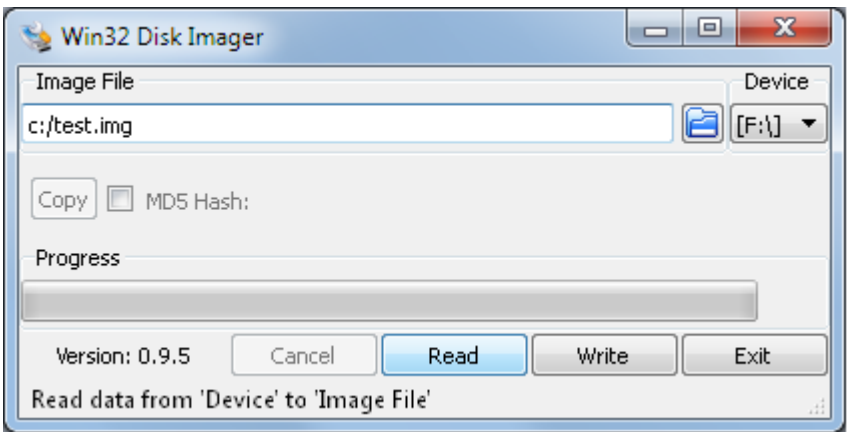


图 25。使用 Win32 磁盘映像器创建 SD 卡映像的步骤

选择“读取”以将当前插入的 Micro SD 卡写入此示例中名为"test.img"的名称图像。该图像已重命名为 TIDA-00204_SD-Card_Image_rev1_0，随后被包含进 TIDA-00204 软件包中。

3.2 基于 TI NDK 的 UDP 数据包传输测试软件

使用德州仪器 NDK 2_24_00_11 软件包开发的千兆以太网通信路径的测试。从该软件包中使用了以下源文件：

- c:\ti\ndk_2_24_00_11\packages\ti\ndk\winapps\testudp.c

有关 testudp 文件的更多信息，请参阅 NDK 文档[8]的第 2.2.3.3 节。

使用原始可执行文件 testudp.exe 时，利用率约为 1%，低于本设计需求；因此进行了修改。原始源码中添加了超线程功能，以实现更高的利用率。编译该源码文件时使用了 MinGW。

NOTE: 要在 Windows PC 上使用超线程功能，请将 libwinpthread-1.dll 添加到可执行文件所在的同一目录中。libwinpthread-1.dll 库文件可从互联网下载。

启用超线程后，单个千兆以太网端口的利用率达到 25%，而千兆以太网端口 1 和端口 2 各为 20%。

以下两个部分将分别说明生成的两种变体。

3.2.1 UDP Throughput Test

The example UDP test file is called "TIDA-00204_UDP_throughput_test.exe". The software flow of the throughput example code can be seen in Figure 26.

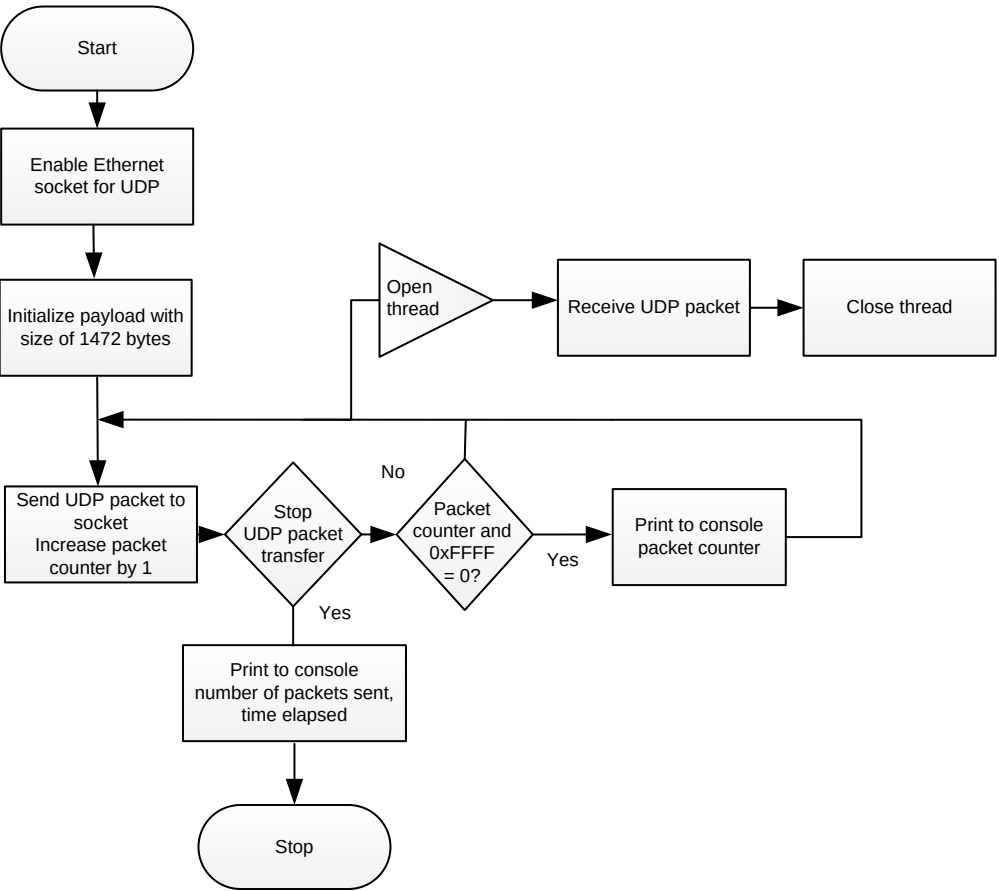


Figure 26. Flowchart of UDP Throughput Test Software

Figure 27 is an example console output can be seen of the example UDP throughput test software.

```

C:\windows\system32\cmd.exe
C:\ethtest>
C:\ethtest>TIDA-00204_UDP_throughput_test 192.168.1.10
UDP throughput test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 65535
UDP packets sent : 131071
Exiting test. Time = 12 seconds
Total packets sent: 139514
C:\ethtest>
  
```

Figure 27. Command Prompt UDP Throughput Print

3.2.1 UDP 吞吐量测试

UDP 测试示例文件名为"TIDA-00204_UDP_throughput_test.exe"。吞吐量示例代码的软件流程如图 26 所示。图 26。

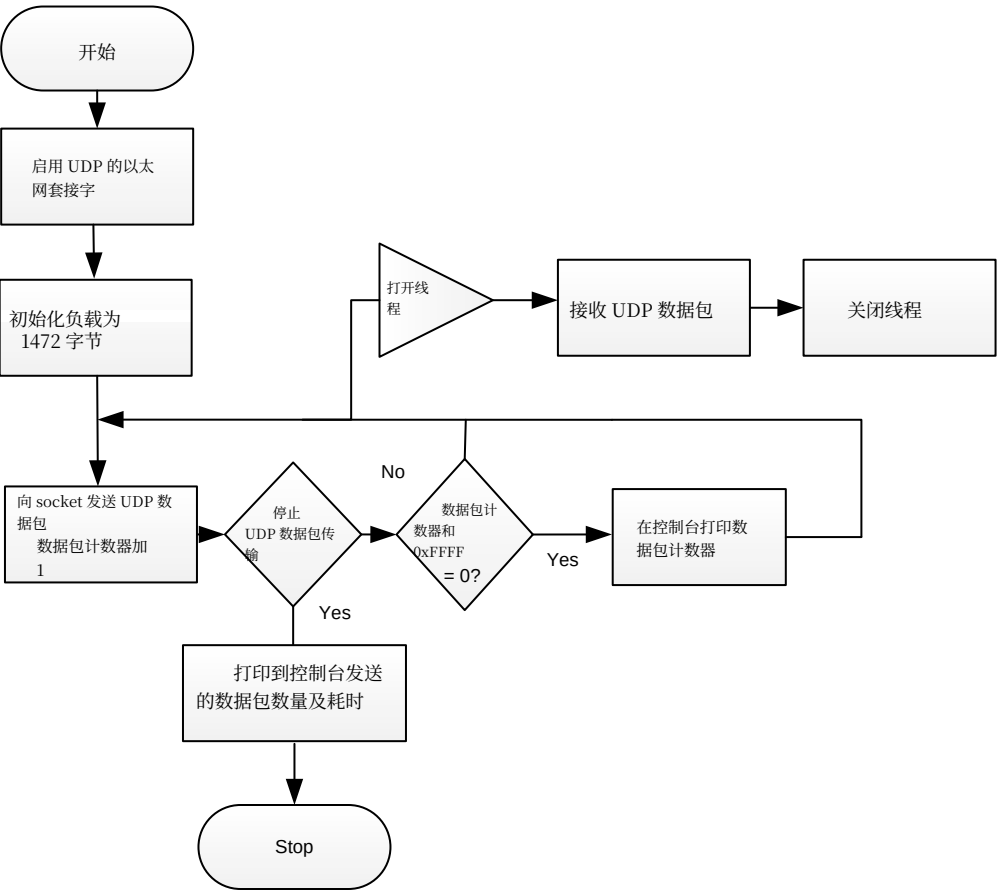


图 26。UDP 吞吐量测试软件流程图

图 27 是示例 UDP 吞吐量测试软件的可见控制台输出示例。

```

C:\windows\system32\cmd.exe
C:\ethtest>
C:\ethtest>TIDA-00204_UDP_throughput_test 192.168.1.10
UDP throughput test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 65535
UDP packets sent : 131071
Exiting test. Time = 12 seconds
Total packets sent: 139514
C:\ethtest>
  
```

图 27. 命令提示符 UDP 吞吐量打印

3.2.2 UDP Packet Error Test

The example UDP test file is called "TIDA-00204_UDP_throughput_test.exe". Figure 28 shows the software flow of the throughput example code can be seen.

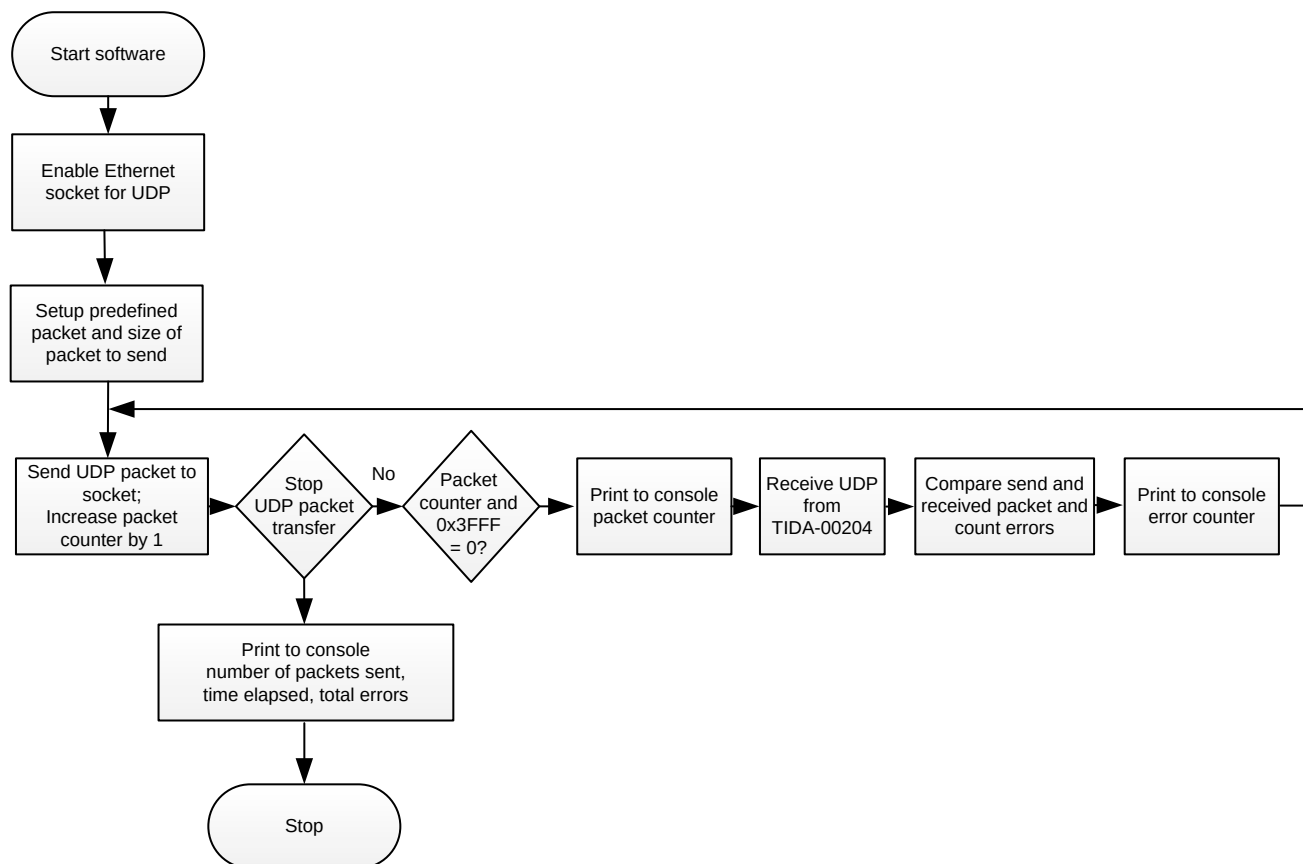


Figure 28. Flowchart of UDP Packet Error Test Software

The above example verifies, if the echoed back packets were corrupted. This functionality is needed check the signal integrity of the connection. Note that with this functionality enabled on the PC, the utilization of the Gigabit Ethernet connection was 5%.

An example console output of the example UDP packet error test can be seen in Figure 29.

```

C:\windows\system32\cmd.exe

C:\Users>cd /

C:\>cd ethtest

C:\ethtest>TIDA-00204_UDP_packet_error_test 192.168.1.10
UDP packet error test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 16383
UDP packets sent : 32767
Exiting test. Time = 49 seconds
Total packets lost: 0
Total packets sent: 33712
C:\ethtest>
  
```

Figure 29. Command Prompt UDP Packet Error Test Print

3.2.2 UDP 数据包错误测试

UDP 测试示例文件名为"TIDA-00204_UDP_throughput_test.exe"。图 28展示了吞吐量示例代码的软件流程。

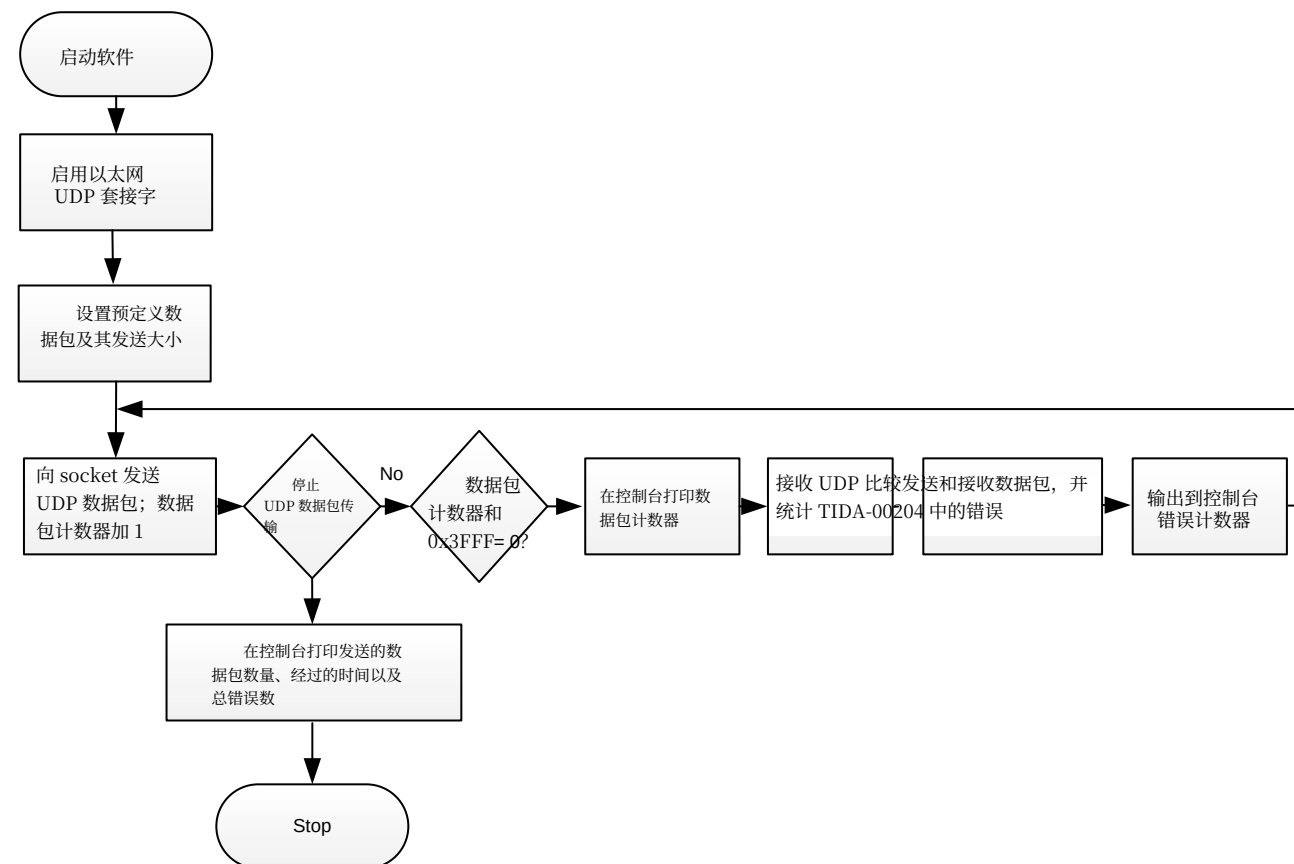


图 28. UDP 数据包错误测试软件流程图

上述示例验证了回发的数据包是否已损坏。此功能用于检查连接的信号完整性。请注意，当在 PC 上启用此功能时，千兆以太网连接的利用率仅为 5%。

示例 UDP 数据包错误测试的示例控制台输出可见于图 29。

```

C:\windows\system32\cmd.exe

C:\Users>cd /

C:\>cd ethtest

C:\ethtest>TIDA-00204_UDP_packet_error_test 192.168.1.10
UDP packet error test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 16383
UDP packets sent : 32767
Exiting test. Time = 49 seconds
Total packets lost: 0
Total packets sent: 33712
C:\ethtest>
  
```

图 29. 命令提示符 UDP 数据包错误测试打印

4 Getting Started

4.1 PCB Overview

A picture of the PCB with key functional blocks is shown in Figure 30.

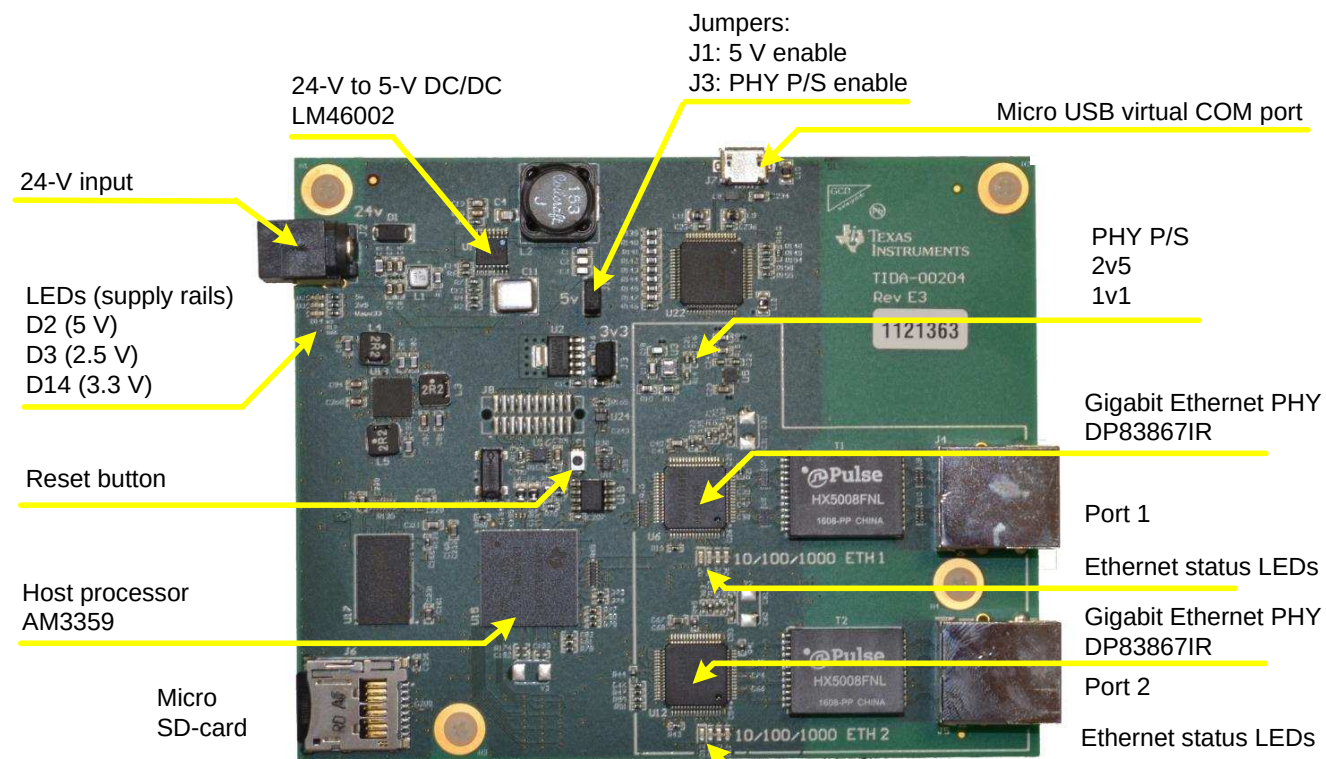


Figure 30. TIDA-00204 Top-Side PCB Photo With Key Functional Blocks

4 快速入门

4.1 PCB 板概览

PCB 板的关键功能模块示意图显示在图 30。

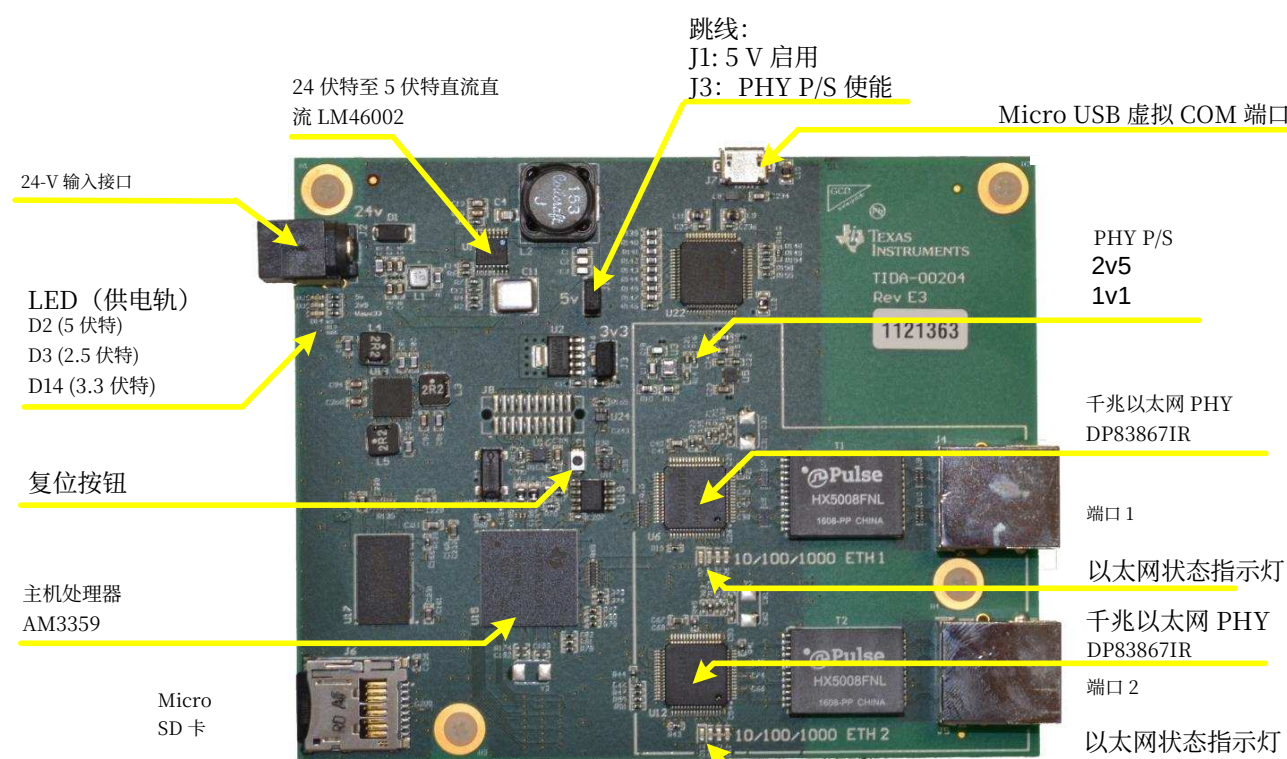


图 30. TIDA-00204 顶层 PCB 板照片, 显示关键功能模块

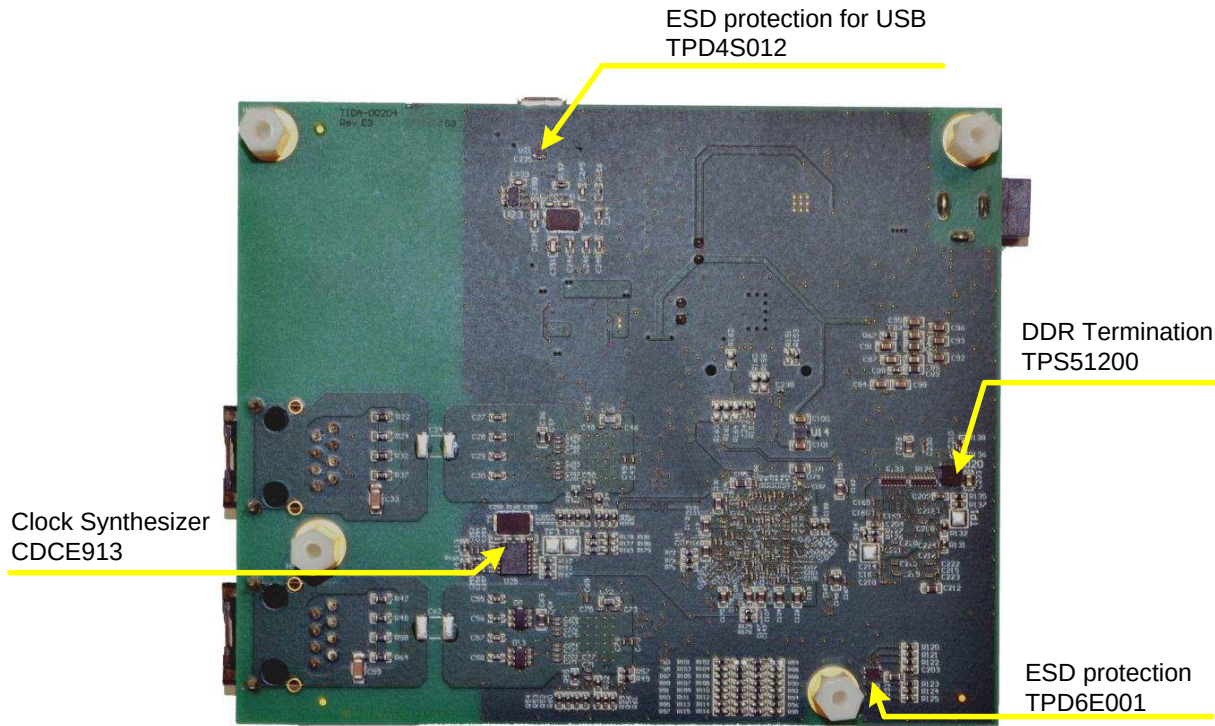


Figure 31. TIDA-00204 Bottom-Side PCB Photo With Key Functional Blocks

4.2 How to Run the Board

The next sections show the prerequisites and the steps to quickly start using the board.

4.2.1 Prerequisites

The following hardware equipment and software is required.

Table 18. Prerequisites

HARDWARE EQUIPMENT OR SOFTWARE	REQUIREMENTS
24-V power supply	24-V output power brick with at least a 250-mA output current Output connector 2.1-mm I.D. × 5.5-mm O.D. × 9.5-mm Female
PC	PC or Laptop with USB port and Gigabit Ethernet port
Ethernet cable	Recommended CAT7 twisted pair cable
Micro USB cable	N/A
Micro SD-card	Big enough to fit the micro SD-card TIDA-00204 image
TeraTerm v4.76 or newer	Download from web, or use any other terminal program of your choice
Win32 Disk Imager v0.9.5 or newer	Download from web
XDS100 v2 PC USB driver	To get this driver, install CCS 6.x
TIDA-00204 software package	Download from the TIDA-00204 web page
libwinpthread-1.dll	Download from web

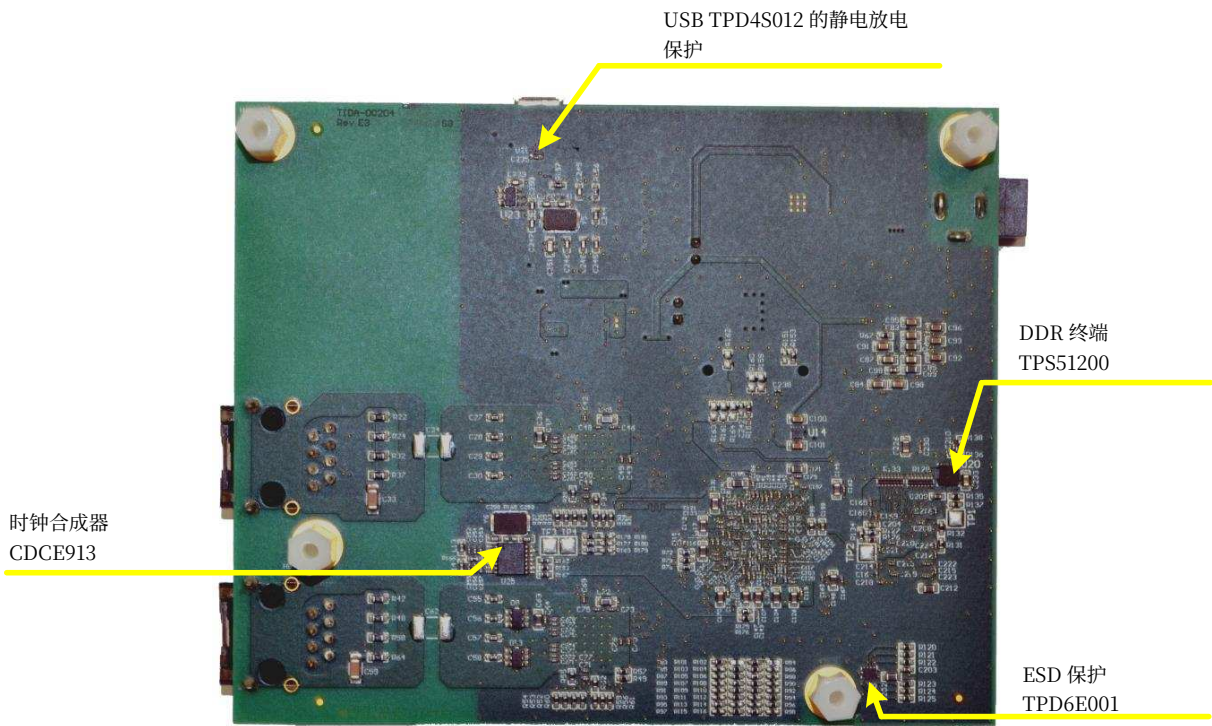


图 31. TIDA-00204 底部 PCB 板照片及关键功能模块

4.2 如何运行董事会

以下章节介绍使用板卡的先决条件和快速入门步骤。

4.2.1 前提条件

需要以下硬件设备和软件。

表 18. 先决条件

硬件设备或软件	需求
24-V 电源供应器	24-V 输出电源适配器，输出电流至少为 250-mA 输出连接器 2.1-mm 内径 × 5.5-mm 外径 × 9.5-mm 母头
PC	带 USB 端口和千兆以太网端口的 PC 或笔记本电脑
以太网电缆	推荐 CAT7 双绞线
Micro USB 线	N/A
Micro SD 卡	足够大，可容纳 TIDA-00204 的 Micro SD 卡镜像
TeraTerm v4.76 或更新版本	从 Web 下载，或使用您选择的任何终端程序
Win32 磁盘映像器 v0.9.5 或更新版本	从网页下载
XDS100 v2 PC USB 驱动程序	要获取此驱动程序，请安装 CCS 6.x
TIDA-00204 软件包	从 TIDA-00204 网页
libwinpthread-1.dll	从网页下载

4.2.2 Jumper Settings

CAUTION

The following two jumpers need to be set: J1 and J3.

The J1 jumper powers the host processor power stage and the J3 jumper enables the Gigabit Ethernet PHY power supply.

4.2.3 Prepare the Micro SD-Card

For easy standalone board operation, a micro SD-Card image is provided. The example image on the SD-Card is configured so the AM3359 reads the boot loader MLO from the micro SD-Card and enables the Sitara AM3359 in a mode. The Sitara AM3359 enables the needed peripherals to communicate with the two PHYs over the MDI and sets those PHYs in autonegotiate 10/100/1000 mode. It also enables a UART functionality where one can read out the PHY registers from the two DP83867 devices over virtual COM port.

Write the image "TIDA-00204_SD-Card_Image_rev1_0.img" to a micro SD-Card. Use Win32 Disk Imager to write the image to the SD-Card (see [Figure 32](#)).

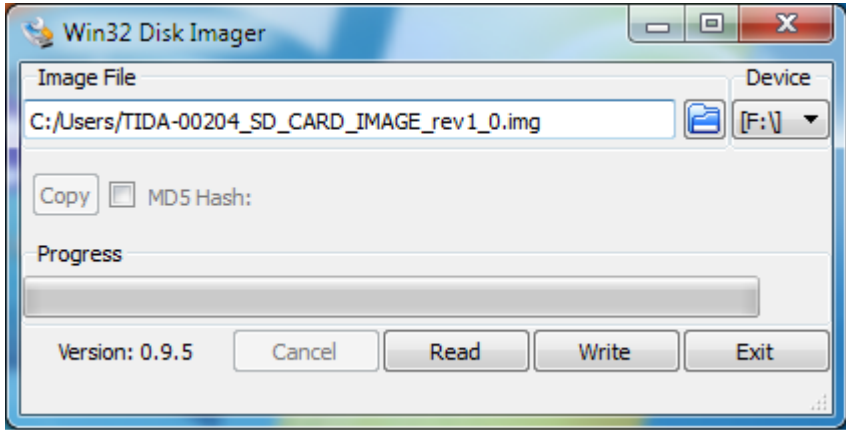


Figure 32. Win32 Disk Imager Image Showing the Write Function

Select the provided TIDA-00204 firmware image and click *Write* to flash the image to the micro SD-Card.

CAUTION

Ensure that the micro SD-Card is selected at the *Device* tab.

If the image is not being written with the onboard PC SD-Card slot, try using an external SD-Card reader.

CAUTION

If the micro SD-Card is not prepared like above, but the individual binary files MLO and app are copied to a preformatted micro SD-Card it might be that the AM3359 does not boot properly. For further details on this topic, see the Section 26.1.7.5.2 of the AM335x technical reference manual[6].

4.2.2 跳线设置

注意

需要设置以下两个跳线：J1 和 J3。

J1 跳线为处理器电源级供电，J3 跳线为千兆以太网 PHY 电源供应器供电。

4.2.3 准备 Micro SD 卡

为方便独立板操作，提供了 Micro SD 卡镜像。SD 卡上的示例镜像配置为：AM3359 从 Micro SD 卡读取引导加载程序 MLO 并启用 Sitara AM3359 模式。Sitara AM3359 通过 MDI 启用所需的外设与两个 PHY 通信，并将这些 PHY 设置为自协商 10/100/1000 模式。它还启用了 UART 功能，可通过虚拟 COM 端口从两个 DP83867 设备中读取 PHY 寄存器。

将图像"TIDA-00204_SD-Card_Image_rev1_0.img"写入 Micro SD 卡。使用 Win32 磁盘映像器将图像写入 SD 卡（参见图 32）。

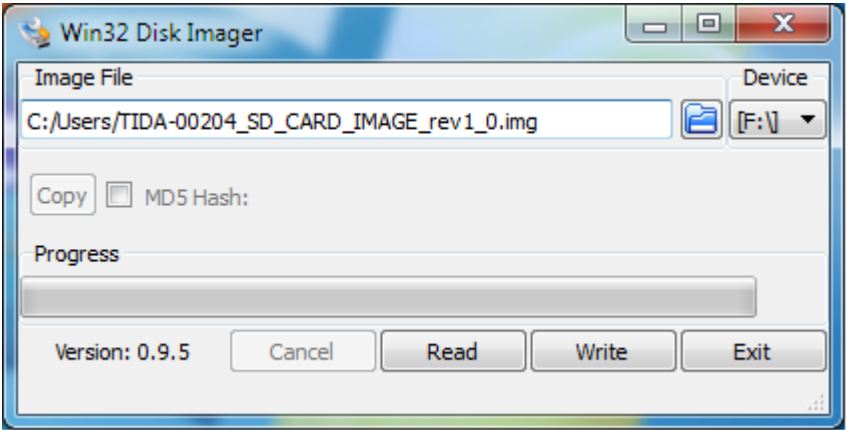


图 32. Win32 磁盘映像器图像显示写入功能

选择提供的 TIDA-00204 固件镜像并点击写入，将镜像写入 Micro SD 卡。

注意

确保在“设备”选项卡中选择 Micro SD 卡。

如果图像不是通过板载 PC SD 卡插槽写入，请尝试使用外部 SD 卡读卡器。

注意

如果 Micro SD 卡未按上述方式准备，而是将单独的二进制文件 MLO 和 app 复制到预格式化的 Micro SD 卡中，则 AM3359 可能无法正常启动。有关此主题的更多详细信息，请参阅 AM335x 技术参考手册第 26.1.7.5.2 节[6]。

4.2.4 Powering the TIDA-00204 Reference Design

Insert the micro SD-Card with TIDA-00204 firmware image into the TIDA-00204 micro SD-Card cage.

Connect a 24-V power supply with at least a 250-mA output current. After that, the three green LEDs (D2, D3, and D14) will turn on. If the LEDs does not turn on, one of the jumpers J1 and J3 are not connected.

Connect the micro USB cable from the TIDA-00204 reference design to the PC. Now the XDS100 v2 PC USB driver is installed on the PC. If PC does not find this driver, install CCS 6 to get the driver.

Invoke a terminal program, like TeraTerm that can connect to the virtual COM port. Setup the terminal program in Serial Console Mode and set the parameters as seen in [Table 19](#).

Table 19. Terminal Program Serial Console Configuration

BAUD RATE	DATA	PARITY	STOP	FLOW CONTROL
115200	8-bit	None	1-bit	None

Afterwards, press the reset button on the board. For the location of the reset button, see [Figure 30](#).

Now on the terminal program, the boot sequence of the TIDA-00204 reference design is printed. See [Figure 33](#) for terminal program printout.

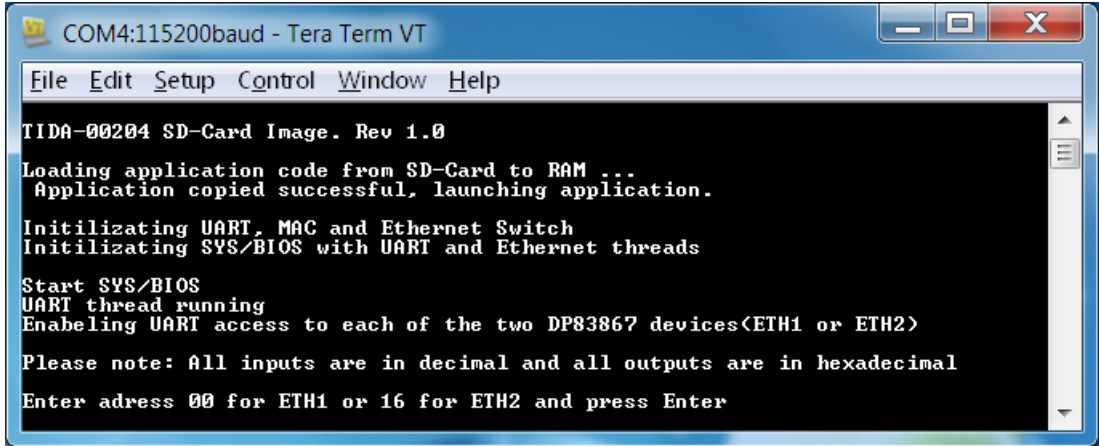


Figure 33. Virtual COM Port Connection During Boot

4.2.4 为 TIDA-00204 参考设计供电

将带有 TIDA-00204 固件图像的 Micro SD 卡插入 TIDA-00204 Micro SD 卡座。

连接一个输出电流至少为 250 mA 的 24-V 电源供应器。之后，三个绿色 LED（D2、D3 和 D14）将点亮。如果 LED 未点亮，则跳线器 J1 和 J3 中有一个未连接。

将 TIDA-00204 参考设计的微 USB 电缆连接到 PC。此时，XDS100 v2 PC USB 驱动程序已安装在 PC 上。如果 PC 未找到该驱动程序，请安装 CCS 6 以获取驱动程序。

调用终端程序（如 TeraTerm），该程序可连接到虚拟 COM 端口。将终端程序设置为串行控制台模式，并设置如表 19 所示的参数。[表 19](#)。

表 19. 终端程序串行控制台配置

波特率	DATA	奇偶校验	STOP	流量控制
115200	8 位	None	1 位	None

随后，按下板上的复位按钮。复位按钮的位置请参见[图 30](#)。现在在终端程序中，TIDA-00204 参考设计的启动序列被打印出来。请参见[图 33](#)中的终端程序打印输出。

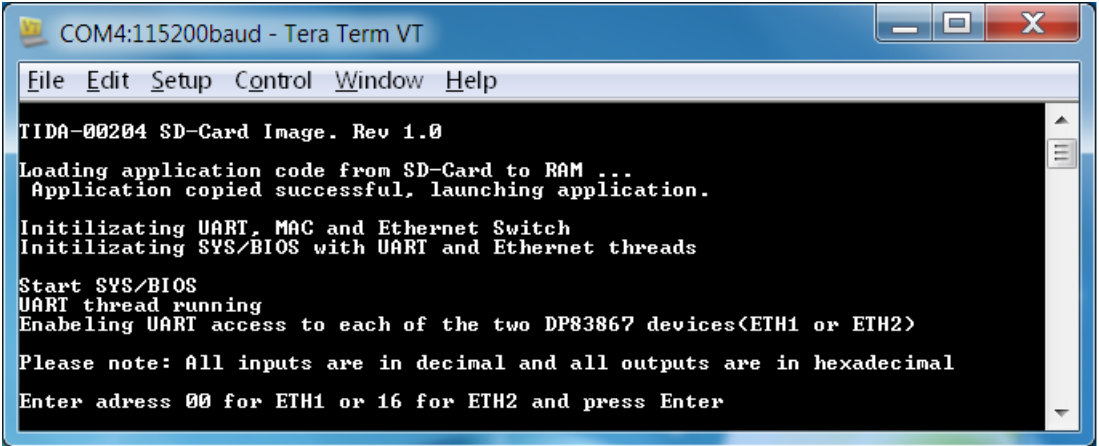


图 33. 启动期间的虚拟 COM 端口连接

4.3 Evaluating the TIDA-00204 Dual-Port Gigabit Ethernet Reference Design

With the board powered up, the next step is to connect the Ethernet link partner.

Connect a link partner to either port 1 or port 2, depending on the link partner autonegotiate will setup 10/100/1000 network capability. Now the Ethernet LEDs will light up showing link status and Ethernet speed.

The status LEDs have the following functionality see [Table 20](#).

Table 20. DP83867 Gigabit PHY Port 1 and Port 2 Status LED Configuration

LED #	D7 / D12	D4 / D9	D5 / D10	D6 / D11
Function	Not assigned to a specific function on TIDA-00204	Receive or transmit activity	1000-Mb link established	Link established

The status LEDs are explained from left to right when looking at the board picture.

The reference design is setup to use the IP address 192.168.1.10, that is a static IP. It does not have DHCP function enabled.

Setup the PC Network connection for static IP in the same address range as the TIDA-00204 reference design with subnet mask 255.255.255.0. For example: 192.168.1.30.

To test the Ethernet connection there are several options: ping, provided UDP test software and a HTTP web server. See examples below.

All Gigabit Ethernet examples demonstrated on both port 1 and port 2.

4.3.1 Ping

With the command prompt, the windows ping function can be used for initial connection test, see [Figure 34](#).

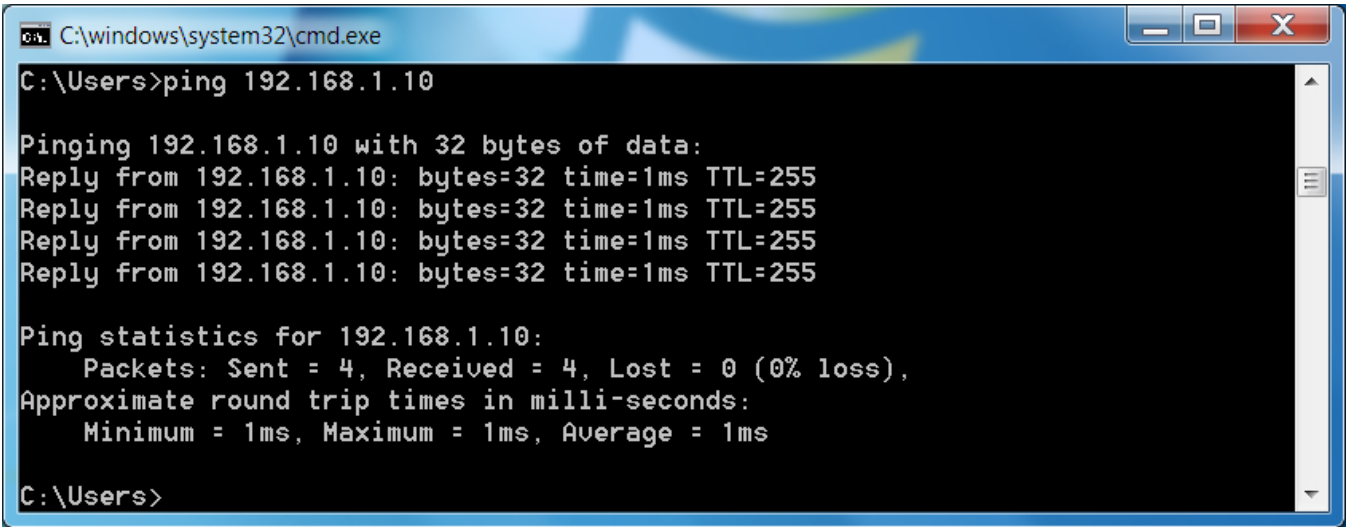


Figure 34. Command Prompt Ping Print

4.3 评估 TIDA-00204 双口千兆以太网参考设计

上电后，下一步是连接以太网链路伙伴。

根据链路伙伴的自动协商能力，将链路伙伴连接到端口 1 或端口 2，以建立 10/100/1000 网络功能。现在，以太网 LED 将点亮，显示链路状态和以太网速度。

状态指示灯具有以下功能，请参阅[表 20](#)。

表 20. DP83867 千兆 PHY 端口 1 和端口 2 状态 LED 配置

LED #	D7 / D12	D4 / D9	D5 / D10	D6 / D11
功能	在 TIDA-00204 上未分配特定功能	接收或发送活动	已建立 1000-Mb 链路	链接已建立

状态指示灯从左至右解释，当查看板子图片时。

参考设计配置为使用 IP 地址 192.168.1.10，这是一个静态 IP。未启用 DHCP 功能。

为 PC 网络连接设置静态 IP，使其与 TIDA-00204 参考设计处于同一地址范围，子网掩码为 255.255.255.0。例如：192.168.1.30。

测试以太网连接有多种选项：Ping、提供的 UDP 测试软件和 HTTP 网络服务器。参见以下示例。

所有千兆以太网示例均在端口 1 和端口 2 上演示。

4.3.1 Ping

使用命令提示符，可以通过 windows ping 函数进行初始连接测试，参见[图 34](#)。

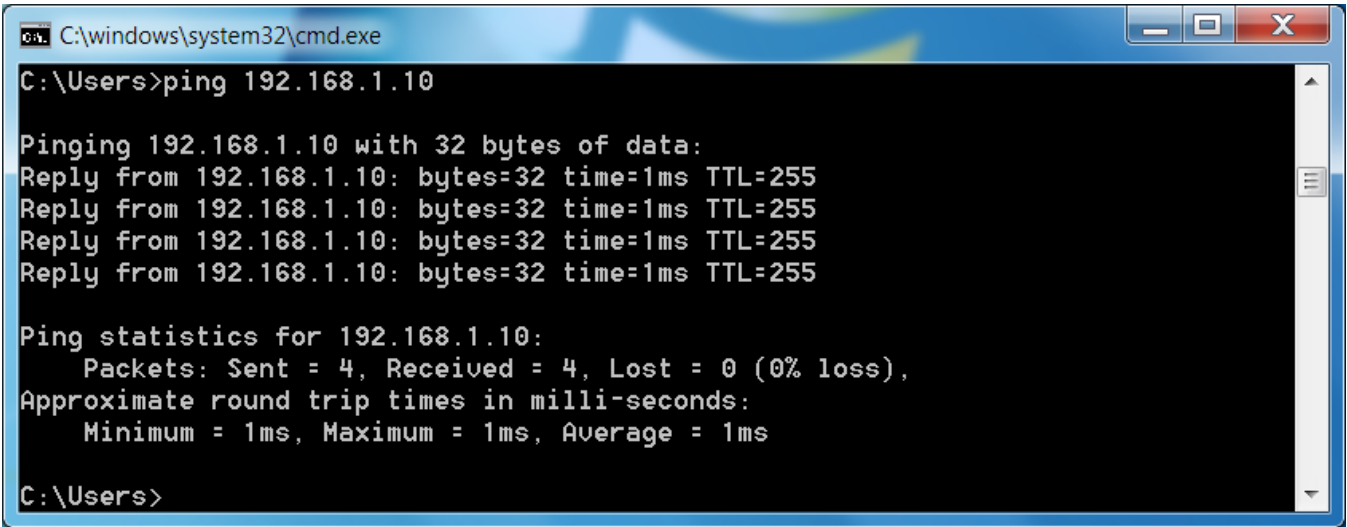


图 34. 命令提示符 Ping 打印

4.3.2 UDP Test Software

The two PC test programs as part of the TIDA-00204 software package can be used.

Insure that the libwinpthread-1.dll is located in the same folder as the TIDA-00204_throughput_test.exe and TIDA-00204_packet_test.exe. This library can be downloaded from the internet.

To invoke these programs, use a Windows PC with the command prompt, see [Figure 35](#) and [Figure 36](#) for example print outs.

```

C:\windows\system32\cmd.exe
C:\ethctest>
C:\ethctest>TIDA-00204_UDP_throughput_test 192.168.1.10
UDP throughput test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 65535
UDP packets sent : 131071
Exiting test. Time = 12 seconds
Total packets sent: 139514
C:\ethctest>

```

Figure 35. Command Prompt UDP Throughput Print

```

C:\windows\system32\cmd.exe
C:\Users>cd /
C:\>cd ethctest
C:\ethctest>TIDA-00204_UDP_packet_error_test 192.168.1.10
UDP packet error test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 16383
UDP packets sent : 32767
Exiting test. Time = 49 seconds
Total packets lost: 0
Total packets sent: 33712
C:\ethctest>

```

Figure 36. Command Prompt UDP Packet Error Test Print

4.3.2 UDP 测试软件

TIDA-00204 软件包中的两个 PC 测试程序可以使用。

确保 libwinpthread-1.dll 位于与 TIDA-00204_throughput_test.exe 和 TIDA-00204_packet_test.exe 相同的文件夹中。该库可从互联网下载。

要调用这些程序，请使用带有命令提示符的 Windows 电脑，参见 [图 35](#)和 [图 36](#) 例如打印输出。

```

C:\windows\system32\cmd.exe
C:\ethctest>
C:\ethctest>TIDA-00204_UDP_throughput_test 192.168.1.10
UDP throughput test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 65535
UDP packets sent : 131071
Exiting test. Time = 12 seconds
Total packets sent: 139514
C:\ethctest>

```

图 35. 命令提示符 UDP 吞吐量打印

```

C:\windows\system32\cmd.exe
C:\Users>cd /
C:\>cd ethctest
C:\ethctest>TIDA-00204_UDP_packet_error_test 192.168.1.10
UDP packet error test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 16383
UDP packets sent : 32767
Exiting test. Time = 49 seconds
Total packets lost: 0
Total packets sent: 33712
C:\ethctest>

```

图 36. 命令提示符 UDP 数据包错误测试打印

4.3.3 HTTP Web Server Example

Open a browser and enter this address: <http://192.168.1.10>. See [Figure 37](#) for a screenshot of the [TIDA-00204 webpage](#).

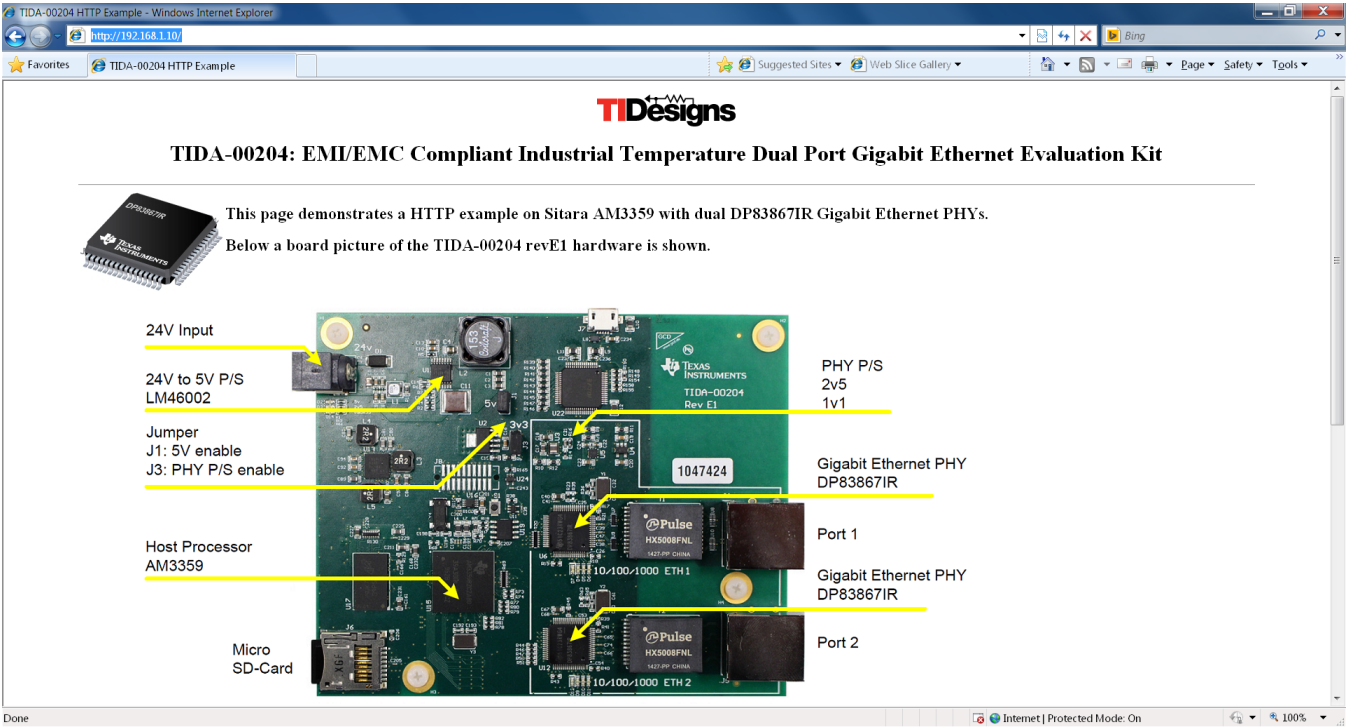


Figure 37. TIDA-00204 HTTP Web Server Example

4.3.3 HTTPWebServer 示例

打开浏览器并输入此地址：<http://192.168.1.10>。查看[图 37](#)中[TIDA-00204 网页](#)的截图。

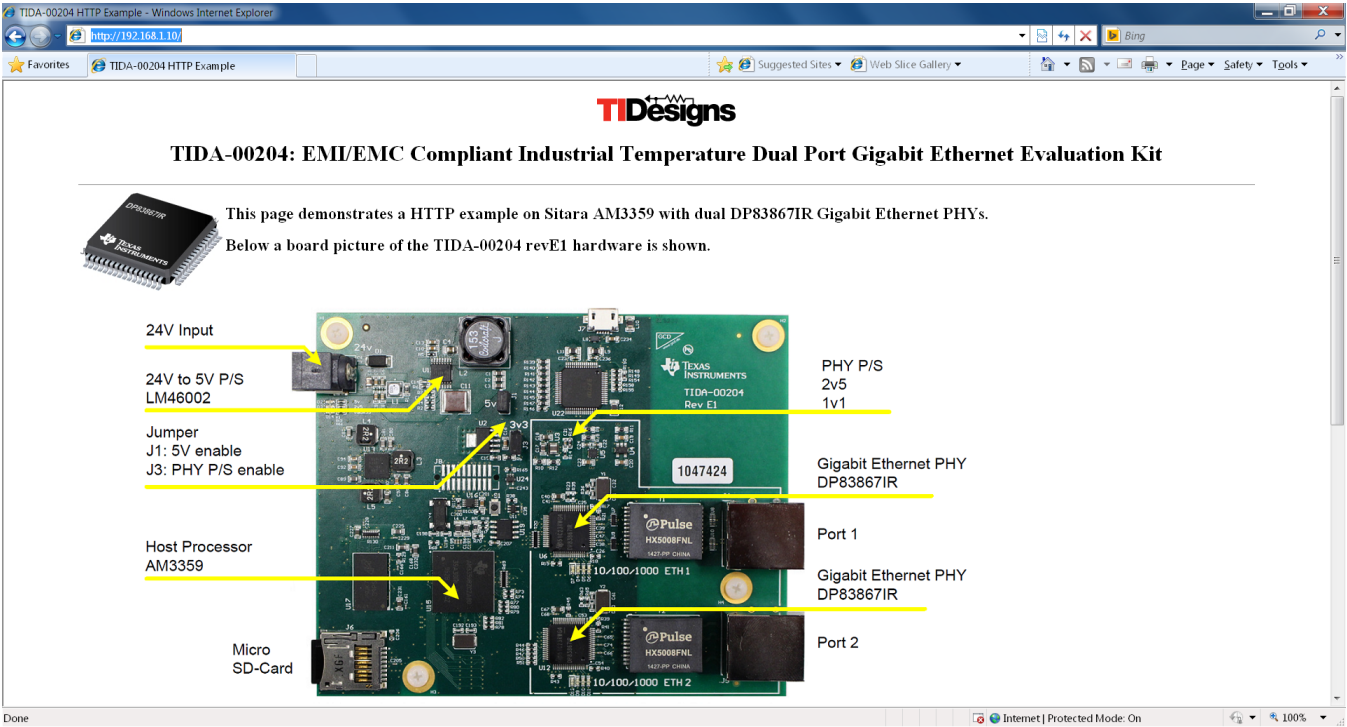


图 37。TIDA-00204 HTTP Web Server 示例

4.3.4 Access to DP83867IR Registers Through USB Virtual COM Port

The virtual COM port connection can be used to read and write to both DP83867IR registers through SMI. With this, it is possible to configure the DP83867IR registers in a different configuration than default for additional test options.

CAUTION

All inputs are in decimal and all outputs are in hexadecimal.

Figure 38 shows an example screenshot for a register read from DP83867IR on Ethernet port 2.

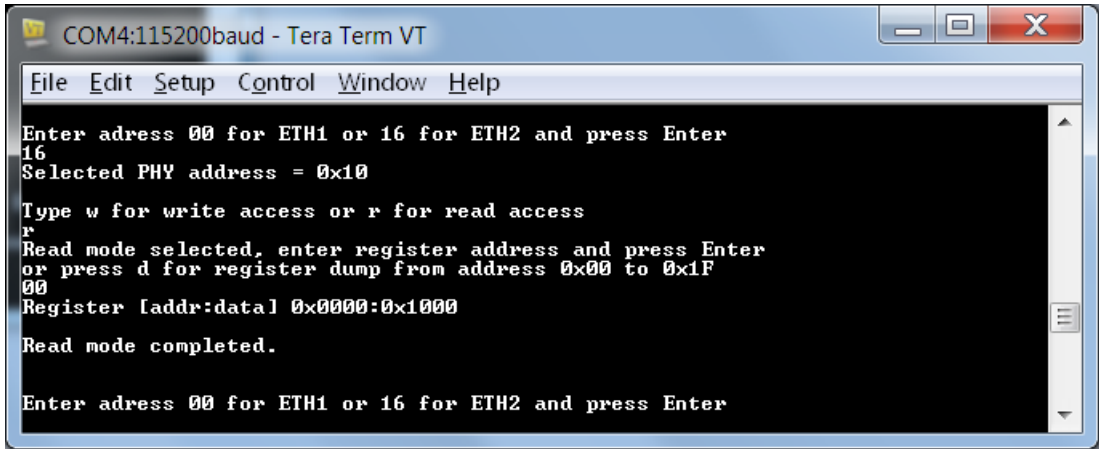


Figure 38. Virtual COM Port Connection for DP83867IR (ETH1) Register Read

To find the Gigabit Ethernet PHY DP83867IR register addresses, see the DP83867IR data sheet[1].

4.3.4 通过 USB 虚拟串口访问 DP83867IR 寄存器

通过 SMI 使用虚拟 COM 端口连接，可以读取和写入 DP83867IR 寄存器。借此，可以将 DP83867IR 寄存器配置为不同于默认值的配置，以支持额外的测试选项。

注意

所有输入均为十进制，所有输出均为十六进制。

图 38展示了从 DP83867IR 读取以太网端口 2 的寄存器的一个示例截图。

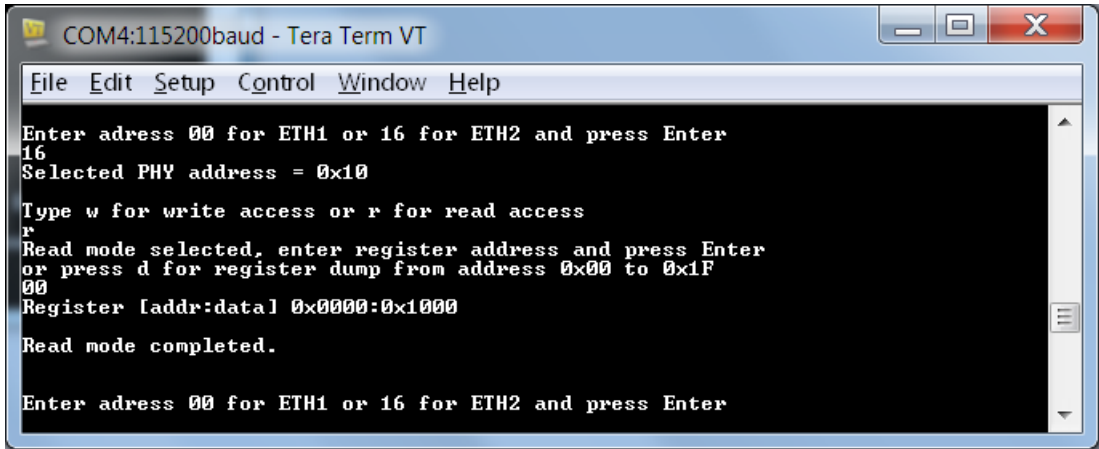


图 38. DP83867IR（ETH1）寄存器读取的虚拟 COM 端口连接

要查找千兆以太网 PHY DP83867IR 的寄存器地址，请参阅 DP83867IR 数据手册[1]。

5 Test Results

For the TIDA-00204 tests, the following test equipment was used:

Table 21. Test Equipment for the TIDA-00204 Gigabit Ethernet Signal Integrity Tests

TEST EQUIPMENT	PART #
24-V power brick	V-Infinity 3A-621DN24
Power supply	Knürr-Heinzinger Polaris 125-5
Gigabit Ethernet adapter for PC	Digitus – USB 3.0 Ethernet adapter
PC	Latitude E6540 with Windows 7 64-bit
Ethernet cable	DRAKA UC900 super screen 27 CAT7 with a CAT6A connector
Oscilloscope	Tektronix TDS794D, P6339A probes Tektronix TDS2024B, P2220 probes
Electronic load	Chroma 63103 with Chroma 6314
Multimeter	FLUKE 179

5.1 Gigabit Ethernet Signal Integrity Tests

5.1.1 RGMII Signals

The RGMII signals between the MAC and the PHY are tested with a 1000-Mb connection on both PHY1 and PHY 2 RX and TX. Figure 39 through Figure 42 show the clock signal and corresponding RX_D1 and TX_D1 data signal. The other data signals D0, D2, and D3 were measured too and exhibited the same waveforms. This was expected because the design was optimized for impedance and matched length of the clock and data lines.

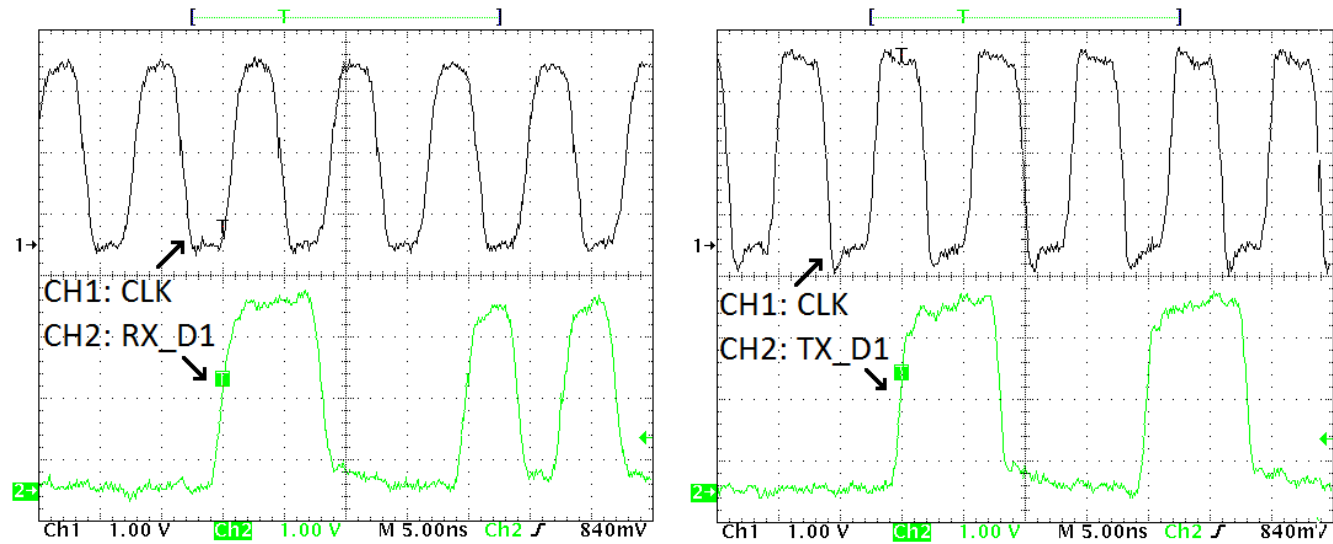


Figure 39. ETH1 RGMII RX Clock and Data Signal During Gigabit Ethernet Connection

Figure 40. ETH1 RGMII TX Clock and Data During Gigabit Ethernet Connection

5 测试结果

对于 TIDA-00204 测试，使用了以下测试设备：

表 21. TIDA-00204 千兆以太网信号完整性测试设备

测试设备	部件 #
24 伏电源适配器	V-无穷大 3A-621DN24
电源供应器	Knürr-Heinzinger Polaris 125-5
PC 千兆以太网适配器	Digitus – USB 3.0 以太网适配器
PC	Latitude E6540 搭配 Windows 7 64 位
以太网电缆	DRAKA UC900 超屏 27 CAT7 配 CAT6A 连接器
示波器	Tektronix TDS794D, P6339A 探头 Tektronix TDS2024B, P2220 探头
电子负载	Chroma 63103 与 Chroma 6314
万用表	FLUKE 179

5.1 千兆以太网信号完整性测试

5.1.1 RGMII 信号

MAC 与 PHY 之间的 RGMII 信号在 PHY1 和 PHY 2 的 RX 和 TX 上均使用 1000-Mb 连接进行测试。图 39 通过图 42 展示了时钟信号以及相应的 RX_D1 和 TX_D1 数据信号。其他数据信号 D0、D2 和 D3 也进行了测量，并表现出相同的波形。这是预期的结果，因为设计针对阻抗进行了优化，并且时钟线和数据线具有匹配的线长。

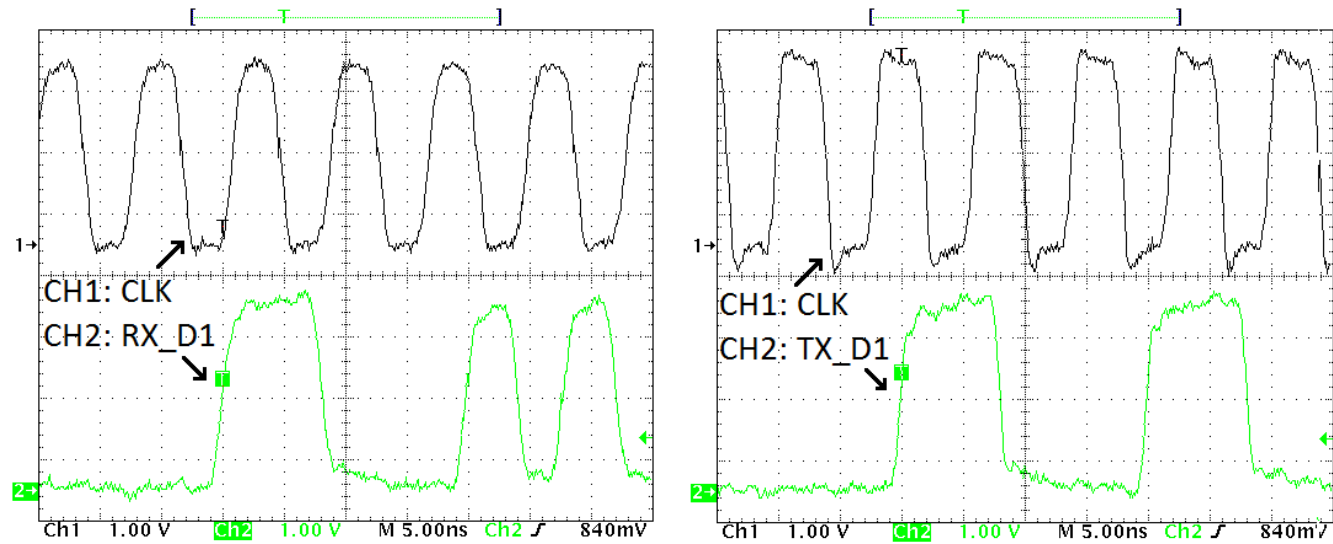


图 39. 千兆以太网连接期间 ETH1 RGMII RX 时钟和数据信号

图 40. 千兆以太网连接期间 ETH1 RGMII TXClock 和数据

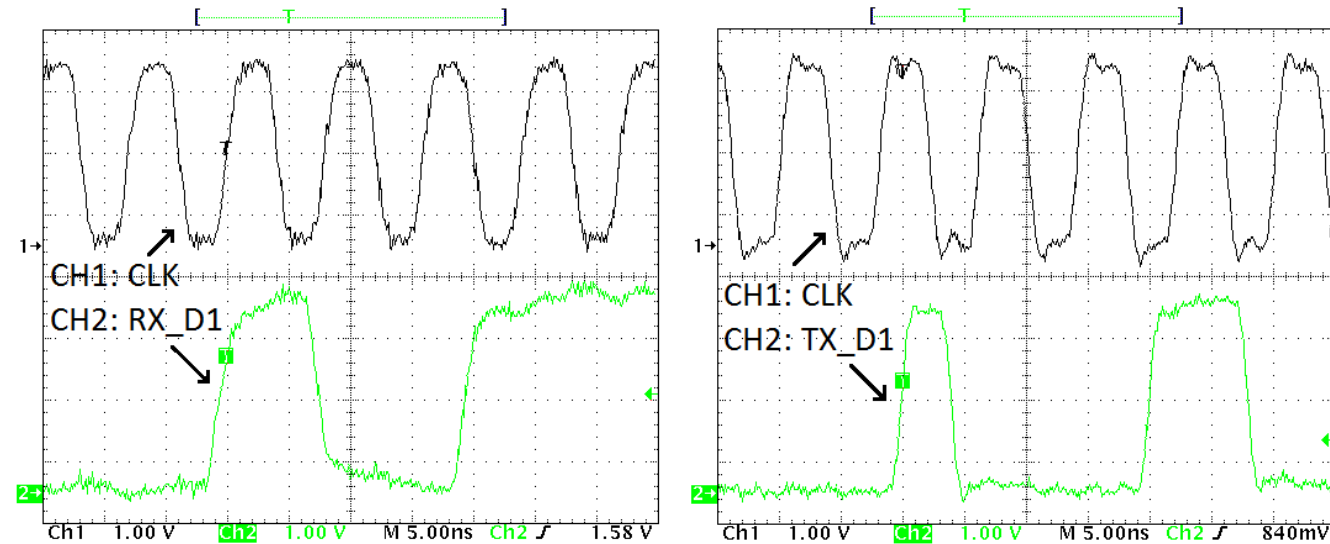


Figure 41. ETH2 RGMII RX Clock and Data Signal During Gigabit Ethernet Connection

Figure 42. ETH2 RGMII TX Clock and Data Signal During Gigabit Ethernet Connection

5.1.2 Serial Management Interface (SMI)

The maximum clock frequency of the SMI of the Sitara AM3359 is 2.5 MHz. This test does not need to have an Ethernet connection running. Figure 43 and Figure 44 show the two signals MDC (Clock) and MDIO (I/O) between the Sitara AM3359 and the two DP83867IR for Gigabit Ethernet ETH1 and ETH2.

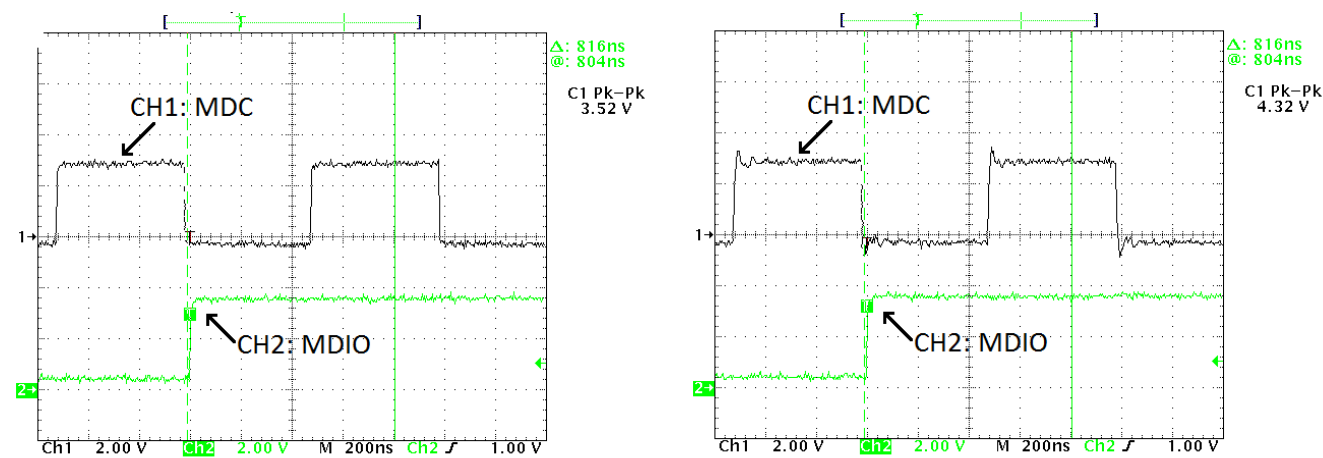


Figure 43. SMI Clock and Data Signal at 2.5 MHz at DP83867IR ETH1

Figure 44. SMI Clock and Data Signal at 2.5 MHz at DP83867IR ETH2

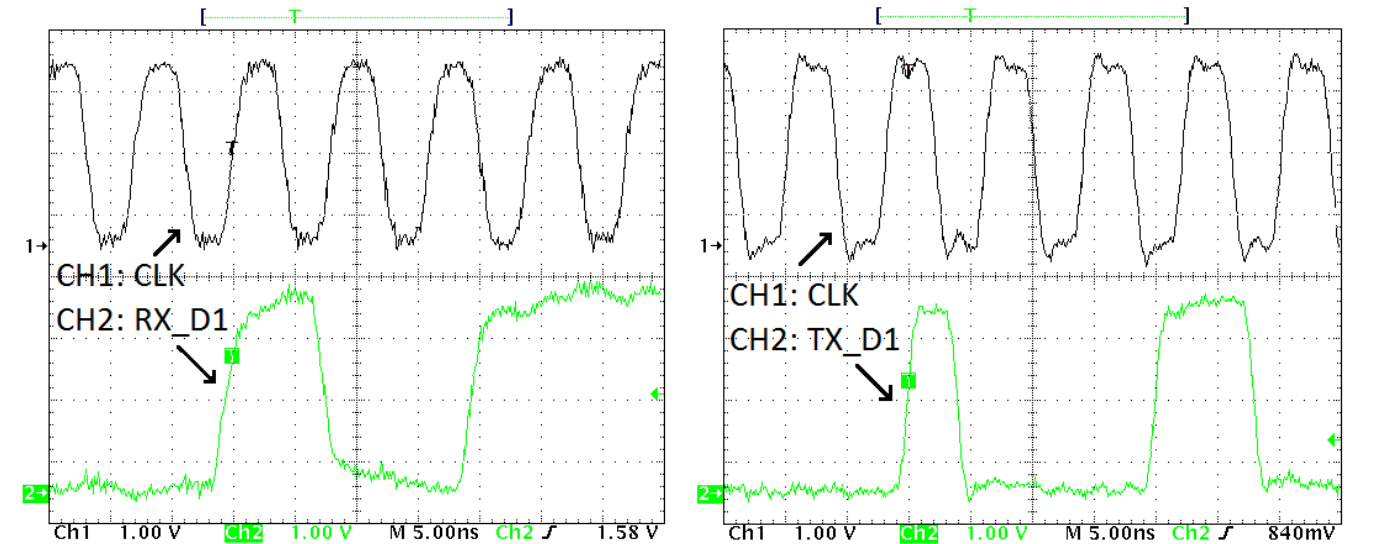


图 41. 千兆以太网连接期间 ETH2 RGMII RX 时钟和数据信号

图 42. 千兆以太网连接时的 ETH2 RGMII 发送时钟和数据信号

5.1.2 串行管理接口 (SMI)

Sitara AM3359 的 SMI 最大时钟频率为 2.5 MHz。本测试无需运行以太网连接。图 43 和图 44 展示了 Sitara AM3359 与千兆以太网 ETH1 和 ETH2 的 DP83867IR 之间的两个信号 MDC (时钟) 和 MDIO (I/O)。

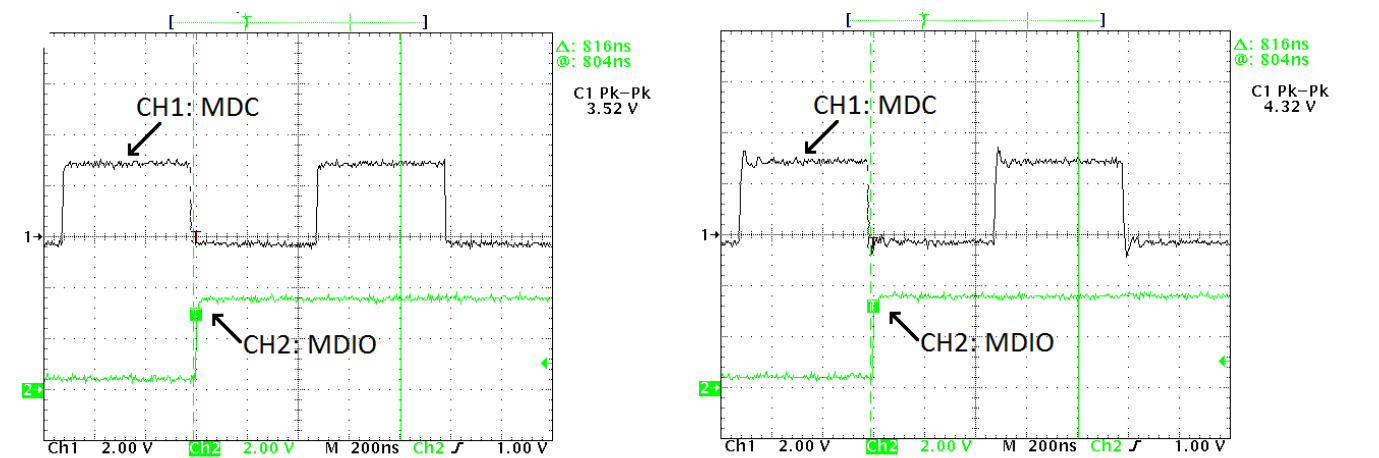


图 43. DP83867IR ETH1 的 SMI 时钟和数据信号，频率为 2.5 MHz

图 44. DP83867IR ETH2 在 2.5 MHz 处的 SMI 时钟和数据信号

5.2 Power Supply Tests

5.2.1 24-V to 5-V DC/DC Power Supply

The LM46002 was configured in this design to work with an input voltage range between 17 and 60 V.

The nominal load current was defined for the following configuration. The Sitara AM3359 running at 600 MHz executing the application code from DDR3, with both DP83867IR connected in Gigabit Ethernet mode with continuous UDP packet transfer at 20% network utilization on both ports. The minimum load current is set to 500 mA, the nominal load to 850 mA, and the maximum load to 1.2 A.

Three tests were done: Output voltage versus input voltage (Figure 45), output voltage versus load current (Figure 46), and the output voltage ripple (Figure 47).

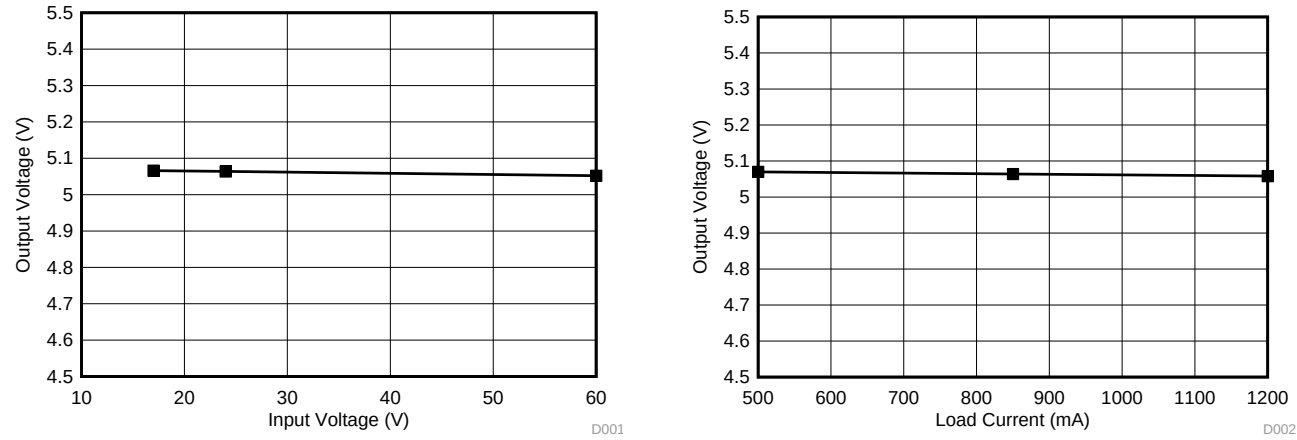


Figure 45. Output Voltage (V_{OUT}) at 850 mA Nominal Load Current With Varying Input Voltage (V_{IN})

Figure 46. Load Regulation: V_{OUT} at 24-V Input Voltage at Varying Loads

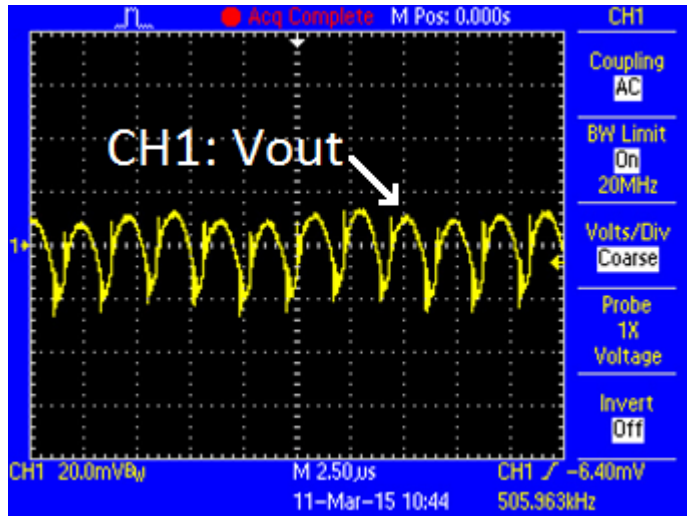


Figure 47. V_{OUT} Output Voltage Ripple at 850 mA Load Current and 24-V V_{IN} Input Voltage

5.2 电源供应器测试

5.2.1 24-V 至 5-V 直流电源供应器

该设计中，LM46002 被配置为在 17 至 60 V 的输入电压范围内工作。

额定负载电流针对以下配置定义：Sitara AM3359 以 600 MHz 运行，从 DDR3 执行应用程序代码，两个 DP83867IR 均连接在千兆以太网模式下，两个端口均以 20% 的网络利用率进行连续的 UDP 数据包传输。最小负载电流设定为 500 mA，额定负载电流为 850 mA，最大负载电流为 1.2 A。

已完成的三项测试：输出电压与输入电压（图 45），输出电压与负载电流（图 46），以及输出电压纹波（图 47）。

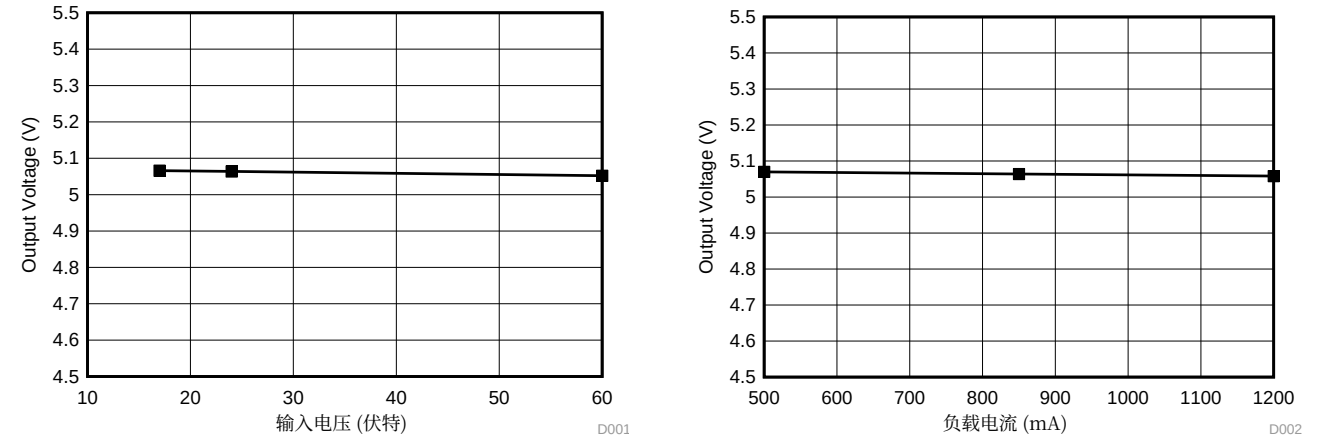


图 45. 在 850 毫安额定负载电流下，随输入电压 (V_{IN}) 变化的输出电压 (V_{OUT})

图 46. 负载调整率：在 24-V 输入电压下，不同负载时的 V_{OUT}

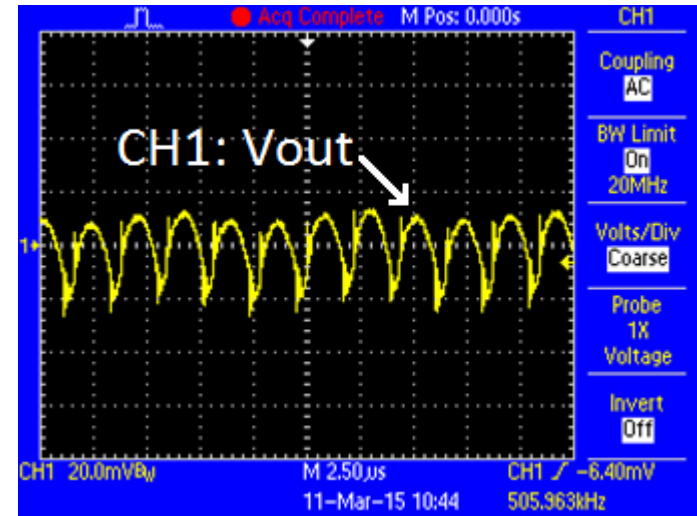


图 47. 850 mA 负载电流和 24-V 输入电压下的 V_{OUT} 输出电压纹波

5.2.2 Power Supply for the DP83867IR Gigabit Ethernet PHYs Tests

The voltages have been verified as shown in Table 22. The startup of the 3v3, 2v5, and 1v1 rails are shown in Figure 48.

Table 22. Measured Voltage Rails of the Gigabit Ethernet PHY

SUPPLY NAME PER SCHEMATICS	SPECIFIED VOLTAGE	MEASURED
3v3	3.3 V	3.31 V
2v5	2.5 V	2.47 V
1v1	1.1 V	1.10 V

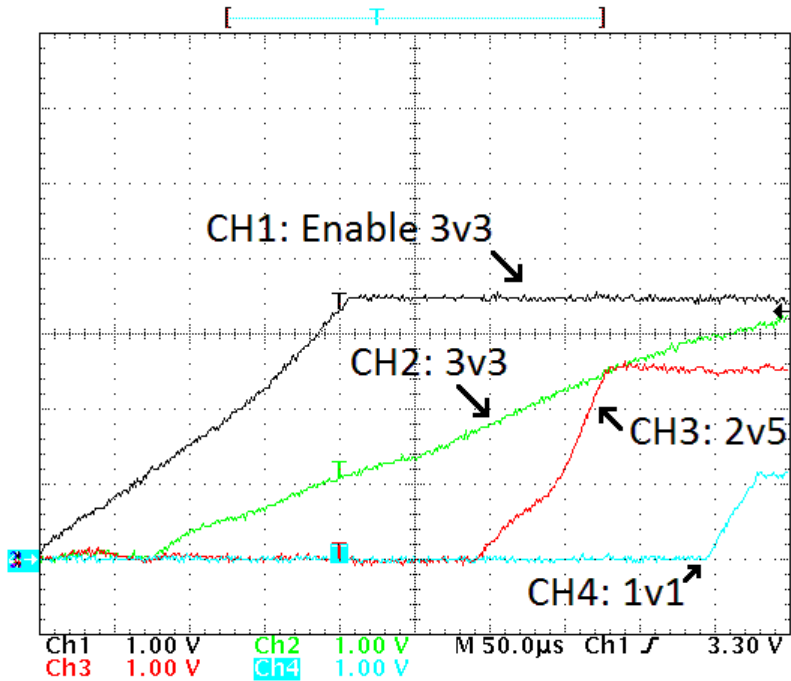


Figure 48. PHY Power Supply Voltage Rails During Startup

5.2.2 DP83867IR 千兆以太网 PHY 电源测试

电压已验证，如表 22 所示。3v3、2v5 和 1v1 轨的启动情况如图 48 所示。

表 22. 千兆以太网 PHY 的实测电压轨

SUPPLY NAME PER SCHEMATICS	指定电压	已测量
3v3	3.3 伏特	3.31 伏特
2v5	2.5 伏特	2.47 伏特
1v1	1.1 伏特	1.10 伏特

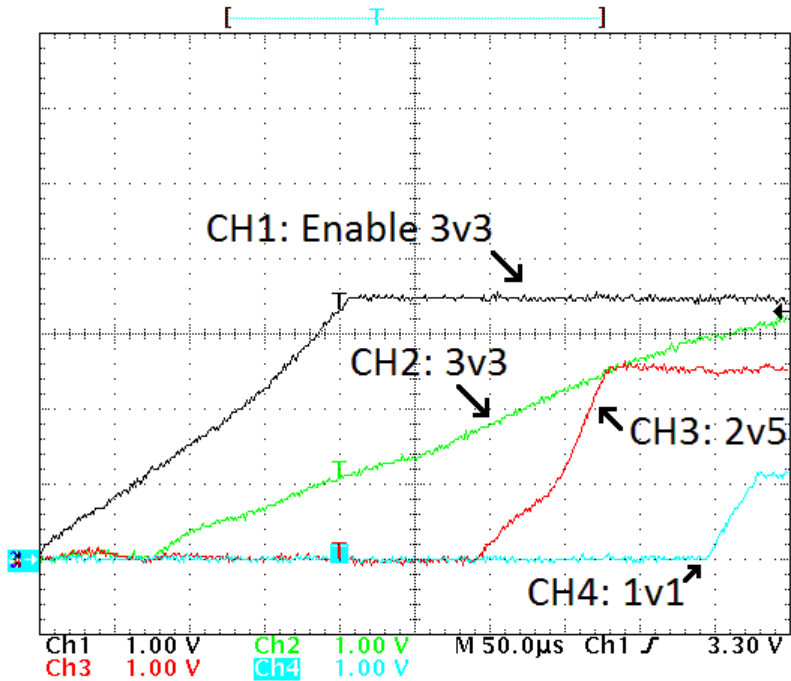


图 48. 启动期间 PHY 电源供应器电压轨

5.2.3 PMIC for Sitara AM3359 Host Processor

The TPS65910A3 output voltages were verified and compared against the expected voltages based on the TPS65910A3 EEPROM pin configuration.

Table 23. Measured Voltages on the PMIC TPS65910A3

SIGNAL	EXPECTED VOLTAGE	MEASURED VOLTAGE
VIO	1.5 V	1.52 V
VDD1	1.1 V	1.11 V
VDD2	1.1 V	1.11 V
VDD3	OFF	N/A
VDIG1	1.8 V	NC. VDIG1 pin was not connected in this design.
VDIG2	1.8 V	1.81 V
VPLL	1.8 V	1.82 V
VDAC	1.8 V	1.80 V
VAUX1	1.8 V	1.82 V
VAUX2	3.3 V	3.31 V
VAUX33	3.3 V	3.33 V
VMMC	3.3 V	3.32 V
VRTC	OFF	N/A

The sequencing has not been tested. Please see *TPS65910Ax User's Guide For AM335x Processors* for further details[7].

5.2.3 Sitara AM3359 主机处理器的电源管理集成电路

已根据 TPS65910A3 EEPROM 引脚配置验证并对比了 TPS65910A3 的输出电压与预期电压。

表 23. PMIC TPS65910A3 上的测量电压

信号	预期电压	测量电压
VIO	1.5 伏特	1.52 伏特
VDD1	1.1 伏特	1.11 伏特
VDD2	1.1 伏特	1.11 伏特
VDD3	OFF	N/A
VDIG1 引脚	1.8 伏特	NC。本设计中未连接 VDIG1 引脚。
VDIG2	1.8 伏特	1.81 伏特
VPLL	1.8 伏特	1.82 伏特
VDAC	1.8 伏特	1.80 伏特
VAUX1	1.8 伏特	1.82 伏特
VAUX2	3.3 伏特	3.31 伏特
VAUX33	3.3 伏特	3.33 伏特
VMMC	3.3 伏特	3.32 伏特
VRTC	OFF	N/A

尚未对序列进行测试。有关详细信息，请参阅《TPS65910Ax 处理器 AM335x 处理器用户指南》[7]。

5.3 Gigabit Ethernet Functional Tests

The following functions have been tested: autonegotiate, ping, UDP transfer, and web server.

Table 24. Functional Tests of TIDA-00204 Reference Design

FUNCTION	TEST RESULT
Autonegotiate	PASS
Ping	PASS
UDP	PASS
Web server	PASS

5.3.1 Autonegotiate

Testing the autonegotiate is done using a network connection which can be programmed into a specific autonegotiate condition. Doing this ensures that the link negotiated is not going into a default setting defined in the IEEE 802.3ab standard. This can happen if one partner of the link is forced and the other is set to autonegotiate.

The default autonegotiate setting of the DP83867IR is set 10/100/1000-Mb half- or full-duplex mode. The DP83867IR recognizes the data rate supported by partner and sets the highest data rate supported by both.

Table 25 shows the different autonegotiate mode results.

Table 25. DP83867IR Autonegotiate Mode Test Results

DP83867IR SETTING	LINK PARTNER SPEED MODE SETTING	LINK PARTNER DUPLEX MODE SETTING	LINK CONNECTION ESTABLISHED
Autonegotiate: 10/100/1000 full duplex or half duplex	10/100/1000	Full duplex	1000 full duplex
	10/100/1000	Half duplex	1000 half duplex
	10/100	Full duplex	100 full duplex
	10/100	Half duplex	100 half duplex
	10	Full duplex	10 full duplex
	10	Half duplex	10 half duplex

5.3 千兆以太网功能测试

已测试以下功能：自动协商、Ping、UDP 传输和 Web 服务器。

表 24. TIDA-00204 参考设计的功能测试

功能	测试结果
自动协商	PASS
Ping	PASS
UDP	PASS
Web 服务器	PASS

5.3.1 自动协商

使用可配置为特定自协商状态的网络连接来测试自协商功能。执行此操作可确保协商出的链路不会进入 IEEE 802.3ab 标准中定义的默认设置。如果链路的其中一个端点被强制设置，而另一个端点设置为自协商，则可能发生这种情况。

DP83867IR 的默认自动协商设置将 10/100/1000-Mb 半双工或全双工模式设置为。DP83867IR 会识别合作伙伴支持的数据速率，并设置双方支持的最高数据速率。

表 25展示了不同的自动协商模式结果。

表 25. DP83867IR 自动协商模式测试结果

DP83867IR 设置	链路合作伙伴速度模式设置	链路伙伴双工模式设置	链接连接已建立
自动协商：10/100/1000 全双工 全双工或半双工	10/100/1000	全双工	1000 全双工
	10/100/1000	半双工	1000 半双工
	10/100	全双工	100 全双工
	10/100	半双工	100 半双工
	10	全双工	10 全双工
	10	半双工	10 半双工

5.3.2 Ping

Ping is invoked from the command prompt using the TIDA-00204 static IP address 192.168.1.10. Figure 49 presents a command prompt printout that shows the ping test.

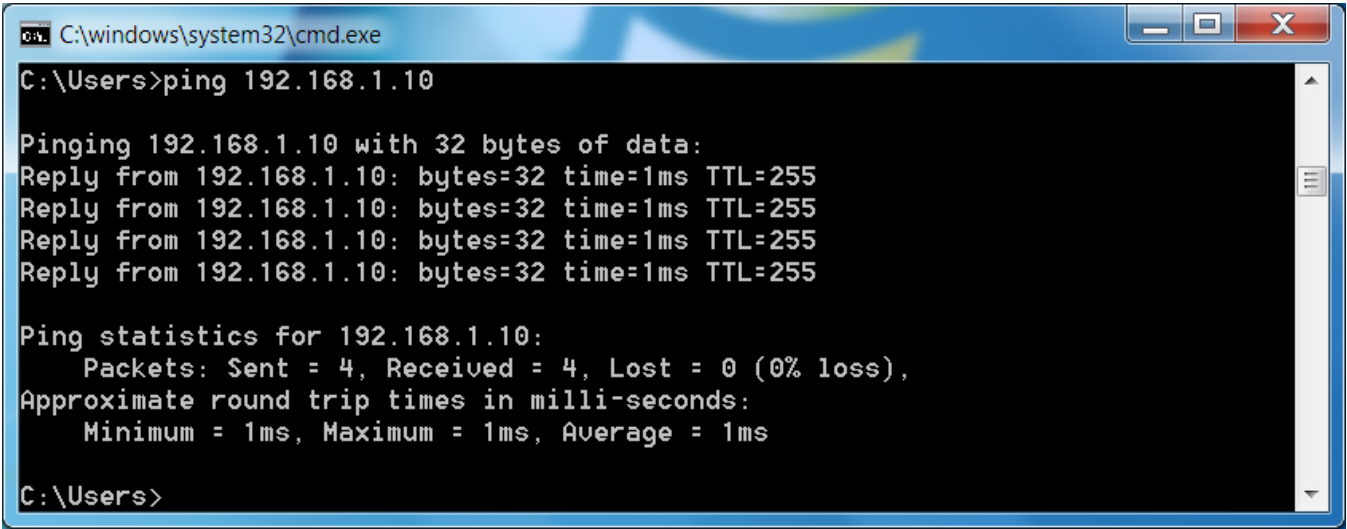


Figure 49. Command Prompt Ping Print

Wireshark was used to log the packet transfer during the ping test. In Figure 50, the Wireshark printout result can be seen.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	ActionSt_12:d9:ed	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.30
2	0.001222000	TexasIns_f5:bb:ef	ActionSt_12:d9:ed	ARP	60	192.168.1.10 is at 88:33:14:f5:bb:ef
3	0.000149000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128
4	0.001081000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=255
5	0.992019000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128
6	0.001234000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=255
7	1.012331000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128
8	0.000812000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=255
9	1.013503000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128
10	0.001241000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=255

Figure 50. Wireshark Log of TIDA-00204 Ping Test

Using a Linux PC as the test machine when running ping tests gives a better indication of the Ethernet utilization. The Linux ping command gives the user more options to define the ping condition as well as the ping speed.

The ping command to use here is:

```
sudo ping 192.168.1.10 -s 1400 -p 5a -f
```

The following result is achieved with this command:

PATTERN: 0x5a

PING 192.168.1.10 (192.168.1.10) 1400(1428) bytes of data.

.^

--- 192.168.1.10 ping statistics ---

952050 packets transmitted, 952050 received, 0% packet loss, time 150052ms

rtt min/avg/max/mdev = 0.100/0.132/0.264/0.013 ms, ipg/ewma 0.157/0.137 ms

This test shows that the TIDA-00204 reference design can send large payload packages with high utilization of the Ethernet port and without any packet errors.

5.3.2 Ping

Ping 使用 TIDA-00204 的静态 IP 地址 192.168.1.10 从命令提示符中调用。图 49展示了显示 Ping 测试结果的命令提示符输出。

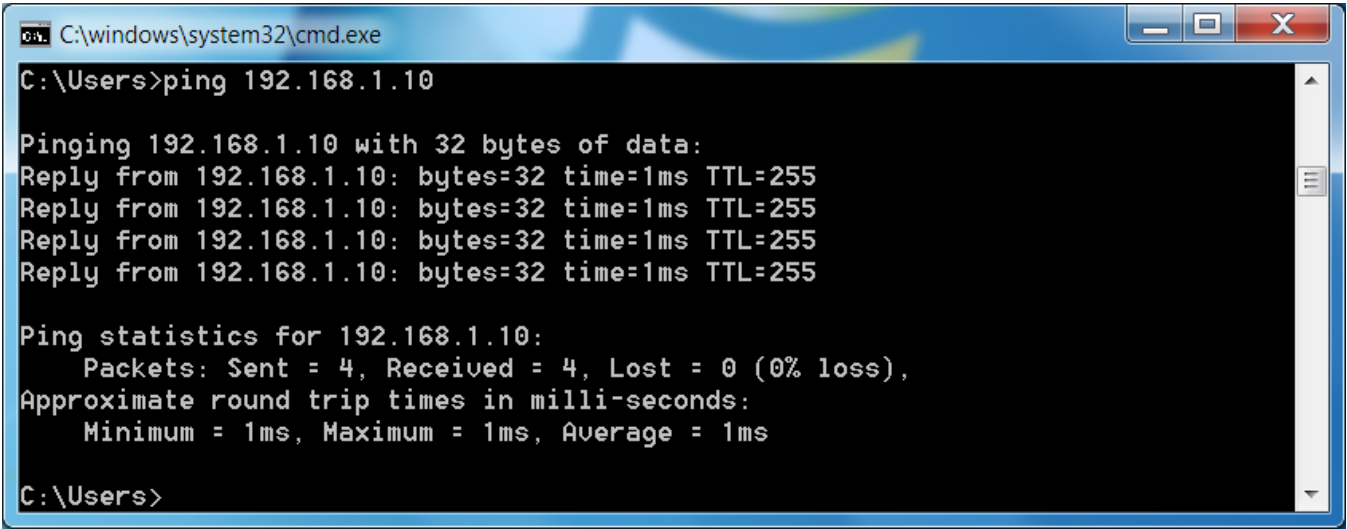


图 49。命令提示符 Ping 打印

Wireshark 被用于记录 ping 测试期间的数据包传输。在 图 50中，可以看到 Wireshark 的打印结果。

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	ActionSt_12:d9:ed	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.30
2	0.001222000	TexasIns_f5:bb:ef	ActionSt_12:d9:ed	ARP	60	192.168.1.10 is at 88:33:14:f5:bb:ef
3	0.000149000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128
4	0.001081000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=255
5	0.992019000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128
6	0.001234000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=255
7	1.012331000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128
8	0.000812000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=255
9	1.013503000	192.168.1.30	192.168.1.10	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128
10	0.001241000	192.168.1.10	192.168.1.30	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=255

图 50. TIDA-00204 Ping 测试的 Wireshark 日志

使用 Linux PC 作为测试机进行 Ping 测试能更好地反映以太网利用率。Linux ping 命令为用户提供了更多选项，以定义 Ping 条件及 Ping 速度。

此处使用的 Ping 命令为：

```
sudo ping 192.168.1.10 -s 1400 -p 5a -f
```

使用此命令可实现以下结果：

PATTERN: 0x5a

PING 192.168.1.10 (192.168.1.10) 1400(1428) bytes of data.

.^

--- 192.168.1.10 Ping 统计信息 ---

952050 个数据包已发送，952050 个已接收，丢包率为 0%，耗时 150052 毫秒

rtt min/avg/max/mdev = 0.100/0.132/0.264/0.013 ms, ipg/ewma 0.157/0.137 ms

该测试表明，TIDA-00204 参考设计能够在以太网端口高利用率的情况下发送大包，且无数据包错误。

5.3.3 UDP

The UDP throughput test is invoked from the command prompt using the TIDA-00204 static IP address 192.168.1.10. Figure 51 presents a command prompt printout that shows the UDP throughput test.

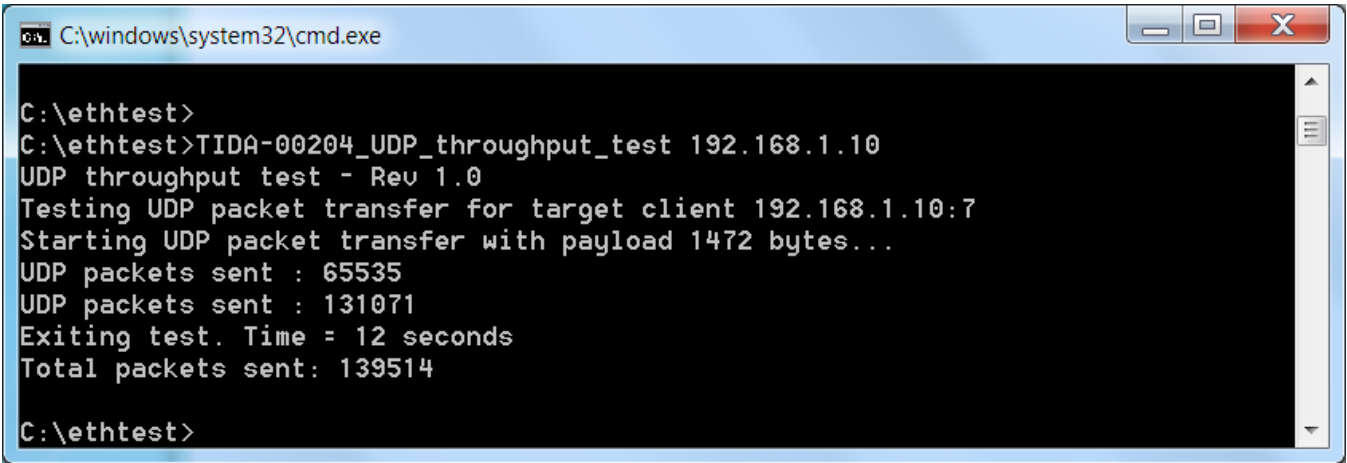


Figure 51. Command Prompt UDP Throughput Print

Windows Task Manager networking was used to log the packet transfer utilization during the UDP throughput test. Figure 52 shows the Windows Task Manager printout.

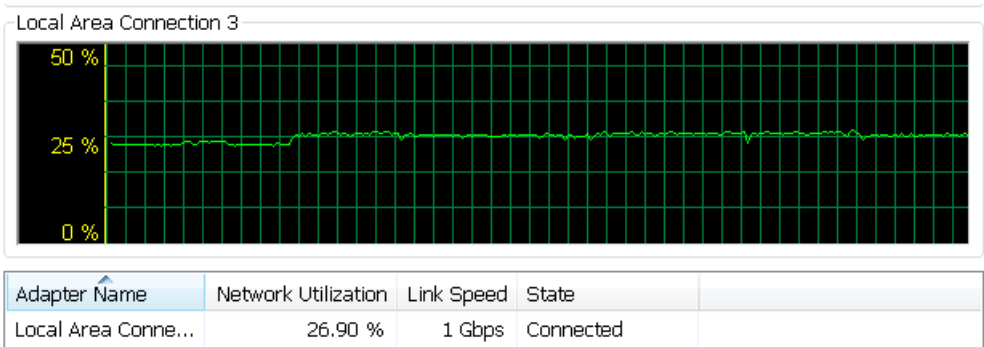


Figure 52. Utilization of One Port Connection With UDP Throughput Test Running

With the throughput test file, it is possible to achieve around 25% utilization of the Gigabit bandwidth. The slight change in the utilization on the curve is due to other network or CPU tasks.

When running this test on both ports simultaneously, the utilization will be 20% on each port.

For the UDP packet error test, the utilization is 5% independent of how many ports are connected, which is a limitation of the UDP packet error test software running on the PC.

5.3.3 UDP

使用 TIDA-00204 静态 IP 地址 192.168.1.10 从命令提示符中调用 UDP 吞吐量测试。图 51 展示了 UDP 吞吐量测试的命令提示符输出。

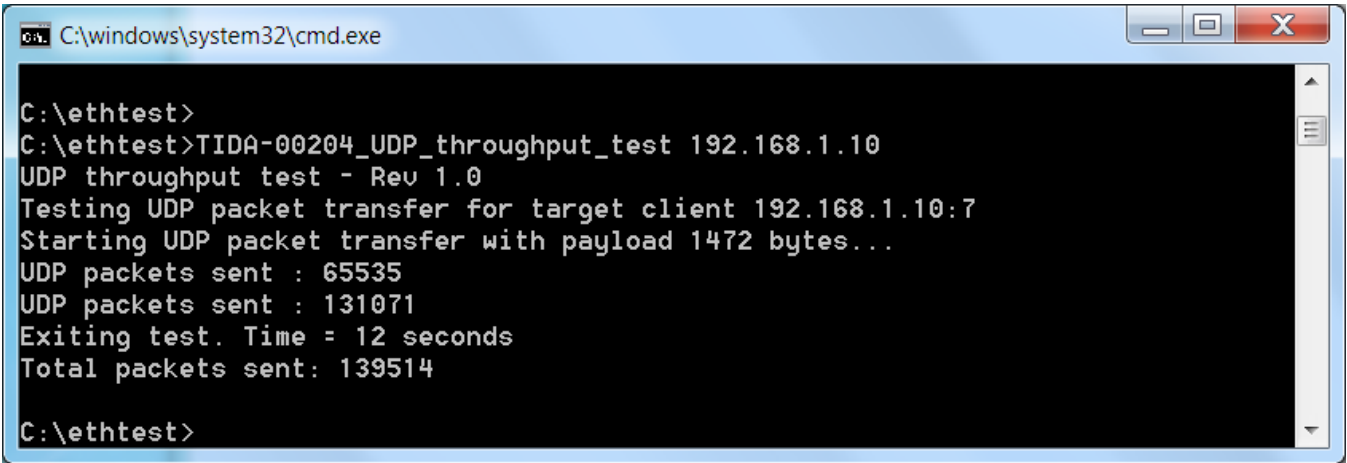


图 51. 命令提示符 UDP 吞吐量打印

Windows 任务管理器网络功能用于记录 UDP 吞吐量测试期间的数据包传输利用率。图 52 显示了 Windows 任务管理器的打印输出。

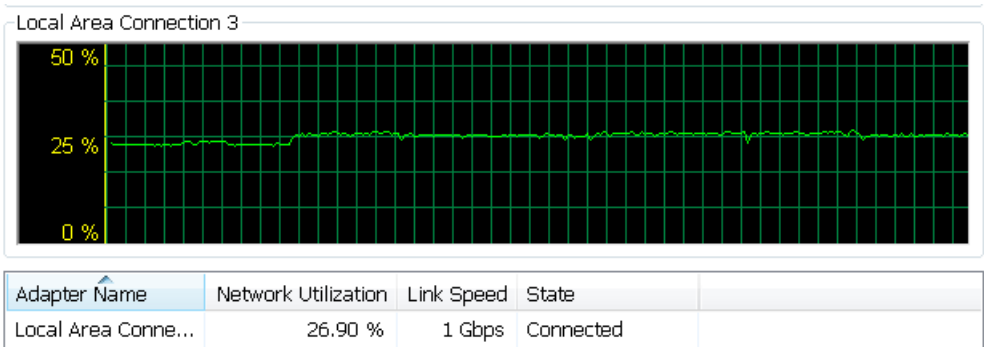


图 52. 使用 UDP 吞吐量测试运行单端口连接利用率

使用 throughputtestfile，可实现约 25% 的千兆带宽利用率。曲线中利用率的微小变化归因于其他网络或 CPU 任务。

当在两个端口上同时运行此测试时，每个端口的利用率将为 20%。

对于 UDP 数据包错误测试，利用率固定为 5%，与连接的端口数量无关，这是运行在 PC 上的 UDP 数据包错误测试软件的局限性。

5.3.4 HTTP Web Server

To test the HTTP web page example, the TIDA-00204's static IP address http://192.168.1.10 is entered into a browser (see Figure 53).

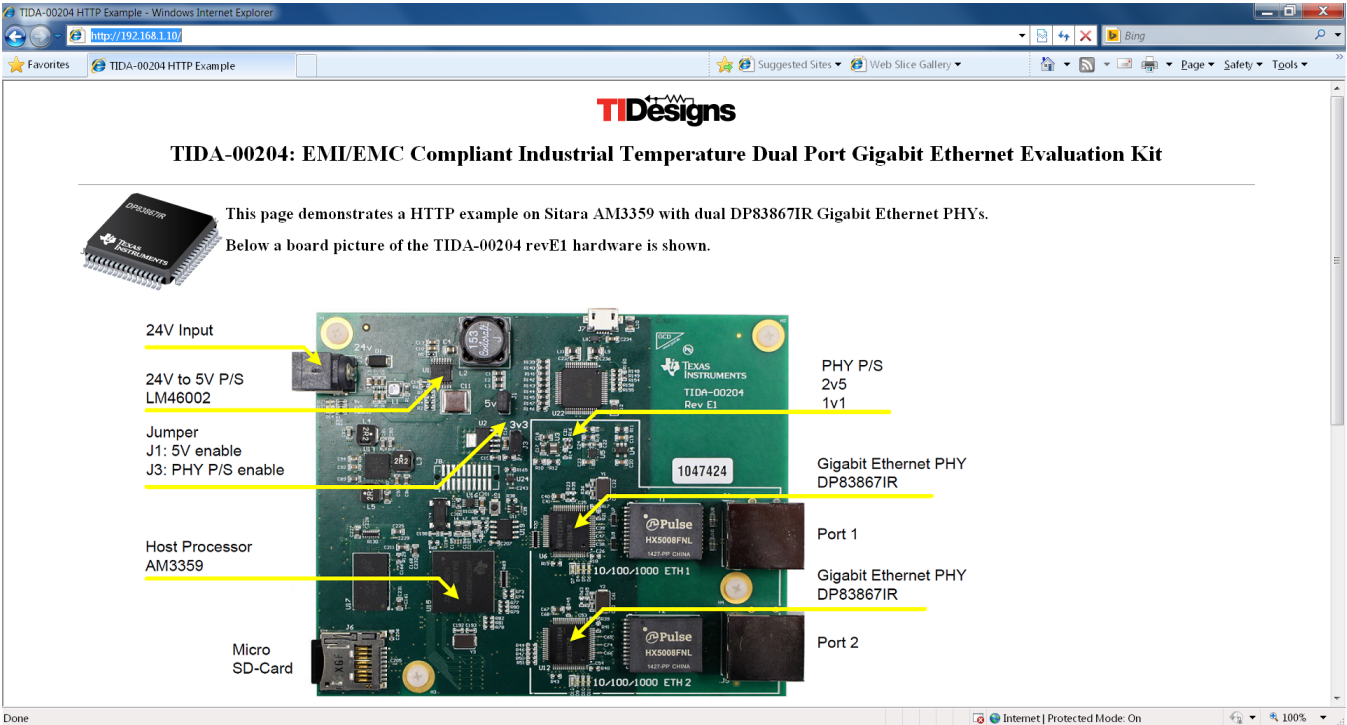


Figure 53. HTTP Web Server Example

For this test, Wireshark was used to log the traffic generated by initializing the web server and the loading of the page.

Figure 54 shows the traffic log from Wireshark.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8e450398
2	1.860607000	ActionSt_12:d9:ed	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.30
3	0.000942000	TexasIns_f5:bb:ef	ActionSt_12:d9:ed	ARP	60	192.168.1.10 is at 88:33:14:f5:bb:ef
4	0.000085000	192.168.1.30	192.168.1.10	TCP	66	64736->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
5	0.000886000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [SYN, ACK] Seq=0 Ack=1 Win=4096 Len=0 MSS=1460
6	0.000099000	192.168.1.10	192.168.1.10	TCP	54	64736->80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
7	0.000254000	192.168.1.30	192.168.1.10	HTTP	565	GET / HTTP/1.1
8	0.000890000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [ACK] Seq=1 Ack=512 Win=3585 Len=0
9	0.000018000	192.168.1.10	192.168.1.30	TCP	97	[TCP segment of a reassembled PDU]
10	0.000067000	192.168.1.10	192.168.1.30	TCP	78	[TCP segment of a reassembled PDU]
11	0.000054000	192.168.1.10	192.168.1.30	HTTP	1127	HTTP/1.0 200 OK (text/html)
12	0.000106000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [FIN, ACK] Seq=1141 Ack=512 Win=4096 Len=0
13	0.000030000	192.168.1.30	192.168.1.10	TCP	54	64736->80 [ACK] Seq=512 Ack=68 Win=64173 Len=0
14	0.000008000	192.168.1.30	192.168.1.10	TCP	54	64736->80 [ACK] Seq=512 Ack=1142 Win=63100 Len=0
15	0.000757000	192.168.1.30	192.168.1.10	TCP	54	64736->80 [FIN, ACK] Seq=512 Ack=1142 Win=63100 Len=0
16	0.000862000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [ACK] Seq=1142 Ack=513 Win=4096 Len=0
17	0.002456000	192.168.1.30	192.168.1.10	TCP	66	64737->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
18	0.000887000	192.168.1.10	192.168.1.30	TCP	60	80->64737 [SYN, ACK] Seq=0 Ack=1 Win=4096 Len=0 MSS=1460
19	0.000113000	192.168.1.30	192.168.1.10	TCP	54	64737->80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
20	0.000041000	192.168.1.30	192.168.1.10	TCP	66	64738->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
21	0.000165000	192.168.1.30	192.168.1.10	HTTP	425	GET /TIDesignsLogo.png HTTP/1.1
22	0.000670000	192.168.1.10	192.168.1.30	TCP	60	80->64738 [SYN, ACK] Seq=0 Ack=1 Win=4096 Len=0 MSS=1460
23	0.000070000	192.168.1.10	192.168.1.30	TCP	60	80->64737 [ACK] Seq=1 Ack=372 Win=3725 Len=0
24	0.000010000	192.168.1.10	192.168.1.30	TCP	112	[TCP segment of a reassembled PDU]
25	0.000057000	192.168.1.10	192.168.1.30	TCP	78	[TCP segment of a reassembled PDU]
26	0.000042000	192.168.1.10	192.168.1.30	TCP	1514	[TCP segment of a reassembled PDU]
27	0.000045000	192.168.1.10	192.168.1.30	HTTP	838	HTTP/1.0 200 OK (PNG)
28	0.000027000	192.168.1.10	192.168.1.30	TCP	60	80->64737 [FIN, ACK] Seq=2327 Ack=372 Win=4096 Len=0
29	0.000027000	192.168.1.30	192.168.1.10	TCP	54	64738->80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
30	0.000008000	192.168.1.30	192.168.1.10	TCP	66	64739->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
31	0.000005000	192.168.1.30	192.168.1.10	TCP	54	64737->80 [ACK] Seq=372 Ack=83 Win=64158 Len=0
32	0.000007000	192.168.1.30	192.168.1.10	HTTP	421	GET /DP83867IR.png HTTP/1.1

Figure 54. Wireshark Log of TIDA-00204 HTTP Web Server Example

5.3.4 HTTP Web 服务器

要测试 HTTP 网页示例，请将 TIDA-00204 的静态 IP 地址 http://192.168.1.10 输入到浏览器中（参见 图 53）。

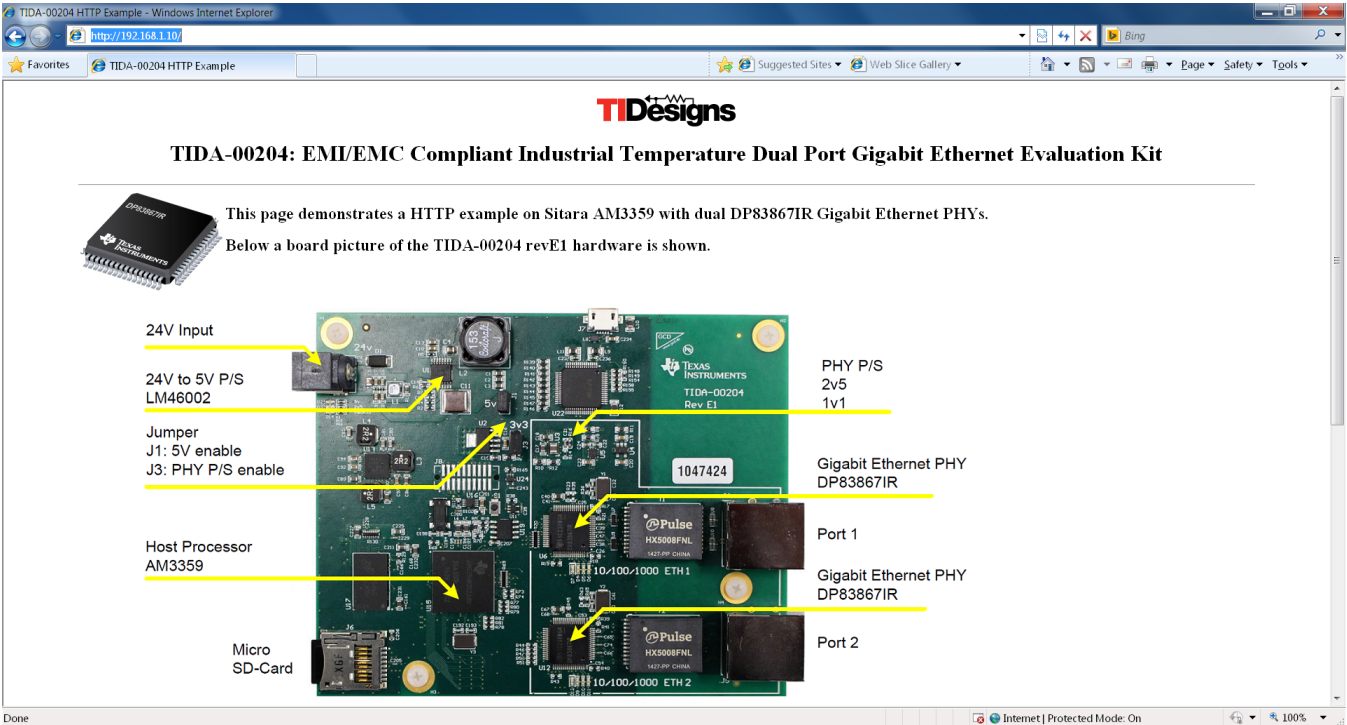


图 53. HTTP Web 服务器示例

本次测试中，使用 Wireshark 记录初始化 Web 服务器及页面加载过程中产生的流量。

图 54显示了 Wireshark 的流量日志。

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8e450398
2	1.860607000	ActionSt_12:d9:ed	Broadcast	ARP	42	Who has 192.168.1.10? Tell 192.168.1.30
3	0.000942000	TexasIns_f5:bb:ef	ActionSt_12:d9:ed	ARP	60	192.168.1.10 is at 88:33:14:f5:bb:ef
4	0.000085000	192.168.1.30	192.168.1.10	TCP	66	64736->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
5	0.000886000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [SYN, ACK] Seq=0 Ack=1 Win=4096 Len=0 MSS=1460
6	0.000099000	192.168.1.10	192.168.1.10	TCP	54	64736->80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
7	0.000254000	192.168.1.30	192.168.1.10	HTTP	565	GET / HTTP/1.1
8	0.000890000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [ACK] Seq=1 Ack=512 Win=3585 Len=0
9	0.000018000	192.168.1.10	192.168.1.30	TCP	97	[TCP segment of a reassembled PDU]
10	0.000067000	192.168.1.10	192.168.1.30	TCP	78	[TCP segment of a reassembled PDU]
11	0.000054000	192.168.1.10	192.168.1.30	HTTP	1127	HTTP/1.0 200 OK (text/html)
12	0.000106000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [FIN, ACK] Seq=1141 Ack=512 Win=4096 Len=0
13	0.000030000	192.168.1.30	192.168.1.10	TCP	54	64736->80 [ACK] Seq=512 Ack=68 Win=64173 Len=0
14	0.000008000	192.168.1.30	192.168.1.10	TCP	54	64736->80 [ACK] Seq=512 Ack=1142 Win=63100 Len=0
15	0.000757000	192.168.1.30	192.168.1.10	TCP	54	64736->80 [FIN, ACK] Seq=512 Ack=1142 Win=63100 Len=0
16	0.000862000	192.168.1.10	192.168.1.30	TCP	60	80->64736 [ACK] Seq=1142 Ack=513 Win=4096 Len=0
17	0.002456000	192.168.1.30	192.168.1.10	TCP	66	64737->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
18	0.000887000	192.168.1.10	192.168.1.30	TCP	60	80->64737 [SYN, ACK] Seq=0 Ack=1 Win=4096 Len=0 MSS=1460
19	0.000113000	192.168.1.30	192.168.1.10	TCP	54	64737->80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
20	0.000041000	192.168.1.30	192.168.1.10	TCP	66	64738->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
21	0.000165000	192.168.1.30	192.168.1.10	HTTP	425	GET /TIDesignsLogo.png HTTP/1.1
22	0.000670000	192.168.1.10	192.168.1.30	TCP	60	80->64738 [SYN, ACK] Seq=0 Ack=1 Win=4096 Len=0 MSS=1460
23	0.000070000	192.168.1.10	192.168.1.30	TCP	60	80->64737 [ACK] Seq=1 Ack=372 Win=3725 Len=0
24	0.000010000	192.168.1.10	192.168.1.30	TCP	112	[TCP segment of a reassembled PDU]
25	0.000057000	192.168.1.10	192.168.1.30	TCP	78	[TCP segment of a reassembled PDU]
26	0.000042000	192.168.1.10	192.168.1.30	TCP	1514	[TCP segment of a reassembled PDU]
27	0.000045000	192.168.1.10	192.168.1.30	HTTP	838	HTTP/1.0 200 OK (PNG)
28	0.000027000	192.168.1.10	192.168.1.30	TCP	60	80->64737 [FIN, ACK] Seq=2327 Ack=372 Win=4096 Len=0
29	0.000027000	192.168.1.30	192.168.1.10	TCP	54	64738->80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
30	0.000008000	192.168.1.30	192.168.1.10	TCP	66	64739->80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=4 SACK_PERM=1
31	0.000005000	192.168.1.30	192.168.1.10	TCP	54	64737->80 [ACK] Seq=372 Ack=83 Win=64158 Len=0
32	0.000007000	192.168.1.30	192.168.1.10	HTTP	421	GET /DP83867IR.png HTTP/1.1

图 54. TIDA-00204 HTTP Web Server 示例的 Wireshark 日志

5.4 EMC/EMI Test Results

The TIDA-00204 TI design has been tested for EMI according to CISPR 11 / EN55011 Class A radiated emissions. For EMC immunity, the design has been tested according to IEC61800-3 and IEC61000-6-2 for ESD, EFT, and Surge with reference to standards IEC61000-4-2, IEC61000-4-4, and IEC61000-4-5, respectively.

The design is compliant to these standards and exceeds the voltage requirements according to IEC61800-3 EMC immunity requirements. A summary is shown in the following tables and more details in the following sections.

Table 26. CISPR 11 / EN55011 Class A Radiated Emission Requirements for Category 2 and Measured Level

REQUIREMENTS			TIDA-00204 MEASUREMENTS	
Phenomenon	Basic standard	Category 2 electric field strength component quasi-peak dB (µV/m)	Measured minimum margin to limit	Test
EMI	CISPR 11 / EN55011 Class A	40 (30 to 230 MHz) 47 (230 to 1000 MHz)	Horizontal: 10.6 dB (125 MHz) Vertical: 4.3 dB (125 Mhz)	PASS

Table 27. IEC61800-3 and IEC61000-6-2 EMC Immunity Requirements for Second Environment and Measured Voltage Levels and Class

REQUIREMENTS					TIDA-00204 MEASUREMENTS		
Port	Phenomenon	Basic standard	Level	Performance (acceptance) criterion	Level	Performance Criterion	Test
Enclosure ports	ESD	IEC61000-4-2	±4-kV CD or 8-kV AD, if CD not possible	B	±6-kV CD	B	PASS (EXCEED)
Ports for control lines and DC auxiliary supplies <60 V	EFT	IEC61000-4-4	±2-kV/5-kHz, capacitive clamp	B	±4 kV	B	PASS (EXCEED)
	Surge 1.2/50 µs, 8/20 µs	IEC61000-4-5	±1-kV; since shielded cable >20-m, direct coupling to shield (2 Ω/500 A)	B	±2 kV	A ⁽¹⁾	PASS (EXCEED)

⁽¹⁾ Class A is considered when there are less than two consecutive UDP packet losses and no major drop in network utilization. One UDP packet loss will be tolerated by the system if the following UDP packet is echoed back successfully. The test has been conducted with a UDP error packet test program at around 5% network bandwidth.

The performance (acceptance) criterion is defined as follows:

Table 28. Performance (Acceptance) Criterion

PERFORMANCE (ACCEPTANCE) CRITERION	DESCRIPTION
A	The module shall continue to operate as intended. No loss of function or performance even during the test.
B	Temporary degradation of performance is accepted. After the test, the module shall continue to operate as intended without manual intervention.
C	During the test, loss of functions accepted, but no destruction of hardware or software. After the test, the module shall continue to operate as intended automatically, after manual restart, or power off, or power on.

5.4 EMC/EMI 测试结果

TIDA-00204 TI 设计已根据 CISPR11 / EN55011 Class A 辐射发射标准进行了 EMI 测试。对于 EMC 抗扰度，该设计已根据 IEC61800-3 和 IEC61000-6-2 标准进行了 ESD、EFT 和浪涌测试，分别参考 IEC61000-4-2、IEC61000-4-4 和 IEC61000-4-5 标准。

该设计符合这些标准，并超过 IEC61800-3 电磁兼容（EMC）抗扰度要求的电压要求。摘要如下表所示，更多详细信息见下文各节。

表 26. CISPR 11 / EN55011 类 A 辐射发射要求（类别 2 和测量电平）

要求			TIDA-00204 测量	
现象	基本标准	2 类电场强度组件准峰值 dB (µV/m)	测量最小余量以限制	Test
EMI	CISPR 11 / EN55011 类 A	40 (30 至 230 MHz) 47 (230 至 1000 MHz)	水平：10.6 dB (125 MHz) 垂直：4.3 dB (125 MHz)	PASS

表 27. 第二环境及测量电压等级的 IEC61800-3 和 IEC61000-6-2 电磁兼容抗扰度要求及类

需求					TIDA-00204 测量		
Port	现象	基本标准	等级	性能（验收）标准	等级	性能标准	Test
外壳端口	ESD	IEC61000-4-2	±4-kV CD 或 8-kV AD，如果 CD 不可行	B	±6-kV CD	B	PASS (超越)
端口控制线和直流辅助电源供应器 <60 伏特	EFT	IEC61000-4-4	±2-kV/5-kHz，容性 clamp	B	±4 kV	B	PASS (超越)
	突发 1.2/ µs, 50 8/20 µs	IEC61000-4-5	±1-kV；由于屏蔽电缆 >20-m，直接耦合至屏蔽 (2Ω/ 500 A)	B	±2 kV	A ⁽¹⁾	PASS (EXCEED)

⁽¹⁾ 当连续 UDP 数据包丢失次数少于两次且网络利用率未出现大幅下降时，考虑 Class A。若系统能成功回显下一个 UDP 数据包，则可容忍一次 UDP 数据包丢失。该测试使用 UDP 错误数据包测试程序，在网络带宽约为 5% 的条件下进行。

性能（验收）标准定义如下：

表 28. 性能（验收）标准

性能（验收）标准	描述
A	该模块应继续按预期运行。即使在测试期间，功能或性能也不应出现任何损失。
B	性能临时降级是允许的。测试后，模块应继续按预期运行，无需人工干预。
C	测试期间，允许功能丢失，但不允许硬件或软件损坏。测试后，模块应自动继续按预期运行，或在手动重启、断电或上电后继续按预期运行。

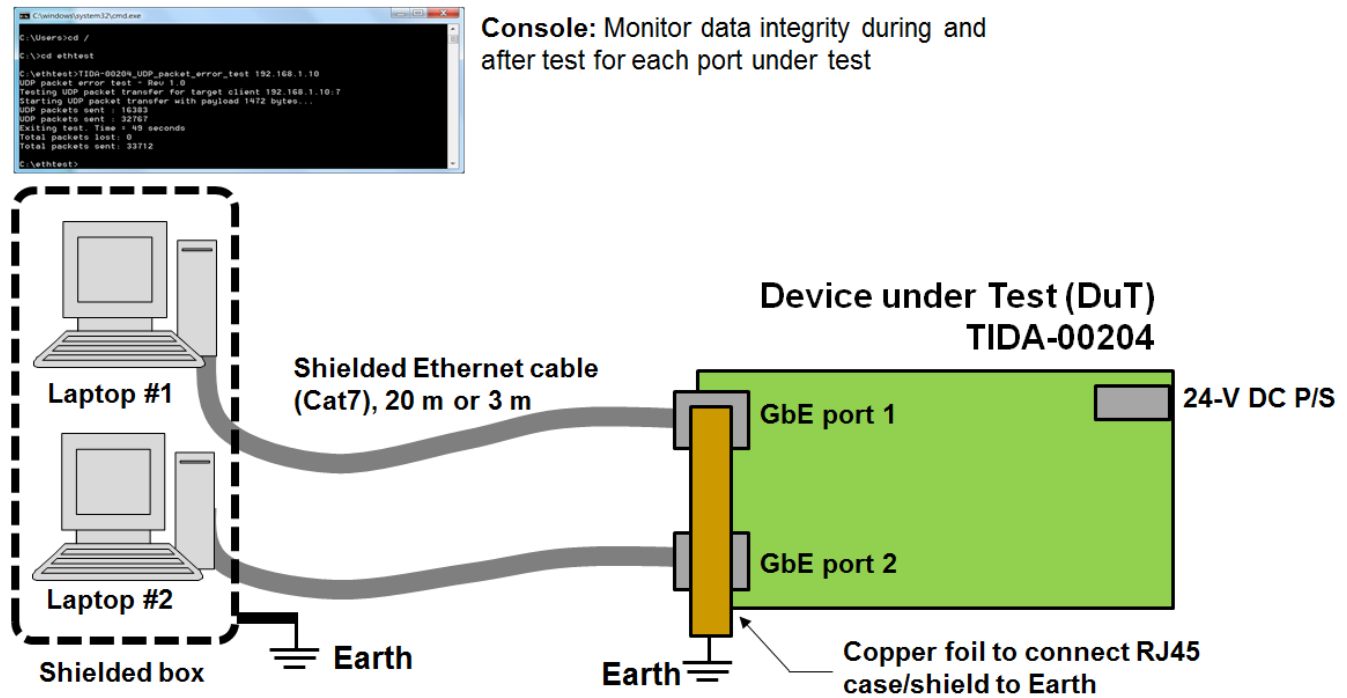
5.4.1 Test Setup

The TIDA-00204 TI design has been tested at the testing laboratory of CSA Group Bayern in Strasskirchen, Germany.

The TIDA-00204 with the test equipment is shown in the below picture. Each of the TIDA-00204 Gigabit Ethernet ports 1 and 2 are connected to a PC or Laptop with an external Gigabit Ethernet to USB3 adapter. Two different test programs ran of the laptops.

The first test program "tida-00204_udp_throughput_test.exe" transmits and receives continuous UDP packages with a payload size of 1472 bytes, yielding the maximum 1514 bytes per IPv4 packet. The firmware of the Sitara AM3359 receives the UDP packets and echoes them back. Each of the two Ethernet ports used around 20% of the network.

The second test program "tida-00204_udp_packet_error_test.exe" implements a UDP packet tester. It compares the transmitted packet with the received packet (echoed back from Sitara AM3359) for the corresponding Gigabit Ethernet port on the TIDA-00204. If a packet error is detected, it prints the number of the packet to the console. The maximum bandwidth/network utilization achievable with this test software was around 5% due to limitation on the PC test software "tida-00204_udp_packet_error_test.exe".



The specific test setups for EMI (Section 5.4.2), ESD (Section 5.4.3), EFT (Section 5.4.4), and Surge (Section 5.4.5).

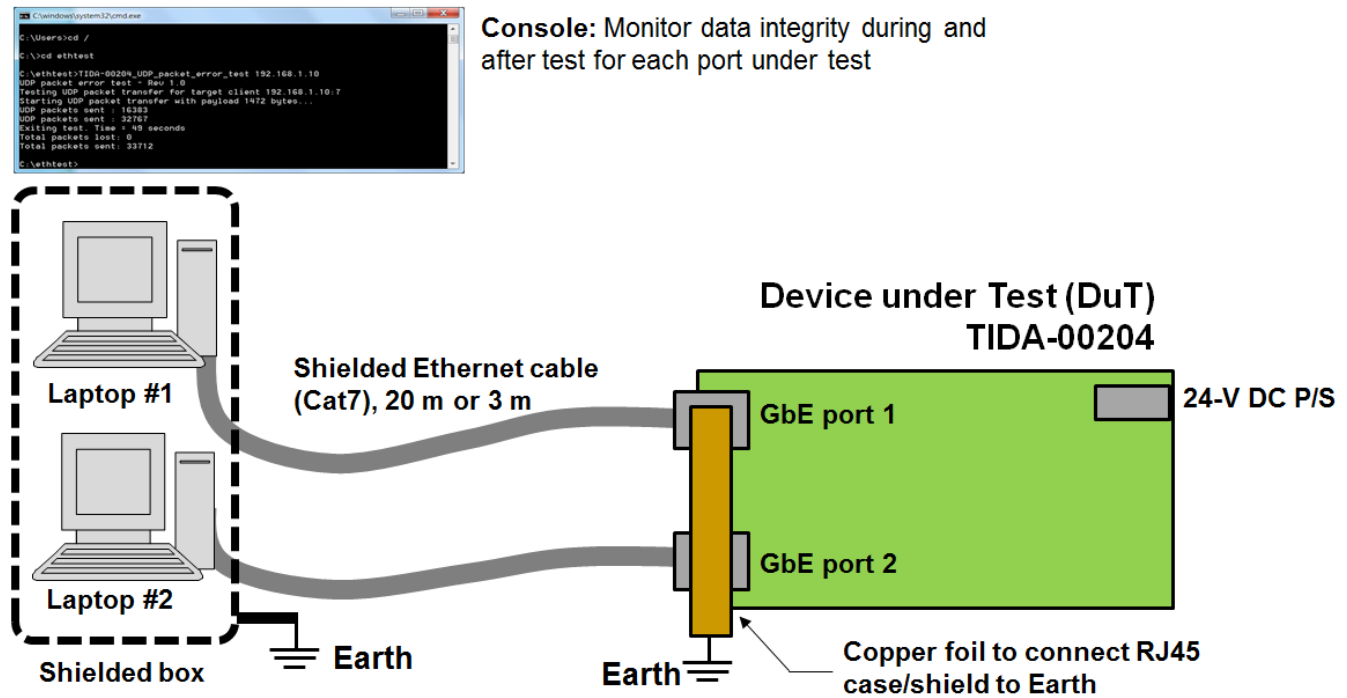
5.4.1 测试环境设置

TIDA-00204 TI 设计已在德国施特拉斯基兴的 CSA Group Bayern 测试实验室进行测试。

下图展示了配备测试设备的 TIDA-00204。TIDA-00204 的千兆以太网端口 1 和 2 分别连接到配备外部千兆以太网转 USB3 适配器的 PC 或笔记本电脑。两台笔记本电脑运行了不同的测试程序。

第一个测试程序"tida-00204_udp_throughput_test.exe"发送和接收连续 UDP 数据包，载荷大小为 1472 字节，每个 IPv4 数据包最大为 1514 字节。Sitara AM3359 的固件接收 UDP 数据包并将其回显。每个使用的以太网端口约占网络的 20%。

第二个测试程序"tida-00204_udp_packet_error_test.exe"实现了一个 UDP 数据包测试器。它比较发送的数据包与从 Sitara AM3359 回显的接收数据包，针对 TIDA-00204 上对应的千兆以太网端口。如果检测到数据包错误，它将打印数据包编号到控制台。由于 PC 测试软件"tida-00204_udp_packet_error_test.exe"的限制，此测试软件可实现的最大带宽/网络利用率约为 5%。



静电干扰（第 5.4.2 节）、静电放电（第 5.4.3 节）、电快速瞬变脉冲群（第 5.4.4 节）和浪涌（第 5.4.5 节）的具体测试设置。

5.4.2 CISPR 11 / EN55011 Radiated Emission Test Results

The TIDA-00204 meets EN55011 / CISPR 11 Class A requirements for category 2 with **at least 4.3 dB** of margin (far-field measurements at 125 MHz, vertical polarization). The minimum margin in horizontal polarization was **10.6 dB** at 125 MHz as well.

An automatic pre-test with near-field measurement (3-m antenna distance to device under test [DuT]) was used to identify the frequencies of maximum EMI in each horizontal and vertical polarization as preparation for the final test with a 10-m antenna distance.

Figure 56 shows the test setup of the final measurement with a 10-m antenna distance to DuT.

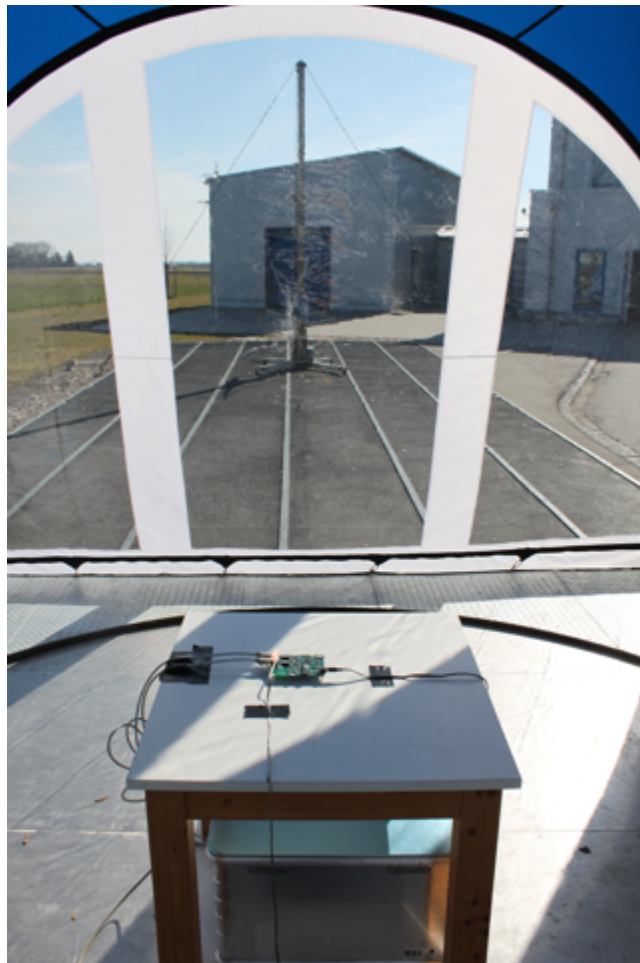


Figure 56. CISPR 11 / EN55011 Test Setup for Far-Field With 10-m Antenna Distance to TIDA-00204 DuT

5.4.2 CISPR 11 / EN55011 辐射发射测试结果

TIDA-00204 满足 EN55011 / CISPR 11 类 A 要求，类别 2，余量至少为 4.3 dB（125 MHz，垂直极化，远场测量）。水平极化下的最小余量在 125 MHz 时也为 10.6 dB。

使用近场测量（3 米天线距离至被测设备 [DuT]）进行自动预测试，以识别水平和垂直偏振下最大电磁干扰频率，为最终测试（10 米天线距离）做准备。

图 56 展示了最终测量中天线与 DUT 之间距离为 10 米的测试设置。

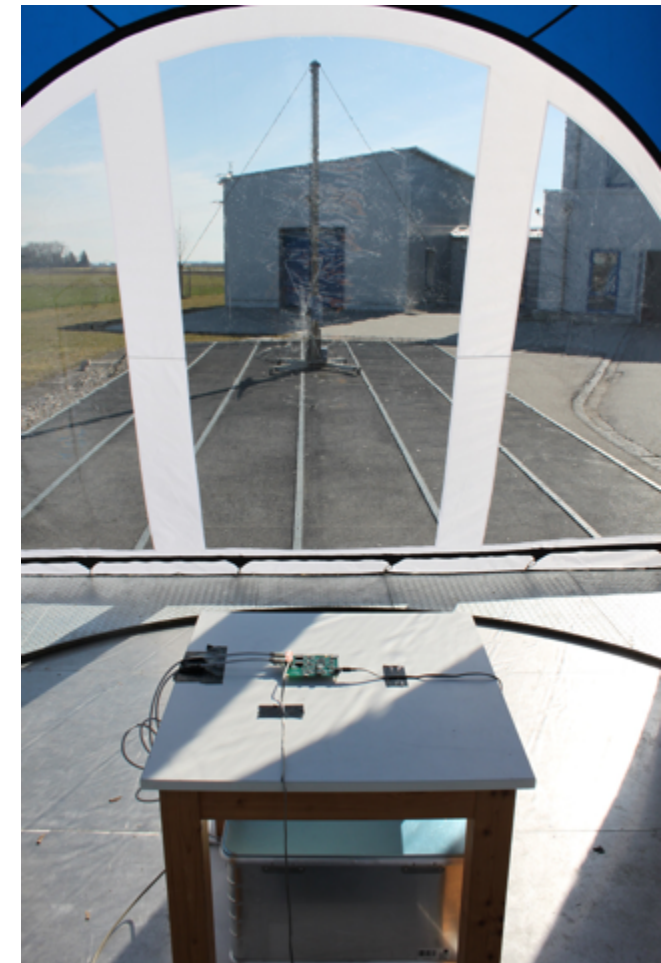


图 56. 针对 TIDA-00204 被测设备（DuT）的远场 CISPR 11 / EN55011 测试配置，天线与设备距离为 10 米

Figure 57 and Figure 58 show the measured spectral density with the final measurement with a 10-m antenna at the critical frequencies identified during the pre-test. As mentioned earlier, the minimum margin was 4.3 dB in vertical and 10.6 dB in horizontal polarization, each at 125 MHz.

Table 29. Measured EMI Spectrum (Quasi-Peak) According to EN55011; 10-m Far-Field, Horizontal Polarization

FREQUENCY (MHz)	READING [dBμV] (QP)	CORRECTION (dB)	VALUES [dBμV/m] (QP)	LIMIT [dBμV/m] (QP)	MARGIN [dB] (QP)
125	18.6	10.8	29.4	40	10.6
250	21.1	15.2	36.3	47	10.7
375	16.6	18.8	35.4	47	11.6
500	12.9	21.8	34.7	47	12.3
875	4.9	27.3	32.2	47	14.8

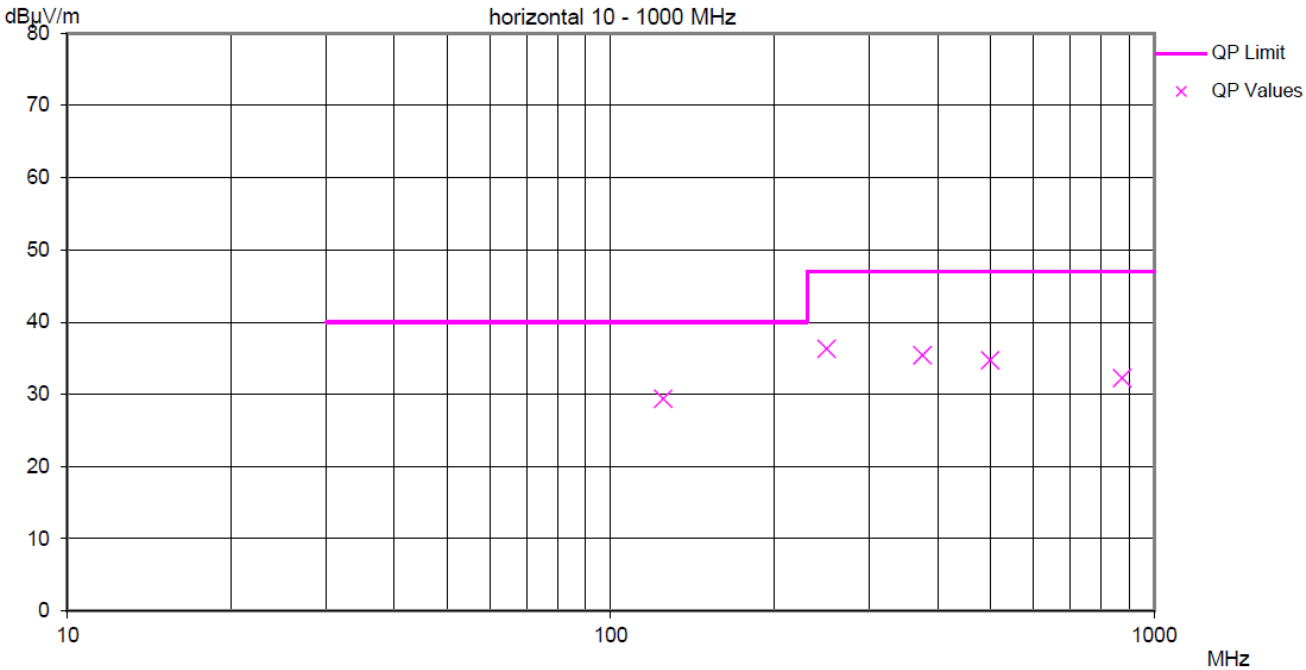


Figure 57. Measured EMI Spectrum (Quasi-Peak) According to EN55011; 10-m Far-Field, Horizontal Polarization

图 57和图 58展示了在预测试中确定的临界频率下，使用 10 米天线进行最终测量时的测量频谱密度。如前所述，垂直极化方向的最小余量为 4.3 dB，水平极化方向为 10.6 dB，均在 125 MHz。

表 29. 根据 EN55011 测量的 EMI 频谱（准峰值）；10 米远场，水平极化

频率 (MHz)	读数 [dBμV](准峰值)	校正 (dB)	VALUES [dBμV/m](准峰值)	LIMIT [dBμV/m](准峰值)	MARGIN [dB](准峰值)
125	18.6	10.8	29.4	40	10.6
250	21.1	15.2	36.3	47	10.7
375	16.6	18.8	35.4	47	11.6
500	12.9	21.8	34.7	47	12.3
875	4.9	27.3	32.2	47	14.8

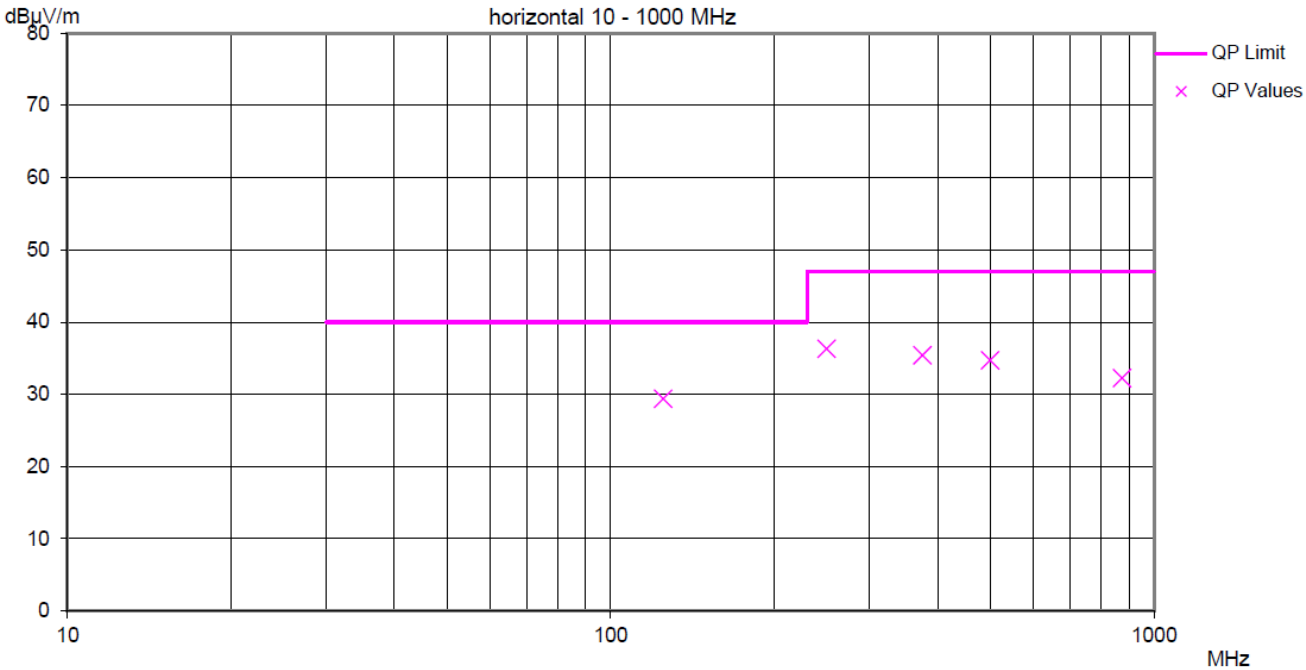


图 57. 根据 EN55011 测量的 EMI 频谱（准峰值）；10 米远场，水平极化

Table 30. Measured EMI Spectrum (Quasi-Peak) According to EN55011;
10-m Far-Field, Vertical Polarization

FREQUENCY (MHz)	READING [dBuV] (QP)	CORRECTION (dB)	VALUES [dBuV/m] (QP)	LIMIT [dBuV/m] (QP)	MARGIN [dB] (QP)
125	24.9	10.8	35.7	40	4.3
250	20.4	15.2	35.6	47	11.4
375	12.6	18.8	31.4	47	15.6
500	8.5	21.8	30.3	47	16.7
875	1.2	27.3	28.5	47	18.5

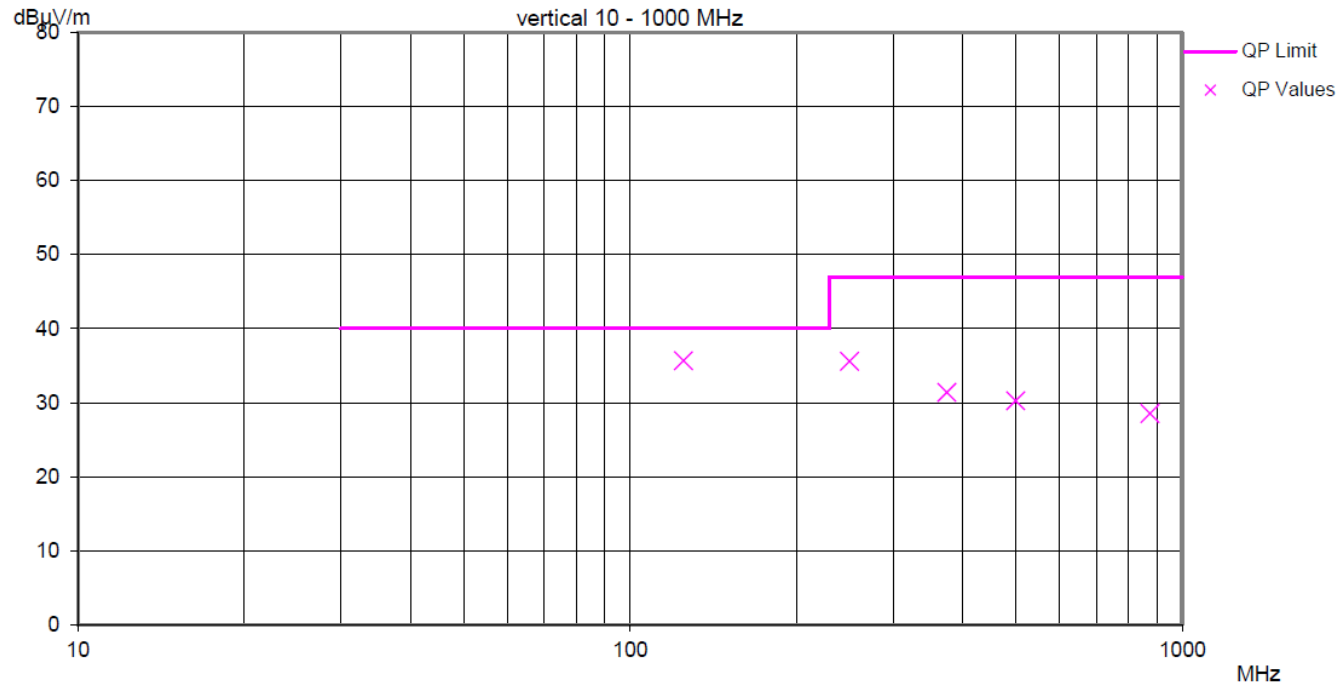


Figure 58. Measured EMI Spectrum (Quasi-Peak) According to EN55011;
10-m Far-Field, Vertical Polarization

表 30. 根据 EN55011 测量的 EMI 频谱（准峰值）；10 米远场，垂直极化

频率 (兆赫)	READING [dBuV] (QP)	校正 (dB)	VALUES [dBuV/m] (QP)	LIMIT [dBuV/m] (QP)	MARGIN [dB](准峰值)
125	24.9	10.8	35.7	40	4.3
250	20.4	15.2	35.6	47	11.4
375	12.6	18.8	31.4	47	15.6
500	8.5	21.8	30.3	47	16.7
875	1.2	27.3	28.5	47	18.5

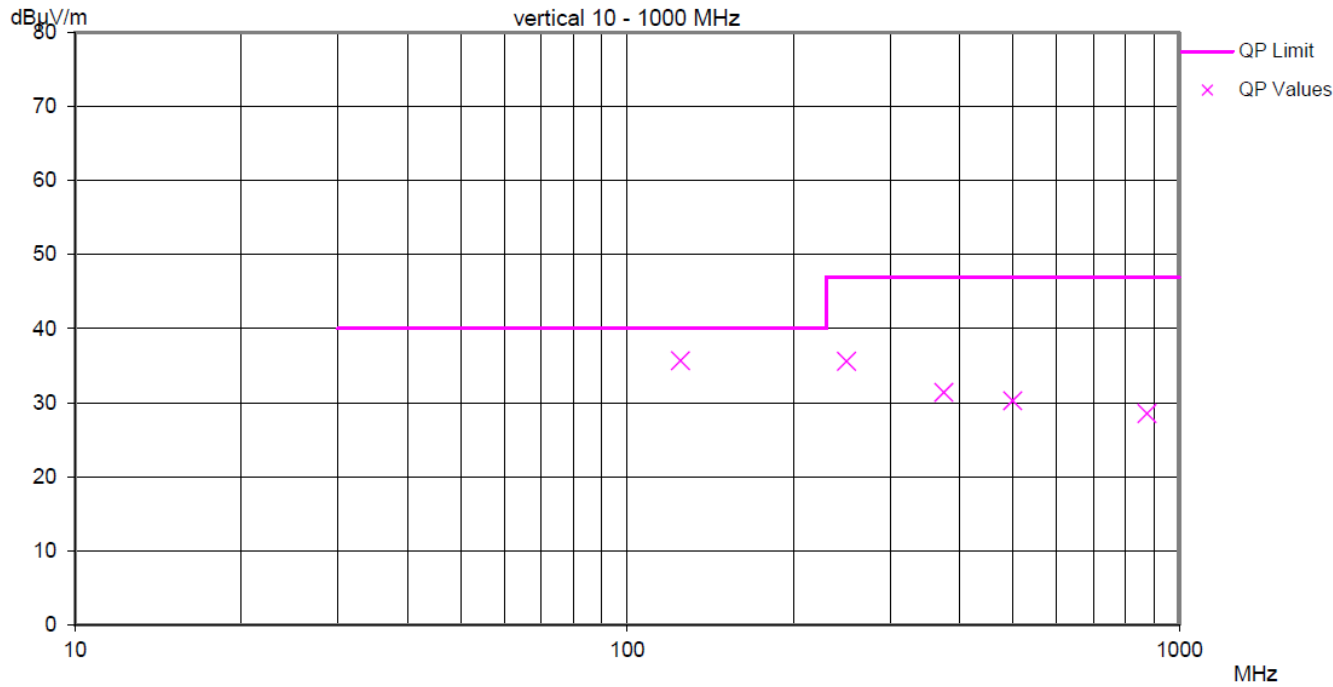


图 58. 根据 EN55011 测量的 EMI 频谱（准峰值）；10 米远场，垂直极化

5.4.2.1 Pre-Test Results With 3-m Antenna Distance

CAUTION

Due to the shorter distance (3 m instead of 10 m), the threshold for EN55011 for radiated EMI is higher by 10 dB as well. The setup is shown in [Figure 59](#).

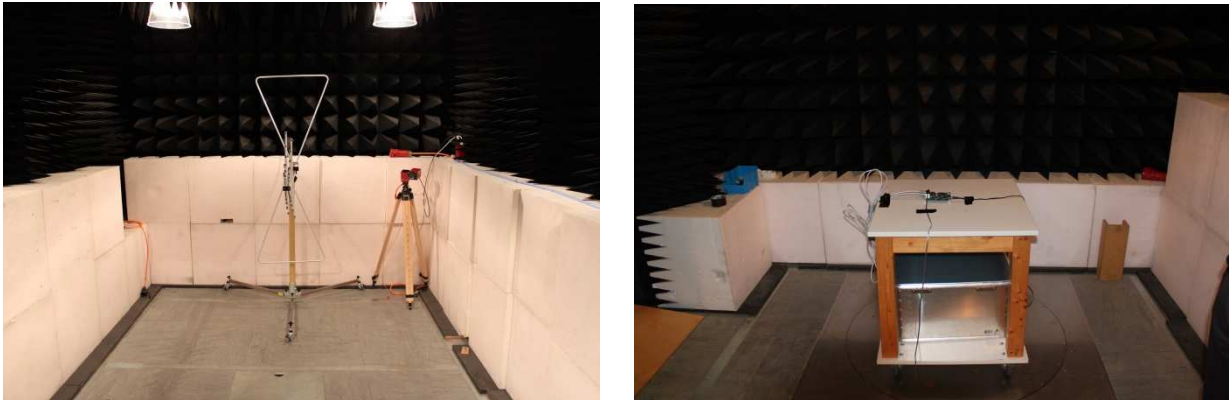


Figure 59. Automatic Pre-Test Setup for EMI With 3-m Antenna Distance to DuT (Left: Antenna, Right: TIDA-00204 DuT and Box With Shielded Laptops)

A pre-test was done to identify the critical frequencies on each polarization. This test has been done in a chamber with a 3-m distance to the antenna.

Table 31. Pre-Test Equipment Setup

TYPE	NAME
Antenna	A5_VULB9168_24-14-007
Cable	A5_Cable_50-13-018
Cable	A5_Cable_50-13-019
Preamplifier	Ohm1_MTS TVV-695_50-01-059
Receiver	Ohm1_FSP7_11-05-002
Turntable	CO1000

5.4.2.1 使用 3-m 天线的预测试结果与天线距离

注意

由于距离更短（3 米而非 10 米），EN55011 的辐射 EMI 阈值也提高了 10 dB。该设置显示在图 59。

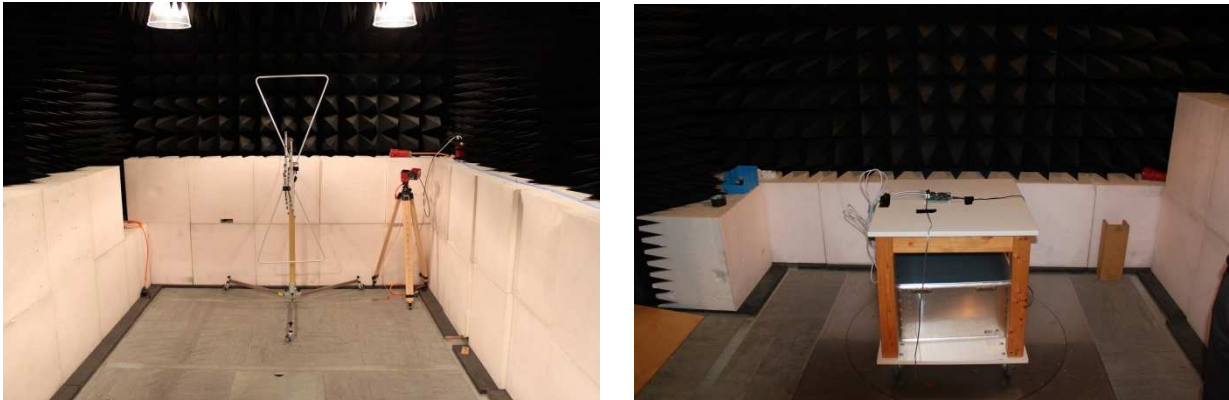


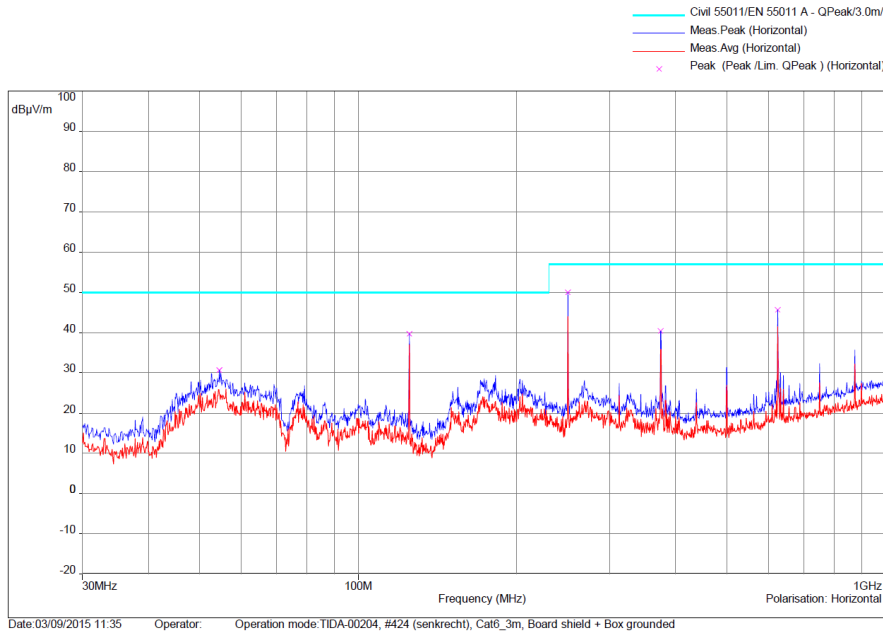
图 59. 使用 3 米天线距离对杜特（DuT）进行自动预测试设置（左侧：天线，右侧：TIDA-00204 杜特和带屏蔽笔记本电脑的盒子）

进行了预测试以识别每个偏振的临界频率。该测试在距离天线 3 米的腔室中进行。

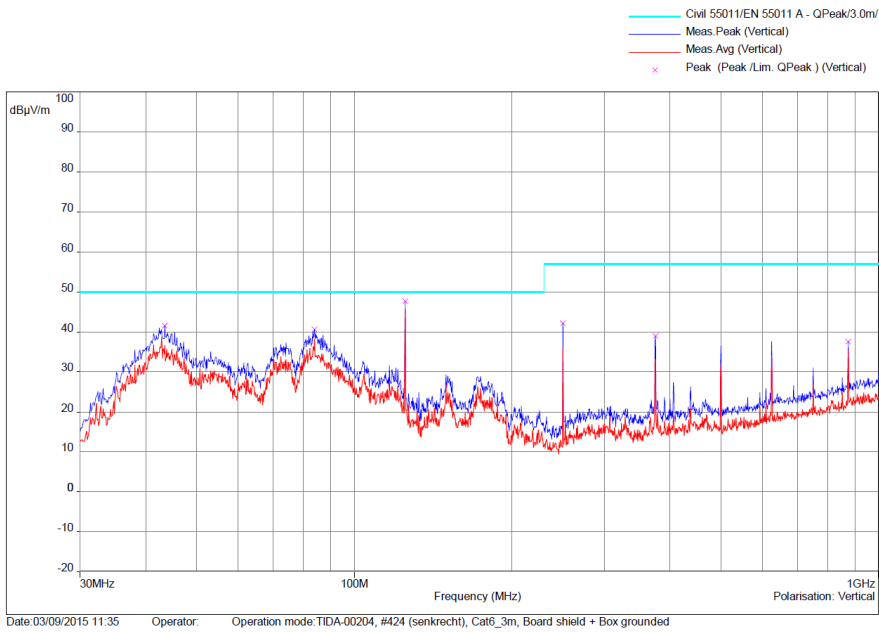
表 31. 测试前设备设置

TYPE	NAME
天线	A5_VULB9168_24-14-007
电缆	A5_Cable_50-13-018
电缆	A5_Cable_50-13-019
前置放大器	欧姆 1_MTS TVV-695_50-01-059
接收器	Ohm1_FSP7_11-05-002
唱机	CO1000

The measured spectrum of the pre-test is shown in [Figure 60](#) for horizontal and in [Figure 61](#) for vertical polarization.



**Figure 60. Measured EMI Spectrum According to EN55011;
3-m Near-Field, Horizontal Polarization (Pre-Test)**



**Figure 61. Measured EMI Spectrum According to EN55011;
3-m Near-Field, Vertical Polarization (Pre-Test)**

预测试的测量光谱如图60所示，水平偏振在图61中，垂直偏振。

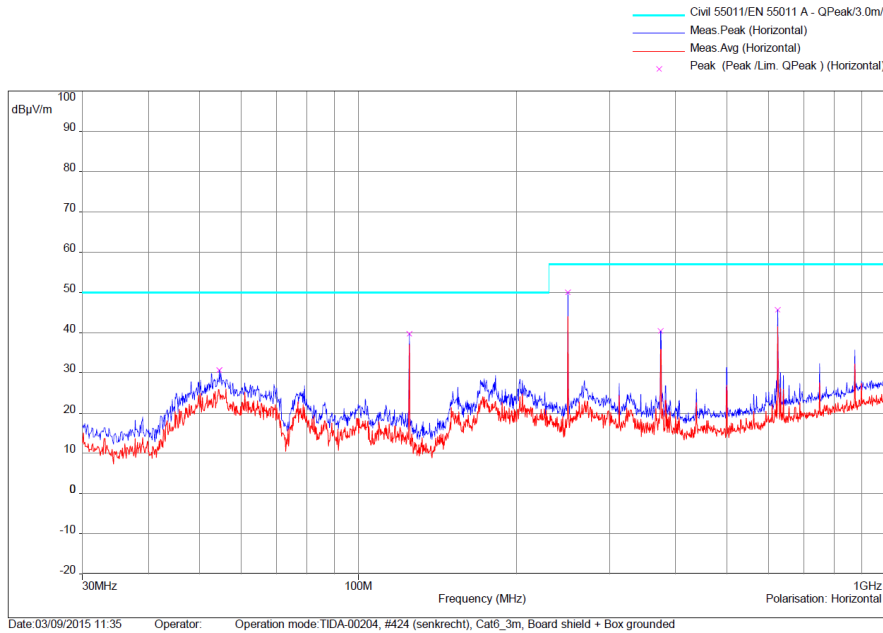


图 60. 根据 EN55011 测量的 EMI 频谱; 3 米近场, 水平极化 (测试前)

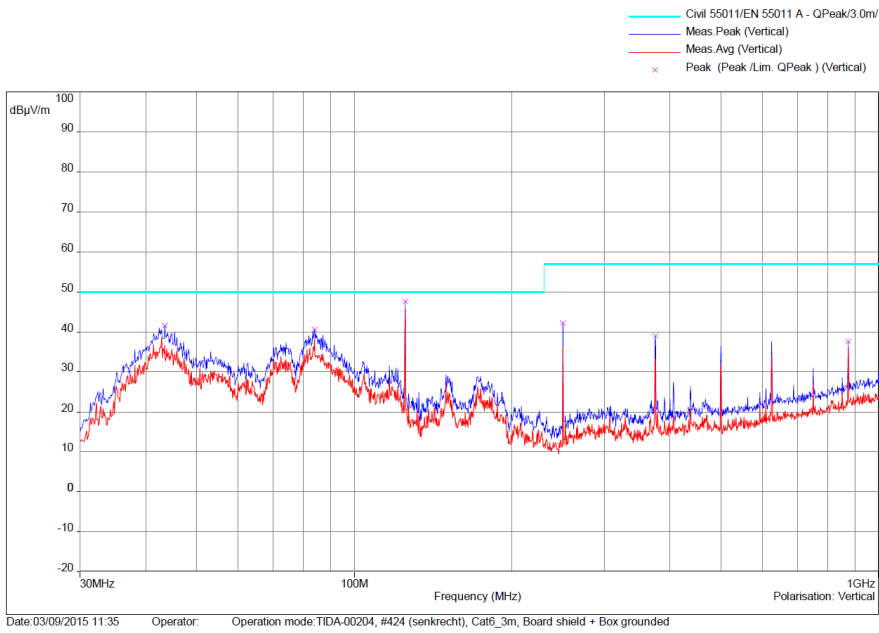


图 61. 根据 EN55011 测量的 EMIS 谱; 3-m 近场, 垂直极化 (预测试)

5.4.3 IEC-61000-4-2 ESD Test Results

Figure 62 shows the ESD test setup. The ESD strike was applied to the RJ45 case or shield of port 1 and port 2. Both RJ45 cases were electrically connected with a conductive copper foil. During the ESD test, both Gigabit Ethernet ports were connected with a 20-m cable to the far-end laptops. A UDP packet transfer was launched on each Gigabit Ethernet link.

The test program "tida-00204_udp_throughput_test.exe" transmits and receives continuous UDP packages with a payload size of 1472 bytes, yielding the maximum 1514 bytes per IPv4 packet. The firmware of the Sitara AM3359 receives the UDP packets and echoes them back. The network utilization was around 20% on each of the two Ethernet ports.



Figure 62. IEC61000-4-2 Test Setup for TIDA-00204 (Picture of 6-kV CD)

5.4.3 IEC-61000-4-2 静电放电测试结果

图 62 展示了静电放电测试装置。静电放电击打在端口 1 和端口 2 的 RJ45 外壳或屏蔽层上。两个 RJ45 外壳均通过导电铜箔进行电气连接。在静电放电测试期间，两个千兆以太网端口均通过 20 米电缆连接到远端笔记本电脑。在每个千兆以太网链路上启动了 UDP 数据包传输。

测试程序"tida-00204_udp_throughput_test.exe"发送和接收连续 UDP 数据包，载荷大小为 1472 字节，每个 IPv4 数据包最大为 1514 字节。Sitara AM3359 的固件接收 UDP 数据包并将其回显。两个以太网端口的网络利用率约为 20%。



图 62. TIDA-00204 的 IEC61000-4-2 测试装置 (6-kV CD 图片)

Table 32 shows the complete ESD test results for contact and air discharge (AD) at voltage levels, which also exceed the requirements per IEC61800-3. This is marked accordingly.

Table 32. IEC-61000-4-2 ESD Test Results for TIDA-00204

PHENOMENON	BASIC STANDARD	LEVEL	TIDA-00204 CONNECTOR UNDER TEST	PERFORMANCE CRITERION ⁽¹⁾	COMMENT
ESD	IEC61000-4-2	±4-kV CD	RJ45 port 1 and 2	B	
ESD	IEC61000-4-2	±6-kV CD	RJ45 port 1 and 2	B	Not required per IEC61800-3
ESD	IEC61000-4-2	±8-kV CD	RJ45 port 1 and 2	C	Not required per IEC61800-3
ESD	IEC61000-4-2	±8-kV AD	RJ45 port 1 and 2	B	
ESD	IEC61000-4-2	±15-kV AD	RJ45 port 1 and 2	B	Not required per IEC61800-3

⁽¹⁾ See Table 28 for performance descriptions.

Class B was claimed when there was a network utilization reduction, but no link loss. Only for 8-kV ESD CD was the link lost and a power-cycle of the boards was required. Figure 63 shows the network utilization during the entire ESD test for ±6-kV CD ESD strikes. It can be seen that the network utilization is reduced due to lost packets, but the link does not drop.

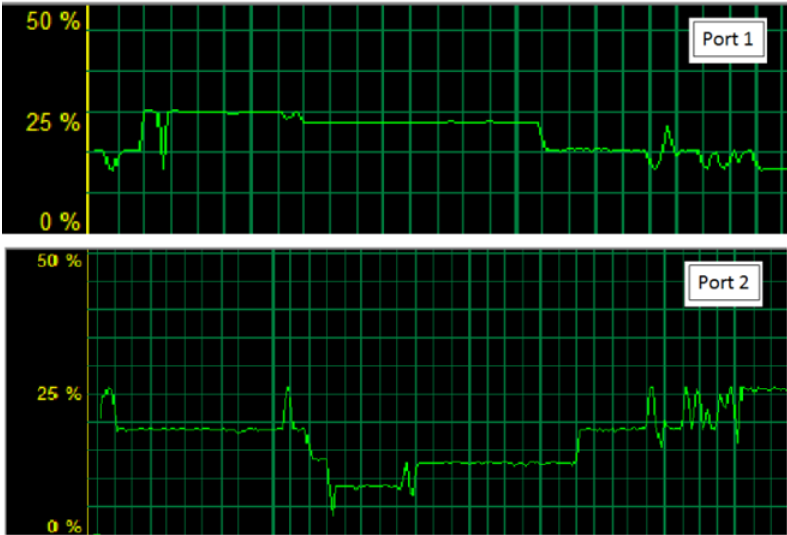


Figure 63. IEC61000-4-2 ±6-kV ESD CD: Network Utilization With UDP Packet Throughput Test on Port 1 and Port 2 (Measured on Two Separate Laptops)

表 32展示了接触和空气放电（AD）在不同电压水平下的完整静电放电（ESD）测试结果，这些结果也超过了 IEC61800-3 标准的要求。相应地进行了标记。

表 32. TIDA-00204 的 IEC-61000-4-2 静电放电测试结果

现象	基本标准	等级	TIDA-00204 被测连接器	性能标准 ⁽¹⁾	注释
ESD	IEC61000-4-2	±4-kV CD	RJ45 端口 1 和 2	B	
ESD	IEC61000-4-2	±6-kV CD	RJ45 端口 1 和 2	B	IEC61800-3 标准无需
ESD	IEC61000-4-2	±8-kV CD	RJ45 端口 1 和 2	C	IEC61800-3 标准无需
ESD	IEC61000-4-2	±8-kV AD	RJ45 端口 1 和 2	B	
ESD	IEC61000-4-2	±15-kV AD	RJ45 端口 1 和 2	B	IEC61800-3 标准未要求

⁽¹⁾ 参见 表 28 了解性能说明。

类 B 是在网络利用率降低但未发生链路丢失时声明的。仅在 8-kV ESD CD 测试中发生链路丢失，需要重启板卡。图 63 展示了整个 ESD 测试期间 ±6-kV CDESD 击穿的 网络利用率。可以看出，由于数据包丢失，网络利用率降低，但链路未断开。

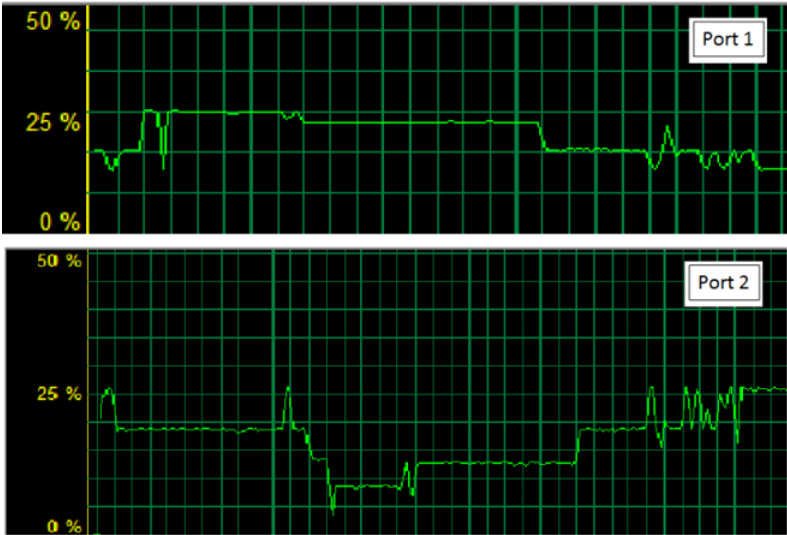


图 63. IEC61000-4-2 ±6-kV 静电放电 CD：端口 1 和端口 2 的 UDP 数据包吞吐量测试（在两台独立笔记本电脑上测量）

5.4.4 IEC-61000-4-4 EFT Test Results

Figure 64 shows the EFT test setup. During the EFT test, both Gigabit Ethernet ports were connected with a 3-m CAT7 cable to far end laptops. The capacitive clamp can be seen in the foreground of Figure 64.

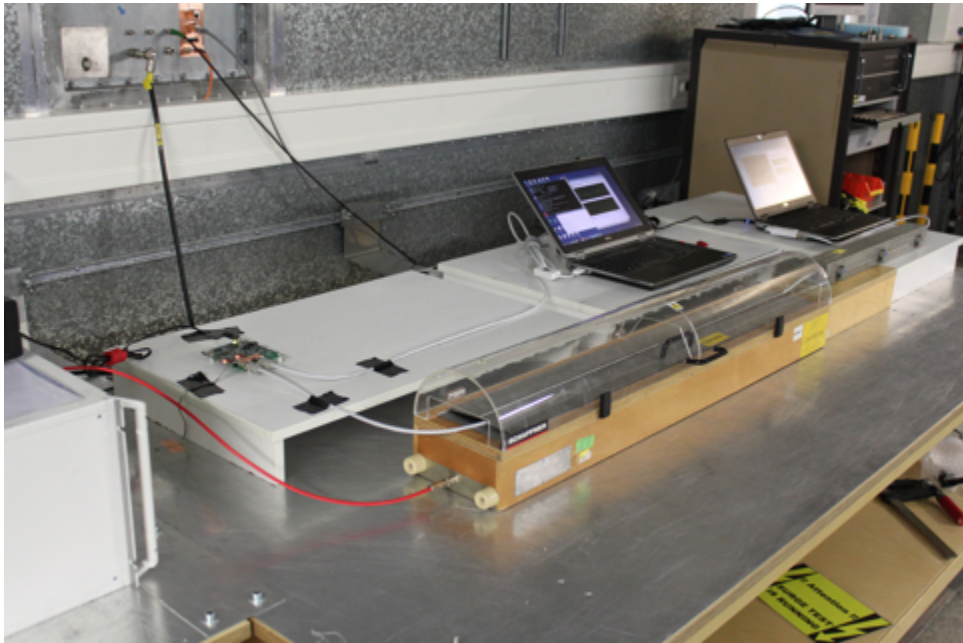


Figure 64. IEC61000-4-4 EFT Test Setup for TIDA-00204

During the EFT tests, a UDP packet transfer was launched on each Gigabit Ethernet port. Each test level and Gigabit Ethernet port was tested with two different PC software programs running on a laptop for analysis, as described earlier. The test results are shown in Table 33.

Table 33. IEC-61000-4-4 EFT Test Results for TIDA-00204

PHENOMENON	BASIC STANDARD	LEVEL	TIDA-00204 CONNECTOR UNDER TEST	PERFORMANCE CRITERION ⁽¹⁾⁽²⁾	COMMENT
EFT	IEC61000-4-4	±2-kV/5-kHz, cap clamp	RJ45 port 1	B	
EFT	IEC61000-4-4	±2-kV/5-kHz, cap clamp	RJ45 port 2	B	
EFT	IEC61000-4-4	±4-kV/5-kHz, cap clamp	RJ45 port 1	B	Not required per IEC61800-3
EFT	IEC61000-4-4	±4-kV/5-kHz, cap clamp	RJ45 port 2	B	Not required per IEC61800-3

⁽¹⁾ See Section 5.4.1 for test setup description.

⁽²⁾ See Table 28 for performance descriptions.

To verify the number of packets lost the program during the EFT tests, a UDP packet transfer was launched on each Gigabit Ethernet port. Each EFT voltage test level and gigabit Ethernet port was tested with two different software programs running on the laptop for analysis of throughput and packet errors during an EFT event.

5.4.4 IEC-61000-4-4EFT 测试结果

图 64展示了 EFT 测试设置。在 EFT 测试期间，两个千兆以太网端口均通过 3 米 CAT7 电缆连接到远端笔记本电脑。电容钳位于图 64。的前景。

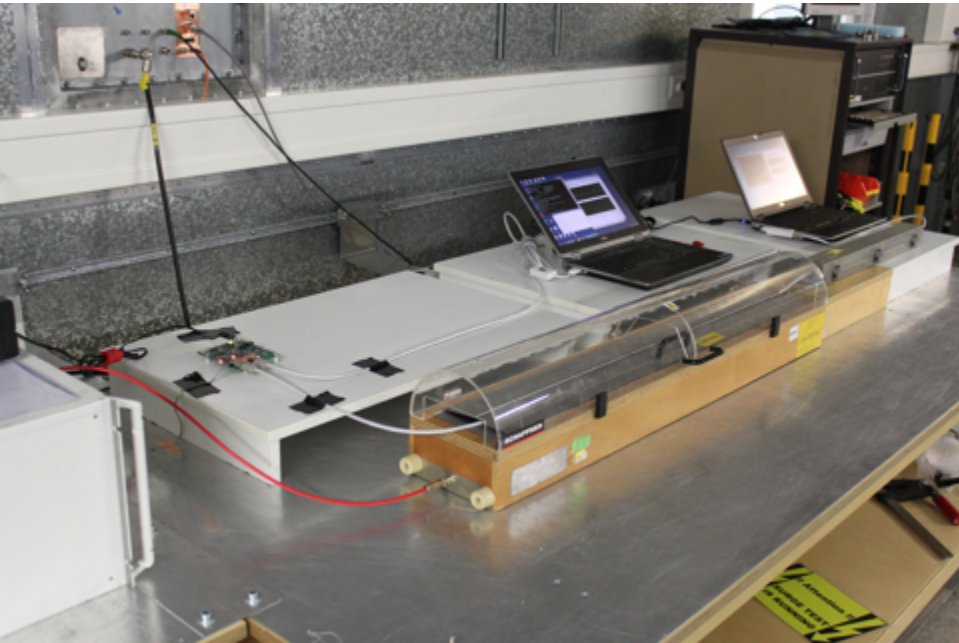


图 64. TIDA-00204 的 IEC61000-4-4 EFT 测试装置

在 EFT 测试期间，在每个千兆以太网端口上启动了 UDP 数据包传输。每个测试级别和千兆以太网端口均使用运行在笔记本电脑上的两种不同的 PC 软件程序进行测试和分析，如前所述。测试结果如表 33。

表 33. TIDA-00204 的 IEC-61000-4-4 EFT 测试结果

现象	基本标准	等级	TIDA-00204 连接器测试中	性能标准 ⁽¹⁾⁽²⁾	注释
EFT	IEC61000-4-4	±2-kV/5-kHz, 电容钳位器	RJ45 端口 1	B	
EFT	IEC61000-4-4	±2-kV/5-kHz, 电容钳位器	RJ45 端口 2	B	
EFT	IEC61000-4-4	±4-kV/5-kHz, 电容钳位器	RJ45 端口 1	B	非必需 IEC61800-3 标准
EFT	IEC61000-4-4	±4-kV/5-kHz, 电容钳位器	RJ45 端口 2	B	IEC61800-3 标准 无需

⁽¹⁾ 参见 第 5.4.1 节以获取测试设置说明。

⁽²⁾ 参见 表 28 的性能描述。

为验证 EFT 测试期间程序丢失的包数量，在每个千兆以太网端口上启动了 UDP 数据包传输。每个 EFT 电压测试等级和千兆以太网端口均使用运行在笔记本电脑上的两个不同软件程序进行测试，以分析 EFT 事件期间的吞吐量和包错误。

Class B was claimed as there was no significant network utilization reduction, from the average of around 20% during the UDP packet throughput test as shown in Figure 65. During the packet error test, the network utilization dropped from 5% to 1% due to a laptop issue when a packet lost was detected.

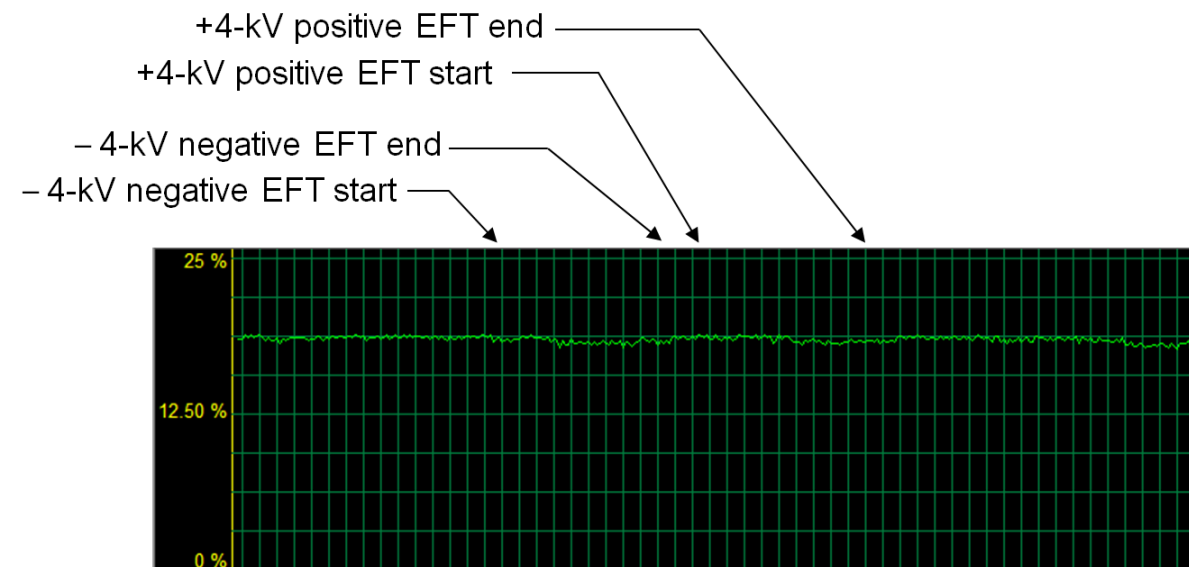


Figure 65. IEC61000-4-4 ± 4 -kV EFT: Network Utilization With UDP Throughput Test on Gigabit Ethernet Port 1

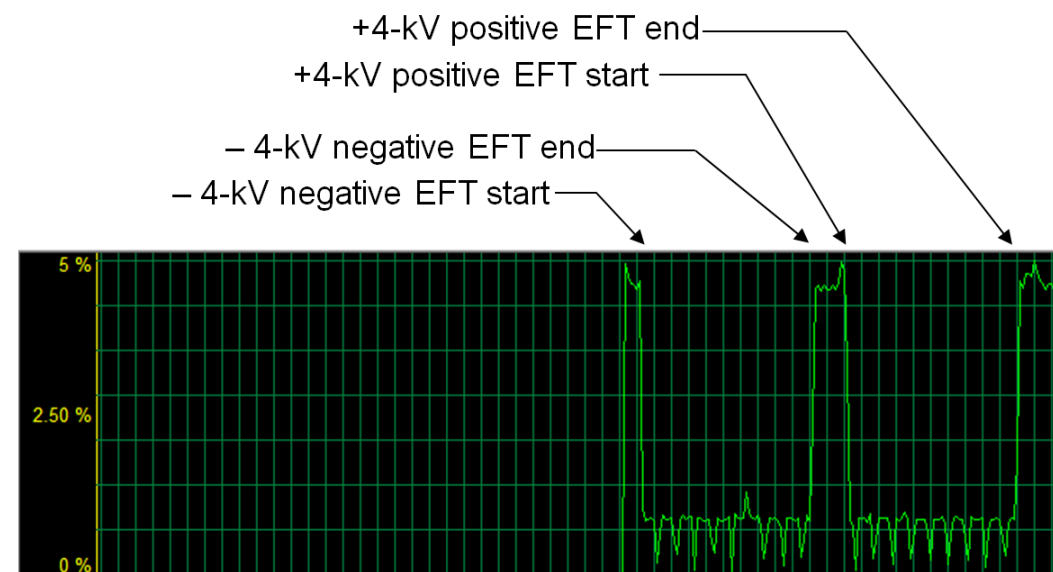


Figure 66. IEC61000-4-4 ± 4 -kV EFT: Network Utilization With UDP Packet Error Test on Gigabit Ethernet Port 1

During the test, the UDP packet error test program printed the packet number for every packet lost as well as the total submitted packets. The total number of submitted packets during the EFT event was 60,680 packets (1526 bytes each). The number of lost packets was 100. However, there was not a single event where two consecutive packets were lost. Therefore, every lost packet was re-submitted successfully.

由于 UDP 数据包吞吐量测试中（如图 65 所示）平均网络利用率约为 20%，并未出现显著降低，因此被归类为 B 类。图 65 在数据包错误测试期间，由于检测到丢包时笔记本电脑出现问题，网络利用率从 5% 降至 1%。

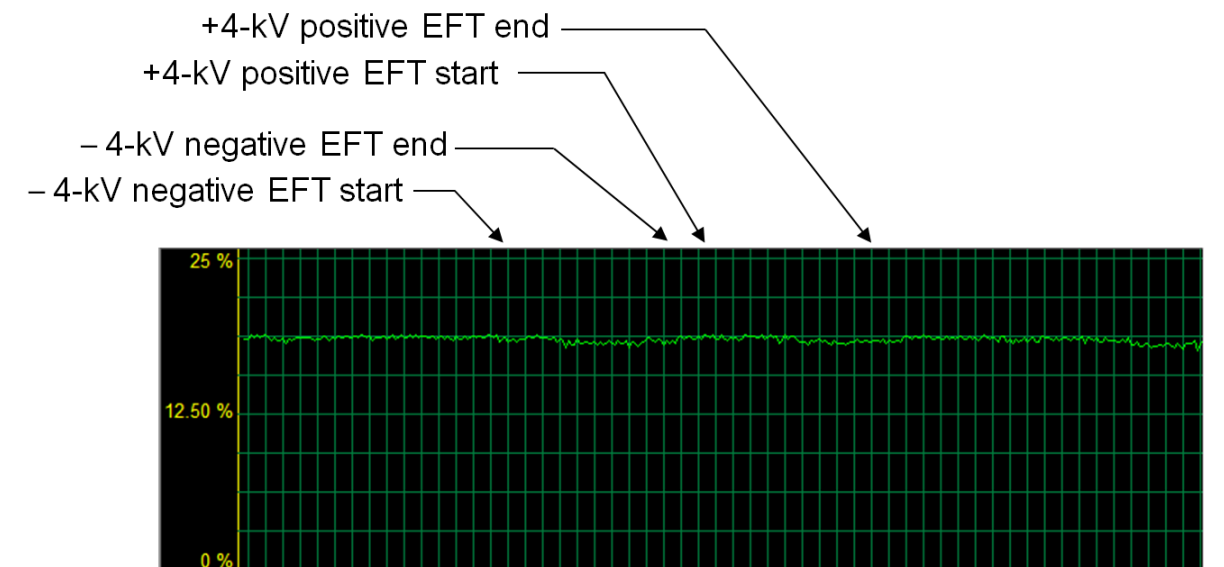


图 65. IEC61000-4-4 ± 4 -kV EFT: 千兆以太网端口 1 上的 UDP 吞吐量测试网络利用率

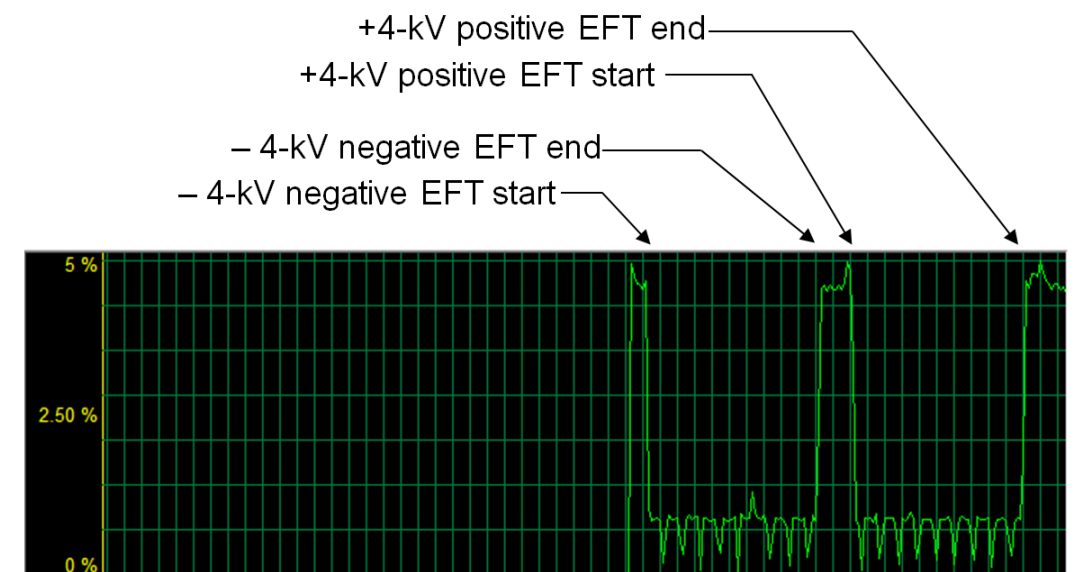


图 66. IEC61000-4-4 ± 4 -kV EFT: 千兆以太网端口 1 上的 UDP 数据包错误测试网络利用率

在测试期间，UDP 数据包错误测试程序打印了每个丢失的数据包的序号以及提交的总数据包数。EFT 事件中提交的总数据包数为 60,680 个（每个 1526 字节）。丢失的数据包数为 100 个。然而，没有任何一次连续两个数据包丢失的事件。因此，每个丢失的数据包都成功重新提交。

5.4.5 IEC-61000-4-5 Surge Test Results

Figure 67 shows the surge test setup. During the surge test, both Gigabit Ethernet ports were connected. The port under test with a 20-m CAT7 cable and the other port, not under test with a 3-m cable to far end laptops, respectively. The surge generator is on the left-hand side outside the picture.

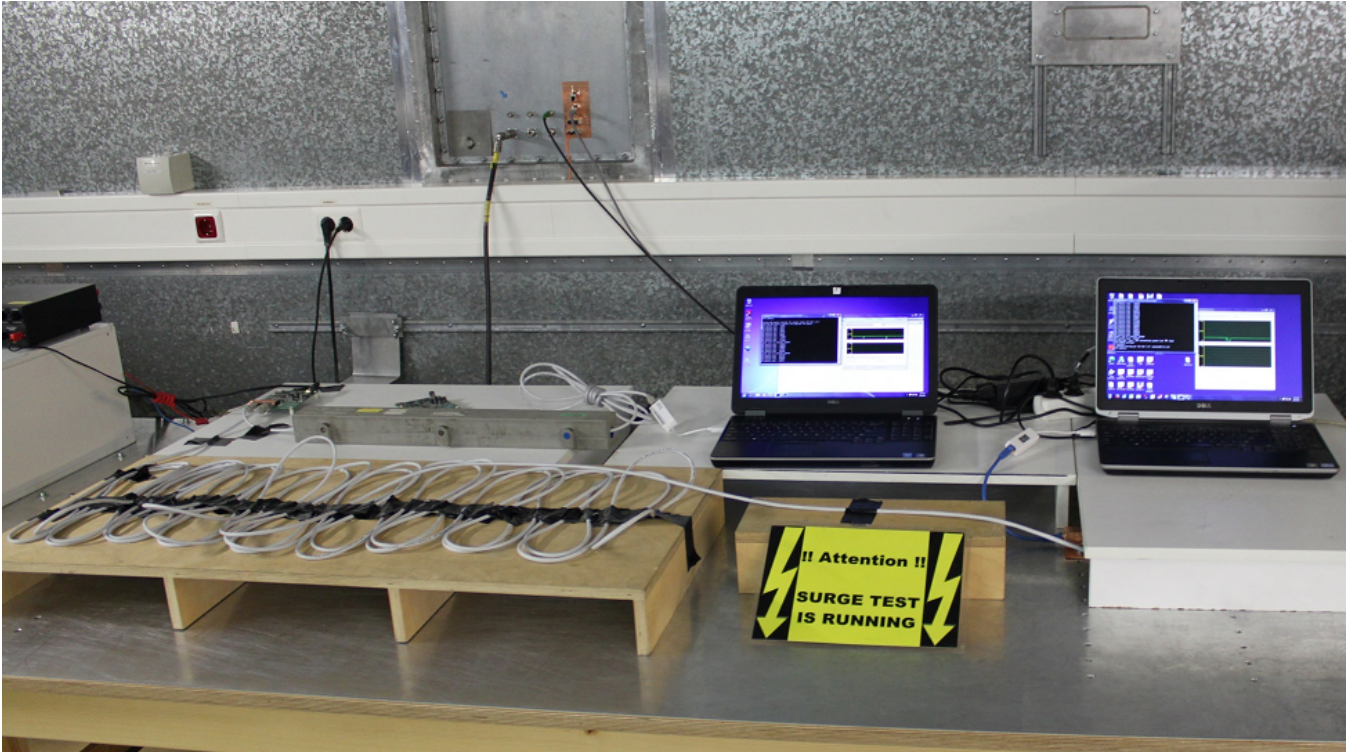


Figure 67. IEC61000-4-5 Surge Test Setup for TIDA-00204

During the surge tests, a UDP packet transfer was launched on each Gigabit Ethernet port. Each test level and Gigabit Ethernet port was tested with two different PC software programs running on a laptop for analysis, as described earlier. The test results are shown in Table 34.

Table 34. IEC-61000-4-5 Surge Test Results for TIDA-00204

PHENOMENON	BASIC STANDARD	LEVEL	TIDA-00204 CONNECTOR UNDER TEST	PERFORMANCE CRITERION ⁽¹⁾⁽²⁾	COMMENT
Surge	IEC61000-4-5	±1-kV, 2-Ω/500-A (20-m shielded CAT7 Ethernet cable)	RJ45 port 1	A	Exceeds IEC61800-3, only Class B required
Surge	IEC61000-4-5	±1-kV, 2-Ω/500-A (20-m shielded CAT7 Ethernet cable)	RJ45 port 2	A	Exceeds IEC61800-3, only Class B required
Surge	IEC61000-4-5	±2-kV, 2-Ω/500-A (20-m shielded CAT7 Ethernet cable)	RJ45 port 1	A	Exceeds IEC61800-3
Surge	IEC61000-4-5	±2-kV, 2-Ω/500-A (20-m shielded CAT7 Ethernet cable)	RJ45 port 2	A	Exceeds IEC61800-3

⁽¹⁾ See Section 5.4.1 for test setup description.
⁽²⁾ See Table 28 for performance descriptions.

5.4.5 IEC-61000-4-5 浪涌测试结果

图 67 展示了浪涌测试装置。在浪涌测试期间，两个千兆以太网端口均连接。被测端口使用 20 米 CAT7 电缆，另一端口（非被测端口）使用 3 米电缆连接至远端笔记本电脑。浪涌发生器位于图片左侧外部。

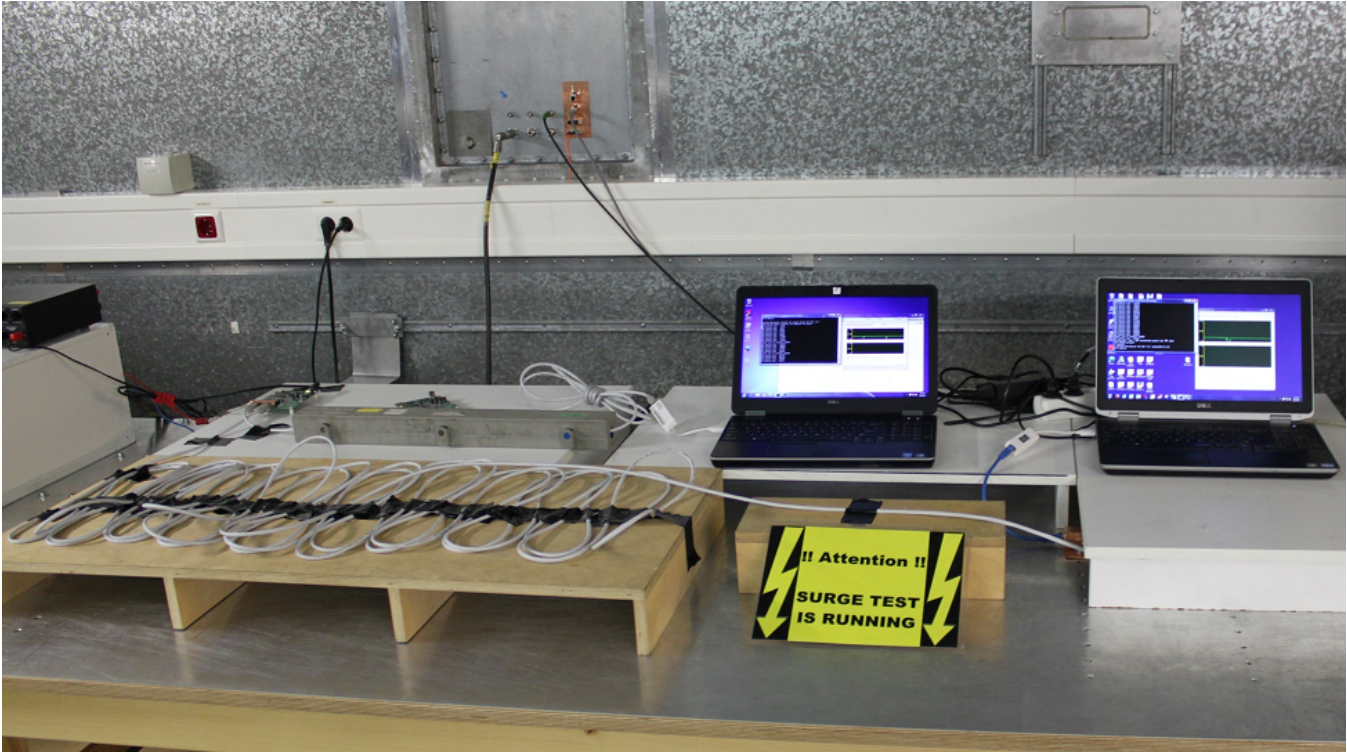


图 67. TIDA-00204 的 IEC61000-4-5 浪涌测试装置

在浪涌测试期间，在每个千兆以太网端口上启动了 UDP 数据包传输。每个测试等级和千兆以太网端口均使用运行在笔记本电脑上的两种不同的 PC 软件程序进行测试和分析，如前所述。测试结果如表 34。

表 34. TIDA-00204 的 IEC-61000-4-5 浪涌测试结果

现象	基本标准	等级	TIDA-00204 被测连接器	性能标准 ⁽¹⁾⁽²⁾	注释
突发	IEC61000-4-5	±1-kV, 2-Ω/500-A (20 米屏蔽 CAT7 以太网电缆)	RJ45 端口 1	A	超过 IEC61800-3 标准, 仅要求 B 类
突发	IEC61000-4-5	±1-kV, 2-Ω/500-A (20-m 屏蔽 CAT7 以太网线)	RJ45 端口 2	A	超过 IEC61800-3 标准, 仅需 Class B 类
突发	IEC61000-4-5	±2-kV, 2-Ω/500-A (20-m 屏蔽 CAT7 以太网线)	RJ45 端口 1	A	超过 IEC61800-3 标准
突发	IEC61000-4-5	±2-kV, 2-Ω/500-A (20-m 屏蔽 CAT7 以太网线)	RJ45 端口 2	A	超过 IEC61800-3 标准

⁽¹⁾ 参见第 5.4.1 节以获取测试设置说明。
⁽²⁾ 参见表 28 的性能描述。

Class A is considered when there are less than two consecutive UDP packet losses and no major drop in network utilization. One UDP packet loss will be tolerated by the system, if the following UDP packet is echoed back successfully. The test has been conducted with a UDP error packet test program at around 5% network bandwidth (see Figure 68).

To verify the number of packets lost the program during the surge tests, a UDP packet transfer was launched on each Gigabit Ethernet port. Each surge voltage test level and Gigabit Ethernet port was tested with two different software programs running on the laptop for analysis of throughput and packet errors during a surge event.

Class A was claimed as there was no significant network utilization reduction, from the average of around 20% during the UDP packet throughput test. During the packet error test, the network utilization remained 5%. The total number of submitted packets during the surge event was 230,000 packets (1526 bytes each). Only one packet was lost. Higher network utilization may yield more packet error losses; however, this was not tested due to lack of test equipment.

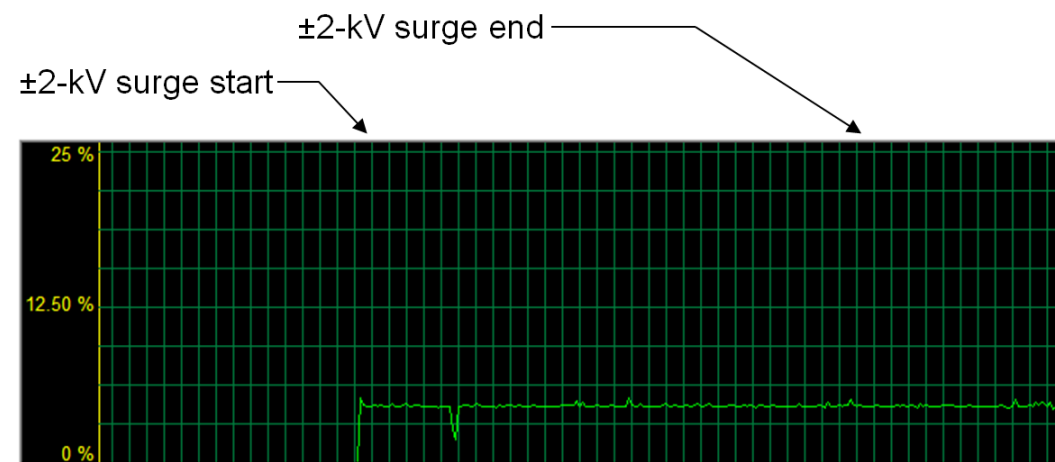


Figure 68. IEC61000-4-5 ±2-kV Surge: Network Utilization With UDP Packet Error Test on Gigabit Ethernet Port 1

当连续 UDP 数据包丢失次数少于两次且网络利用率未出现大幅下降时，会考虑 A 类。系统可容忍一次 UDP 数据包丢失，前提是随后的 UDP 数据包能成功回显。该测试是在约 5% 的网络带宽下使用 UDP 错误数据包测试程序进行的（参见图 68）。

为验证浪涌测试期间程序丢包数量，在每个千兆以太网端口上启动了 UDP 数据包传输。每个浪涌电压测试等级和千兆以太网端口均使用运行在笔记本电脑上的两个不同软件程序进行测试，以分析浪涌事件期间的吞吐量和包错误。

类 A 被认定为有效，因为在 UDP 数据包吞吐量测试期间，网络利用率平均约为 20%，未出现显著降低。在数据包错误测试中，网络利用率保持在 5%。突发事件期间提交的数据包总数为 230,000 个（每个 1526 字节）。仅丢失 1 个数据包。更高的网络利用率可能导致更多的数据包错误丢失，但由于缺乏测试设备，未对此进行测试。

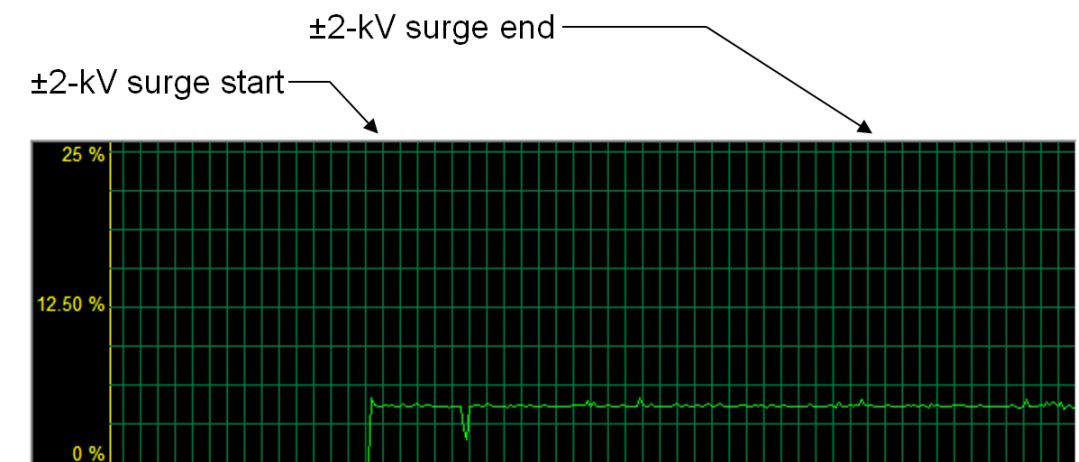


图 68. IEC61000-4-5 ±2-kV 浪涌：千兆以太网端口 1 上的 UDP 数据包错误测试网络利用率

```
UDP packet error test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 16383
UDP packets sent : 32767
UDP packets sent : 49151
UDP packets sent : 65535
Packet lost on transmit 77363
UDP packets sent : 81919
UDP packets sent : 98303
UDP packets sent : 114687
UDP packets sent : 131071
UDP packets sent : 147455
UDP packets sent : 163839
UDP packets sent : 180223
UDP packets sent : 196607
UDP packets sent : 212991
UDP packets sent : 229375
UDP packets sent : 245759
UDP packets sent : 262143
UDP packets sent : 278527
UDP packets sent : 294911
UDP packets sent : 311295
UDP packets sent : 327679
UDP packets sent : 344063
UDP packets sent : 360447
UDP packets sent : 376831
UDP packets sent : 393215
UDP packets sent : 409599
UDP packets sent : 425983
UDP packets sent : 442367
UDP packets sent : 458751
UDP packets sent : 475135
UDP packets sent : 491519
Exiting test. Time = 263 seconds
Total packets lost: 1
Total packets sent: 499038
```

Figure 69. IEC61000-4-5 ±2-kV Surge: Packet Error Analysis, Console Printout on Gigabit Ethernet Port 1

Additional tests were done with a modified board that did not have a TVS diode between the PHY and transformer. During ±1-kV surge (as per IEC61800-3), no significant changes were observed with or without TVS diode. However, at ±2-kV (exceeding the level required by IEC61800-3), the packet error losses significantly increase by a factor of 10. Therefore, if higher surge immunity is desired the TVS diode is recommend to be populated between the PHY and the transformer as per the TIDA-00204 schematics. The TVS diode between the transformer and the RJ45 connector on ETH1 (port 1) was an initial test option and not needed. It is marked DNP (not fitted) on the TIDA-00204 schematics.

```
UDP packet error test - Rev 1.0
Testing UDP packet transfer for target client 192.168.1.10:7
Starting UDP packet transfer with payload 1472 bytes...
UDP packets sent : 16383
UDP packets sent : 32767
UDP packets sent : 49151
UDP packets sent : 65535
Packet lost on transmit 77363
UDP packets sent : 81919
UDP packets sent : 98303
UDP packets sent : 114687
UDP packets sent : 131071
UDP packets sent : 147455
UDP packets sent : 163839
UDP packets sent : 180223
UDP packets sent : 196607
UDP packets sent : 212991
UDP packets sent : 229375
UDP packets sent : 245759
UDP packets sent : 262143
UDP packets sent : 278527
UDP packets sent : 294911
UDP packets sent : 311295
UDP packets sent : 327679
UDP packets sent : 344063
UDP packets sent : 360447
UDP packets sent : 376831
UDP packets sent : 393215
UDP packets sent : 409599
UDP packets sent : 425983
UDP packets sent : 442367
UDP packets sent : 458751
UDP packets sent : 475135
UDP packets sent : 491519
Exiting test. Time = 263 seconds
Total packets lost: 1
Total packets sent: 499038
```

图 69。IEC61000-4-5 ±2-kV 浪涌：千兆以太网端口 1 的包错误分析，控制台打印输出

使用未配备 PHY 与变压器之间 TVS 二极管的修改版电路板进行了额外测试。在符合 IEC61800-3 标准的 ±1-kV 浪涌下，无论是否安装 TVS 二极管，均未观察到显著变化。然而，在 ±2-kV（超过 IEC61800-3 规定的浪涌水平）时，数据包错误丢失量显著增加 10 倍。因此，若需要更高的浪涌耐受能力，建议按照 TIDA-00204 原理图，在 PHY 与变压器之间安装 TVS 二极管。变压器与 ETH1（端口 1）RJ45 连接器之间的 TVS 二极管仅为初始测试选项，并非必需。该位置在 TIDA-00204 原理图中标记为 DNP（未安装）。

6 Design Files

6.1 Schematics

To download the schematics, see the design files at [TIDA-00204](#).

6.2 Bill of Materials

To download the bill of materials (BOM), see the design files at [TIDA-00204](#).

6.3 Layer Plots

To download the layer plots, see the design files at [TIDA-00204](#).

6.4 PCB Layout Guidelines

6.4.1 Layer Stack

Ensure that the differential signal layers have a reference ground and that the PCB differential signal traces are matched to 100-Ω impedance. For the single-ended traces, the board is matched to 50 Ω.

A layer size of 0.018 mm was chosen for the top and mid layers. This size can be defined as the base thickness. When actually building the board, the mid layers will decrease in size and the outer layers will increase in size. This means that the expected top layer thickness would be 0.040 mm and a mid layer would have an expected thickness of 0.012 mm. This decrease and increase in thickness varies on the manufacturer. This variation is why it is very important when working with high-speed differential signals to define the layer stack and double check with the PCB manufacturer to ensure that the expected impedances are achieved.

Stack Up		Layer Stack				Calculated Values (by ISOLA MultiCal 6.0)			
Layer	Board Layer Stack	Name	Material	Thickness	Constant	Thickness Base / Expected	Zdiff=100R Line Width/Separation	Zdiff=90R Line Width/Separation	Z0=50R Line Width
1		Top Paste							
2		Top Overlay							
3		Top Solder	Solder Resist	0.015mm	3.9	0.015mm			
4	■ ■ ■ ■ ■ ■ ■ ■	TopLayer - S1 (.GTL)	Copper	0.018mm		0.018 / 0.040 mm	0.101 / 0.215mm	0.130 / 0.210 mm	0.125 mm
5	■ ■ ■ ■ ■ ■ ■ ■	Dielectric1	FR4, IS420ML, 1x 1080 (Prepreg)	0.085mm	4.36	0.089 mm			
6	■ ■ ■ ■ ■ ■ ■ ■	MidLayer1 - S2 (.G1)	Copper	0.018mm		0.018 / 0.012 mm	REF GND	REF GND	REF GND
7	■ ■ ■ ■ ■ ■ ■ ■	Dielectric2	FR4, IS420ML, 1x 2116 (Core)	0.101mm	4.72	0.101mm			
8	■ ■ ■ ■ ■ ■ ■ ■	MidLayer2 - S3 (.G2)	Copper	0.018mm		0.018 / 0.012 mm	0.101 / 0.210 mm	-	0.119 mm
9	■ ■ ■ ■ ■ ■ ■ ■	Dielectric3	FR4, IS420ML, 3x 2157 (Prepreg)	0.440mm	4.75	0.401mm			
10	■ ■ ■ ■ ■ ■ ■ ■	MidLayer3 - S4 (.G3)	Copper	0.018mm		0.018 / 0.012 mm		GND	
11	■ ■ ■ ■ ■ ■ ■ ■	Dielectric4	FR4, IS420ML, 1x 2116 (Core)	0.101mm	4.72	0.101mm			
12	■ ■ ■ ■ ■ ■ ■ ■	MidLayer4 - S5 (.G4)	Copper	0.018mm		0.018 / 0.012 mm		Power	
13	■ ■ ■ ■ ■ ■ ■ ■	Dielectric5	FR4, IS420ML, 3x 2157 (Prepreg)	0.440mm	4.75	0.401mm			
14	■ ■ ■ ■ ■ ■ ■ ■	MidLayer5 - S6 (.G5)	Copper	0.018mm		0.018 / 0.012 mm	0.101 / 0.210 mm	-	0.119 mm
15	■ ■ ■ ■ ■ ■ ■ ■	Dielectric 6	FR4, IS420ML, 1x 2116 (Core)	0.101mm	4.72	0.101mm			
16	■ ■ ■ ■ ■ ■ ■ ■	MidLayer6 - S7 (.G6)	Copper	0.018mm		0.018 / 0.012 mm	REF GND	REF GND	REF GND
17	■ ■ ■ ■ ■ ■ ■ ■	Dielectric 7	FR4, IS420ML, 1x 1080 (Prepreg)	0.085mm	4.36	0.089 mm			
18	■ ■ ■ ■ ■ ■ ■ ■	BottomLayer - S8 (.GBL)	Copper	0.018mm		0.018 / 0.040 mm	0.101 / 0.215mm	-	0.125 mm
19		Bottom Solder	Solder Resist	0.015mm	3.9	0.015mm			
20		Bottom Overlay							
21		Bottom Paste							
Height : 1.527mm									

Figure 70. Layer Stack

For a less complex processor design, the PCB layers for DP83867IR routing can be reduced to four layers. See the DP83867IR data sheet for examples[1].

6 设计文件

6.1 原理图

要下载原理图，请查看 [TIDA-00204](#)。

6.2 物料清单

要下载物料清单（BOM），请查看[TIDA-00204](#)。

6.3 分层图

要下载层图，请查看 [TIDA-00204](#)。

6.4 PCB 布局指南

6.4.1 层堆栈

确保差分信号层具有参考地，且 PCB 差分信号走线匹配至 100-Ω 阻抗。对于单端走线，板级匹配至 50 Ω。

顶层和中层选择了 0.018 mm 的层厚。该尺寸可定义为基准厚度。在实际制作电路板时，中层厚度会减小，而外层厚度会增加。这意味着顶层的预期厚度为 0.040 mm，中层预期厚度为 0.012 mm。这种厚度变化因制造商而异。正是这种差异，使得在处理高速差分信号时，定义层叠结构并与 PCB 制造商进行双重确认变得至关重要，以确保达到预期的阻抗值。

Stack Up		Layer Stack				Calculated Values (by ISOLA MultiCal 6.0)			
Layer	Board Layer Stack	Name	Material	Thickness	Constant	Thickness Base / Expected	Zdiff=100R Line Width/Separation	Zdiff=90R Line Width/Separation	Z0=50R Line Width
1		Top Paste							
2		Top Overlay							
3		Top Solder	Solder Resist	0.015mm	3.9	0.015mm			
4	■ ■ ■ ■ ■ ■ ■ ■	TopLayer - S1 (.GTL)	Copper	0.018mm		0.018 / 0.040 mm	0.101 / 0.215mm	0.130 / 0.210 mm	0.125 mm
5	■ ■ ■ ■ ■ ■ ■ ■	Dielectric1	FR4, IS420ML, 1x 1080 (Prepreg)	0.085mm	4.36	0.089 mm			
6	■ ■ ■ ■ ■ ■ ■ ■	MidLayer1 - S2 (.G1)	Copper	0.018mm		0.018 / 0.012 mm	REF GND	REF GND	REF GND
7	■ ■ ■ ■ ■ ■ ■ ■	Dielectric2	FR4, IS420ML, 1x 2116 (Core)	0.101mm	4.72	0.101mm			
8	■ ■ ■ ■ ■ ■ ■ ■	MidLayer2 - S3 (.G2)	Copper	0.018mm		0.018 / 0.012 mm	0.101 / 0.210 mm	-	0.119 mm
9	■ ■ ■ ■ ■ ■ ■ ■	Dielectric3	FR4, IS420ML, 3x 2157 (Prepreg)	0.440mm	4.75	0.401mm			
10	■ ■ ■ ■ ■ ■ ■ ■	MidLayer3 - S4 (.G3)	Copper	0.018mm		0.018 / 0.012 mm		GND	
11	■ ■ ■ ■ ■ ■ ■ ■	Dielectric4	FR4, IS420ML, 1x 2116 (Core)	0.101mm	4.72	0.101mm			
12	■ ■ ■ ■ ■ ■ ■ ■	MidLayer4 - S5 (.G4)	Copper	0.018mm		0.018 / 0.012 mm		Power	
13	■ ■ ■ ■ ■ ■ ■ ■	Dielectric5	FR4, IS420ML, 3x 2157 (Prepreg)	0.440mm	4.75	0.401mm			
14	■ ■ ■ ■ ■ ■ ■ ■	MidLayer5 - S6 (.G5)	Copper	0.018mm		0.018 / 0.012 mm	0.101 / 0.210 mm	-	0.119 mm
15	■ ■ ■ ■ ■ ■ ■ ■	Dielectric 6	FR4, IS420ML, 1x 2116 (Core)	0.101mm	4.72	0.101mm			
16	■ ■ ■ ■ ■ ■ ■ ■	MidLayer6 - S7 (.G6)	Copper	0.018mm		0.018 / 0.012 mm	REF GND	REF GND	REF GND
17	■ ■ ■ ■ ■ ■ ■ ■	Dielectric 7	FR4, IS420ML, 1x 1080 (Prepreg)	0.085mm	4.36	0.089 mm			
18	■ ■ ■ ■ ■ ■ ■ ■	BottomLayer - S8 (.GBL)	Copper	0.018mm		0.018 / 0.040 mm	0.101 / 0.215mm	-	0.125 mm
19		Bottom Solder	Solder Resist	0.015mm	3.9	0.015mm			
20		Bottom Overlay							
21		Bottom Paste							
Height : 1.527mm									

图 70。层叠结构

对于更简单的处理器设计，DP83867IR 走线的 PCB 层数可减少至四层。有关示例，请参阅 DP83867IR 数据手册[1]。

6.4.2 Layout RGMII Signals

Looking at the layout, each RX or TX group of the RGMII signals are routed on the same layer with matched trace length as outlined in [Section 2.3.1](#).

The goal for routing the signals is to minimize the trace length of both RGMII 1 and RGMII 2 interfaces, which is done using the following considerations.

The RGMII interface is a 125-MHz signal, which gives a full clock cycle of 8 ns.

The typical propagation delay in a FR4 stripline is 7.087 ns/mm, which means that the length of the RGMII interface introduces a delay that should be kept as small as possible. For the RGMII signal below 0.5 ns, this should be a delay that can be ignored. So the signal can be around 63.5-mm long, maximum.

A second important point is that when changing layers, a GND via is added next to the signal via to ensure a good matching as well offer a current return path close to the signal.

The PCB signals are impedance controlled to 50 Ω , and have series line termination. For the series termination at both the PHYs, 0- Ω resistors were chosen due to the DP83867IR internal 50- Ω impedance. These resistors were only added for test and debug. For the AM3359, 22- Ω series termination resistors were chosen.

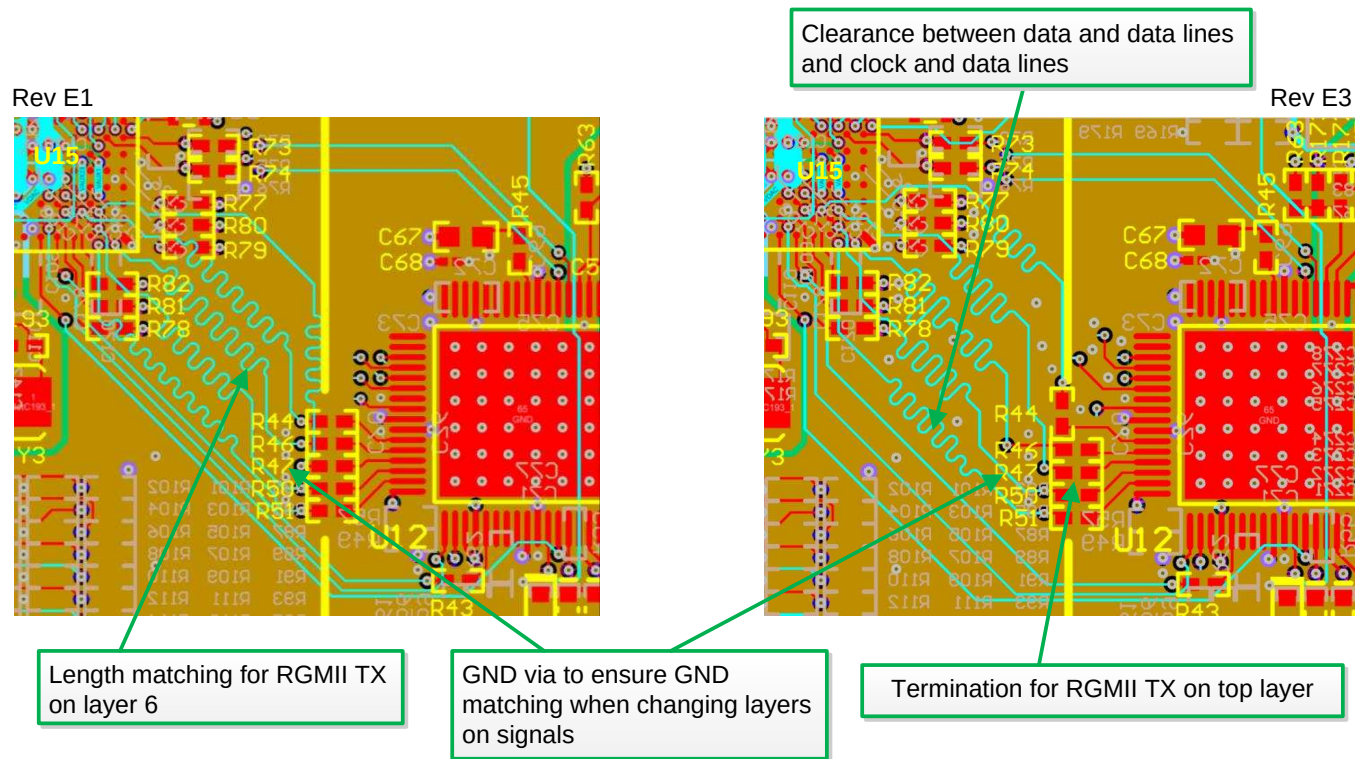


Figure 71. Layout Guidelines for RGMII TX Signals Comparing E1 and E3

6.4.2 布局 RGMII 信号

查看布局，RGMII 信号的每个 RX 或 TX 组均在同一层上布线，且走线长度匹配，如第 2.3.1 节所述。

信号路由的目标是最小化 RGMII 1 和 RGMII 2 接口的走线长度，这通过以下考虑来实现。

RGMII 接口是一个 125-MHz 信号，这提供了 8 ns 的完整时钟周期。

FR4 微带线中的典型传播延迟为 7.087 ns/mm，这意味着 RGMII 接口的长度引入的延迟应尽可能小。对于低于 0.5 ns 的 RGMII 信号，该延迟可忽略不计。因此，信号长度最大约为 63.5 mm。

另一个重要点是，在切换层时，会在信号过孔旁边添加一个 GND 过孔，以确保良好的匹配，并为信号提供近似的电流回路路径。

PCB 信号阻抗控制在 50 Ω ，并带有串联线端接。对于两个 PHY 的串联端接，由于 DP83867IR 内部 50- Ω 阻抗，选择了 0- Ω 电阻。这些电阻仅用于测试和调试。对于 AM3359，选择了 22- Ω 串联端接电阻。

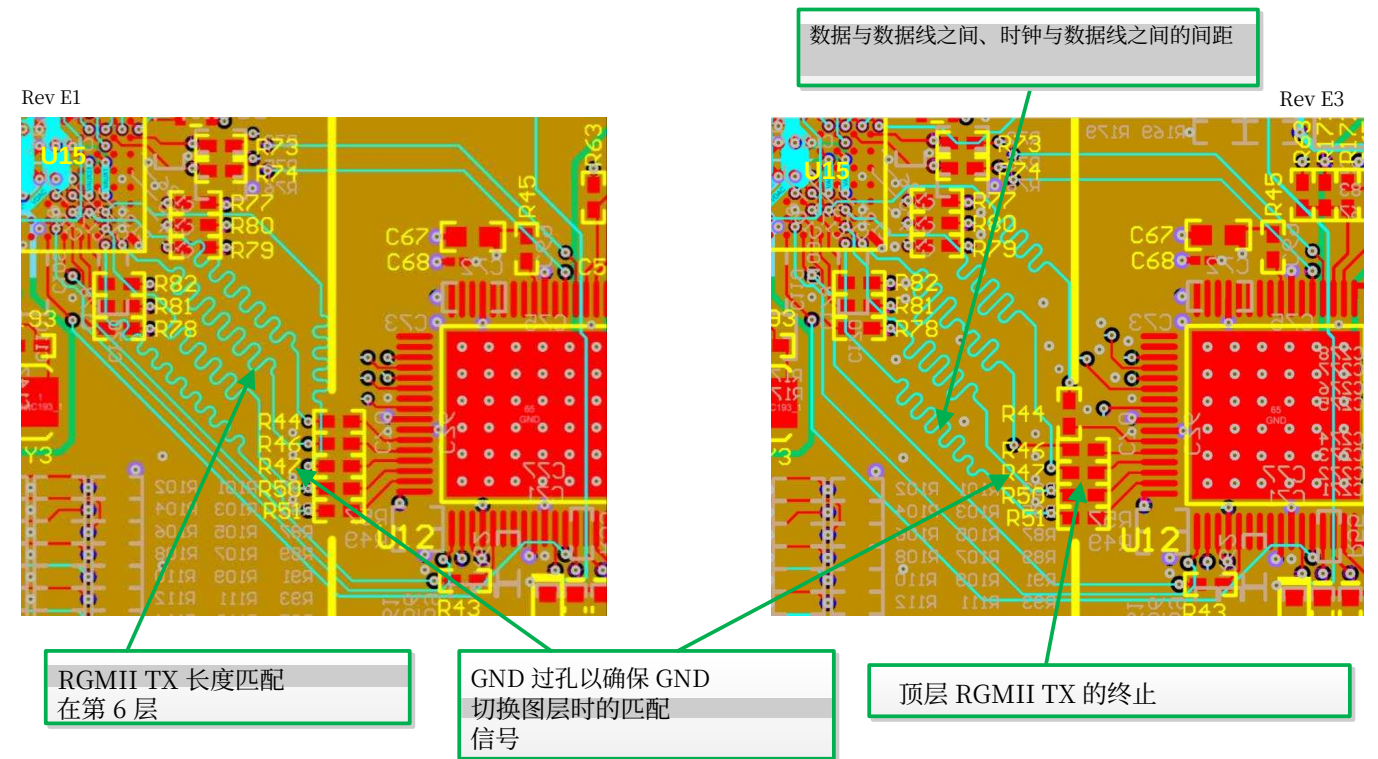


图 71. RGMII TX 信号布局指南：比较 E1 和 E3

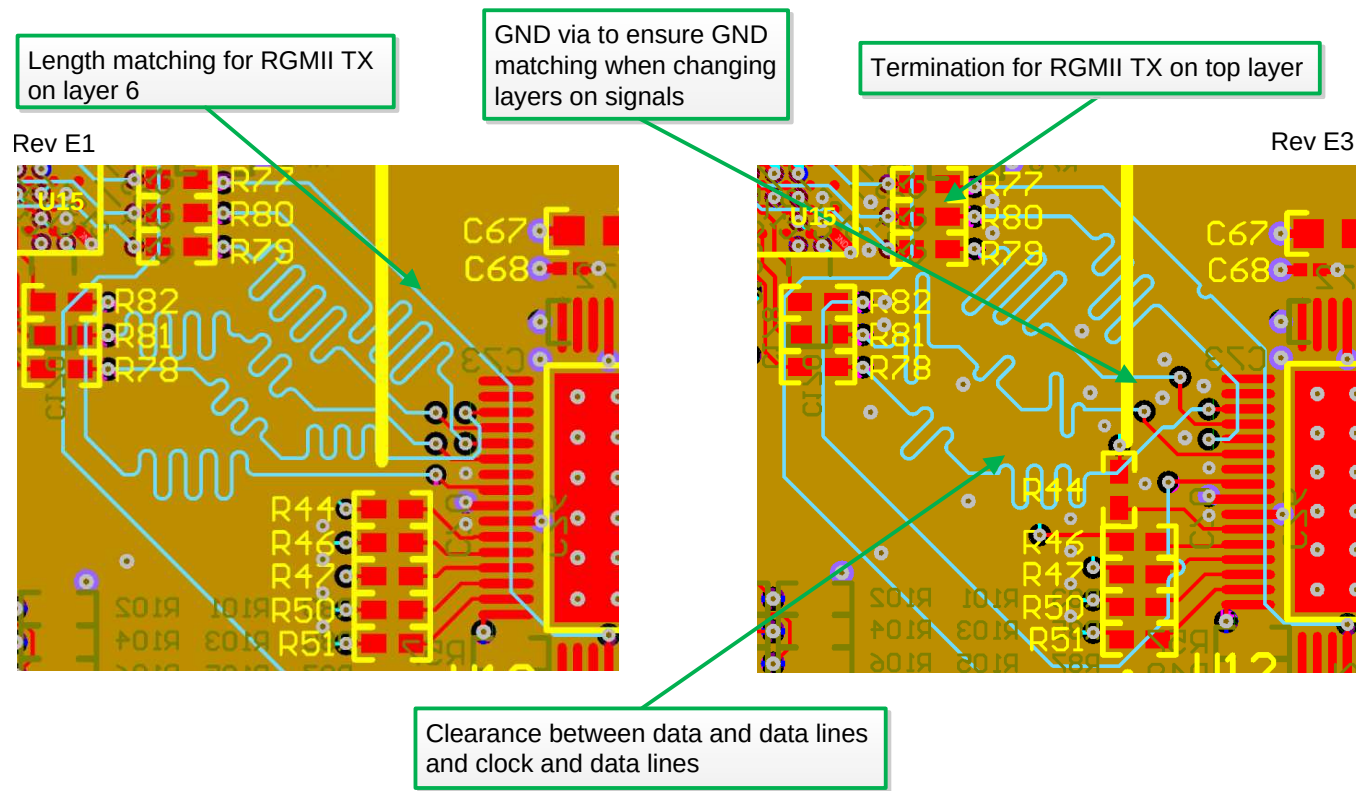


Figure 72. Layout Guidelines for RGMII RX Signals Comparing E1 and E3

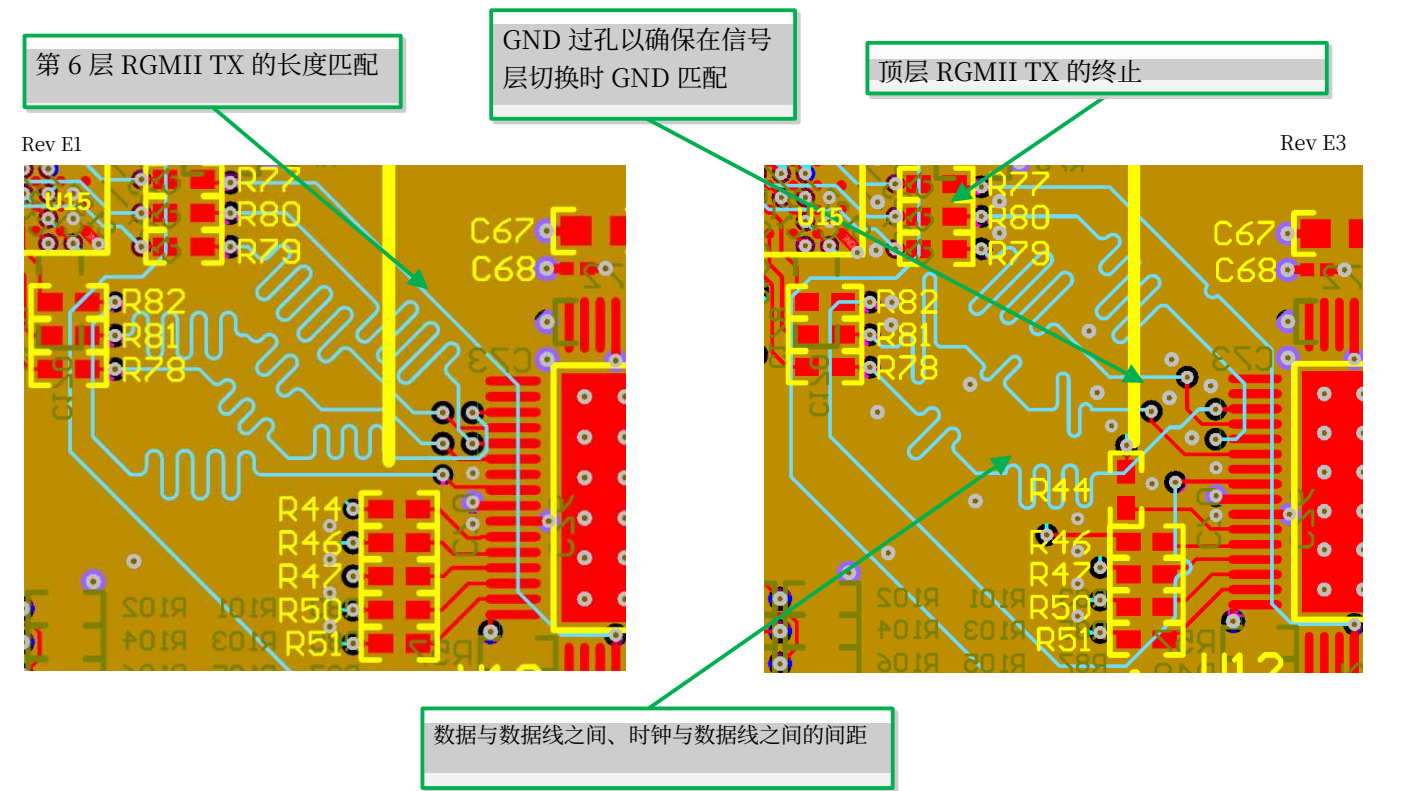


图 72. RGMII RX 信号布局指南：比较 E1 和 E3

6.4.3 Layout for Differential Gigabit Ethernet Signals

From the PHY to the magnetics the differential pairs are routed on the top layer with differential impedance control matching of $100\ \Omega$. The reference GND is on mid layer 1. This is achieved by ensuring that isolation between top layer and mid layer 1 is smaller than the surrounding planes.

From the magnetics to the RJ45 the differential pairs A, C, and D are routed on the top layer while pair B is routed on mid layer 2. This is due to crossing the signals on the RJ45 Ethernet jack. Again, the differential pairs are impedance matched to $100\ \Omega$ with reference to GND on mid layer 1. Due to the GND being mid layer 1, there is no need for GND vias added around the B signal.

This routing achieves the minimum possible trace lengths from PHY to magnetics, magnetics to RJ45, and here a maximum delta of the trace lengths of 1 mm between the differential pairs.

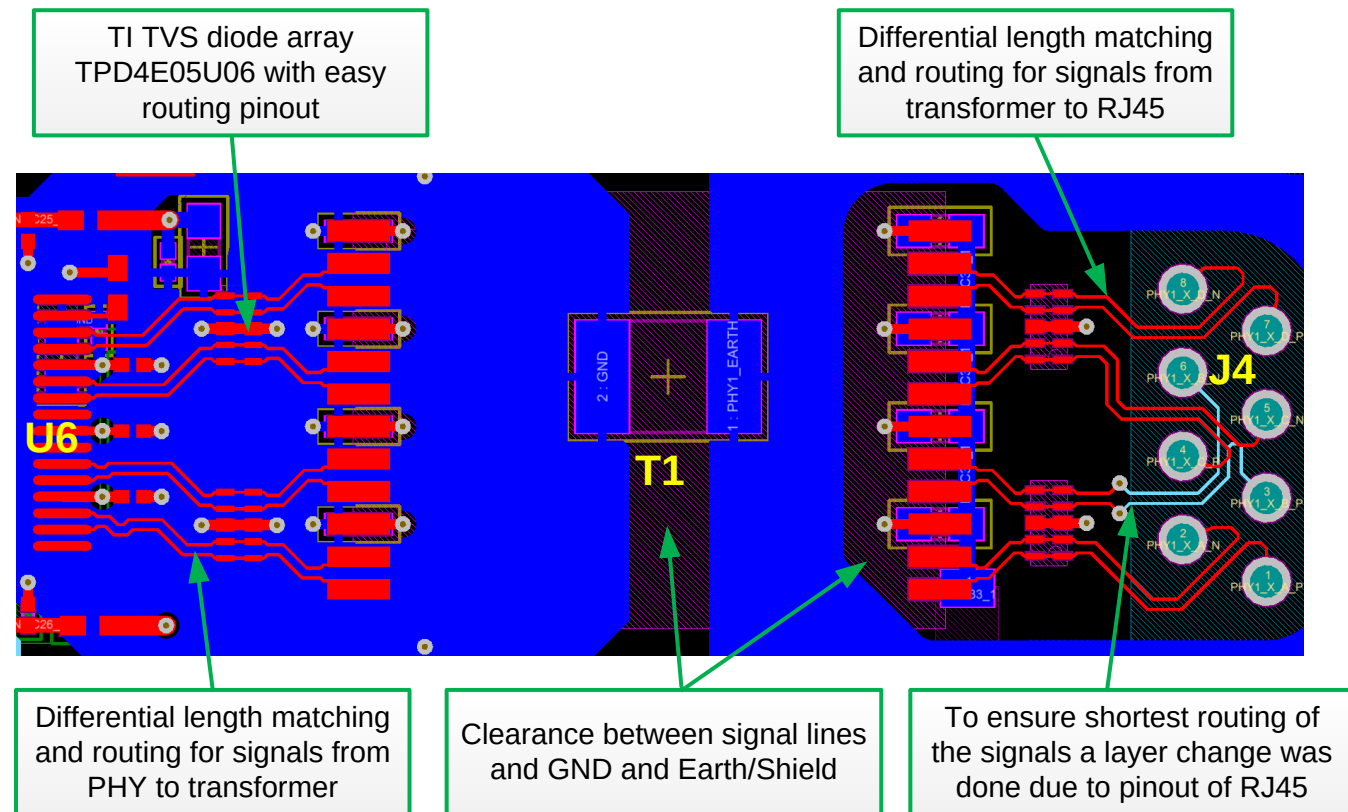


Figure 73. Layout Guidelines for PHY Differential Signals

6.4.3 差分千兆以太网信号布局

从 PHY 到磁性元件，差分对在顶层走线，差分阻抗控制匹配为 $100\ \Omega$ 。参考地 GND 位于中层 1。这是通过确保顶层与中层 1 之间的隔离小于周围平面来实现的。

从磁件到 RJ45，差分对 A、C 和 D 走顶层，而 B 对走第 2 中层。这是由于在 RJ45 以太网接口处信号交叉所致。同样，差分对相对于第 1 中层的 GND 阻抗匹配至 $100\ \Omega$ 。由于 GND 位于第 1 中层，因此无需在 B 信号周围添加 GND 过孔。

该布线方案实现了从 PHY 到磁性元件、从磁性元件到 RJ45 的最小可能走线长度，且差分对之间的走线长度最大偏差为 1 mm。

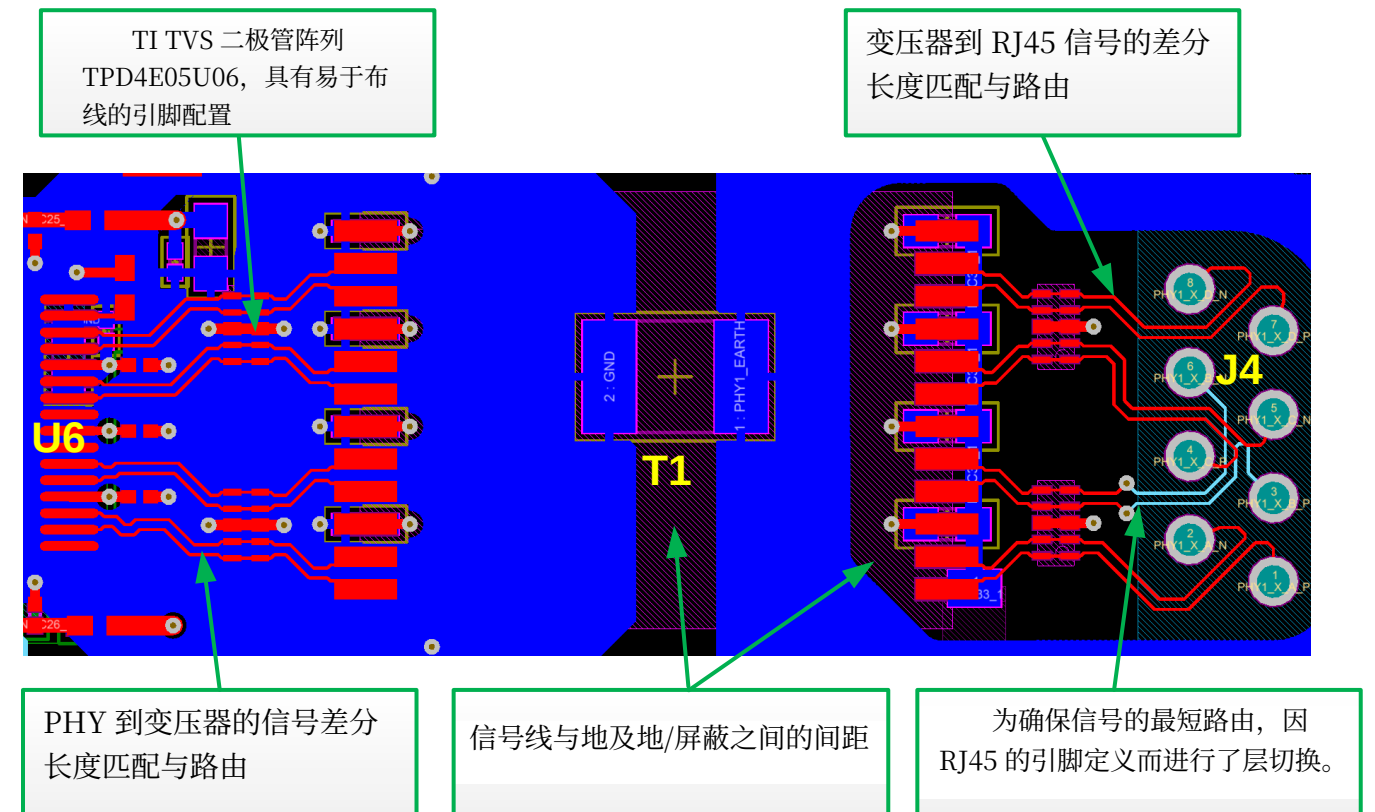


图 73。PHY 差分信号的布局指南

6.4.4 Layout PHY Power Supply

For the 2.5-V LMZ rail, add a keep-out GND cut in mid layer 1 to decrease noise coupling. To additional thermal management, add as many vias as possible on the powerPAD of the 1.1-V and 1.8-V LDO and add an additional thermal release plane on the bottom side of the board.

Ensure wide enough traces when adding the 1.1-V and 2.5-V power rails. The 1.1-V rail is added as a solid supply plane below each PHY. Figure 74 shows the PCB layout of TIDA-00204.

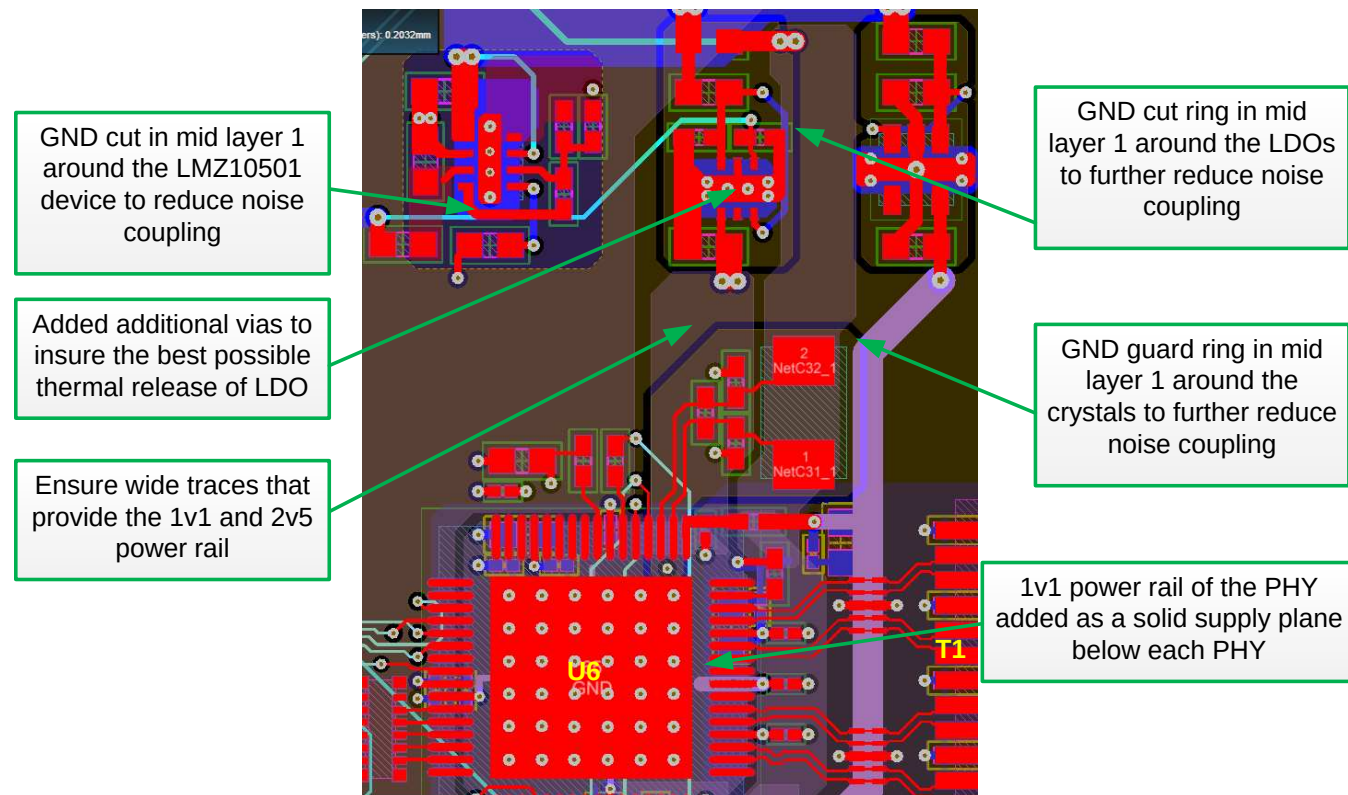


Figure 74. Layout Guidelines for PHY Power Supply

6.4.4 布局电源供应器

对于 2.5-V LMZ 轨，请在中间层 1 添加 GNDcut 以保持隔离，以降低噪声耦合。为增强热管理，请在 1.1-V 和 1.8-V LDO 的 powerPAD 上尽可能多地添加过孔，并在电路板底部添加额外的散热释放层。

在添加 1.1-V 和 2.5-V 电源轨时，请确保走线足够宽。1.1-V 电源轨作为每个 PHY 下方的实心供电平面添加。图 74 展示了 TIDA-00204 的 PCB 布局。

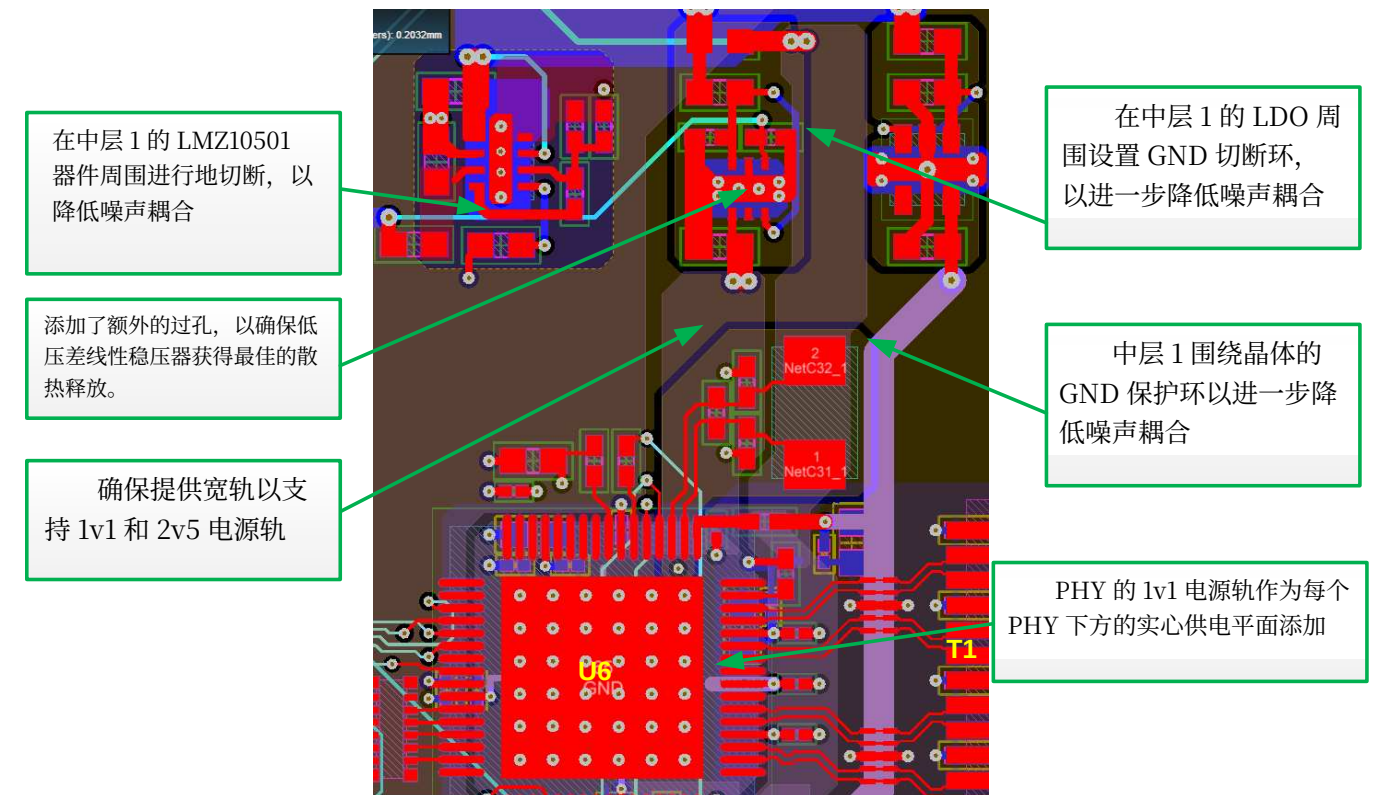


图 74。PHY 电源供应器布局指南

6.4.5 Layout of the 24-V to 5-V DC/DC Buck (LM46002)

Follow the layout guidelines from the LM46002 data sheet. Cut out layer 1 under LM46002 to avoid coupling into other layers. Ensure that current loop of the device is maintained as small as possible. The example layout of the LM46002 on TIDA-00204 is shown in Figure 75.

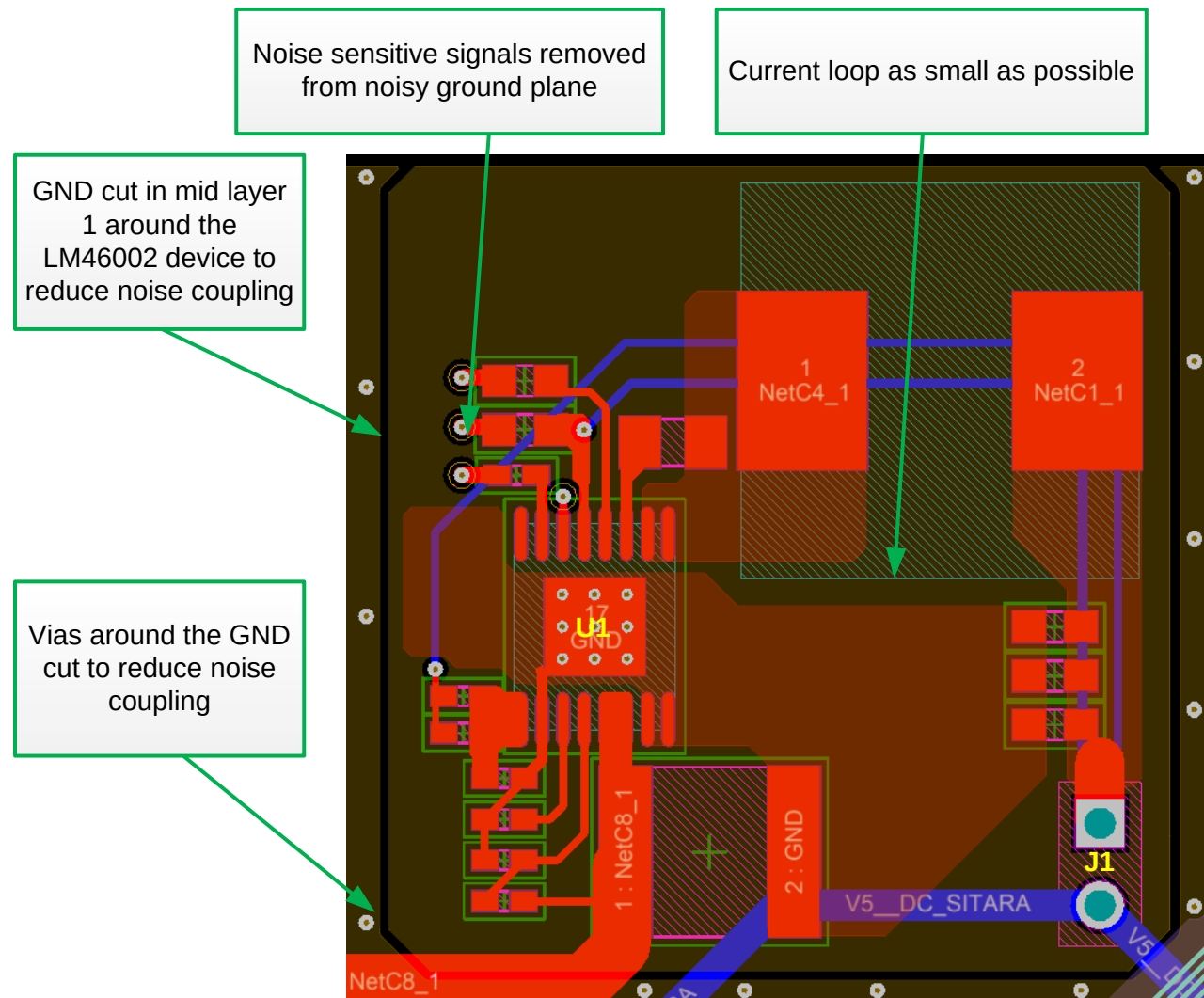


Figure 75. Layout Guidelines for 24-V to 5-V DC/DC Buck With LM46002

6.4.6 Layout Sitara Power Supply

For the Sitara power supply, the following guidelines were considered.

Routing the power tracks to the Sitara each power track as a GND plane as reference GND, mid layer 4 used as supply layer with supply planes for each high current power track. VMMC, VAUX33, and VAUX1 power track added in mid layer 2 and VDIG2 and VPLL power track added in mid layer 6.

VDAC intentionally does a u-trace as the DAC functionality is not used. If this functionality is needed, add additional traces for this power rail.

6.4.7 Layout DDR3

Layer 6 is GND reference for DDR3 (90-μm dielectric distance layer to layer). The other layers are 400 μm away. All DDR3 signals are on one layer.

6.4.5 24-V 至 5-V DC/DC 降压 (LM46002) 布局

请遵循 LM46002 数据表中的布局指南。在 LM46002 下方切割第一层，以避免耦合到其他层。确保器件的电流回路尽可能小。TIDA-00204 上 LM46002 的示例布局如图 75 所示。图 75。

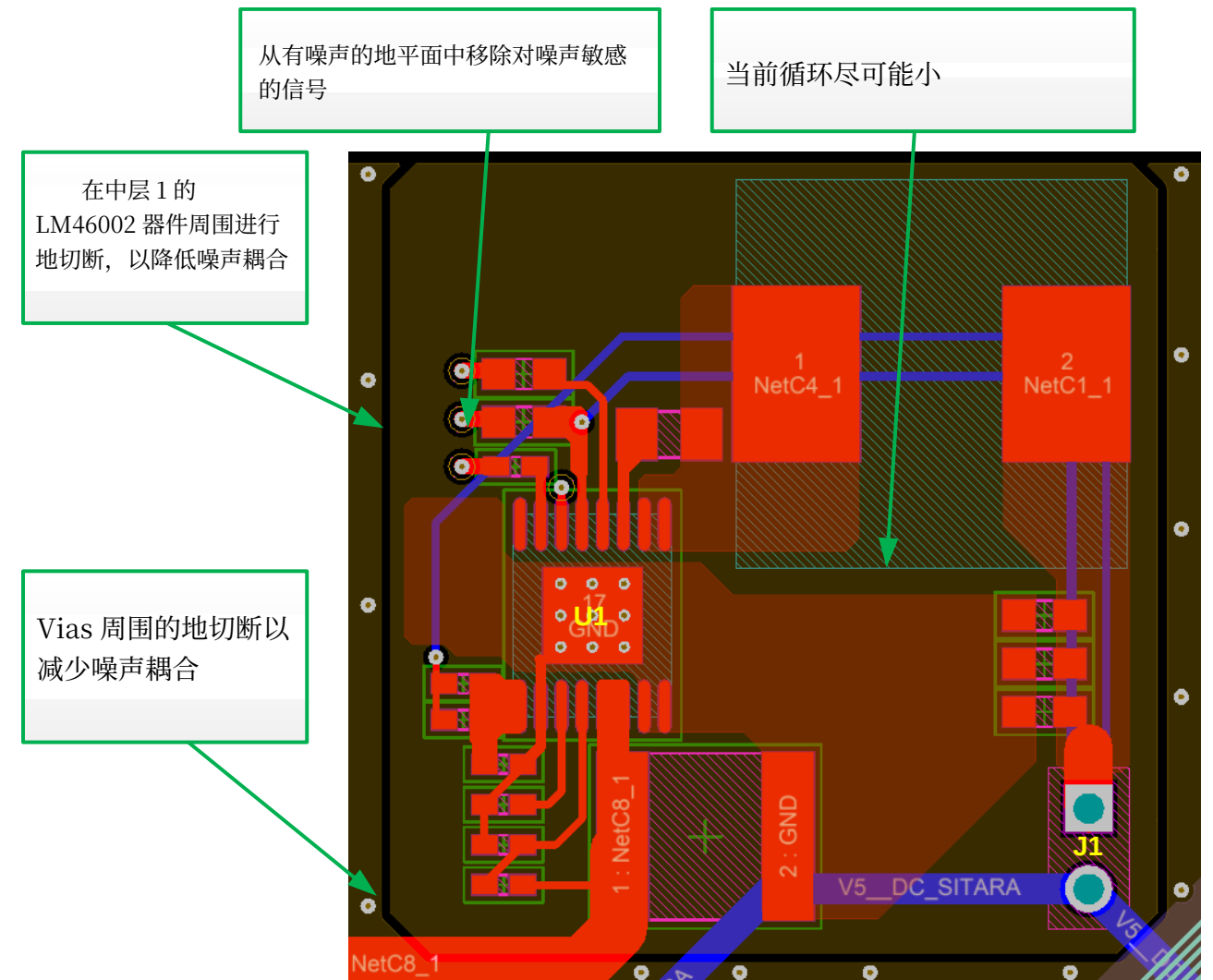


图 75. LM46002 的 24 V 至 5 V DC/DC 降压转换器布局指南

6.4.6 布局 Sitara 电源供应器

针对 Sitara 电源供应器，考虑了以下指南。

将电源走线路由至 Sitara，每条电源走线均作为 GND 平面作为参考 GND，中层 4 用作电源层，并为每条大电流电源走线提供电源平面。VMMC、VAUX33 和 VAUX1 电源走线添加至中层 2，VDIG2 和 VPLL 电源走线添加至中层 6。

VDAC 故意不执行 u-trace，因为 DAC 功能未被使用。如果需要此功能，请为该电源轨添加额外的走线。

6.4.7 布局 DDR3

第 6 层是 DDR3 的 GND 参考（90 微米介电距离层到层）。其他层距离为 400 微米。所有 DDR3 信号均位于同一层。

6.5 Altium Project

To download the Altium project files, see the design files at [TIDA-00204](#).

6.6 Gerber Files

To download the Gerber files, see the design files at [TIDA-00204](#).

7 Software Files

To download the software files, see the design files at [TIDA-00204](#).

8 Related Documentation

1. Texas Instruments, [DP83867IR Robust, Low Power 10/100/1000 Ethernet Physical Layer Transceiver](#), DP83867 Data Sheet (SNLS484)
2. Texas Instruments, [AN-1469 PHYTER Design & Layout Guide](#), Application Report (SNLA079)
3. Texas Instruments, [EN55011-Compliant, Industrial Temperature 10/100-Mbps Ethernet PHY Brick](#), TI Design (TIDU515)
4. Texas Instruments, [PCB Design Guidelines For Reduced EMI](#), Application Report (SZZA009)
5. Texas Instruments, [High-Speed Interface Layout Guidelines](#), Application Report (SPRAAR7)
6. Texas Instruments, [AM335x Sitara™ Processors](#), AM335x Data Sheet (SPRS717)
7. Texas Instruments, [AM335x Sitara™ Processors](#), AM335x Technical Reference Manual (SPRUH73)
8. Texas Instruments, [TPS65910Ax User's Guide For AM335x Processors](#), TPS65910Ax User's Guide (SWCU093)
9. Texas Instruments, [TI Network Developer's Kit \(NDK\) v2.24](#), NDK User's Guide (SPRU523)
10. Texas Instruments, [Sitara Wiki Page](#) (<http://processors.wiki.ti.com/index.php/Sitara>)
11. Texas Instruments, [AM3359 Industrial Communications Engine](#), Evaluation Module, Part # TMD5ICE3359 (<http://www.ti.com/tool/tmdsice3359>)
12. Texas Instruments, [LM46002 SIMPLE SWITCHER® 3.5 V to 60 V 2 A Synchronous Step-Down Voltage Converter](#), LM46002 Data Sheet (SNVSA13)
13. IEEE, [802.3-2012 – IEEE Standard for Ethernet](#), Section Three (<http://standards.ieee.org>)
14. HP, [Reduced Gigabit Media Independent Interface \(RGMI\) 4/1/2002 Version 2.0: Reduced Pin-count Interface for Gigabit Ethernet Physical Layer Devices](#) (http://www.hp.com/rnd/pdfs/RGMIIv2_0_final_hp.pdf)
15. Dirk S. Mohl, [IEEE 1588 Precision Time Protocol](#) (<http://www.ieee1588.com>)
16. CISPR11 / EN 55011+A1:2010-01-31, *EN 55011: Industrial, scientific and medical equipment - Radio-frequency disturbance characteristics - Limits and methods of measurement*
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18. IEC, *Amendment 1 - Adjustable speed electrical power drive systems - Part 3: EMC requirements and specific test methods*, IEC 61800-3-am1 ed2.0: 2011-11-14
19. Micron, *TN-41-08: Design Guide for Two DDR3-1066 UDIMM Systems*
20. [Wireshark](#), v1.12.3 (<http://www.wireshark.org>)
21. [TeraTerm](#), v4.76 (<http://ttssh2.sourceforge.jp/>)
22. [GCD-Printlayout](#) GmbH (<http://www.gcd-printlayout.de/>)
23. MinGW, [Minimalist GNU for Windows](#) (<http://www.mingw.org/>)
24. SourceForge, [Win32 Disk Imager](#) (<http://sourceforge.net/projects/win32diskimager/>)

6.5 Altium 项目

要下载 Altium 项目文件，请参阅设计文件[TIDA-00204](#)。

6.6 Gerber 文件

要下载 Gerber 文件，请查看 [TIDA-00204](#)。

7 软件文件

要下载软件文件，请查看[TIDA-00204](#)。

8 相关文章

1. 德州仪器, [DP83867IR 鲁棒、低功耗 10/100/1000 以太网物理层收发器](#), DP83867 数据表 (SNLS484)
2. 德州仪器, [AN-1469 PHYTER 设计与布局指南](#), 应用报告 (SNLA079)
3. 德州仪器, [EN55011 合规, 工业温度 10/100-Mbps 以太网 PHY 砖](#), TI 设计 (TIDU515)
4. 德州仪器, [降低 EMI 的 PCB 设计规范](#), 应用报告 (SZZA009)
5. 德州仪器, [高速接口布局指南](#), 应用报告 (SPRAAR7)
6. 德州仪器, [AM335x Sitara™ 处理器](#), AM335x 数据手册 (SPRS717)
7. 德州仪器, [AM335x Sitara™ 处理器](#), AM335x 技术参考手册 (SPRUH73)
8. 德州仪器, [TPS65910Ax 用户指南 \(适用于 AM335x 处理器\)](#), TPS65910Ax 用户指南 (SWCU093)
9. 德州仪器, [TI 网络开发人员套件 \(NDK\) v2.24](#), NDK 用户指南 (SPRU523)
10. 德州仪器, [Sitara Wiki 页面](#)(<http://processors.wiki.ti.com/index.php/Sitara>)
11. 德州仪器, [AM3359 工业通信引擎](#), 评估模块, 部件号 TMD5ICE3359 (<http://www.ti.com/tool/tmdsice3359>)
12. 德州仪器, [LM46002 SIMPLE SWITCHER®3.5 V 至 60 V 2 A 同步降压转换器](#), LM46002 数据表 (SNVSA13)
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14. HP, [千兆以太网物理层设备减少引脚数接口 \(RGMI\) 4/1/2002 版本 2.0: 千兆以太网物理层设备减少引脚数接口](#)(http://www.hp.com/rnd/pdfs/RGMIIv2_0_final_hp.pdf)
15. Dirk S. Mohl, [IEEE 1588 高精度时间协议](#)(<http://www.ieee1588.com>)
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17. IEC, 可调速电气驱动系统 - 第 3 部分: 电磁兼容要求和特定测试方法, IEC 61800-3 ed2.0: 2004-08-12
18. IEC, 修正案 1 - 可调速电气驱动系统 - 第 3 部分: 电磁兼容性要求和特定测试方法, IEC 61800-3-am1 ed2.0: 2011-11-14
19. Micron, TN-41-08: DDR3-1066 UDIMM 双通道系统设计指南
20. [Wireshark](#), v1.12.3 (<http://www.wireshark.org>)
21. [TeraTerm](#), v4.76 (<http://ttssh2.sourceforge.jp/>)
22. [GCD-Print 布局](#)GmbH (<http://www.gcd-printlayout.de/>)
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24. SourceForge, [Win32 磁盘映像器](#)(<http://sourceforge.net/projects/win32diskimager/>)

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9 About the Authors

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Revision History	
NOTE: Page numbers for previous revisions may differ from page numbers in the current version.	
Changes from A Revision (April 2015) to B Revision	Page
• Added CDCE913 to Resources section	1
• Changed listed feature from "Hardware Support for Start of Frame (SOF) Detect Allows Implementation of IEEE 1588 PTP" to "Reduces Jitter and Phase Shift from Clocks by Using Clock Synthesizer to Generate System Clocks"	1
• Changed front-page board image to updated figure.....	1
• Changed front-page block diagram to updated figure	1
• Changed location of content from <i>TI Design Overview</i> section to System Description	2
• Changed title of <i>Design Features</i> section to Key System Specifications and moved under System Description	2
• Added new section hierarchy (System Overview) to house Section 2.2.1 , Section 2.2.2 , and Section 2.2.3	4
• Changed location of Block Diagram to under System Overview	4
• Changed Figure 1 to updated block diagram.....	4
• Changed location of Section 2.2.1 , Section 2.2.2 , and Section 2.2.3 to under Design Considerations	4
• Added new section System Design Theory	7
• Changed location of Circuit Design and Component Selection to under System Design Theory	7
• Changed number of MAC interface symbols for GMII from 24 to 25	10
• Changed Table 7 to updated title <i>DP83867 RGMII Design Rules on TIDA-00204 PCB</i> and updated the listed rules and distances	10
• Changed Figure 7 to updated schematic	11
• Added detail on clock option to sync all clocks using a clock distribution network for DP83867IR Input Clock Selection	12
• Added detail on clock option to sync all clocks using a clock distribution network for AM3359 Clocking Options	17
• Changed Figure 17 to updated figure from <i>CDCE913 Clock Distribution Circuit Example</i> to <i>Crystal Phase Shift Diagram</i>	27
• Changed listed advantage of "Smaller component area used for clocking" to "High accurate clocking with minimum frequency and phase shift"	27
• Changed listed advantage of "Possibility to enable spread spectrum (SSC)" to "Smaller component area used for clocking".....	27
• Changed Figure 19 to updated image	28
• Added Figure 20	29
• Changed Figure 30 to updated image	37
• Added Figure 31	38
• Added details about using the ping command during ping testing	51
• Deleted all embedded images from Schematics . Schematics have been updated; download the most recent from TI.com	68
• Deleted BOM table from Bill of Materials . BOM has been updated; download the most recent version from TI.com	68
• Deleted all embedded images from Layer Plots . Layer plot files have been updated; download the most recent from TI.com	68
• Changed Figure 71 to updated image	69
• Changed Figure 72 to updated image	70
• Added reference for High-Speed Interface Layout Guidelines	74

Changes from Original (March 2015) to A Revision	Page
• Changed from preview page.....	1

修订历史	
注意： 先前修订版本的页码可能与当前版本的页码不同。	
从 A 修订版（2015 年 4 月）到 B 修订版的变更 页	
• 已添加 CDCE913 到 资源 部分. 1	
• 将 “硬件支持帧起始（SOF）检测，允许实现 IEEE 1588 PTP"改为 “通过使用时钟合成器生成系统时钟，从而减少时钟抖动和相位偏移".....	1
•	1 将首页板图像更改为更新后的图表.将首
•	1 页框图更改为更新后的图表.
•	2 将内容位置从` <i>TI Design Overview</i>
•	2 ` 部分移至` System Description ` 部分.将` <i>Design Features</i> ` 部分标题改为` Key System Specifications `，并将其移至` System Description ` 下方.....
• 新增了一个章节层级（ 系统概述 ）用于容纳 2.2.1 节， 2.2.2 节，以及 2.2.3 节	4
•	4 将 块图 的位置更改为 系统概述 下方.将 图
•	4 更改为更新的块图.
•	2 2 1 ， 2 2 2 ， 2 2 3
•	4 将 Section ..Section .. 和 Section .. 的
•	7 位置移至 Design Considerations 之下.新增章节 System Design Theory .
• 更改了 电路设计与组件选择 的位置，使其位于 系统设计理论 . 7	
• 将 GMII 的 MAC 接口符号数量从 24 改为 25. 10	
• 已更改 表 7 为更新后的标题 TIDA-00204 PCB 上的 DP83867 RGMII 设计规则 并更新了所列规则及间距. 10	
• 已更改 图 7 至更新后的电路图. 11	
• 添加了时钟选项的详细信息，以便使用时钟分配网络同步所有时钟 DP83867IR 输入时钟选择 12	
• 为 AM3359 添加时钟选项的详细信息，以便通过时钟分配网络同步所有时钟 AM3359 时钟选项	17
• 已更改 图 17 为更新后的图，源自 CDCE913 时钟分配电路示例 至 晶振相位偏移图. 27	
• 将 “用于时钟的更小组件面积” 的列出的优势更改为 “具有最小频率和相移的高精度时钟” . 27	
• 将"启用扩频（SSC）的可能性” 列为优势改为"时钟电路占用的组件面积更小".	27
•	19 28 将 Figure 更新为最新图片.
• 添加 图 20 . 29	
•	30
•	31 Changed Figure to updated
• 添加了关于在 Ping 测试期间使用 Ping 命令的详细信息. 51	38
• 已从 原理图中删除了所有嵌入的图像 。原理图已更新；请从 TI.com 下载最新版本. 68	
• 已从 中删除物料清单 物料清单 。物料清单已更新；请从 TI.com 下载最新版本.....	68
• 已从 图层图 中删除所有嵌入图像。图层图文件已更新；请从 TI.com 下载最新版本.	68
•	71 69 已将 图 更改为更新后的图像.
•	72 70 将 Figure 更改为更新后的图片.
• 为 High-Speed Interface Layout Guidelines 添加了参考.	74

从 2015 年 3 月原版到修订版 页码	
• 已从预览页更改. 1	

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